AWI–ESM3 (v3.4, CORE3 / TCO95) PI–control tuning campaign

`Tuning_test_06*` — progress report

Evaluation via `release_evaluation_tool2` (part2 / part8 / part4)

2026–06–30 (rev. **2026–08–09**)

Abstract

AWI–ESM3 v3.4 (OpenIFS 48r1 / FESOM2 / LPJ-GUESS, TCO95L91) is being tuned for a pre-industrial spin-up against two faults: a positive TOA and surface radiative imbalance, and a NH high-latitude boreal-summer cold bias severe enough to lose the boreal forest. An AMIP testbed (§11) runs the atmosphere alone so the vegetation–albedo feedback is open and the root cause can be hunted without the coupled runaway closing over it.

Where we are (2026–08–09). K1 + P5 is the carried-forward configuration: the sub-grid snow depletion, rebuilt on 36 492 Russian snow-course surveys after the `tanh` form was falsified outright, delivers Siberian JJA +1.149 K with *every* season inside its own threshold — DJF −0.060 against the `tanh`'s −2.783 — and beats the scheme-off baseline against 105 RIHMI stations (§36). The more aggressive P3 buys +1.341 but at −0.584 DJF and a 2.4× worse autumn cover. Coupled, 11B reproduces the `tanh` disaster on a controlled pair (−18.99 K of DJF soil) and the coupled branch is one run — 11D at SWEMIN = 15 with K1 — from being level with the atmosphere-only configuration (§38.1).

What is left. Three routes are closed by measurement rather than opinion: transpiration→cloud (§42.2), snow-melt albedo (§27.2), and land surface albedo, which buys at most one sixth of the universal bias (§30.3). Skin conductivity is falsified for both the winter route (§28.1) and the boreal organic layer (§30.1). **Round 22 relocated the two remaining targets** (§38): the Southern Ocean is confirmed as the largest regional term and grows period-cleanly to +7.46 W m^{-2}, but two thirds of it is missing cloud *area* that no lever — including the adopted INP branch — has ever moved; and the **global tropospheric cold bias** is now located, at the *extratropical tropopause* rather than in tropical convection, where the atmosphere is both 1.6 K too cold and 10 % too dry, two large errors that cancel in OLR and so cannot be fixed one at a time.

Read §42 for the causal web and what is still dark; §25 for what the observations can and cannot settle; §44 for every mechanism this campaign has killed, with the measurement that killed it. Parts I–III are the coupled work that preceded the AMIP testbed; Part IV is the atmosphere-only campaign in the order it happened.

Front-matter note — new-sea-ice branch `Tuning_test_09A/09B/09C`

The dated account of this branch is §12 in the timeline; what follows is the run-identity reference only.

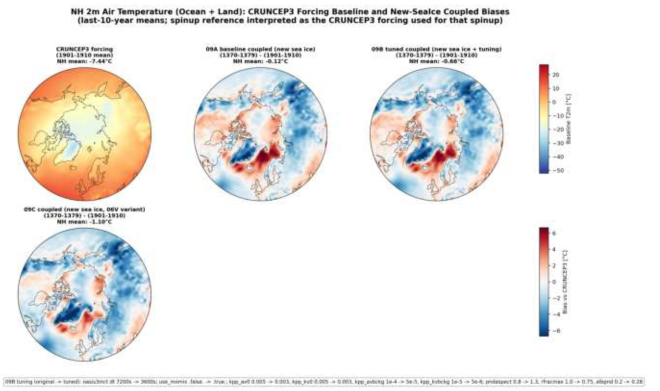
This report now includes the new three-way coupled comparison branch: `09A` (baseline + new sea ice), `09B` (new sea ice + tuned stack), and `09C` (new sea ice, 06V variant). All three completed the same 30-year segment and are compared on the same last-10-year window (model years 1370–1379).

Run identities

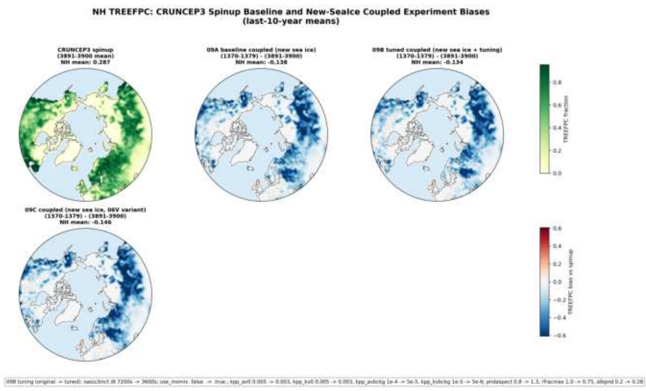
- `09A`: `Tuning_test_09A_lpjguess_Baseline_coupled_fromCRUNCEP_newSeaIce`
- `09B`: `Tuning_test_09B_06T_1hCPL_MOSPP_KPPLOW_CRUNCEPinit_newSeaIce`
- `09C`: `Tuning_test_09C_06V_CRUNCEPinit_newSeaIce`

09B tuning block documented on the figures (original → tuned)

- `oasis3mct.time_step`: 7200 s → 3600 s
- `use_momix`: `.false.` → `.true.`
- `kpp_av0`: 0.005 → 0.003, `kpp_kv0`: 0.005 → 0.003
- `kpp_avbckg`: 1e-4 → 5e-5, `kpp_kvbckg`: 1e-5 → 5e-6
- `pndaspect`: 0.8 → 1.3, `rfracmax`: 1.0 → 0.75, `albpnd`: 0.2 → 0.28



(a) NH T2m bias vs CRUNCEP3 spinup reference (ocean + land).



(b) NH TREEFPC bias vs CRUNCEP3 spinup reference.

Figure 1: New-sea-ice branch update: common last-10-year means (1370–1379) for `09A/09B/09C`.

Part I

Tuning away the high TOA imbalance

We set out to cut the AWI–ESM3 PI-control's $+1.5$ *W/m*^{2} *TOA imbalance. Every effective lever does it by* ***cooling*** *the model, and the aggressive coupling/ocean-mixing levers cut the imbalance hardest — but push the planet too cold against ERA5. This part documents that trade-off across 22 runs and traces the aggressive lever to a Southern-Ocean ocean-convection mechanism.*

1 Goal & setup

Reduce the net top-of-atmosphere (TOA) radiative imbalance of the AWI–ESM3 pre-industrial control so it equilibrates faster, *without* degrading the CMIP6-normalised performance index (CMPI, `part4`) or inflating the 2 m-temperature RMSE vs ERA5 (`part8`). 22 experiments, each a 30-year segment (model years 1350–1379) of the CORE3/TCO95 configuration (FESOM2 CORE3 mesh, OpenIFS TCO95/A096, LPJ-GUESS), all branched from one **Baseline**. The baseline PI-control still drifts with a positive TOA imbalance of $\approx +1.5\,\mathrm{W/m^2}$ (energy *into* the system — the signature of unequilibrated spin-up and/or ongoing ocean heat uptake). **A caution on the target.** The tuning levers change the model's *equilibrium*, not its rate of approach, so "lower imbalance" is a useful *ranking* objective but not automatically "more equilibrated": a low-imbalance-but-cold state is not obviously better than a high-imbalance-but-unbiased one. This is why the campaign's real aim (§9) is a *warm-floor* state where the imbalance is cut by a *moderate* lever, not by cooling into a cold bias — i.e. we treat imbalance and 2 m-T bias as joint targets, not imbalance alone.

Structural note. Almost every non-baseline run also carries a *common melt-pond shift* (`albpnd` $0.2 \to 0.28$, `rfracmax` $1.0 \to 0.75$, `pndaspect` $0.8 \to 1.3$). Differences below are each run's *distinguishing* change on top of that.

Four parts. **(I)** evaluates the 22 imbalance-tuning runs and exposes the imbalance↔cold-bias trade-off, tracing the aggressive lever to a Southern-Ocean mechanism. **(II)** shows that the cold bias which makes that lever costly is largely *not* the tuning but a **structural** LPJ-GUESS boreal-forest collapse (§6) — a *counter-lever*: fixing it warms the NH high latitudes, buying the RMSE headroom to spend the aggressive global cooling on the imbalance. Feedback-free spin-ups then quantify the sensitivity to forcing and isolate the comparatively small contribution from replacing net radiation alone. **(III)** records the original warm-floor work plan. **(IV)** supersedes its ordering where newer evidence requires it: first fix the atmospheric forcing on AMIP, then re-spin LPJG and tune only the residual (§15, §16).

2 Headline: the imbalance ↔ cold-bias trade-off

Every lever that meaningfully lowers the TOA imbalance does so by *cooling* the model, which worsens the cold bias against ERA5. Two families separate cleanly: the sea-ice/albedo/melt-pond changes move gently down-left (small imbalance gain, small bias cost); the coupling + ocean-mixing changes cut the imbalance hard (30–57 %) but push the planet 1.7–2.1 K too cold.

The cold-bias axis is two things at once. It mixes (i) a *shared structural floor* — every run, including the baseline and the "clean" melt-pond runs, carries a strong Siberia/N. Canada cold bias that is *not* a tuning artefact but an LPJ-GUESS boreal-forest-cover bug (§6); and (ii) the *differential*, tuning-induced global cooling that the aggressive runs add on top. The structural floor is the same for all points and is independently fixable; the tuning only moves runs vertically relative to it. This is why the "warming lever" in §9 is the LPJG fix, not another atmospheric retune.

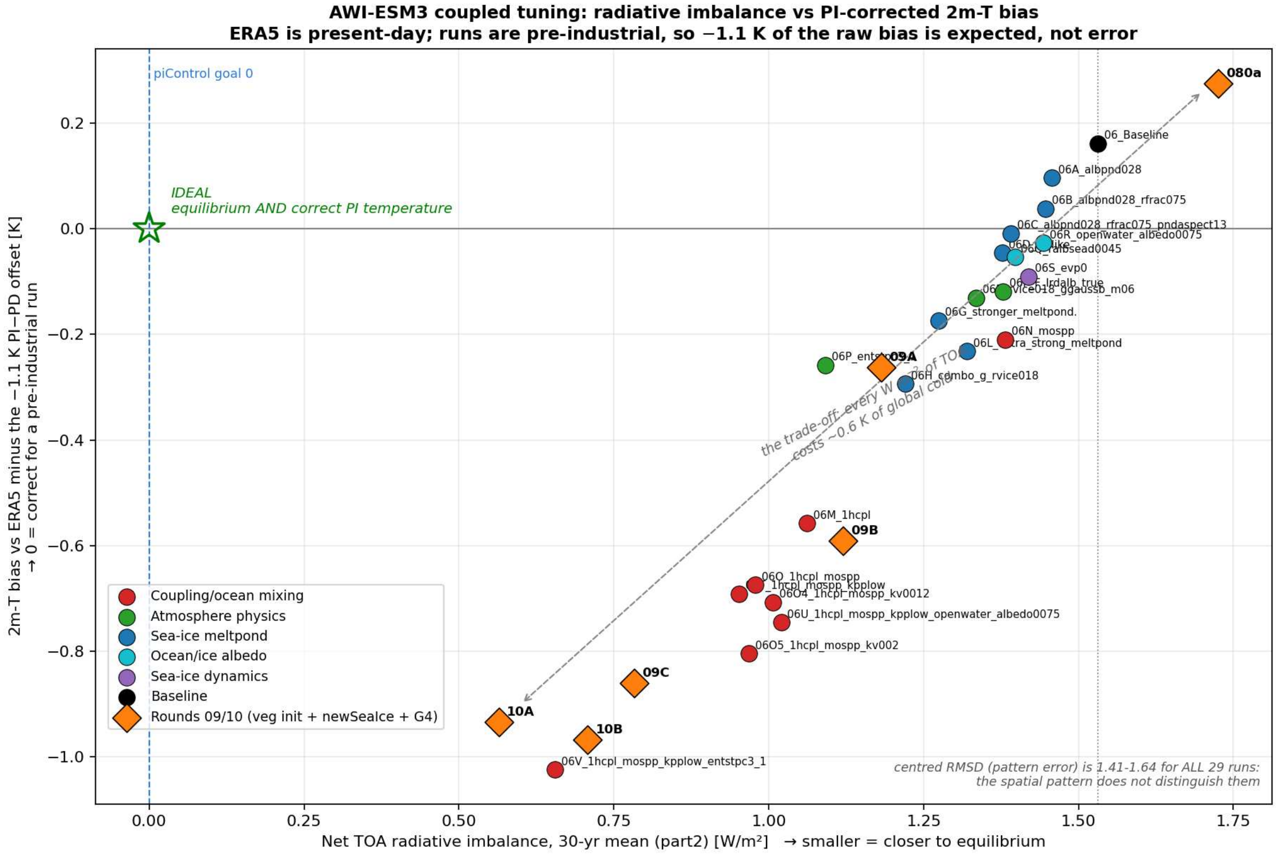


Figure 2: Net TOA imbalance (30-yr mean, `part2`) against the 2 m-T bias vs ERA5 *corrected for the reference period*, now including rounds 09 and 10 (orange diamonds, common window 1354–1379). **ERA5 is present-day and the runs are pre-industrial, about** 1.1 **K apart, so that much of the raw cold bias is expected rather than error**; the y axis subtracts it, and 0 is correct for a PI run. Ranking by raw RMSD penalises exactly the runs that are correctly cold, which is why the earlier version of this figure was misleading. Dashed blue = the corrected piControl goal of 0 (§37); the star marks the ideal, which nothing occupies. **The runs lie on a line, not in a cloud**: every $\mathrm{W\,m^{-2}}$ of TOA improvement costs ≈0.6 K of global cold, from 080a at (1.73, +0.28) to 10A at (0.57, −0.93). And the *pattern* error is not what distinguishes them — centred RMSD, $\sqrt{\mathrm{RMSD}^2 - \mathrm{bias}^2}$, which removes any uniform offset, is 1.41–1.64 K across all 29 runs, a ±8 % spread against a raw-RMSD spread of 64 %. Colour = parameter family.

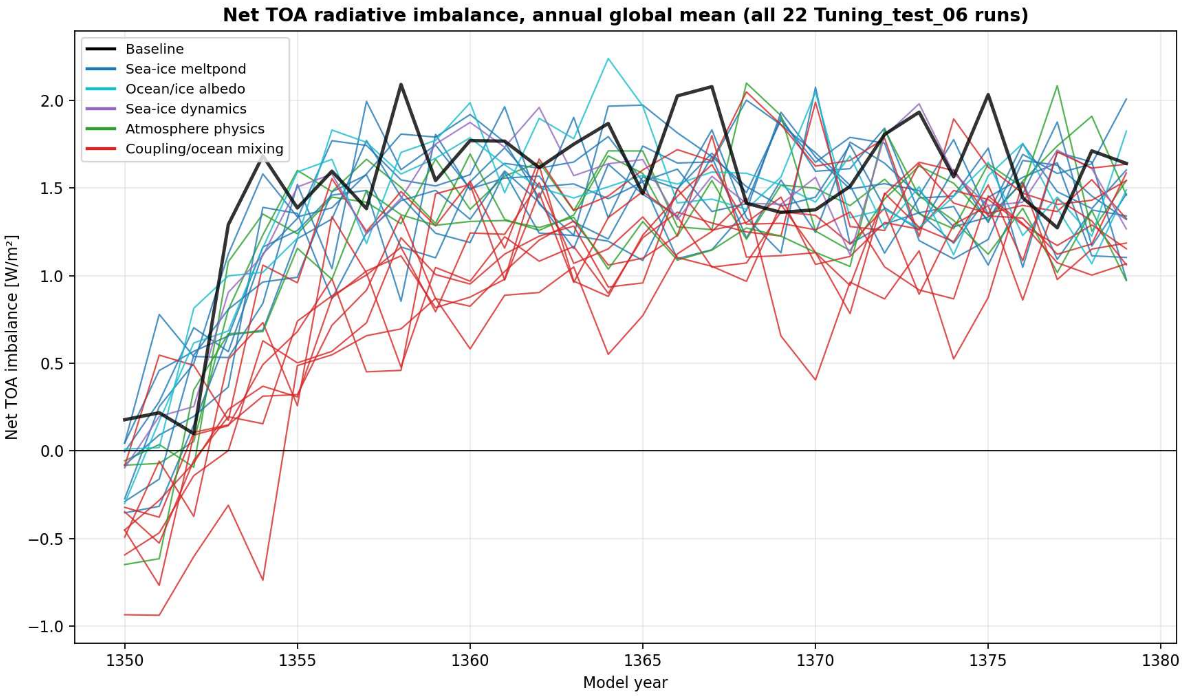


Figure 3: Annual global-mean net TOA imbalance, all 22 runs (independent CDO calculation; matches `part2`). None has flattened within 30 yr, so treat the mean level as a *ranking*, not a final equilibrium value.

3 Master metric table (coupled runs, sorted by TOA imbalance)

Run	TOA imb. [W/m^2]	vs base	2m RMSD [K]	2m bias [K]	CMPI
06V_1hcpl_mospp_kpplow_entstpc3_1	0.655	−57%	2.671	−2.124	0.803
06T_1hcpl_mospp_kpplow	0.952	−38%	2.385	−1.791	0.773
06O5_1hcpl_mospp_kv002	0.968	−37%	2.511	−1.904	0.788
06O_1hcpl_mospp	0.978	−36%	2.389	−1.774	0.772
06O4_1hcpl_mospp_kv0012	1.007	−34%	2.387	−1.807	0.781
06U_1hcpl_mospp_kpplow_openw0075	1.020	−33%	2.422	−1.845	0.776
06M_1hcpl	1.062	−31%	2.315	−1.658	0.773
06P_entstpc3_1	1.091	−29%	2.076	−1.358	0.759
06H_combo_g_rvice018	1.220	−20%	2.082	−1.394	0.757
06G_stronger_meltpond	1.274	−17%	1.990	−1.274	0.753
06L_extra_strong_meltpond	1.320	−14%	2.050	−1.332	0.751
06E_rvice018_ggaussb_m06	1.335	−13%	1.969	−1.231	0.754
06D_HRlike	1.377	−10%	1.892	−1.146	0.750
06F_lrdalb_true	1.378	−10%	1.962	−1.219	0.752
06N_mospp	1.382	−10%	2.011	−1.310	0.751
06C_albpnd028_rfrac075_pndaspect13	1.391	−9%	1.884	−1.109	0.751
06Q_ralbsead0045	1.397	−9%	1.891	−1.154	0.748
06S_evp0	1.419	−7%	1.935	−1.191	0.745
06R_openwater_albedo0075	1.443	−6%	1.860	−1.127	0.744
06B_albpnd028_rfrac075	1.447	−5%	1.847	−1.062	0.741
06A_albpnd028	1.457	−5%	1.811	−1.004	0.748
06_Baseline	**1.531**	–	**1.717**	−0.939	**0.745**
Rounds 09/10 (added 2026–08–07), same window 1354–1379.					
Bias column also given PI-corrected (raw +1.1 K); cRMSD = pattern error.					
10A (06V + G4)	**0.565**	−63%	2.522	−2.034 (−0.93)	cR 1.49
10B (10A + `tanh` depletion)	0.708	−54%	2.547	−2.067 (−0.97)	cR 1.49
09C (06V + newSeaIce)	0.783	−49%	2.475	−1.960 (−0.86)	cR 1.51
09B (06T + newSeaIce)	1.120	−27%	2.238	−1.691 (−0.59)	cR 1.47
09A (baseline + newSeaIce)	1.182	−23%	2.042	−1.363 (−0.26)	cR 1.52
080a (CRUNCEP veg init)	1.726	+13%	**1.629**	−0.825 (+0.28)	cR 1.41

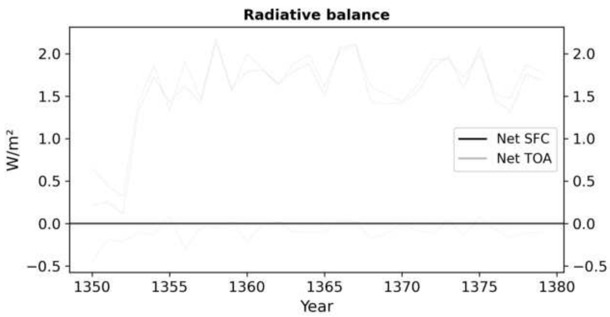
The 09/10 block is the campaign's central trade-off in one place, but read the bias column with the reference period in mind: ERA5 is present-day and these runs are pre-industrial, ≈1.1 K apart, so the PI-corrected value in brackets is the one that means anything. On that basis 10A is 0.93 K too cold where 06_Baseline is 0.16 K too warm — a real gap, but a third of what the raw numbers suggest. **And the spatial pattern is not what changes**: centred RMSD is 1.41–1.64 across all 29 runs, so the entire raw-RMSD spread is the global mean, counted twice. Nothing yet occupies the low-TOA, correct-temperature corner. Over their own longer window (1370–1399) 10A reads 0.802/2.565 and 10B 0.916/2.383, so the ranking is window-dependent and only the common window is tabulated. CMPI is left blank for these rows rather than guessed — it needs `part4_cmpi.py` and its own reference set. TOA = 30-yr global-mean net TOA from `part2_rad_balance.py`. RMSD/bias from `part8` methodology (cos-lat weighted, model−ERA5, yrs 1354–1379); cross-checked vs the tool's on-figure values (baseline 1.717 vs 1.721; 06V 2.671 vs 2.678). CMPI from `part4_cmpi.py`: < 1 = better than CMIP6 ensemble-mean error; lower is better.

The per-run distinguishing parameter change (which knob each `06*` run alters on top of the common melt-pond shift) is catalogued in App. A.

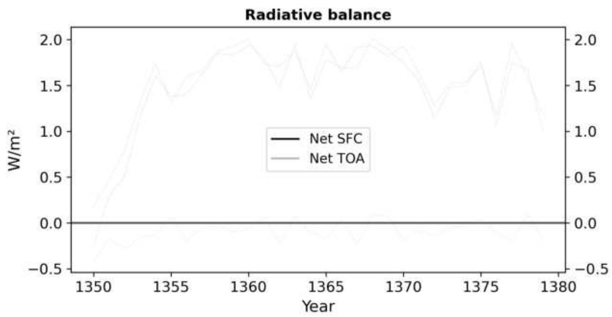
4 Core diagnostics: radiation, 2 m temperature, CMPI

4.1 Radiation budget (`part2`)

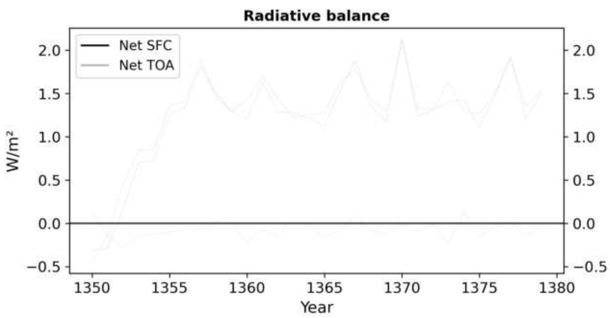
Surface net flux tracks TOA to within $\approx 0.1\,\mathrm{W/m^2}$ in every run — i.e. the atmospheric column stores essentially no energy on these multi-year means, as expected. Note this does *not* by itself localise the *cause*: a TOA $\approx$ surface match is equally consistent with a genuine TOA radiative error and with ocean heat uptake / unequilibrated spin-up drift passing cleanly through the surface (§5, §6.3). What it rules out is an atmosphere-only bookkeeping leak, so the imbalance and the surface energy budget can be read interchangeably. Baseline $\approx +1.53\,\mathrm{W/m^2}$; the coupling+mixing stack drives it down to $+0.66\,\mathrm{W/m^2}$ (06V).



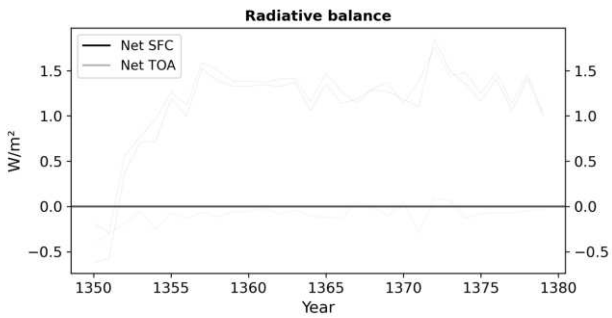
Baseline (TOA 1.53)



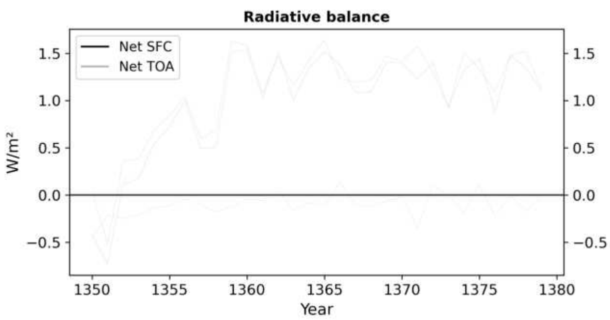
06A albpnd0.28 (1.46)



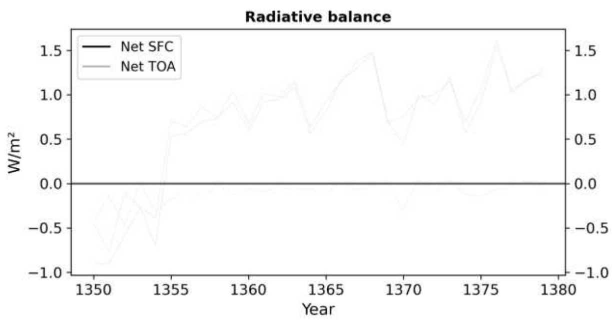
06H meltpond+rvice (1.22)



06P entstpc3 (1.09)



06O 1hcpl+mospp (0.98)



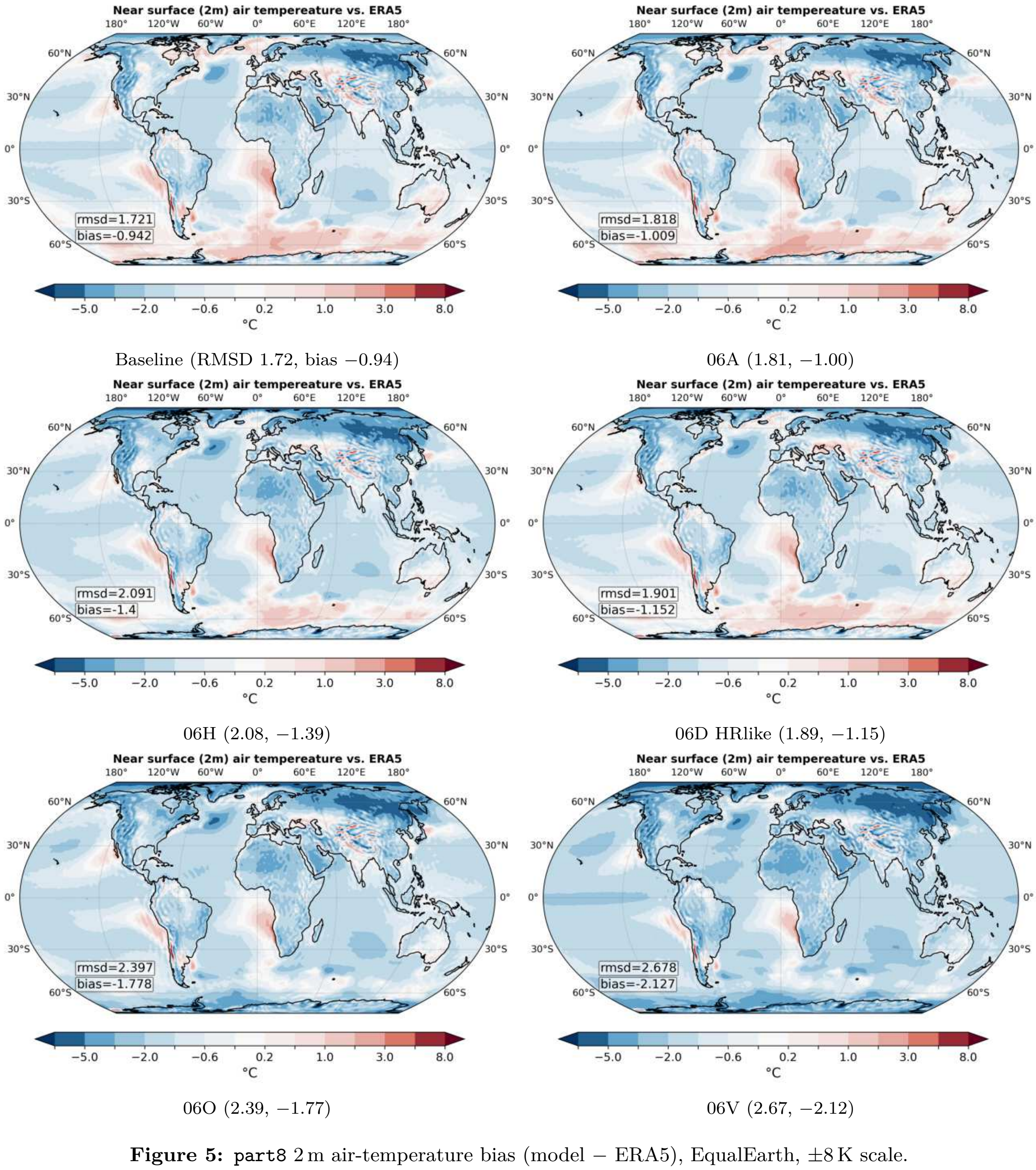
06V (most aggressive, 0.66)

Figure 4: `part2` radiation budget: grey = net TOA, blue = net surface, dark = their difference ($\mathrm{W/m^2}$).

4.2 2 m temperature bias vs ERA5 (`part8`)

Baseline: cold over NH continents, warm over the Southern Ocean and tropical upwelling. The aggressive coupling runs remove the Southern-Ocean warm bias but turn the whole planet cold (06V is cold almost everywhere). *Caveat:* this is PI-control vs present-day ERA5, so ≈ 0.5–0.7 K of global cold offset is physically expected; the coupling runs' -1.7 to -2.1 K is clearly too much, the melt-pond family (-1.0 to -1.4 K) is more defensible.

The deep **Siberia / N. Canada cold lobe is present in *every* panel** — baseline and the clean melt-pond runs included — and barely changes between them: it is the structural LPJ-GUESS forest-cover signal (§6), not a product of the imbalance tuning. The aggressive coupling runs add a broad, near-global cooling *on top* of that fixed lobe.



Baseline (RMSD 1.72, bias −0.94)

06A (1.81, −1.00)

06H (2.08, −1.39)

06D HRlike (1.89, −1.15)

06O (2.39, −1.77)

06V (2.67, −2.12)

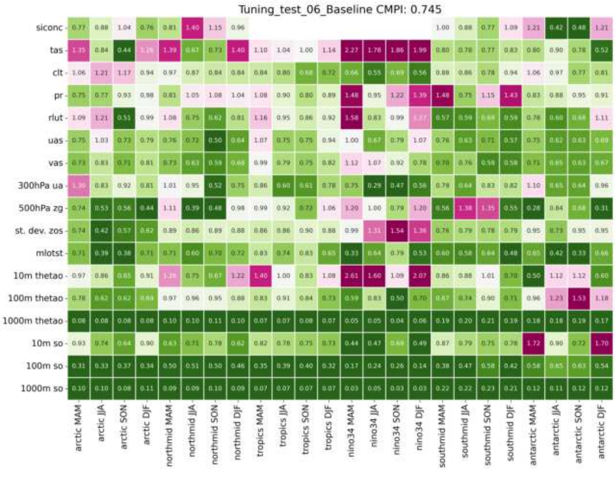
Figure 5: `part8` 2 m air-temperature bias (model − ERA5), EqualEarth, ±8 K scale.

4.3 CMPI (`part4`)

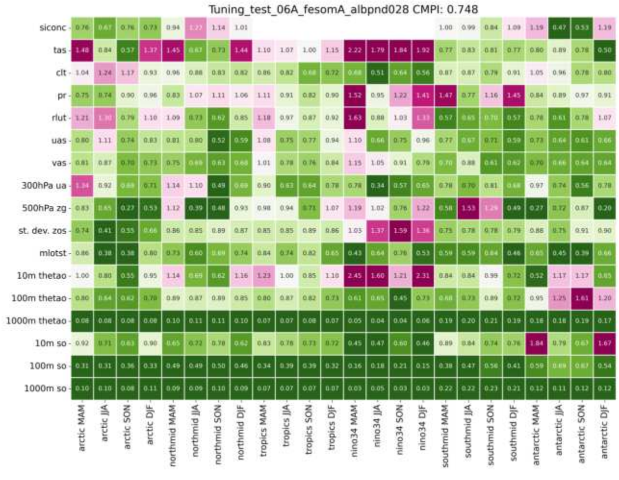
CMPI integrates 13 fields (siconc, tas, clt, pr, rlut, uas, vas, ua@300, zg@500, zos, mlotst, thetao, so) over regions/seasons, normalised against 30 CMIP6 models; < 1 = **better than the CMIP6 ensemble-mean error**. Run on a subset spanning the trade-off front (now complete for all 22 runs).

How to read this (important). CMPI here compares a *PI-control* run to *present-day* observations, so a moderate global cold offset — and hence some CMPI rise — is *expected* and is not by itself a disqualifier. The informative question is not "did CMPI go up?" but "*where* did the error change, and is that consistent with the parameter that was altered?" A sea-ice-albedo change should move `siconc` in the polar regions and little else; if it instead degrades the tropics, that is a red flag. The per-variable / per-region breakdown below is the tool for that judgement; the global number is a summary, not a verdict.

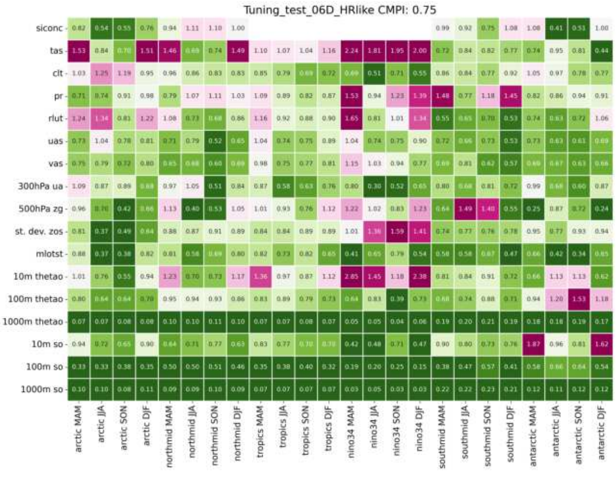
Run	CMPI	TOA imb.	2m RMSD
06_Baseline	0.745	1.531	1.717
06A_albpnd028	0.748	1.457	1.811
06D_HRlike	0.750	1.377	1.892
06H_combo_g_rvice018	0.757	1.220	2.082
06O_1hcpl_mospp	0.772	0.978	2.389
06T_1hcpl_mospp_kpplow	0.773	0.952	2.385
06V_.._entstpc3_1	0.803	0.655	2.671



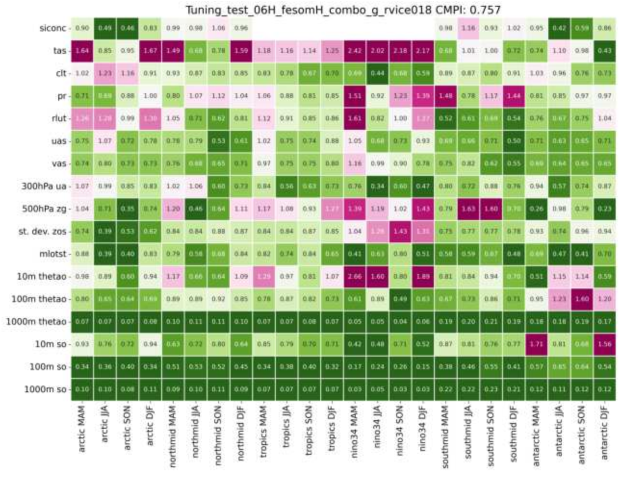
Baseline (CMPI 0.745)



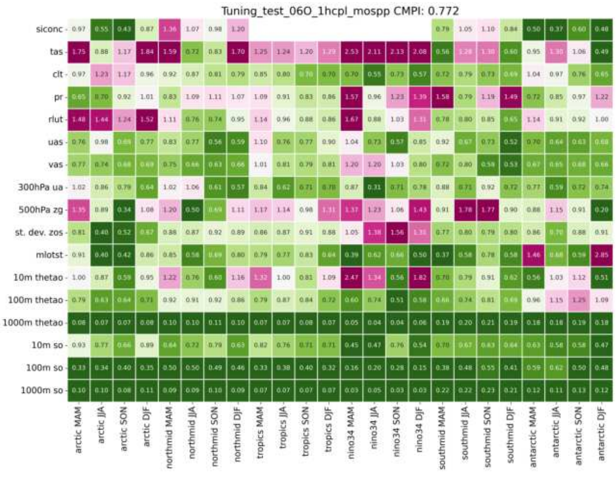
06A (CMPI 0.748)



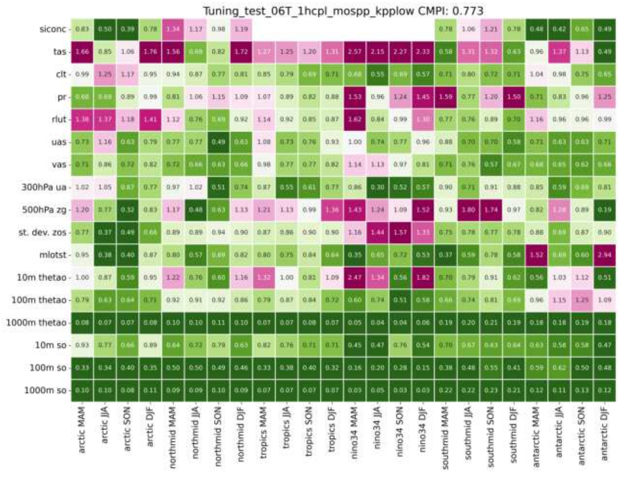
06D HRlike (CMPI 0.750)



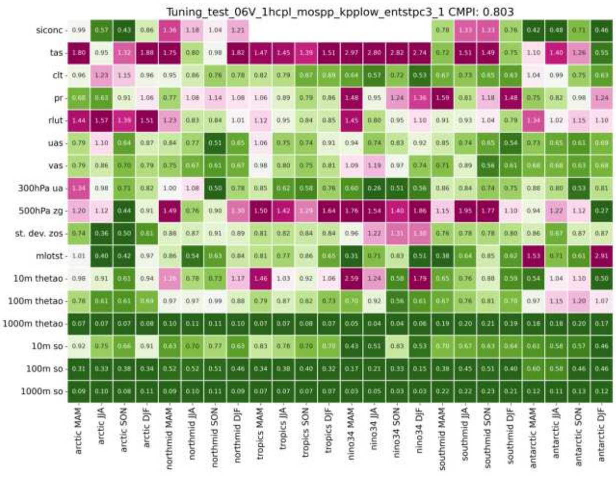
06H (CMPI 0.757)



06O (CMPI 0.772)



06T (CMPI 0.773)



06V (CMPI 0.803)

Figure 6: `part4` CMPI heat maps (per variable/region/season, normalised to CMIP6).

4.3.1 Where does the error change? (per-variable / per-region)

The maps below show each run's CMPI fractional error *minus baseline* (red = worse, blue = better). This is where the PI-vs-PD caveat earns its keep: the global CMPI rises monotonically (0.745 → 0.803), but the *pattern* of change tells a much more favourable story for the targeted tunings.

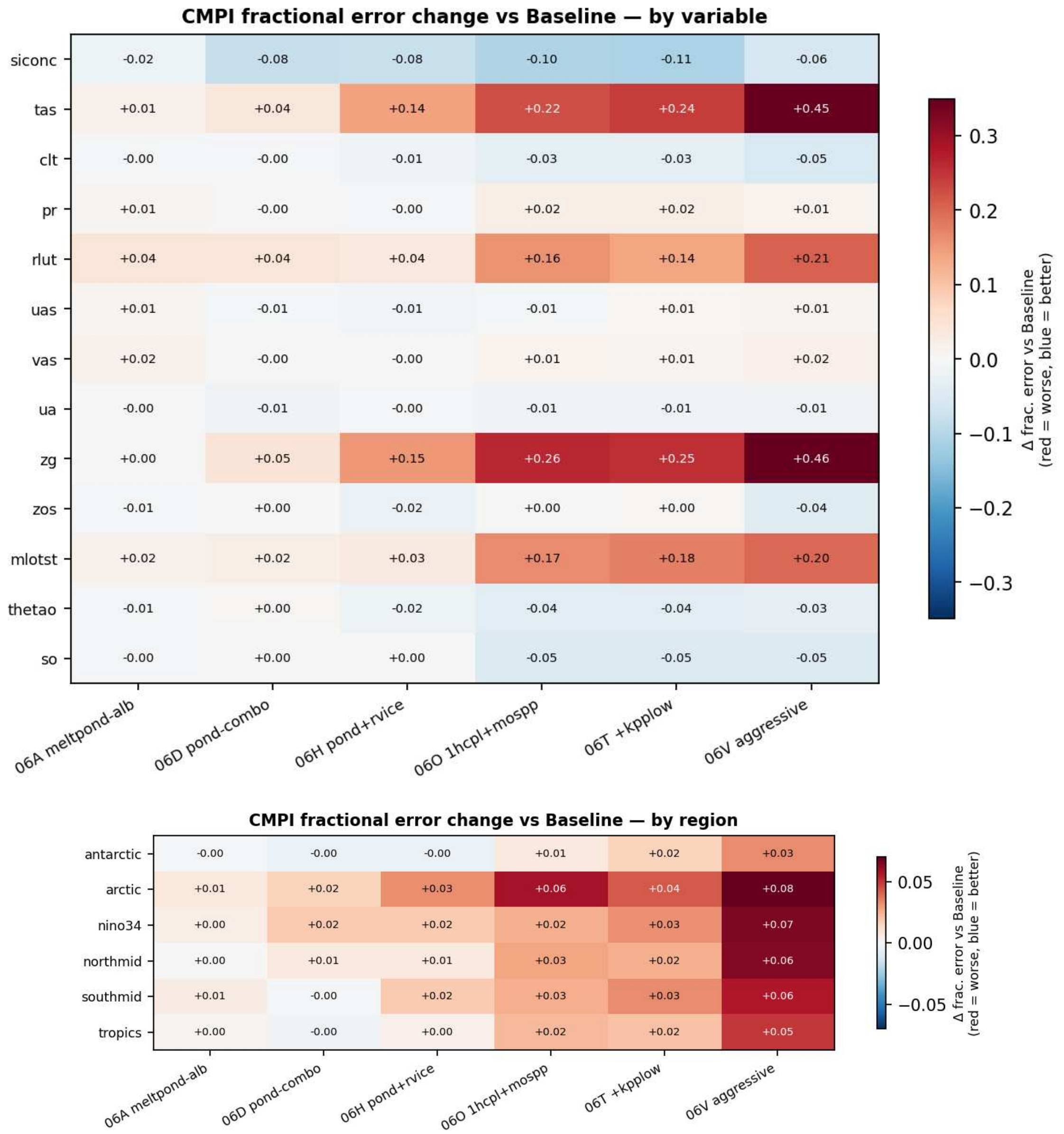


Figure 7: Change in CMPI fractional error vs Baseline, by variable (top) and region (bottom).

Reading the breakdown:

- **06A / 06D (melt-pond):** improve `siconc` (−0.02/−0.08) with essentially *zero* collateral elsewhere and a flat region map — a cleanly targeted change. These should be regarded as "no degradation where it matters," despite the tiny global CMPI uptick.
- **06O / 06T (mospp + kpplow):** genuinely *improve* the ocean interior (`thetao`, `so`: ≈ −0.04/−0.05), `siconc` and `clt` — exactly where ocean-mixing should act. But they *degrade* `tas`/`zg`/`rlut` (+0.16 to +0.26) and even their own target `mlotst` (+0.17), and the region map reddens *everywhere incl. tropics and Niño-3.4* — i.e. the cooling leaks well beyond where an ocean-mixing parameter has any business acting. *That* is the warning sign, more than the global number.
- **06V (+entstpc3):** largest, most widespread atmospheric degradation (`tas` +0.45, `zg` +0.46); the deep-convection change destabilises the whole atmospheric column.

5 Southern-Ocean mechanism (adapted from a270234's notebooks)

To see *why* the ocean-mixing/coupling levers act as they do, two of a270234's Weddell-Sea diagnostics (`mld_sic_sit_compare`, `temp_fesom_core3_WWS`) were re-pointed at the coupled `Tuning_test_06` runs (same CORE3 mesh). Panels show the Baseline absolute field plus each run *minus* Baseline, austral winter (September), yr1370–79. (The `temp_fesom_core3_WWS` tripyview transect could not decode these runs' pre-1582 `gregorian` calendar, so an equivalent pyfesom2 depth–latitude section was used.)

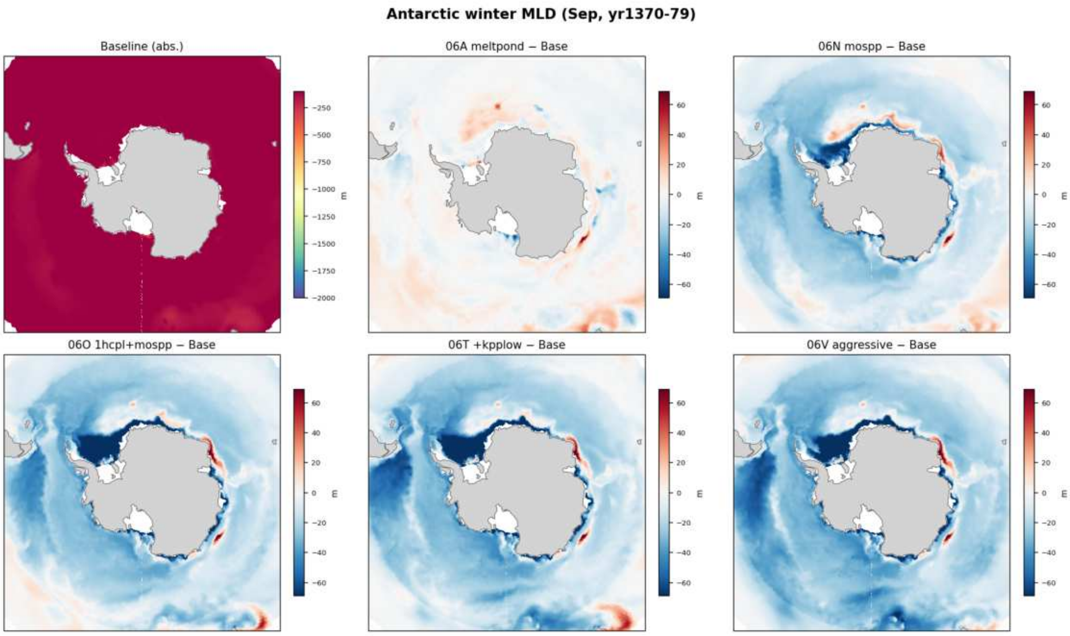


Figure 8: Antarctic September mixed-layer depth (MLD3): Baseline *absolute* (top-left, own scale), others = *anomaly* vs Baseline on a symmetric ±70 m diverging scale (98th-pctile auto-scale; the intense Weddell/Ross convection core saturates it). The mixing runs (06N/O/T/V) restructure the Weddell/Ross winter mixing relative to Baseline, whereas melt-pond (06A) barely changes it. Deep open-ocean winter mixing there is FESOM "open-ocean convection" that ventilates warm deep water — a known driver of the Southern-Ocean warm bias (Heuzé et al. 2013; Heuzé 2021) and of TOA imbalance.

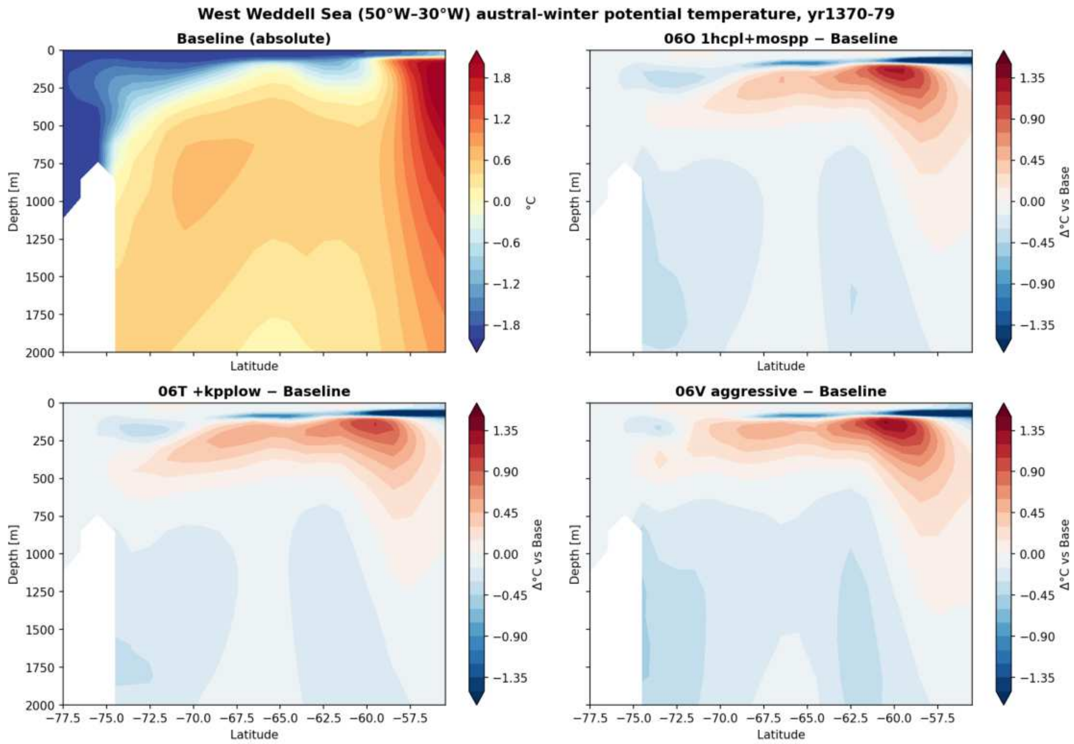


Figure 9: West Weddell Sea (50–30°W) September potential-temperature section: Baseline absolute and anomalies. The ventilated mid-column cools in the mixing runs (less convective entrainment of warm deep water to the surface), consistent with suppressed open-ocean convection; note that suppressing convection should leave the *deepest* water trapped/warmer rather than colder, and a near-surface edge warm anomaly is also present — i.e. the signal is a redistribution between layers, not uniform subsurface cooling.

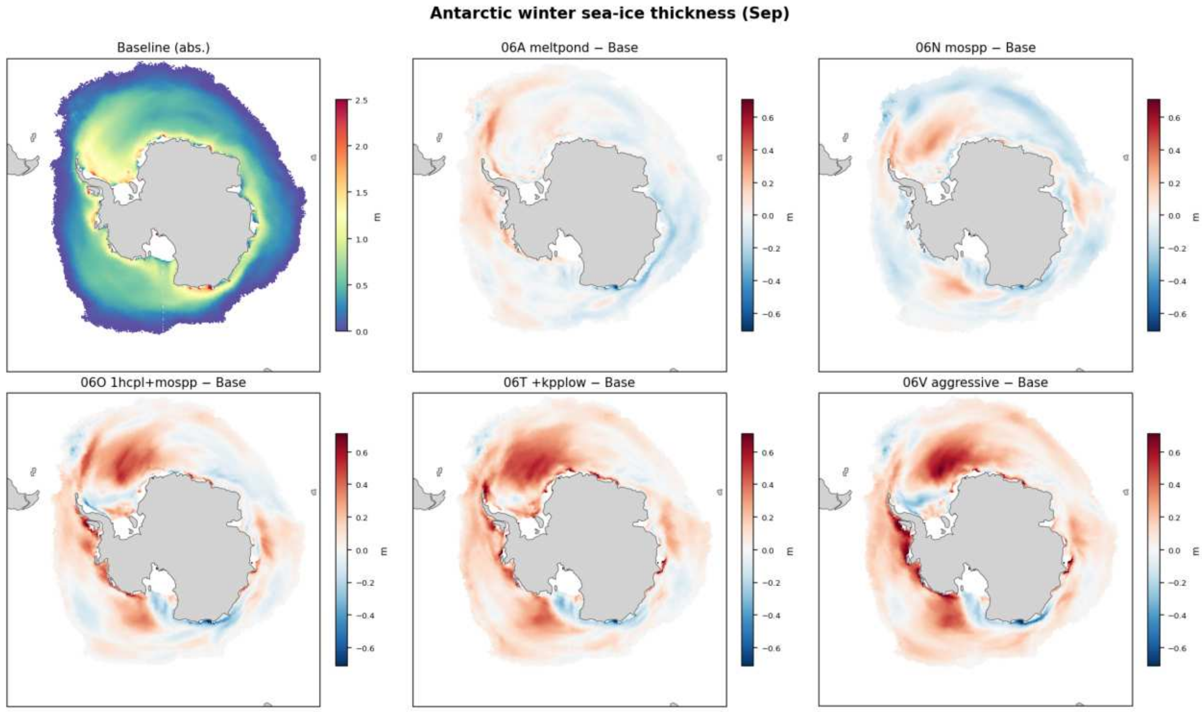


Figure 10: Antarctic September sea-ice *thickness* (Baseline absolute, top-left; others = anomaly vs Baseline). Thickness — not concentration — is the honest growth diagnostic here: winter concentration is already near 1 over most of the pack, so it *saturates* and cannot register added ice, whereas thickness is unbounded. The mixing runs (06O/06T/06V) thicken the pack broadly (positive/red anomaly over the Weddell and Ross sectors), consistent with suppressed open-ocean convection insulating and stratifying the surface; melt-pond (06A) barely changes it. (The concentration version, `sic_winter_antarctic.png`, shows only the ice-edge shifts for the reason above.)

Mechanism. The mospp / weaker-KPP changes reduce Weddell-Sea open-ocean winter convection, cooling the subsurface and letting sea ice grow — improving `thetao/so/siconc` (consistent with the CMPI breakdown) and removing the Southern-Ocean surface warm bias. Spurious open-ocean deep convection in the Weddell/Ross gyres is the canonical CMIP Antarctic bias (Heuzé et al. 2013; Heuzé 2021, which evaluates AWI–CM in the multi-model set; Mohrmann et al. 2021 for the polynya surface expression), and open-ocean Weddell convection is a real-but-rare mode that models over-express (de Lavergne et al. 2014) — so damping it is physically the right direction. The *proximate radiative* closure is defensible: more winter ice → higher albedo → less absorbed SW → lower imbalance. We are more cautious about the "reduced ocean heat uptake" half of the chain: suppressing convection also *traps* heat at depth, which can *increase* deep ocean-heat-content drift, so the net uptake change is not established here — it should be pinned to an OHC-tendency diagnostic before being asserted. The cost is that the surface cooling propagates globally into the atmosphere (`tas/zg`), the broad cold bias seen in `part8`.

Counter-literature (a compensating-error risk this campaign must own). §5 attributes the SO warm bias and its removal almost entirely to *ocean* convection, but Hyder et al. (2018, *Nat. Commun.*) trace the leading 40–60°S warm-SST bias primarily to *atmospheric* net surface-heat-flux errors dominated by shortwave/cloud biases, and warn that a small *net* flux bias is often achieved by *compensating* component errors. The two coexist at different scales — broad atmospheric-SW warmth vs. localised Weddell/Ross deep-convection anomalies — and our mixing levers act only on the ocean branch. If part of the bias is atmospheric-SW in origin, the ocean-mixing lever is partly compensating for an atmospheric error, which is *exactly* the "cooling leaks beyond where the parameter should act" failure mode flagged for these runs in §4.3 (the region map reddens into the tropics/Niño-3.4). This tension is the report's own evidence pointing at Hyder; it strengthens, not weakens, the case for treating aggressive ocean-mixing tunings as provisional.

Part II

Counter-tuning the NH cold bias — the structural floor

*The cold bias that makes Part I's aggressive lever expensive is largely **not** the tuning: it is a structural LPJ-GUESS boreal-forest collapse that every run inherits. That makes it a **counter-lever** — fixing it warms the NH high latitudes at essentially unchanged imbalance, buying the RMSE headroom to spend the aggressive global cooling on the imbalance. This part attributes the bias with a clean WITH-vs-WITHOUT-LPJG experiment, resolves its two mechanisms and a residual. Feedback-free spin-ups then show that the* source *of the low-tree state is strongly forcing-sensitive: the complete CRUNCEP–AMIP transfer matters greatly, whereas replacing CRUNCEP net radiation by CERES alone has only a small equilibrium effect.*

6 The NH high-latitude cold bias is a structural LPJ-GUESS problem

The runs carry a severe near-surface **cold bias over Siberia and northern Canada** (visible in every `part8` map). This chapter documents a working investigation into whether the coupled dynamic vegetation (LPJ-GUESS) is responsible — motivated by the idea that fixing it could be a *warming* lever to offset the imbalance-tuning cold bias, and by prior EC-Earth experience that LPJG drives such boreal biases.

6.1 Symptom: boreal needleleaf replaced by grass

Dominant-PFT maps (tool `part26`) for the HTESSEL spin-up (the reference, run with the same atmosphere but *without* atmosphere↔LPJG feedback) vs the coupled runs show boreal needleleaf giving way to C3 grass over Siberia / N. Canada. Tree-dominant fraction north of 50°N: **20.1 % (spin-up) → 16.8 % (baseline) → 11.4 % (06V)** — progressively worse with the more aggressive (colder) runs. *Caveat on the prior:* DGVMs disagree substantially on boreal tree-vs-grass partitioning, and real boreal forest is genuinely open with substantial non-tree ground cover (Jiang et al. 2012), so "grass over Siberia" is not *automatically* a bug. What makes the establishment-failure reading defensible here is the *magnitude*: the coupled state reaches cvh ≈ 0.02–0.05 with ~~no closed canopy anywhere (FORESTFPC= 0)~~[1] (§10), an outlier well below the ∼ 0.4–0.7 closed-taiga range (MODIS VCF), not merely a grass-leaning partition.

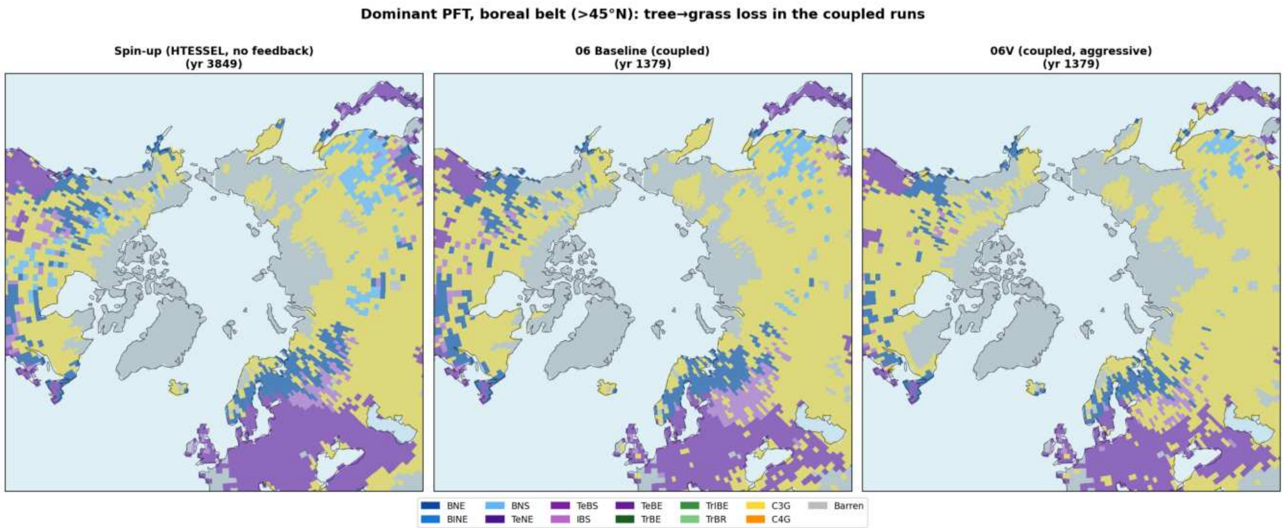


Figure 11: Dominant PFT, boreal belt: HTESSEL spin-up (reference) vs coupled baseline vs 06V. Needleleaf (blues) thins and C3 grass (yellow) expands across central/eastern Siberia in the coupled runs.

[1] **Void diagnostic.** `FORESTFPC` sums only PFTs with `landcover==FOREST` (`commonoutput.cpp`:1143), i.e. the *managed-forest* land-use class; natural trees are counted in `TREEFPC` via the `NATURAL` branch. In this PI configuration nothing is allocated to FOREST (`Natural_sum`= 0.684, `Total`= 0.665), so `FORESTFPC`≡ 0 *by construction* while `TREEFPC`= 0.323. It is a land-use bookkeeping artifact and says nothing about canopy closure.

6.2 Is it grass growing, or trees dying? And is it a year-1 jump or a slow drift?

Belt-mean (55–75°N) needleleaf LAI is essentially *conserved* over the baseline run (0.40→0.37, trend ≈0), while C3 grass LAI *rises* (0.65→0.83, +0.10/dec): the signal is **grass growing**, with needleleaf loss only regional (E. Siberia) and offset by gains in the milder west. The coupled run is initialised directly from the spin-up's year-3850 LPJG state, and baseline year-1350 boreal LAI (tree 0.407, grass 0.649) is *identical* to the spin-up end-state (0.401 / 0.669) — so there is **no wholesale year-1 collapse**; the degraded, grass-heavy state is *inherited from the spin-up*. Within 1350–1379 the boreal 2 m temperature is at its (cold) level from year one and does *not* drift down (trend +0.42 K/dec, noisy) — consistent with a memory-less atmosphere that equilibrates to the surface state within a year rather than a slow vegetation runaway. (These are belt-mean *LAI* diagnostics; as §6.3 shows, the quantity that actually matters is not LAI-per-tile but the forest *cover fraction*.)

6.3 The decisive test: a clean WITH-vs-WITHOUT-LPJG pair (CORE3, 30 yr)

The earlier control was confounded by the ocean (core2 vs core3). We removed that confound: a purpose-built **no-LPJG AWI-CM3-develop reference** — *same* CORE3 ocean, *same* TCO95 atmosphere and matched component branches, HTESSEL + prescribed satellite vegetation, a full 30-yr PI-control (model years 1350–1379, `awicm3_noLPJG_CORE3_30y`) — differs from the LPJG Baseline in **LPJG alone**, so the 2 m-T difference is now a clean attribution. Its net TOA imbalance is +1.73 W/m^2 (`part2`), near the warm/high-imbalance end of Fig. 1 — confirming the LPJG signal is a near-surface *land* effect, orthogonal to the TOA tuning of §2. **Crucially, it beats the LPJG Baseline on *both* skill metrics: global 2 m-T RMSD vs ERA5 1.514 K (bias −0.76 K, `part8`) vs 1.717/−0.94, and CMPI 0.738 (`part4`) vs 0.745** — at higher (warmer) imbalance. Removing LPJG improves the global temperature fit *and* the multivariate skill score *while* the imbalance changes by only +0.2 W/m^2 (+1.73 vs +1.53) — within the inter-run drift of these unequilibrated 30-yr segments (§8), so the warming lever is real *and costs essentially no imbalance to within that uncertainty* (not, strictly, "zero"). Over the final decade, boreal DJF:

- **Forest cover collapses.** Siberian high-veg cover `cvh`: **0.63 (no LPJG) → 0.02 (LPJG)** (pan-boreal 0.49 → 0.07). LPJG passes essentially no forest to the atmosphere.
- **And it is genuinely colder — confound gone.** With the ocean held fixed, the LPJG run is unambiguously −3.1 **K** colder over Siberia in DJF (−30.4 vs −27.4°C).
- **Snow-masking (spring albedo)** operates as before: forest removal brightens the snow season (MAM **+0.077**). This is the canonical boreal-forest climate mechanism — a masked snowpack lowers albedo and warms high latitudes, and its removal cools most in spring (Bonan, Pollard & Thompson 1992; Betts 2000; Bonan 2008), with tree/canopy cover setting the snow-albedo-feedback magnitude that many models misrepresent (Loranty et al. 2014; Chapin et al. 2005) — so a model that loses its forest gets exactly this feedback wrong. But it is a *sunlight* effect and cannot explain the equally-cold polar night (§6.4).

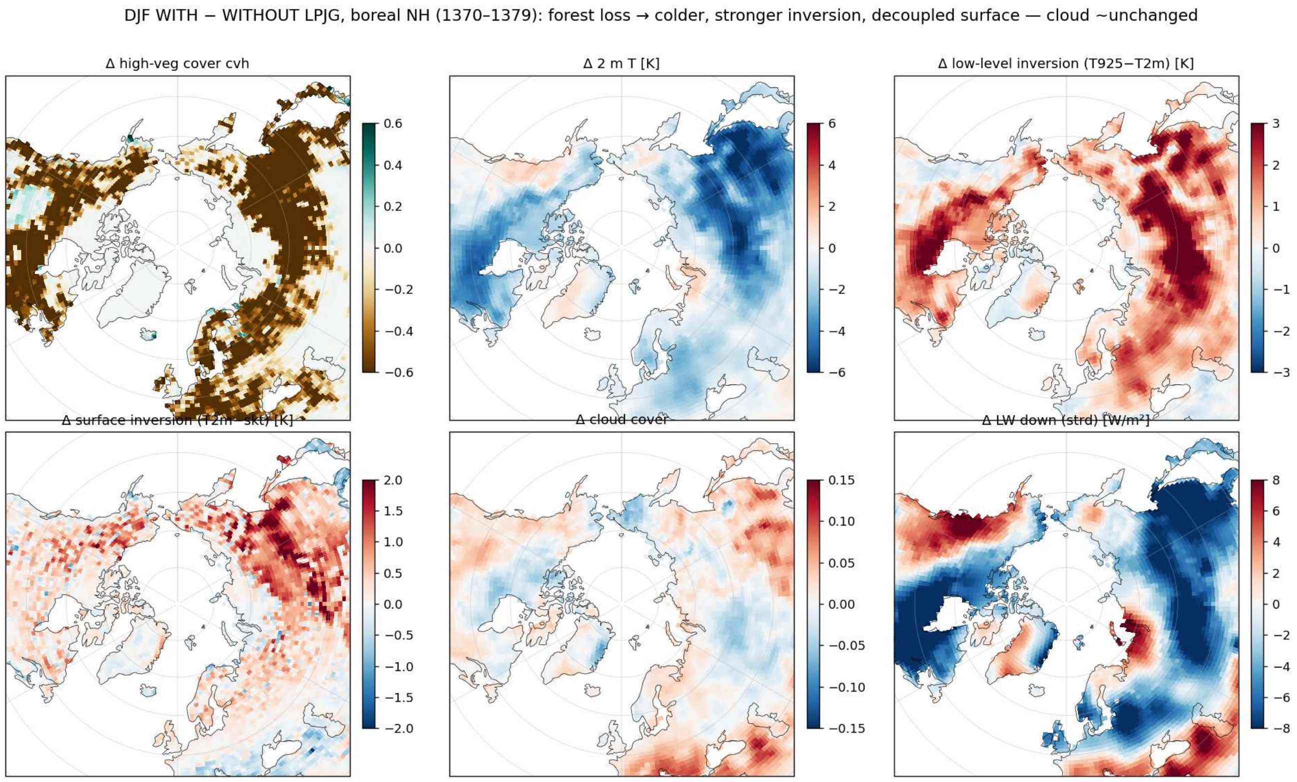


Figure 12: DJF WITH−WITHOUT LPJG, boreal NH (final decade, land-masked). Forest-cover collapse (`cvh`, top-left, brown) is spatially co-located with the cold anomaly (top-mid), a *stronger* low-level inversion ($T_{925}-T_{2m}$, top-right) and surface decoupling ($T_{2m}-T_{skt}$, bottom-left), at *unchanged* cloud (bottom-mid). LW-down (bottom-right) is modestly reduced over the cold interior.

6.4 Seasonality: which months carry the cold effect

Resolving the same clean CORE3 WITH−WITHOUT pair by *month* (boreal box 55–70°N, Siberia + N. America, final decade) shows the LPJG cold effect is **present in every month except high summer**. The WITH−WITHOUT 2 m-T difference is negative from September through June and only vanishes in July–August (−0.1/−0.2 K), peaking in the *shoulder/dark* seasons: **November** −3.9 **K, April** −3.3 **K**, with DJF −2.4 K and MAM −2.5 K (annual box-mean −2.0 K). **This is the key seasonal fingerprint**: a pure snow-albedo (sunlit) mechanism would confine the cold to the bright spring, yet the effect is *as large in near-sunless November/December* — which is exactly what motivates the winter roughness→inversion mechanism of §6.5. The near-zero July–August signal also confirms the growing-season atmosphere is unaffected (consistent with JJA gates being cleared, §6.3). Overlaid is the independent **OIFS–AMIP** run (§10.2: observed-SST, no-LPJG, PI 1870–79) — it tracks the *warm* WITHOUT curve (annual −7.6 vs −6.3 °C; DJF −25.6 vs −24.9; MAM even milder), confirming that a feedback-free, realistically-forced boreal climate is warm, and that the forcing we drive the decisive offline re-spin with (§10.2) is *not* itself cold-biased.

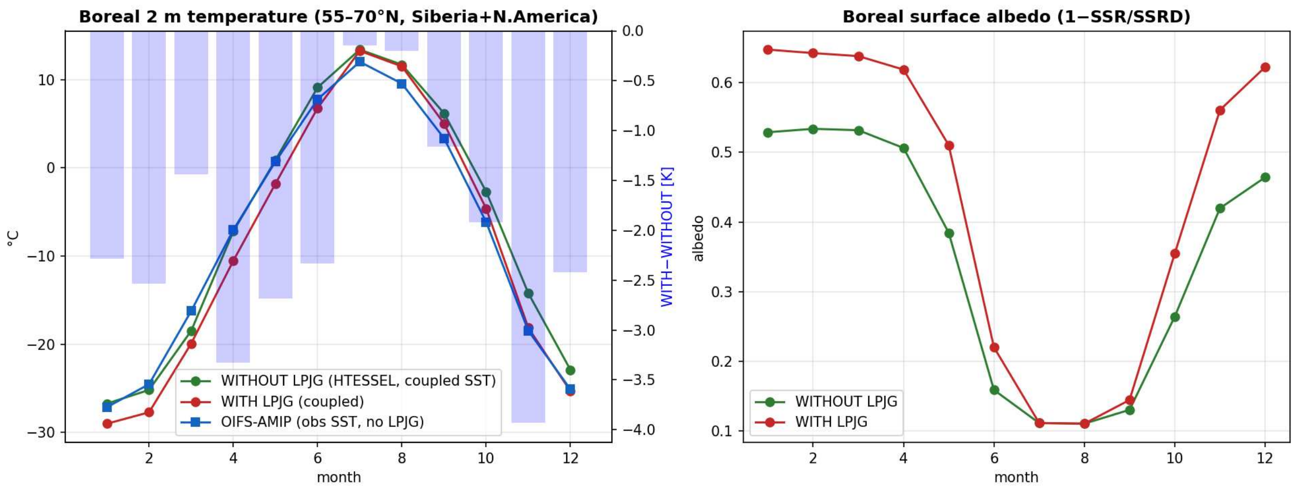


Figure 13: Boreal (55–70°N, Siberia+Canada) monthly climatology. *Left:* 2 m-T annual cycle — WITHOUT LPJG (green, clean CORE3 reference), WITH LPJG (red, coupled), and the OIFS–AMIP obs-SST no-LPJG forcing run (blue); blue bars = WITH−WITHOUT [K, right axis], negative (LPJG colder) in every month bar July–August, strongest in Nov/Apr. *Right:* surface albedo (1−SSR/SSRD); LPJG brightens the snow season (spring/autumn), the sunlit part of the bias. (AMIP emits no down-welling SW, so it appears on the temperature panel only. WITH/WITHOUT are model years 1370–79; AMIP is obs-SST 1870–79 — both ∼PI.)

6.5 A second, winter mechanism: roughness → inversion

Snow-albedo needs sunlight, yet the cold bias is as strong in DJF as in MAM. The missing forest acts a second way, in the dark. OpenIFS roughness length is cover-weighted by `cvh`, and needleleaf forest ($z_{0m} = 2.0\,\mathrm{m}$) is 20–60× rougher than the tundra/grass (0.034–0.10 m, `susveg_mod.F90`) that replaces it. A smoother surface couples less efficiently to the atmosphere; in polar night the stable boundary layer (SBL) decouples and the surface radiatively cools, deepening the near-surface inversion. That surface–atmosphere coupling strength in the SBL controls screen temperature and inversion depth is well established (Holtslag et al. 2013; Sandu et al. 2013); we assemble the specific vegetation-roughness → SBL-decoupling → screen-T chain from the land-surface and SBL literatures rather than a single published result, so it is presented as a mechanism the data then test, not a settled citation. **The data show exactly this fingerprint** (DJF Siberia, LPJG − no-LPJG): a **+2.1 K stronger** low-level inversion ($\mathrm{T}_{925}-\mathrm{T}_{2m}$), a **+0.7 K stronger** surface inversion ($\mathrm{T}_{2m}-\mathrm{T}_{skt}$), **unchanged cloud** ($\Delta \approx 0.00$) and -4.7 $\mathbf{W/m^2}$ **less** down-welling longwave — a decoupled, radiatively-cooling stable surface, not a cloud/radiation artefact. Spatially over boreal land the forest-loss pattern *is* the inversion pattern: area-weighted $r(\Delta\texttt{cvh}, \Delta\mathrm{inv}) = -0.60$, $r(\Delta\texttt{cvh}, \Delta\text{sfc-decouple}) = -0.68$, $r(\Delta T_{2m}, \Delta\mathrm{inv}) = -0.83$, while the negative control $r(\Delta\texttt{cvh}, \Delta\mathrm{cloud}) = -0.05$ is ≈ 0 — a real mechanism, not a shared trend. Regression lever: **restoring `cvh` by** +0.4 **warms DJF screen-T by** $\sim +1.2\,\mathbf{K}$ through roughness alone and weakens the inversion by $\sim 0.7\,\mathrm{K}$ — additive to the spring/albedo warming. This slope-based estimate inherits the scatter of the underlying correlation ($r = 0.63$, so $< 40\%$ of grid-cell variance explained); read it as $\sim +1.2 \pm 0.5\,\mathrm{K}$, an order-of-magnitude lever rather than a precise prediction. The *direction* and existence of the effect rest on the WITH−WITHOUT experimental design (§6.3), not on this regression.

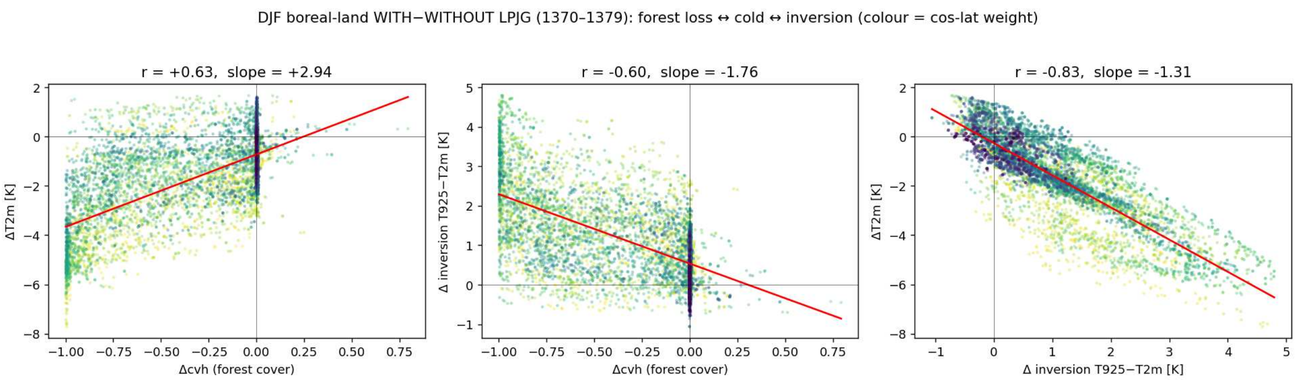


Figure 14: DJF boreal-land, WITH−WITHOUT LPJG (final decade). Forest loss (negative Δ`cvh`) tracks the cold (ΔT_{2m}, left), the stronger inversion (middle) and, transitively, cold ↔ inversion (right, $r = -0.83$). Colour = cos-lat weight.

6.6 How much does the LPJG fix buy? (vs ERA5)

Against ERA5 the current Baseline is −4.5 **K** too cold over Siberia in DJF. The lever helps but does not close it: a realistic partial fix (+1 K) leaves −3.5 K, and *full* forest restoration (the no-LPJG run itself — the ceiling) reaches −1.4 **K**. So vegetation/roughness accounts for ∼3 K, roughly two-thirds of the Siberian DJF cold bias, and boreal-land RMSD falls 4.4 → 3.3 K — but a residual ∼1.5 K survives *even with perfect forest*.

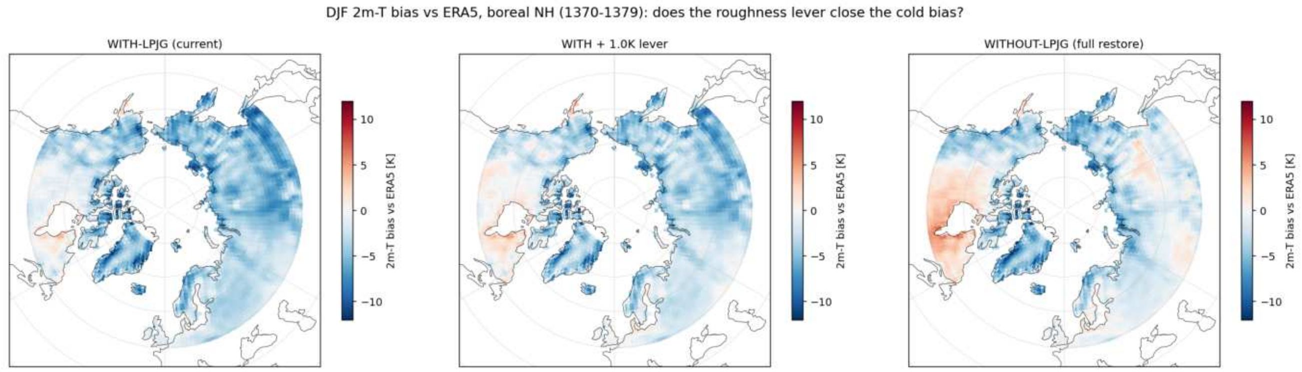


Figure 15: DJF 2 m-T bias vs ERA5, boreal NH (final decade): current LPJG Baseline (left, model output), an **illustrative** +1 K roughness lever (middle — a uniform +1 K offset applied to the Baseline, *not* a simulation), and full forest restoration = the no-LPJG run (right, model output). The lever panel only visualises the magnitude of a plausible partial fix; the load-bearing comparison is left vs right, which shows a residual cold survives even full forest restoration.

6.7 The residual driver: an over-strong stable-boundary-layer inversion

Diagnosing that residual in the *full-forest* run (vegetation thereby ruled out) against ERA5 (DJFM, boreal land) points to a single process. The model's low-level inversion is **+1.4 K too strong** ($T_{925}-T_{2m}$: model +5.6 vs ERA5 +4.2 K), and the excess is **entirely surface-trapped** — the 925–1000 hPa inversion bias is +0.07 K (≈ 0), so all the extra cooling sits in the lowest layer. Spatially this inversion-excess explains $r = +0.76$ of *where* the model is too cold; the Siberian High (MSLP bias +0.15 hPa, $r = 0.27$) and cloud-*fraction* (bias +0.07, $r = 0.00$) are ruled out. A too-strong wintertime polar inversion is a real, observationally-benchmarked model bias (Medeiros et al. 2011; Zhang et al. 2011), which supports reading the "+1.4 K too strong" residual as genuine. That surface-trapped signature, with circulation and cloud-fraction eliminated, is also the textbook fingerprint of a **stable boundary layer that under-mixes** — the IFS-family "too-stable" winter cold bias (Sandu et al. 2013). Its lever is the OpenIFS stable-BL turbulent mixing — the "long-tail" stability functions / minimum stable diffusion whose lineage runs Louis (1979) → Beljaars & Viterbo (1998) — *orthogonal* to the vegetation fix.

Two counters this diagnosis must own (both align with the report's own "compensating error" worry). *(a) The mixing fix is a documented compensating error:* enhanced long-tail stable mixing is stronger than LES/observations justify and is retained only because reducing it degrades the large-scale flow — it improves 2 m temperature but deepens cyclones and harms low cloud (Sandu et al. 2013; Holtslag et al. 2013), so Phase 1B must check the storm-track/low-cloud climatology (which CMPI/`part8` will not catch cleanly). *(b) The inversion may be a cloud-phase problem:* Pithan et al. (2014) attribute much of the modelled Arctic wintertime cold bias to missing mixed-phase low cloud (a LW, cloud-*phase* effect) — our null control ruled out cloud *fraction* but not phase, so we scope the residual claim to "given cloud fraction is unbiased" and flag a cloud-phase / LW↓ test as a prerequisite. Likewise the $r = +0.76$ residual-to-inversion match is suggestive, not proof that the "fourth lever" is truly orthogonal.

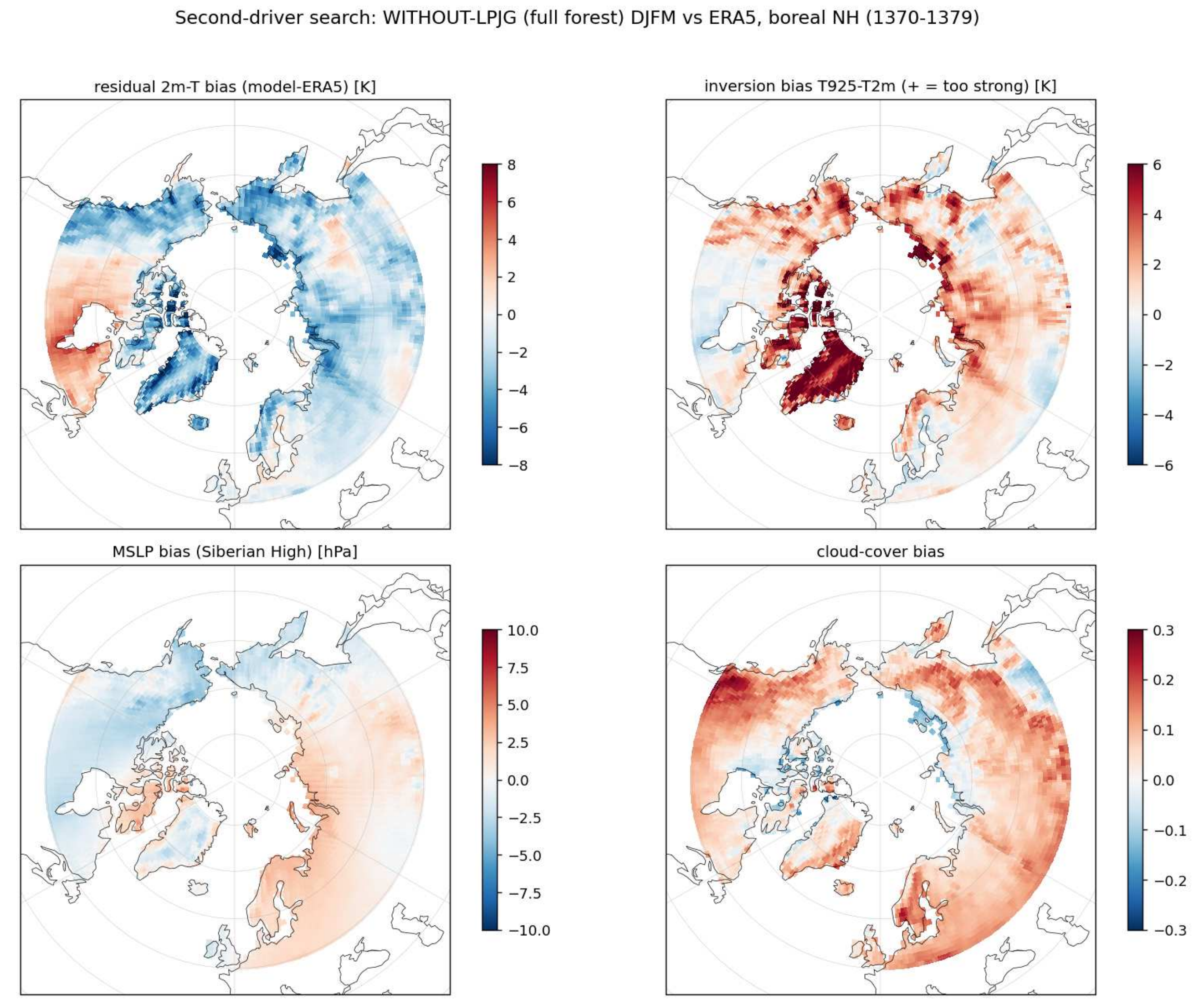


Figure 16: Second-driver search: full-forest (no-LPJG) run DJFM vs ERA5, boreal NH. Residual cold (top-left) matches the inversion-excess (top-right, $r = +0.76$); MSLP (bottom-left) and cloud (bottom-right) biases are diffuse and uncorrelated.

Near-complete attribution. The two mechanisms close the boreal DJF cold bias between them: of the $\approx -4.5\,\mathrm{K}$ Siberian bias vs ERA5, the LPJG forest-cover collapse (snow-albedo in spring, roughness in winter) contributes $\approx -3\,\mathrm{K}$, and the over-strong stable-BL inversion the remaining $\approx -1.4\,\mathrm{K}$. Two *orthogonal* levers — restoring a forcing-consistent forest state and enhanced stable-BL mixing — therefore target essentially the entire bias. (The veg cooling is also a *stable* offset, not a runaway: over 20 yr the LPJG run carries more/thicker Arctic ice as a bounded mean offset with no amplifying sea-ice drift.)

6.8 Working conclusion & lever

LPJ-GUESS sends essentially no boreal *forest cover* to OpenIFS (cvh $\sim 0.6 \to 0.02$ over Siberia, §6.3), so the canopy no longer masks snow (spring albedo, §6.3) *nor* roughens the surface (winter inversion, §6.5) — a physically robust driver of the Siberia/N. Canada cold bias, consistent with the EC-Earth family in which LPJ-GUESS is interactively coupled (Döscher et al. 2022) and in which dynamic vegetation-cover / effective-LAI improves high-latitude spring climate through the snow-albedo coupling (Alessandri et al. 2017). Source trace (verified): LPJG computes the high-veg fraction and exports it over OASIS as `GUE_FRAH`→`cvh` (`framework.cpp::computeDailyLAIandFPCforIF` `namcouple` line 120). With the AWI setting `iftreefracca=1` (`global.ins`:151), cvh is the summed tree *crown area*×density (no 0.05 clamp — that constant is dead code; `natural_frac`= 1 over PI land), so cvh≈0.05 means boreal tree cover is genuinely ∼5%. The establishment gates (`gdd5min_est=500`, `twmin_est=5`) are *not* the blocker — they are cleared across the boreal — so the degraded state is **inherited from the LPJG spin-up**, not a coupled runaway. It was initially read as a tree-vs-grass competition failure, but the controlled forcing-transfer test (§14.3) reproduces most of the boreal loss without changing LPJG parameters; competition must therefore be diagnosed only after a forcing-consistent re-spin. Seasonally (§6.4) the LPJG cold effect spans September–June

(DJF −2.4, MAM −2.5 K over the boreal box) and vanishes only in July–August — i.e. it is a cold-season, not a growing-season, effect. The co-equal *sunless* late-autumn/winter part — which pure snow-albedo cannot explain — is now resolved (§6.3–6.7): it is the roughness→inversion effect of the lost forest (§6.5) plus a model-intrinsic over-strong stable-BL inversion (§6.7).

Fixes: (1) correct the atmospheric forcing, regenerate the daily LPJG forcing and re-do the spin-up with one verified binary; only the residual against that forcing may justify tuning `greff_min`, C3-grass competitiveness or `pstemp_low` (§14.3, §16); (2) if a structural residual remains, revisit `iftreefracca`/crown allometry; (3) a woody stem-area floor on the snow-masking fraction in OpenIFS (`surfbc_ctl_mod.F90`:389–391 / `surfrad_ctl_mod.F90`:650–655) — but this only bites once (1) raises cvh, since the masked-snow tile is weighted by cvh itself. Restoring forest cover is the physically-grounded *warming lever* of §9 and buys ∼3 K (§6.6), but its cause must not be prejudged as a parameter error. **A fourth, orthogonal lever** attacks the residual ∼1.4 K (§6.7): *enhanced stable-boundary-layer mixing* in OpenIFS (VDF stability functions / minimum stable diffusion), which would relax the over-strong surface inversion. Together (1)–(4) target essentially the entire boreal DJF cold bias.

6.9 Feedback-free AMIP spin-up — substantially qualified by §14.3 and §7

Read this section with the later controlled tests. The argument below turns on a standalone LPJG re-spin driven by the OIFS–AMIP forcing under-vegetating despite the establishment gates being cleared, which was read as an LPJG-internal competition failure. A later controlled experiment (§14.3) shows that *moving between the CRUNCEP and AMIP forcings alone*, with byte-identical parameters, costs −54 % of Siberian and −66 % of East-Siberian `TREEFPC`. The under-vegetation seen here is therefore largely what the forcing change predicts on its own, and cannot by itself demonstrate a competition failure. The four-arm radiation test (§7) further shows that replacing only CRUNCEP `rsns`/`rlns` by CERES removes only a small part of the NH contrast: the strong full-forcing effect is not a net-radiation effect alone. `FORESTFPC` is in fact a *void diagnostic* here: it counts only the managed-forest land-use class, which is empty in this PI configuration, so it is identically zero whatever the vegetation does (§14.3). There is no separate closed-canopy failure.

Is the boreal forest deficit a *climate* problem (fix the atmosphere) or an *LPJG-internal* tree-vs-grass competition problem (fix the vegetation)? We answered it directly with a **feedback-free** experiment: a clean OpenIFS–AMIP run (prescribed observed SST, no coupled ocean/LPJG cold feedback; ∼PI 1870–79) generated a realistic daily forcing, which drove a 2000-yr standalone LPJG re-spin on a *dense* 11 538-cell gridlist (build detail in §10.2). Fig. 17 puts that forcing (top) beside the equilibrium vegetation it produced (bottom). **The forcing is *not* growing-season-cold**: over the Siberian box the warmest-month T2m is 14.7 °C (100% of cells clear the `twmin_est`= +5 gate) and GDD5 is 730 (76% clear 500) — establishment is climatically permitted essentially everywhere trees should be. **Yet the spin-up under-vegetates and grass wins**: cvh= 0.21 (far below the 0.4–0.7 closed-taiga range), `GRASSFPC`= 0.37 > `TREEFPC` = 0.21, and ~~no closed canopy forms anywhere (FORESTFPC= 0)~~[2], with grass filling exactly the East-Siberian interior. On a dense gridlist this cannot be dismissed as under-sampling. **Verdict: clearing the establishment gates does not prove that competition is the dominant error.** The controlled transfer test shows that the complete forcing change reproduces most of the regional tree loss (§14.3), while the new four-arm experiment shows that replacing net radiation alone explains only a small part of the NH contrast (§7). Competition remains a possible *residual* target, but it must be measured after a forcing-consistent re-spin rather than assumed from this AMIP arm. One climate co-limit visible here is the deep-winter cold (−28.3 °C box mean, < −40 in NE Siberia) that gates out evergreens on ∼ 40% of cells — but climatically-permitted *larch* (no cold gate) fails there too (BNS cover 0.035), so the winter warming of Phase 1B is only a *secondary* aid.

[2]**Void diagnostic.** `FORESTFPC` sums only PFTs with `landcover==FOREST` (`commonoutput.cpp`:1143), i.e. the *managed-forest* land-use class; natural trees are counted in `TREEFPC` via the `NATURAL` branch. In this PI configuration nothing is allocated to FOREST (`Natural_sum`= 0.684, `Total`= 0.665), so `FORESTFPC`≡ 0 *by construction* while `TREEFPC`= 0.323. It is a land-use bookkeeping artifact and says nothing about canopy closure.

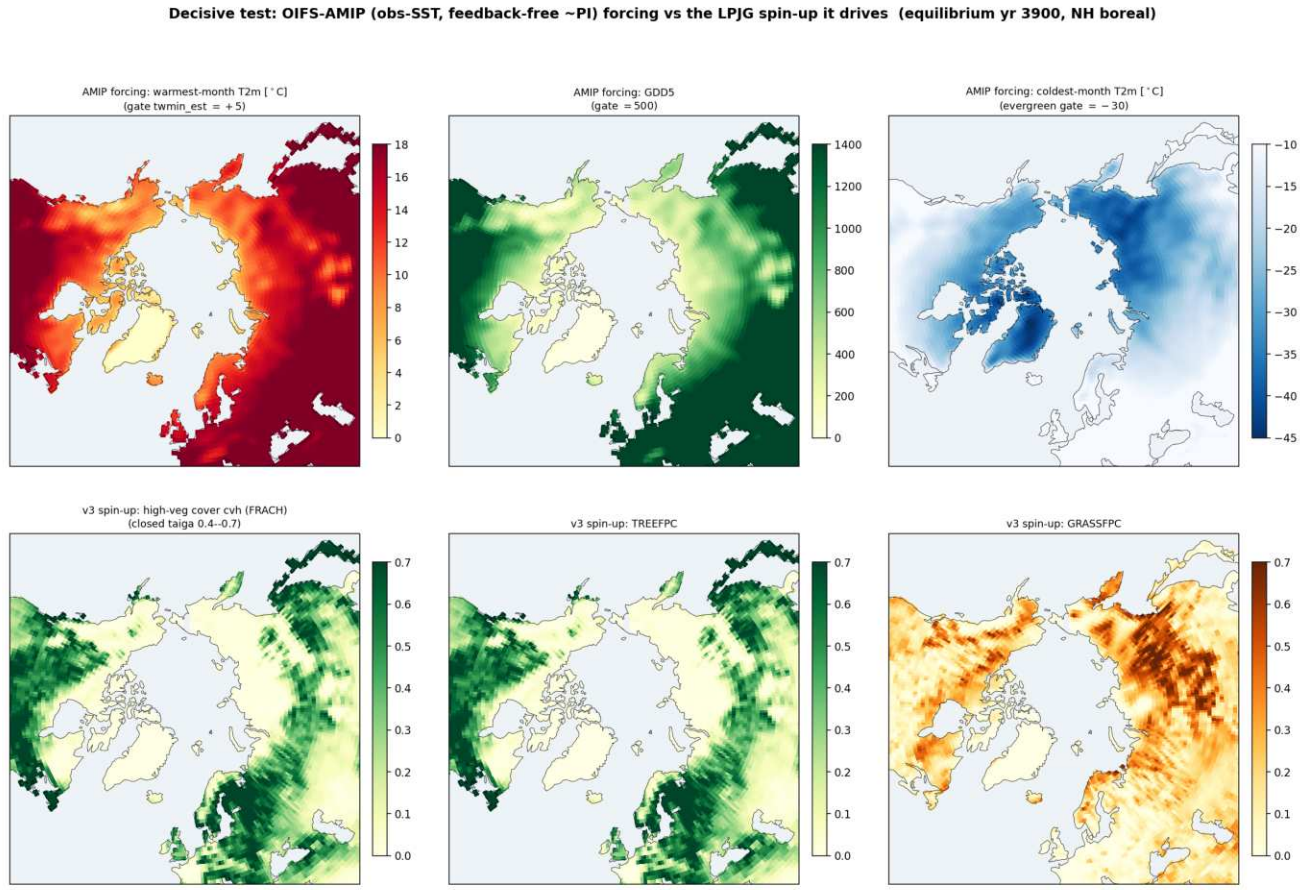


Figure 17: Decisive climate-vs-competition test (NH boreal, equilibrium yr 3900). *Top:* the feedback-free obs-SST ∼PI OIFS–AMIP forcing that drove the standalone LPJG re-spin — warmest-month T2m (vs `twmin_est`= +5), annual GDD5 (vs 500), coldest-month T2m (vs evergreen `tcmin_est`= −30). *Bottom:* the equilibrium vegetation it produced — cvh, `TREEFPC`, `GRASSFPC`. The growing-season gates are largely cleared, yet the AMIP-forced equilibrium has cvh≈ 0.2 and extensive East-Siberian grass. This panel diagnoses the AMIP-arm state; the controlled transfer test (§14.3) shows that it cannot by itself attribute the state to LPJG competition.

7 Forcing sensitivity: CRUNCEP radiation constrained by CERES EBAF

Question and controlled design. The full forcing-transfer test in Part IV shows that moving from CRUNCEP to AMIP strongly reduces equilibrium boreal tree cover, but it does not isolate which forcing fields cause that response. We therefore completed four 2000-yr standalone LPJ-GUESS spin-ups on TCO95 and compared their final year (model year 3900). The two additional arms isolate the narrower effect of replacing only CRUNCEP surface net shortwave (`rsns`) and net longwave (`rlns`) by CERES-constrained radiation:

experiment	daily forcing cycle	radiation treatment
AMIP	OpenIFS–AMIP, 1870–1879	corrected native OpenIFS `rsns`/`rlns`
CRUNCEP v3	calibrated CRUNCEP v7, 1901–1910	calibrated CRUNCEP net radiation
CRUNCEP + CERES direct	CRUNCEP v3 non-radiation fields	CERES EBAF Ed4.1 monthly 2001–2010, bilinear to TCO95 and linearly interpolated between month midpoints
CRUNCEP + CERES daily variability	CRUNCEP v3 non-radiation fields	CERES official 12-month climatological means, retaining CRUNCEP relative daily SW and additive daily LW anomalies

In both CERES experiments `tas`, `tasmin`, `tasmax`, precipitation, humidity and wind are byte-for-byte unchanged from CRUNCEP v3. Runtime files carry the generic name `AM04_atm_cmip6_1d_1990-1999_fast_lpjgforcing.nc`; checksums and full variable comparisons verified that they are exact runtime copies of the intended direct and daily-variability source files.

CERES provenance and what the comparison means. The observational reference is NASA Langley CERES EBAF Edition 4.1 on its native 1° grid, specifically `sfc_net_sw_all_clim` from `CERES_EBAF_Ed4.1_Subset_CLIM01-CLIM12.nc`. It is the official July 2005–June 2015 climatology (product DOI). Figure 18 puts all forcings on a common 1° land grid and shows JJA plus the Siberian seasonal cycle. AMIP–CERES and CRUNCEP–CERES are *forcing differences across products and epochs*, not strict pre-industrial biases. The small differences for the two CERES-derived forcings are *construction checks*, not independent validations: the direct case uses 2001–2010 monthly CERES data, whereas the daily-variability case is explicitly constrained to the same climatology used as the reference.

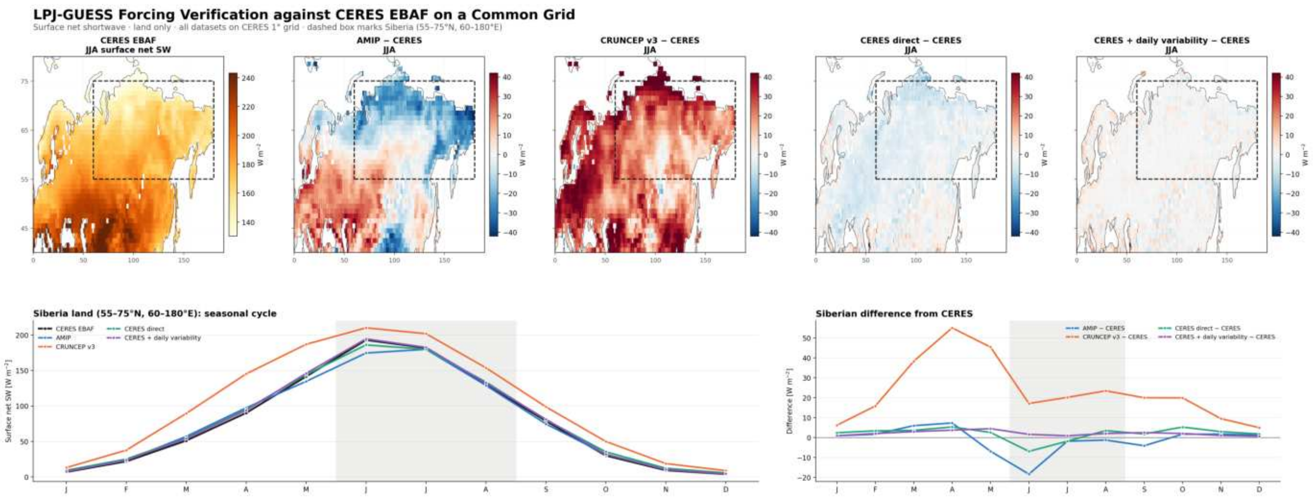


Figure 18: Surface net-shortwave forcing check against CERES EBAF Ed4.1. Top: CERES JJA net SW and each forcing minus CERES after common-grid remapping over land. Bottom: Siberian land seasonal cycles and their differences from CERES (55–75°N, 60–180°E). Near-zero CERES-derived panels confirm the forcing construction; they are not an independent observational test.

Equilibrium vegetation response. In the annual `treeFrac` diagnostic, the NH-land display-grid mean is 0.186 under AMIP, 0.269 under CRUNCEP v3, 0.262 under direct CERES radiation and 0.263 when CRUNCEP daily variability is retained. Thus the full CRUNCEP–AMIP difference is +0.083, whereas the two CERES substitutions remain +0.076 above AMIP. Replacing CRUNCEP radiation reduces its mean tree fraction by only 0.006–0.007 and leaves about 92% of the CRUNCEP–AMIP difference. The almost identical direct and daily-variability results show that this choice of monthly-to-daily radiation treatment has negligible influence on the hemispheric mean at equilibrium, although regional differences remain visible.

Interpretation. Net radiation alone is not the dominant cause of the AMIP–CRUNCEP equilibrium tree-cover difference: changing only `rsns/rlns` removes roughly 8% of this hemispheric `treeFrac` contrast. Together with the regional `TREEFPC` transfer result of §14.3, this localises most of the full-forcing penalty to other meteorology (especially the documented temperature/GDD5 change), field covariability and nonlinear LPJG response. It does *not* establish a residual competition error. The appropriate conclusion is: *the vegetation equilibrium is strongly forcing-sensitive, but within the CRUNCEP family a CERES net-radiation correction alone is insufficient; competition parameters should be considered only after a forcing-consistent same-binary re-spin.*

Part III

The path forward: raise the floor, then tune

*Parts I–II give the strategy: the imbalance can only be cut by cooling, and cooling is only affordable once the structural NH cold floor is raised. So first fix the floor (vegetation competition + stable-BL mixing), **then** spend a moderate cooling lever to reach surface balance — now without the RMSE blow-up. First the caveats that condition this, then the recommendation and a phased plan.*

8 Caveats

1. **06D "HRlike" has no HR-specific namelist change** — only the common pond shift (identical to 06C); its metrics confirm this. If a resolution/timestep/viscosity change was intended, it is not in the namelists.

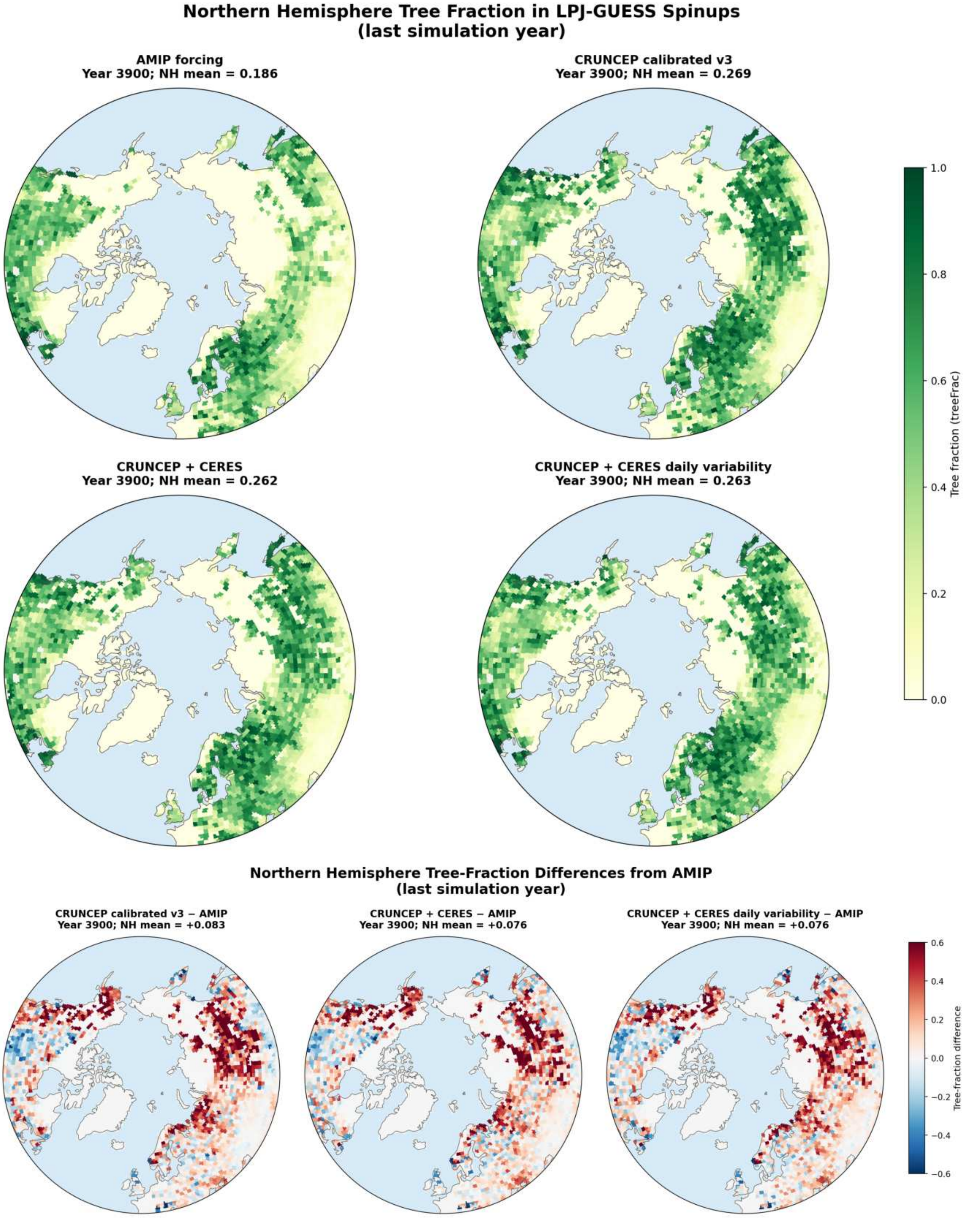


Figure 19: Final-year tree fraction in the four 2000-yr offline spin-ups. Top: absolute year-3900 tree fraction. Bottom: CRUNCEP-family differences from AMIP. CERES-constrained radiation modestly reduces CRUNCEP tree cover, but does not remove the much larger AMIP–CRUNCEP contrast.

2. **Filename convention.** These runs write `atm_remapped_1m_<var>_YYYY-YYYY.nc` (no `_1m_` infix). I added backward-compatible fallbacks to `part2`, `part8` and the CMPI preprocessing; existing configs are unaffected.
3. **Accumulation period** for these XIOS runs is **3600 s** (not the 21600 s of the old HR config); configs use `accumulation_period=3600` (verified: tsr $\to\approx 240\,\mathrm{W/m^2}$).
4. None of the runs is at equilibrium within 30 yr — numbers are a ranking, not final values.
5. **CERES is a modern, monthly constraint**, not a pre-industrial daily observation (§7). The CERES-derived panels test that the forcing construction worked; only the AMIP–CERES and CRUNCEP–CERES panels compare independent products, and even those span different epochs.

9 Recommendation & synthesis

Reading global CMPI/RMSD as a guide rather than a gate (the PI-vs-PD offset is expected), the picture is:

- The **melt-pond / albedo family (06A–06D, etc.)** are *clean*: they improve sea ice where they should and leave everything else essentially untouched. Their imbalance gain is modest (5–20%) but it comes with no unexpected collateral — safe building blocks.
- The **ocean-mixing / coupling family (06O/T)** delivers the only large imbalance reductions (30–38%) *and* genuinely improves the ocean interior and sea ice — but the cooling leaks into the atmosphere globally (incl. tropics), which is degradation in places the parameter should not reach. That makes them promising but not yet "free."
- **06V** pushes too far: the `ENTSTPC3` convection change degrades `tas/zg` everywhere — not recommended as-is.

Best-balanced single steps: **06P** (−29%, well-localised) and **06H** (−20%).

Synthesis — and where the warming lever comes from. The two arcs join here. The imbalance can only be cut substantially by cooling levers (ocean mixing/coupling), so every combo tried so far overshoots into cold bias. But §6 shows the shared Siberia/N. Canada cold lobe is *not* a tuning artefact, and a purpose-built clean no-LPJG CORE3 reference (§6.3) now *quantifies* it: of the $\approx -4.5\,\mathrm{K}$ Siberian DJF bias vs ERA5, $\approx -3\,\mathrm{K}$ is the LPJG forest-cover collapse (via snow-albedo in spring *and* surface roughness in winter, §6.5) and $\approx -1.4\,\mathrm{K}$ is a model-intrinsic over-strong stable-BL inversion (§6.7). The clean run is *better* than the Baseline globally (RMSD 1.514 vs 1.717 K) at *near-unchanged* imbalance ($+0.2\,\mathrm{W/m^2}$, within drift) — indicating both are *warming levers largely orthogonal to* the TOA tuning. That reframes the path forward:

1. First correct the atmospheric forcing and re-spin LPJG with a verified common binary (§14.3, §16). The CERES experiment (§7) shows that net-radiation correction alone is insufficient; tune tree–grass competition only if a residual remains under the corrected forcing. Do not relax the already-adequate summer establishment gates.
2. Attack the residual with *enhanced stable-BL mixing* in OpenIFS (§6.7) — an independent knob that relaxes the surface-trapped winter inversion.
3. On that warmer baseline, apply a *moderate* mixing/coupling setting (06O-like, not 06V) for the imbalance, plus the clean melt-pond changes — the cooling it adds is then offset rather than additive.
4. Re-judge on the *pattern* of CMPI change (§6.3 method), not the global value, since the PI-vs-PD offset is expected.

In short: don't chase the imbalance with ever-colder atmospheres; remove the structural cold bias first, then the moderate cooling needed for the imbalance becomes affordable.

9.1 2026–07–24 coupled follow-up from the CRUNCEP3-initialized branch: the year-50 land–ocean split refines the path

After the earlier report structure was drafted, the coupled CRUNCEP3-initialized branch (alias "080a") reached 50 years, which adds an important constraint on the tuning strategy. The new result is *not* a simple hemisphere-wide cold drift. By year 50 the system shows a marked **warm-ocean / cold-land split**, while boreal tree cover continues to retreat — most strongly over Siberia / East Siberia.

Two points matter:

1. The older LPJG story remains necessary: the spin-up / competition state is still part of the problem, and the vegetation levers are real.
2. But the 50-year coupled adjustment shows that a *regional continental-land* climate failure also develops, so vegetation-only retuning is not sufficient.

The finished year-50 diagnostics make the asymmetry explicit. Relative to year 0, NH45+ `TREEFPC` declines from 0.336 to 0.215, but the strongest loss is concentrated over Siberia / East Siberia. In the Siberian box (55–75°N, 60–180°E) `TREEFPC` drops by −0.188, and in East Siberia (55–75°N, 90–160°E) by −0.232, accompanied by a strong drop in `AGDD5` (−89 and −133 respectively). At the same time, NH45+ aggregate `AGDD5` actually *increases*, and Scandinavia / Canada also warm in thermal-growing metrics. So the problem is not “the NH is too cold”; it is that **boreal continental land, especially East Siberia, remains too cold relative to the oceanic state**. This is exactly what one sees in the year-50 seasonal-temperature maps against the CRUNCEP3 reference: ocean-adjacent sectors are warm-biased while large boreal land sectors remain cold.

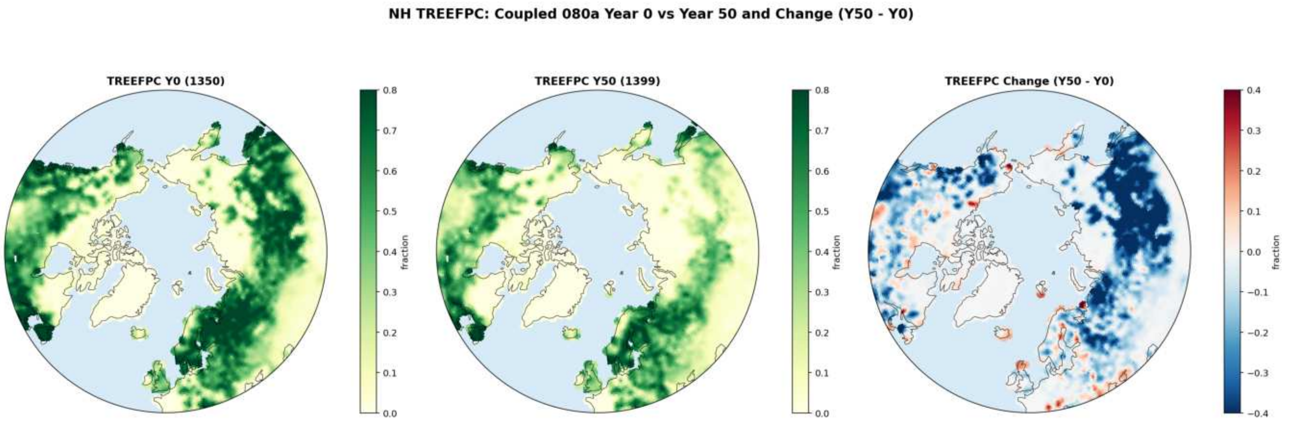
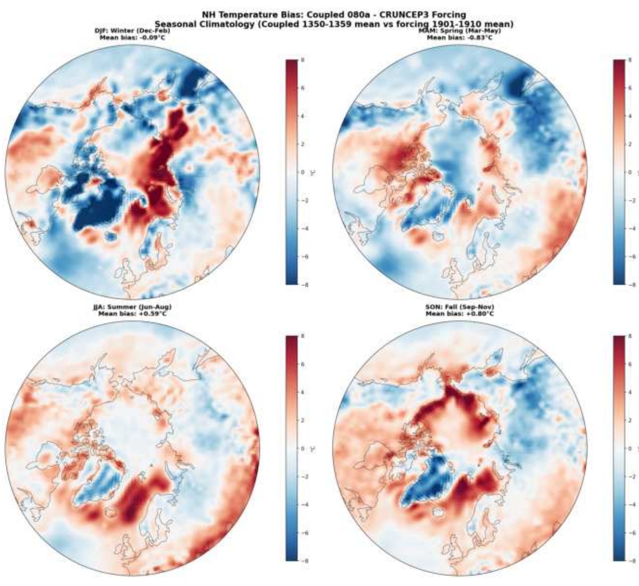
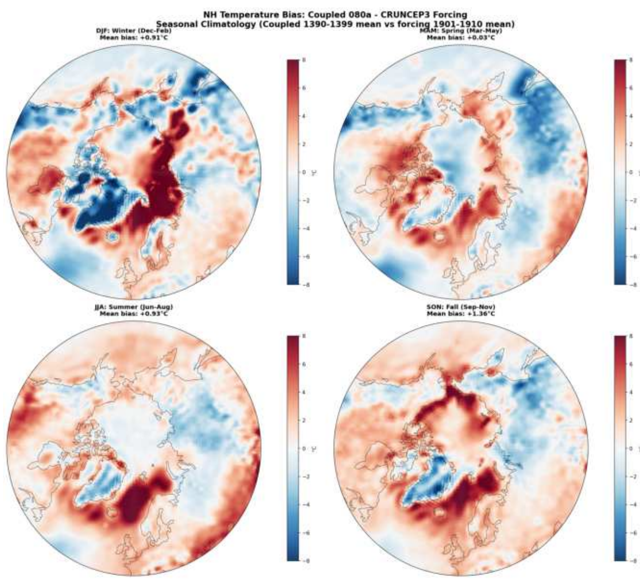


Figure 20: Coupled CRUNCEP3-initialized branch year-50 follow-up: NH `TREEFPC` at year 0 (1350), final year (1399) and change (Y50−Y0). The retreat is broad, but the largest loss is concentrated over Siberia / East Siberia.



(a) Early coupled decade: 1350–1359 vs CRUNCEP3 forcing



(b) Final coupled decade: 1390–1399 vs CRUNCEP3 forcing

Figure 21: Seasonal NH T2m bias of the coupled CRUNCEP3-initialized branch against CRUNCEP3 forcing. The year-50 state is not a uniform NH cooling; instead it shows a land–ocean split, with boreal continental land still cold while oceanic sectors are relatively warm.

This follow-up changes the recommendation in one practical way: the next runs should **fix the physical coupled branch first, and only then add LPJ compensation**. In other words, the "warm-floor" concept of this report still holds, but the relevant floor is now more specific: **a warmer boreal land floor**, not merely a warmer NH mean. The cleanest next matrix is therefore:

1. **06T-only** physical branch: `oasis3mct.time_step=3600`, `use_momix=True`, `momix_kv=0.01`, `kpp_av0=0.003`, `kpp_kv0=0.003`, `kpp_kvbckg=5e-06`.
2. **06V-only** physical branch: 06T plus `ENTSTPC3=1`.
3. Best of 06T/06V plus **07A** (`BNS greff_min=0.035`).
4. Best of 06T/06V plus **07A+07C** (`C3G pstemp_low=12`).

The year-50 evidence also explains why **LPJ-only tuning should not be the first move**. The all-tuned coupled NH bias panels (seasonal and annual) show that the coupled atmosphere response already differs strongly among branches even before the LPJ compensation is evaluated. If the physical branch leaves East Siberia cold, then LPJ-only changes risk compensating the wrong climate state rather than fixing it. Conversely, if a 06T/06V-type branch can hold the global net-radiation target while reducing the East-Siberian land cold pool, then the LPJ levers can be used for their intended role — retaining boreal trees — rather than masking a regional coupled-climate defect.

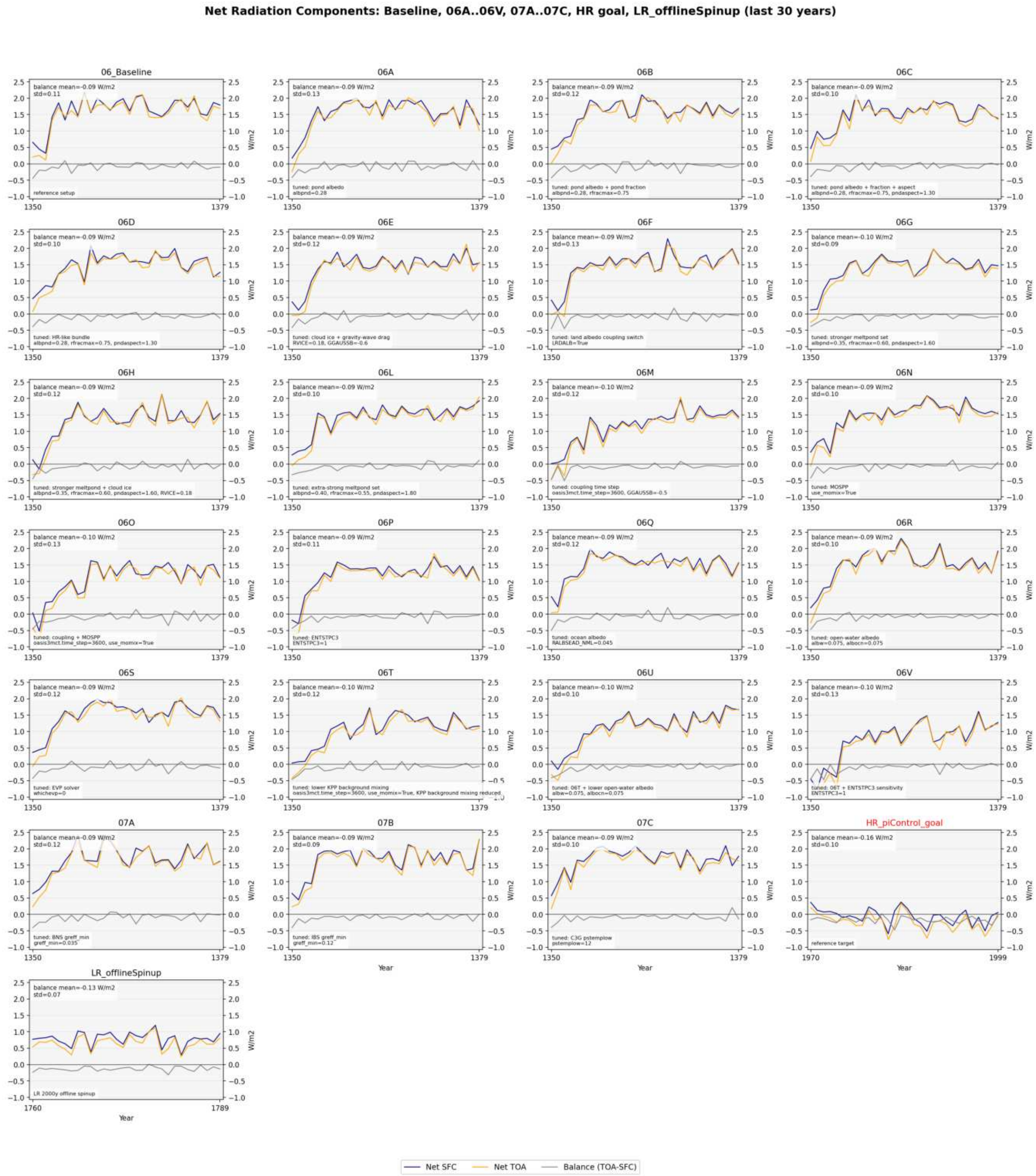
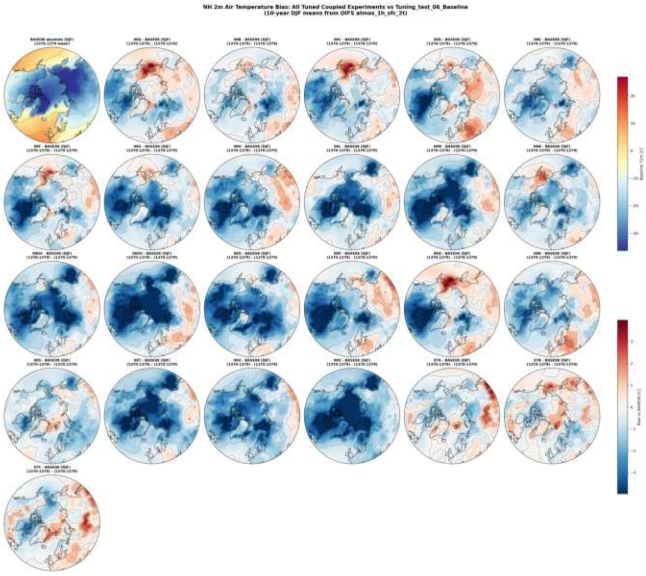
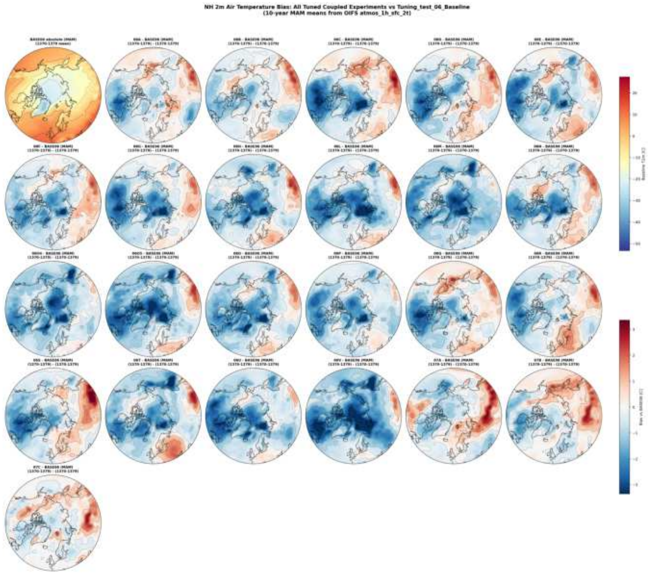


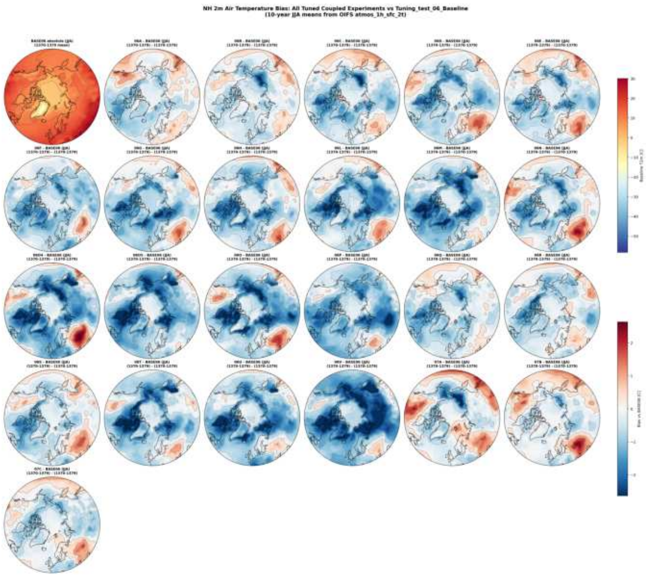
Figure 22: Global net-radiation history used to define the next physical branch. The preferred candidates remain the late 06-series branches that move closest to the target while staying within a plausible tuning envelope; the follow-up now adds the requirement that they must also improve boreal land climate, not only the global radiation budget.



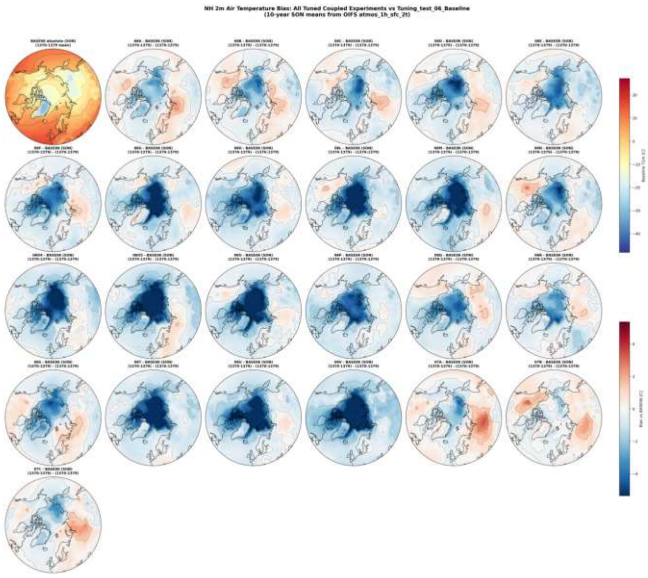
(a) DJF



(b) MAM



(c) JJA



(d) SON

Figure 23: All tuned coupled experiments relative to 06_Baseline, resolved by season (10-year means). These panels help separate branches that reduce the global radiation problem by globally cooling everything from branches that have a more usable boreal-land signal.

10 Work plan: a systematic path to a warm-floor, surface-balanced PI control

Two revised targets. (i) net *surface* energy balance $|F_{\mathrm{sfc}}| \lesssim 0.3\,\mathrm{W/m^2}$ — *not* TOA. Surface and TOA net flux track to within $\approx 0.1\,\mathrm{W/m^2}$ in every run (§4.1); they are one problem, so a surface-zero target is equivalent to TOA $\approx -0.1\,\mathrm{W/m^2}$, marginally stricter than TOA$=0$. (ii) global 2 m-T bias vs ERA5 $\approx -1.3\,°\mathrm{C}$ — the physically-expected PI-minus-present-day offset, *not* zero: a run that reaches $-1.3\,°\mathrm{C}$ is *correct*, not cold-biased.

Ordering principle. The imbalance can only be cut by cooling, and every cooling lever over-cools if applied to today's cold-biased base. So we *first* raise the warm floor with two structural fixes that are *orthogonal to the TOA balance* (they change the land surface and BL mixing, not the radiation budget — §6.3 showed the no-LPJG floor sits at *near-unchanged* imbalance, $+0.2\,\mathrm{W/m^2}$ within drift), *then* spend a *moderate* cooling lever to reach surface balance, now affordable because the floor is ~ 3–$4\,\mathrm{K}$ warmer where it was worst. Phases 1A/1B are independent (parallel); Phase 2 depends on both; Phase 3 gates.

10.1 Phase 1A — Restore boreal tree cover (forcing first; LPJG residual second)

Ordering updated by Part IV. This phase was originally written as an LPJG-competition tuning step. The controlled transfer test now requires the opposite order: correct/regenerate the atmospheric forcing, re-spin with one LPJG binary, and tune competition only against the remaining vegetation residual (§14.3, §16).

Literature & observational basis — what realistic trees require (so we don't green the tundra). The observed arctic/boreal treeline sits near the July 10 °C isotherm (Köppen), refined by Körner & Paulsen (2004) to a growing-season mean of $\sim$ 6–7 °C over $\sim$ 90 days, with the northern boreal limit at $\lesssim$ 30 days $>$ 10 °C. **Crucially, LPJG's current gates are already the standard, treeline-calibrated values** (Smith et al. 2001; Sitch et al. 2003 — byte-identical to the canonical `global.ins`): BNE/BINE `twmin_est`$=$ 5 (already *below* the 10 °C isotherm) and `gdd5min_est`$=$ 500 sit essentially *at* the observed treeline — so loosening them risks trees north of the real treeline. Two further realism facts constrain the fix: (i) NE Siberia's forest is ~80 % *larch* (deciduous `BNS`), because evergreen conifers cannot survive its extreme winters (Kharuk et al.; Herzschuh 2020, *Glob. Ecol. Biogeogr.* 29, 198–206, the deciduous-vs-evergreen distribution paper) — LPJG already encodes this (BNE/BINE `tcmin_surv`$=$ -31 excludes them; BNS has *no* cold gate and `gdd5min_est`$=$ 350), so the realistic tree there is larch, and it is *already permitted*. (ii) Closed taiga tree cover is $\sim$ 0.4–0.7 (MODIS VCF), falling below 0.2 at the taiga–tundra ecotone — the spatial shape cvh must reproduce.

Larch caveat. The realistic NE-Siberian tree is *deciduous* larch, whose leafless winter canopy masks little snow — so the spring *albedo* mechanism is genuinely weaker over the larch zone (though it retains full force over the evergreen west Eurasia / N. America). This only re-weights the two mechanisms, it does not soften the conclusion: the *roughness* mechanism survives (even leafless, larch is $\gg$ rougher than grass/tundra), and the seasonality fingerprint (§6.4) shows the LPJG cold effect is co-equal in the *sunless* Nov–Jan months — i.e. the dark-season roughness/inversion part is what carries the larch zone.

The coupled winter cold is a self-inflicted trap, not the trigger — *for the evergreens.* **This paragraph applies to BNE/BINE only**: as §31.1 shows, larch (BNS) has no cold gate at all, clears its GDD gate on 84 % of NE-Siberian cells, and is lost to *growing-season competition with grass* (5.8$\times$ its LAI) rather than to any gate. Since NE Siberia is ~80 % larch, the mechanism below does not explain the region that matters most. At the cells where prescribed veg has forest but LPJG does not (1785 boreal-land cells), the growing-season gates are already cleared in every run (JJA $\geq$ 12 °C, GDD5 $>$ 500); only the winter `tcmin_est`$=$ -30 evergreen gate binds. But against the *forested* WITHOUT-LPJG climate it excludes evergreens on just 23% of cells ($\approx$ERA5's 22%), vs 43% in the degraded WITH-LPJG run — the missing forest's own -3.3 °C winter cooling doubles the exclusion. So it is a **bistable trap** seeded by the under-forested spin-up, not a climate trigger.

The feedback-free tests now constrain the order. The full CRUNCEP–AMIP forcing change reproduces most of the regional loss with unchanged parameters (§14.3); the four-arm experiment shows that CERES net-radiation replacement removes only a small part of the NH `treeFrac` contrast (§7). Trees do establish and fire is not involved, but those facts do not isolate a competition error. Boreal `greff_min`, C3-grass competitiveness and cold-tree `pstemp_low` remain candidate levers only for a residual measured after a forcing-consistent re-spin. Phase-1B winter warming (§10.3) remains an independent test on the coldest evergreen-gated cells.

Why not just relax the gates? The summer gates are not binding (above), so loosening them cannot help; and the *one* binding gate — the winter `tcmin_est` — correctly excludes evergreens from the extreme-cold larch zone, so relaxing it would plant *unrealistic* evergreens where even ERA5 says larch. The fix is therefore the winter *climate*, not the limits. (Keep `iftreefracca 1`, `global.ins`:150: cvh is the summed tree *crown area*, so it tracks establishment success directly.)

Screen & deploy. First run the CRUNCEP and corrected-AMIP arms with the *same* LPJG binary and unchanged parameters. Only then screen any competition knobs on the cheap standalone spin-up, targeting the MODIS-VCF pattern ($\sim$ 0.4–0.7 closed taiga, $<$ 0.2 at the ecotone, larch winning the cold NE) and ~~a closed canopy (FORESTFPC> 0)~~[3] — tree NPP $>$ grass NPP — *not* merely a higher global mean. On a passing screen, re-do the coupled LPJG spin-up, regenerate the restart, and carry it into Phase 2.

10.2 Execution: the clean AMIP–noLPJG forcing generator

The decisive test (§6.9) needs a *feedback-free* climate to drive it — reusing coupled-LPJG output would carry the very cold bias we are testing. We therefore built a dedicated **OpenIFS–AMIP** generator at TCO95 (`oifsamip-cy48`, git-aligned to `awiesm3-develop`): AMIP prescribes observed SST/sea-ice, so the atmosphere has no coupled ocean/LPJG feedback. Building and running it required fixing three esm-tools bugs (a FESOM-less crash, missing XIOS/coupling compile flags, a missing WAM date-preprocessing recipe — all committed &

[3] **Void diagnostic.** `FORESTFPC` sums only PFTs with `landcover==FOREST` (`commonoutput.cpp`:1143), i.e. the *managed-forest* land-use class; natural trees are counted in `TREEFPC` via the `NATURAL` branch. In this PI configuration nothing is allocated to FOREST (`Natural_sum`$=$ 0.684, `Total`$=$ 0.665), so `FORESTFPC`$\equiv$ 0 *by construction* while `TREEFPC`$=$ 0.323. It is a land-use bookkeeping artifact and says nothing about canopy closure.

pushed; detail in App. B). A 10-yr ∼PI decade (1870–79 observed SST, GHG fixed 1850) produced the daily forcing. *One methodological catch worth recording:* the first re-spin came out barren everywhere because the XIOS field-def de-accumulated fluxes over the 6 h `freq_op` cadence instead of the true `NFRHIS`×`TSTEP`= 1 h period, making `rsns/rlns/pr` exactly 6× too low (net SW 28 vs 166 W/m^2) — fixed by a validated ×6 patch and, at root, by templating the divisor from esm-tools. **Expected payoff** of the whole Phase 1A (§6.6): ≈ −3 K off the Siberian DJF bias, ≈ −0.2 K global RMSD, at near-unchanged imbalance.

10.3 Phase 1B — Enhance stable-boundary-layer mixing (OpenIFS)

Change. The residual ≈ 1.4 K surface-trapped inversion (§6.7) is relaxed by strengthening stable-BL turbulent mixing in the EDMF K-diffusion scheme via the "long-tail" (LTG) stability functions — lengthening the tail keeps mixing non-zero at large Richardson number and weakens the over-strong surface inversion. There is no namelist for the stable branch, so this is ∼2 single-line OpenIFS source edits (minimum stable mixing length `ZLMIN` and the LTG coefficient `ZCB`) plus recompile; the exact file:line targets and secondary companions are in App. B. *Bear the documented side-effects in mind* (§6.7): long-tail mixing beyond the observational range can deepen cyclones and degrade low cloud, so verify the storm-track/low-cloud climatology, and pre-test cloud phase / LW↓ (Pithan et al. 2014) since the residual may be partly a cloud-microphysics rather than turbulence error.

Test. Change *one* knob first (`ZLMIN`) to keep attribution clean; scan 2–3 values with short (1–5 yr) coupled sensitivity runs of the no-LPJG reference. **Gate:** the DJFM low-level inversion bias (`djf_second_driver.py`) moves from +1.4 K toward ERA5's +4.2 K level and the residual boreal cold shrinks, *without* eroding legitimate inversions or warming the tropics (judge globally via `part8`/CMPI pattern). **Risk:** over-mixing warms too broadly / degrades low cloud — mitigate with small steps and global CMPI-pattern judging.

10.4 Phase 2 — Moderate imbalance tuning on the warm floor

With the floor raised (1A+1B) and global skill already better than Baseline (RMSD 1.514, CMPI 0.738; §6.3), spend a *moderate* cooling lever to bring the *surface* budget to zero. Build the combined config in layers, cleanest-first:

layer	lever(s)	ΔTOA [W/m^2]	raw bias [K]	rationale
base	melt-pond (06A–06C; opt. 06G)	−0.08 to −0.26	−1.0 to −1.3	collateral-free
localised	06P (`ENTSTPC3`)	−0.44 (−29%)	−1.36	best efficiency ∼ 1.05 W/m^2/K
moderate	06O (`1hcpl`+`mospp`)	−0.55 (−36%)	−1.77	improves ocean/ice; *not* 06V
(spare)	+`kpplow` = 06T	−0.58 (−38%)	−1.79	if more cut needed

Why this stops short of 06V. 06V (`entstpc3`+`kpplow` all stacked) gives the biggest cut (−57%) but overshoots to −2.1 K and degrades `tas/zg/rlut` in *every* region including the tropics. The base+06P+06O stack cuts ∼ 1.0–1.1 W/m^2 of surface imbalance (reference surface +1.52 →∼ 0.4–0.5; iterate the coupling strength or add `kpplow` to close the last ∼ 0.4). **Cold-bias accounting:** the ∼ −1.7 K raw cooling of the coupling lever, applied on a floor lifted ∼ 3–4 K where the bias was worst (and ∼ 0.2–0.4 K globally), lands the global mean near the target −1.3 °C rather than the −1.8 to −2.1 °C the same lever produces today.

10.5 Phase 3 — Verification protocol and decision gates

Run the combined config 30 yr (evaluate the last decade). Judge with the `reval` toolchain *plus* the DJF driver diagnostics:

target	metric (tool)	accept
surface balance	`part2` "Surface total"	$\lvert F_{\mathrm{sfc}} \rvert \lesssim 0.3$ W/m^2 (TOA ≈ −0.1)
cold bias	`part8` 2 m bias vs ERA5	−1.3 ± 0.3 °C; RMSD ≤ ref. 1.51
multivariate skill	`part4` CMPI	≤ Baseline 0.745; *pattern*, not just global
no new regionals	CMPI by region (esp. tropics)	no family-typical global cooling leak
driver check	`djf_second_driver.py / ...roughness...`	inversion bias → ERA5; cvh restored

Iteration & critical path. Each axis adjusts independently (the floor fixes are orthogonal to the imbalance lever): too cold → back off coupling; imbalance too high → add `kpplow`; inversion too strong → raise `ZLMIN`. Phases 1A and 1B run in parallel now (both cheap/offline); 1A's coupled re-spin is the long pole, and the Phase-2 run starts once the re-spun restart and the chosen stable-BL build are both in hand.

The submission copy of all scripts, data, figures and this report lives in `/work/ab0995/a270270/analysis/investigation_awiesm3_high_lat_cold_bias_global_toa_positive/`: `data/MASTER_metrics.csv`, `data/cmpi_by_variable.csv`, `data/cmpi_by_region.csv`, `configs/`, `plots/`, `run_figures/<run>/`, `notebooks/`. Tool: `release_evaluation_tool2` (reval env); per-run diags via `part2_rad_balance.py` / `part8_t2m_vs_era5.py` / `part4_cmpi.py`.

Part IV

The atmosphere alone — attribution, and where the remaining error lives

*Parts I–III argued from coupled runs, in which atmosphere, ocean and vegetation errors are inseparable. This part removes that ambiguity with a **feedback-free OpenIFS–AMIP reference**: prescribed observed SST/sea-ice, prescribed vegetation, same OIFS branch as the coupled campaign. Two results follow. **(i)** With SST prescribed, the model still runs +0.67 W/m^2 out of TOA balance — that fraction of the imbalance cannot be ocean heat uptake and is a genuine atmospheric radiative error, which places a* floor *beneath what any ocean-side lever can achieve. **(ii)** The boreal* summer *cold bias survives in full without ocean or LPJ-GUESS, so it is the atmosphere's — and it is a surface* shortwave *deficit, not the boundary-layer problem of §6.7. Together these redirect the work plan: the next tuning should happen on AMIP, where both remaining errors live and where an iteration costs ∼7× less than a coupled run.*

11 The feedback-free AMIP reference (2026–07–27/28)

`amip_pi_base`, `/work/bb1469/a270092/runtime/oifsamip-cy48/`: OpenIFS TCO95L91, GHG/ozone/ aerosol fixed at 1850 (`NCMIPFIXYR=1850`), driven by the earliest observed AMIP SST decade 1870–1879 — a self-consistent ∼PI atmosphere directly comparable to the PI-control coupled runs. 10 yr complete in 2:44 on 14 nodes.

Two configuration decisions make it a valid counterpart to the coupled model, and both were wrong in the earlier `_lpjgforcing` run that Part II used (§10.2):

- **OIFS branch.** The oifsamip tree was on `local_combined_fixes` @ `1303a90`; the coupled campaign runs esm-tools version `48r1v5` = `movcav-landice+co2-concdriven`. The branches differ in *atmospheric physics*, not merely coupling plumbing (`suecozv`, `updecozv`, `suecrad`, `suphec`, `srfi_mod`, `surfece`, `voskin_mod`). Rebuilt at `f3ccacb` — the last non-debug commit, and the same state as our coupled tree.
- **Sea albedo.** The old runscript carried `TMPRALBSEAD`= 0.075. The coupled runs never set it, so they use the source default `RALBSEAD`= 0.06 (`susrad_mod.F90`:107); OASIS exchanges only the *ice* albedo (`sia_feom`→`A_Ice_albedo`) so OpenIFS computes open-water albedo itself in coupled mode too. A 0.015 albedo offset over ocean is worth order 1 W/m^2 of reflected SW — comparable to the quantity being attributed. Removed.

11.1 Which budget the target applies to: snow enthalpy

`part2_rad_balance.py`:255–268 computes

$$F_{\text{sfc}} = \text{ssr} + \text{str} + \text{sshf} + \text{slhf} - \text{sf} \cdot \rho L_f, \qquad F_{\text{TOA}} = \text{tsr} + \text{ttr}.$$

The snowfall-enthalpy term enters the *surface* budget only, and correctly so: it is an internal atmosphere↔surface transfer and cancels at the top of the atmosphere. Every net-TOA number quoted in this report is (tsr+ttr)/Δt_{acc}, i.e. exactly part2's TOA total.

	TOA	SFC w/o snow	snow enthalpy	SFC (part2)	TOA−SFC
AMIP 1872–79	0.667	1.441	0.925	**0.516**	+0.152
09C 1370–79	0.918	1.897	0.944	**0.953**	−0.035
080a 1370–79	1.682	2.588	0.823	**1.765**	−0.083

The term is 0.82–0.94 W/m^2 — *larger than the entire remaining gap to target.* Without it the surface budget misses TOA by ≈ 1 W/m^2; with it, 080a closes at −0.083, reproducing the ≈ −0.09 quoted in §4.1. **Convention warning:** the work plan's acceptance target is on the *surface* (§10), where the same runs read +0.52 (AMIP), +0.95 (09C), +1.77 (080a). State which convention is in use whenever these numbers are compared. *Open item:* AMIP's closure (+0.152) is 2–4× worse than the coupled runs'. Small against the +0.67 signal, so the attribution is unaffected, but unexplained — suspect the prescribed SST/sea-ice surface tile. Chase it before quoting the AMIP floor to better than ∼ 0.15 W/m^2.

11.2 Result I — roughly 45 % of the imbalance is atmospheric

AMIP net TOA = +0.67 $\mathbf{W/m^2}$ (1872–79, after discarding two soil spin-up years; per-year 0.13–0.92; `tsr`= 240.3 W/m^2 confirms the 3600 s accumulation divisor). With SST prescribed this *cannot* be ocean heat uptake.

Against a coupled baseline of +1.44 (080a), that is ≈ 45 % of the coupled imbalance. The best branch in the entire campaign (09C, +0.92) now sits only 0.25 W/m^2 above this floor: of the 1.08 W/m^2 still separating it from the HR goal, ≈ 0.83 is atmospheric and only ≈ 0.25 is reachable by ocean-side tuning. **This is the campaign's most consequential number** — it says the lever family of Part I is nearly exhausted.

Caveat. AMIP runs over *observed* SST/ice while the coupled model runs over its own biased state, so +0.67 is "what this atmosphere does over a correct ocean", not a strict lower bound on the coupled value. But for a ∼PI configuration, where the real world was near equilibrium, the expected AMIP TOA is ≈ 0 — so most of the +0.67 is model error.

11.2.1 Where does it live? Nowhere in particular — the global budget is already right

Decomposing the +0.67 against CERES-EBAF (`amip_toa_decomp.py`) was expected to localise it. It does the opposite:

AMIP global	model	CERES	diff	band	net TOA	contribution
absorbed SW	240.13	240.5	−0.37	90S–60S	−90.5	−6.06
OLR	239.46	240.2	−0.74	60S–30S	−22.1	−4.04
planetary albedo	0.2947	0.293	+0.0017	tropics 30S–30N	+42.6	+21.30
SW CRE	−44.40	−45.4	+1.00	30N–60N	−21.8	−3.98
LW CRE	+24.88	+25.8	−0.92	60N–90N	−97.7	−6.55
net CRE	**−19.52**	**−19.6**	**+0.08**	**sum**		**+0.67**

The cloud radiative effect matches observations to 0.08 $\mathbf{W/m^2}$ and the planetary albedo to 0.002. The AMIP and 09C zonal profiles lie almost on top of one another: the coupled model's radiation *is* the atmosphere's.

~~The band contributions reproduce the textbook meridional structure with no band carrying an anomalous imbalance, so the Hyder et al. atmospheric-SW hypothesis does not appear as a gross cloud error and there is no region to fix.~~ **That conclusion was an artifact of the comparison.** The bands above were compared to *each other*, and only the *global* CRE to CERES. Differencing each band *against CERES* (§11.2.2) shows the global agreement is a **cancellation of large regional errors**, and that the Southern Ocean carries most of the imbalance.

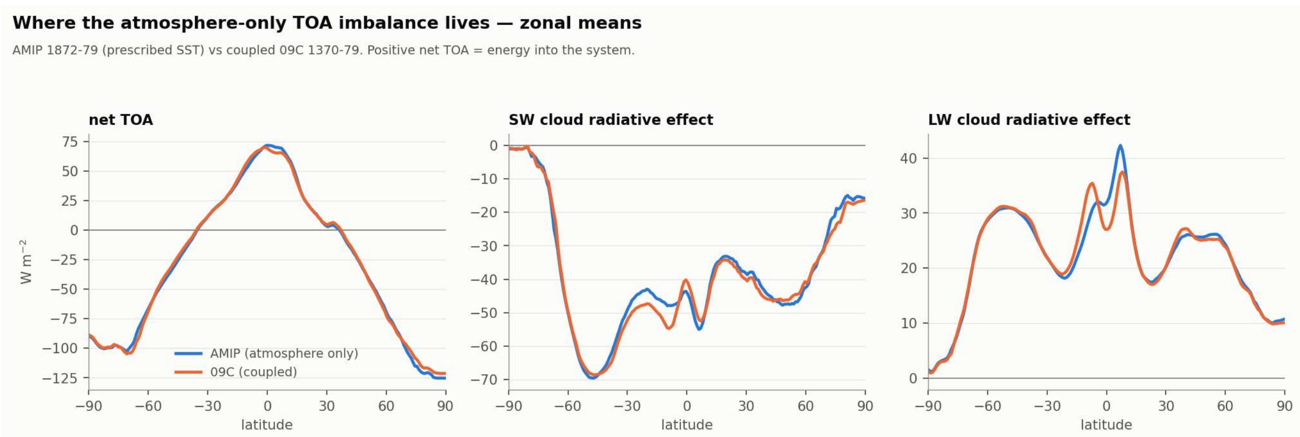


Figure 24: Zonal-mean net TOA and cloud radiative effect, AMIP (atmosphere only) vs coupled 09C. The curves are nearly indistinguishable, and the imbalance is the small residual of a large tropical surplus against extratropical deficits — not a localised error.

11.2.2 Differenced against CERES per band: the Southern Ocean carries it

band	model	CERES	diff	area	**contribution**
90S–65S	−97.0	−104.0	+7.03	4.8 %	**+0.34**
SO 65S–45S	−49.5	−55.3	**+5.81**	9.8 %	**+0.57**
45S–30S	−5.6	−1.4	−4.20	10.4 %	−0.44
tropics 30S–30N	+42.6	+45.1	−2.51	50.0 %	−1.26
30N–45N	−4.8	−5.7	+0.89	10.4 %	+0.09
45N–65N	−50.1	−54.1	+4.02	9.8 %	+0.40
65N–90N	−106.3	−106.1	−0.15	4.8 %	−0.01

The global budget is right by cancellation, not by being right. The Southern Ocean (65–45°S) runs +5.8 W/m^2 too much absorbed energy and contributes **+0.57** W/m^2 to the global mean — on its own, comparable to the entire +0.67 imbalance. The cause is a cloud *deficit*: cloud area 83.1 % against CERES's 89.7 % (−6.6 pp) with SW CRE −60.3 against −68.1 (+7.8 W/m^2 too little reflection). **This is precisely the Hyder et al. (2018) signature**, and it vindicates the compensating-error worry of §5: the ocean-mixing levers of Part I have been removing a Southern-Ocean warm bias that is substantially *atmospheric* in origin.

Consequence for the plan. The +0.67 is not irreducible after all — but neither is it fixable by a global knob, since the global mean is already correct. It is fixable *regionally*, and the two largest regional errors have *opposite* sign at TOA, which is what makes the round-10 design work (§16). Re-framing the campaign arithmetic:

	net TOA	= atmosphere +	coupled-state
080a (untuned coupled)	1.44	0.67 +	**0.77**
09C (06V + newSeaIce)	0.92	0.67 +	**0.25**

The ocean-side tuning has already delivered about two-thirds of what was available to it. Whether the HR piControl goal of −0.16 is even the right target for this configuration should be revisited, given that ≈ 0.67 of any coupled value is an atmosphere matching observations. *Revisited and answered on 2026–08–07: it is not the right target, and the goal is now* 0 *(§37).*

11.3 Result II — the boreal *summer* cold bias is atmospheric

Monthly 2 m-T bias against CRUNCEP3 (1901–10), land only, cos-lat weighted:

series	Boreal JJA/ann	Siberia JJA/ann	E. Siberia JJA/ann
AMIP (atmosphere only)	−1.64 / −1.05	**−2.16** / −0.89	**−2.24** / −0.49
080a (baseline, old sea ice)	−0.36 / −0.90	−1.41 / −1.36	−1.80 / −1.48
09A (baseline + newSeaIce)	−1.24 / −2.06	−2.19 / −3.00	−2.47 / −2.89
09B (06T + newSeaIce)	−1.56 / −2.33	−2.50 / −3.11	−2.72 / −3.04
09C (06V + newSeaIce)	−1.93 / −2.64	−2.54 / −3.08	−2.69 / −3.05

The atmosphere alone reproduces essentially all of 09A's Siberian JJA bias (−2.16 vs −2.19) and most of 09B/09C's. **Removing the ocean and LPJ-GUESS does not remove the summer cold bias.** The cold-season half behaves oppositely — AMIP's Siberian *annual* bias is only −0.89 against −3.0 coupled — so September–April remains the coupled/vegetation problem of Part II, and JJA is now a separate, atmospheric one.

11.3.1 Correction (2026–07–29), itself superseded 2026–07–30: the reference period matters, but far less than estimated

Read §17.2 first. The indirect estimate in this subsection put the Siberian JJA period offset at ≈ 1.12 K and concluded the genuine boreal bias was ≈ −1.0 K. The `amip_presentday` run has since *measured* that offset: **+0.42 K**. The conclusions below are therefore **withdrawn** — the boreal bias is ≈ −2.0 K, close to the original figure. The reasoning is retained because the method error is instructive: it applied an *observed* epoch change to a model that does not reproduce that change. The number above is scored against **ERA5 1990–2014**, while these runs use **1870s observed SST with 1850 GHG**. Roughly 130 years of greenhouse warming therefore sits inside a quantity the campaign has been treating as model error — and tuning against. Measured from the DKRZ ERA5 pool (`scripts/analysis/era5_period_offset.sh`), Siberian JJA over the same box used for evaluation:

ERA5 period	offset vs 1990–2014
1979–1989	$-\mathbf{0.670}$ K
1940–1969	$-\mathbf{0.790}$ K

The interannual sd is 0.578 K, giving SE = 0.209 K on the 1979–89 vs 1990–2014 difference ($t \approx 3.2$): the offset is real. ERA5 stops at 1940, so chaining through HadCRUT5's global series (1870s → 1990–2014 = +0.756 K) scaled by ERA5's own Siberian-JJA-to-global amplification over the overlapping window (×1.48) implies a Siberian JJA period offset of ≈ 1.12 K, leaving a **genuine model cold bias of ≈ −1.0 K rather than −2.2 K**.

This changes the target, not the diagnosis. The attribution above is unaffected — it is a model-versus-model comparison (AMIP against coupled 09A) in which any reference offset cancels, so "the JJA bias is atmospheric" stands. What changes is *how much* of it we should try to remove: closing the full 2.2 K would manufacture a ≈ 1 K *warm* bias in the coupled pre-industrial state.

Caveats. The ERA5 box is unmasked whereas the model figures are land-masked; land warms faster than ocean, so the true land-only offset is if anything larger and this correction is conservative. The ×1.48 amplification is extrapolated back from a single window, and HadCRUT5's 1870s global mean rests on sparse coverage. **The same flaw applies to the CERES-based energy targets** (§13.1, §11.2.2): CERES EBAF here is the 07/2005–06/2015 climatology scored against an 1870s model, so the +8.08 W/m^2 SO SW CRE gap and the +0.53 W/m^2 imbalance carry an analogous, *as yet unquantified*, period offset. The decisive fix for both is a single present-day AMIP leg (1989–2015, transient GHG) scored directly against ERA5 1990–2014 and CERES 2005–2015; the AMIP SST forcing already covers 187001–201512.

11.3.2 Attribution (2026–07–29): the boreal forest is a *temperature* problem

The forcing-transfer test (§14.3) showed that swapping CRUNCEP → AMIP forcing alone costs −54 % of Siberian `TREEFPC`, but could not say *which* variable carried it: radiation and temperature both feed NPP, and therefore both feed the `greff_min` growth-efficiency mortality that §6.9 identifies as the actual killer (§6.9). A colleague's spin-ups separate them by holding CRUNCEP temperature fixed and swapping *only* the radiation to CERES (`LPJG-SPINUP_2000Y_TCO95_CORE3_CRUNCEPandCERES`). **`AGDD5` is identical between those two arms** (775.8 Siberia, 748.0 E. Siberia, 1157.5 NH) — the control confirming that only radiation moved.

region	radiation only (CRUNCEP→CERES)	full swap (CRUNCEP→AMIP)	radiation share
NH 45N+	−1.7 %	−25.1 %	**7** %
Siberia	−8.9 %	−54.3 %	**16** %
E. Siberia	−10.5 %	−66.0 %	**16** %

The radiation arm moves −21 W/m^2 while the full swap moves ≈ −32, so scaling linearly puts radiation at ≤ 25 %. The daily-variability variant is indistinguishable (−7.8 vs −8.9 %), so it is not a sub-daily distribution effect either. **Temperature carries ≈ 75–85 % of the boreal tree collapse.**

Consequence: prong A's radiation work will not restore the forest. Closing the *entire* Siberian SW deficit recovers at most a quarter of the tree loss. The forest depends on the ≈ 1.0 K temperature bias established in §11.3.1, not on the Southern-Ocean/energy target. These are two separate jobs and should stop being described as one.

The period mismatch also sits on the LPJG side. CRUNCEP3 spans **1901–2015** — twentieth-century climate — yet was used to spin up the *pre-industrial* vegetation state that the coupled 1850 model restarts from. Part of the 131-`GDD5` Siberian gap is therefore a real PI-versus-20thC climate difference rather than model error: 131 `GDD5` over a ≈ 120-day season is ≈ 1.1 K, of which the 1870s → 1901–2015 offset plausibly accounts for 0.3–0.45 K, leaving 0.6–0.8 K genuine — consistent with the ≈ 1.0 K derived against ERA5 1990–2014, since CRUNCEP's period is cooler than ERA5's. *Caveat:* the AMIP arm used the superseded forcing build, ≈ 0.9 K warmer over Siberian JJA, so the temperature share is if anything understated.

Does this shift the residual onto LPJ-GUESS's CRUNCEP calibration? Partly, but the corroboration attempted on the *temperature* axis fails: NCEP2 Siberian JJA is −0.118 K *colder* than ERA5 over 1990–2014 (paired by year, $t = -2.6$) — real but negligible, and the wrong sign for a story in which LPJ-GUESS grew its trees under an over-warm forcing. That is consistent with CRUNCEP3's construction: temperature derives from CRU, bias-corrected against stations, while *radiation* derives from NCEP, which is precisely where the +21 W/m^2 excess against CERES was measured (§13.1). The calibration mismatch sits on the radiation axis. The direct evidence remains the forcing-transfer test (§14.3), which does not depend on which variable carries it.

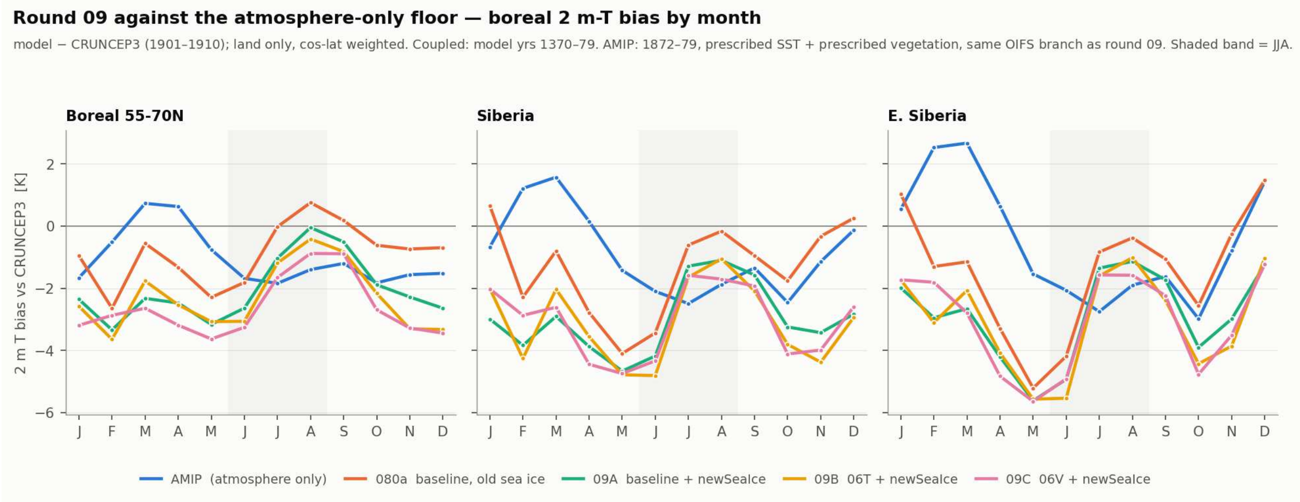


Figure 25: Boreal 2 m-T bias by month against CRUNCEP3, land-masked. The atmosphere-only AMIP curve (blue) sits at or below the coupled runs through JJA (shaded) but far above them in winter: **the summer bias is atmospheric, the cold-season bias is not**. *Caveat:* the "Boreal 55–70°N" panel is contaminated by the Greenland ice sheet (JJA snow depth 320 mm w.e. in that box vs 5.5 mm for Siberia); the Siberia and E. Siberia panels are clean and carry the argument.

Superseded numbers. ~~AMIP Siberia JJA −1.23, E. Siberia −1.59, i.e. indistinguishable from the coupled baseline, so coupling adds nothing in summer.~~ Those came from the earlier `amip_pi_forcing` run, built on the wrong OIFS branch and with `TMPRALBSEAD`= 0.075. The corrected build is ≈ 0.9 K colder in Siberian JJA; the conclusion strengthens (the atmosphere is now *colder* in summer than the coupled baseline) but the numbers changed.

12 Round 09 — new sea-ice thermodynamics (2026–07–27/28)

FESOM 2.7 `awiesm3-implicit-ice-surftemp` @ `db7dd9af` makes the ice surface-temperature solve implicit (dQ/dT-linearised about the last transmitted `ist`) and hands ownership of the ice surface energy balance to OpenIFS, which now runs a slab-conduction ice tile using FESOM-supplied thickness; FESOM ice growth becomes $Q_{\text{atmice}} = -a2ihf$. The pairing is verifiable at runtime in the OASIS `namcouple`: the 09 runs carry a sixth o2a field, `sit_feom`→`A_Ice_thickness`, absent from every 08 run. Run identities and the 09B parameter block are catalogued in the 2026–07–28 update at the head of this report; the quantitative comparison is below. `09C` was built and run on our own tree (`Tuning_test_09C_06V_CRUNCEPinit_newSeaIce`) to close the missing 06V corner.

run	net TOA [W/m^2]	global T2m [°C]	Siberia JJA [K]	Siberia ann [K]
080a baseline, old sea ice	1.442	13.606	−1.41	−1.36
09A baseline + newSeaIce	1.286	13.148	−2.19	−3.00
09B 06T + newSeaIce	1.171	12.794	−2.50	−3.11
08B 06V, old sea ice	0.935	12.347	−3.11	−4.09
09C 06V + newSeaIce	**0.919**	12.492	**−2.54**	**−3.08**

On the clean old-vs-new 06V pair (08B → 09C) the radiation barely moves (0.935 → 0.919) but the boreal **warms by +1.0 K in the Siberian annual mean**, +0.57 K in JJA, +0.15 K globally. **09C dominates 08B on both objectives simultaneously** — the first time in the campaign that has happened, and the "free warming" sought in §9. But the effect is *not* an offset: on the baseline (080a → 09A) the same code change goes the other way, cooling 0.46 K globally and 1.6 K in the Siberian annual mean. The sign depends on the ocean-mixing branch; no mechanism is established here, and pinning it down needs the surface energy budget over ice.

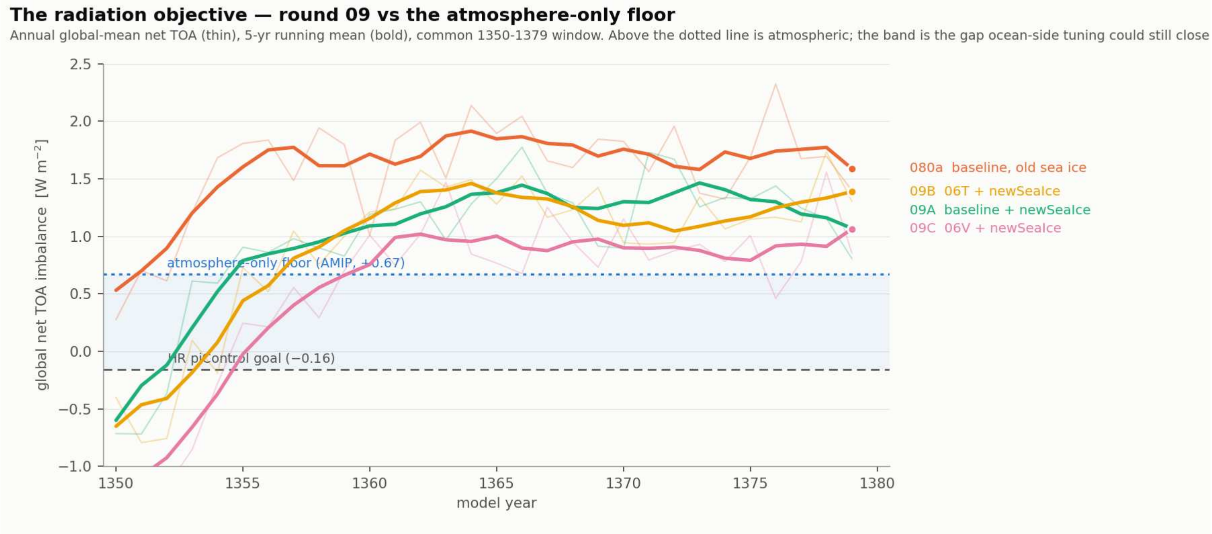


Figure 26: Annual global-mean net TOA (thin) with 5-yr running mean (bold), common 1350–1379 window. The dotted line is the **atmosphere-only floor** measured by AMIP (+0.67 W/m^2); the shaded band is the only part of the remaining gap that ocean-side tuning can still close. Everything above the dotted line is atmospheric.

Bookkeeping caveat. `sacct` shows a `Tuning_test_08A_06T_1hCPL_MOSPP_KPPLOW_CRUNCEPinit` completing all three chunks on 24–25 July, but no 08A directory exists on disk. 08A was the 06T-only physical control — Run 1 of the 2026-07-24 design note's own sequence — so within round 08 every surviving 06T-family entry (08C, 08E, 08G) carries an LPJ lever and the intended 06T-vs-06V comparison cannot be made cleanly. Round 09 supersedes this, but conclusions drawn from round 08 alone inherit the hole.

13 Why the summer bias exists — a shortwave deficit (2026–07–28)

Siberia JJA surface energy budget, land-masked, cos-lat weighted (AMIP 1872–79 vs CRUNCEP3 1901–10, which carries `rsns`/`rlns` and so permits a direct radiative comparison):

	AMIP	CRUNCEP3	Δ
2 m temperature	9.74 °C	11.92 °C	**−2.18** K
SW net at surface	155.1	188.3	**−33.2** W/m^2
LW net at surface	−49.2	−64.5	+15.2 W/m^2
net radiation			**−18** W/m^2

Supporting AMIP diagnostics: SW↓ 188.6, surface albedo 0.178 (`fal` 0.172), LW↓ 315.6, SH/LH −24.6/ − 56.8 (**Bowen** 0.43), cloud `tcc` 0.778 / `lcc` 0.505 / `mcc` 0.386, `skt−2t` +0.43 K, `swvl1` 0.343, snow depth 5.5 mm w.e.

The LW term is *compensation* (a colder, cloudier surface loses less longwave), not cause. It is *not* albedo: a realistic boreal-summer 0.13–0.15 instead of the model's 0.178 buys only 5–9 W/m^2. So SW↓ itself is too low. This is a different mechanism from the winter stable-BL inversion of §6.7, and it is not addressed by any lever proposed so far.

13.1 Re-measured against CERES: cloud, not clear-sky — and three times smaller

CRUNCEP3 is a reanalysis-derived forcing product, so the deficit was re-measured against **CERES EBAF Ed4.1** (satellite), which supplies all-sky *and* clear-sky net SW at the surface and so permits the decomposition $\Delta(\mathrm{SW_{net}}) = \Delta(\mathrm{SW_{net,clear}}) + \Delta(\mathrm{SW\ CRE_{sfc}})$ with no assumption about surface albedo or the direct/diffuse split.

JJA	Siberia land			NH 55–75°N ocean		
	model	CERES	diff	model	CERES	diff
SW net at surface, all-sky	155.1	166.3	**−11.2**	142.4	155.4	−13.0
SW net at surface, clear-sky	246.6	241.4	+5.3	241.9	252.6	−10.7
SW CRE at surface	−91.6	−75.1	**−16.5**	−99.5	−97.3	**−2.2**
cloud area [%]	77.8	69.6	**+8.2**	84.6	83.0	**+1.6**

Three results, all of which revise §13:

(i) It is unambiguously a cloud error. The decomposition closes exactly ($-11.2 = +5.3 - 16.5$), and clear-sky surface SW is if anything slightly *too bright*. Aerosol, water vapour and surface albedo are therefore all exonerated — and since PI aerosol is lower than present-day, the PI-vs-CERES period mismatch works *against* this finding, making it conservative.

(ii) The deficit is 11 $\mathbf{W/m^2}$, not 33. ~~Surface net SW over Siberian land in JJA is 33 W/m^2 too low.~~ That figure was measured against CRUNCEP3, which is itself $\approx 21\,W/m^2$ brighter than satellite — **CRUNCEP3 was the outlier, not the model**. An $11\,W/m^2$ summer deficit remains the right order to explain a $\approx 2\,K$ land cold bias, so the mechanism survives; only its magnitude changes.

Verified independently, because the original comparison had assembled the two sides on different grids with different land masks — a way to manufacture exactly this kind of difference. Redone with all three sources on the CERES 1° grid under a single land mask (`verify_ceres_cruncep.py`): CERES 166.7, AMIP 155.8 (-10.8), CRUNCEP3 187.6 ($+21.0$). Raising the land-mask threshold $0.5 \rightarrow 0.8$ changes these by $< 1\,W/m^2$, and CERES's own JJA-mean file gives 168.4 against 166.7 from the monthly climatology. The grid inconsistency was therefore not the cause.

Why CRUNCEP3 is bright splits cleanly by season: in JJA it is the downward flux ($234\,W/m^2$ in July against CERES's 209, the NCEP high-latitude brightness bias), whereas in spring it is the forcing file's fixed albedo (§14.1). CRUNCEP3 blends CRU *station* temperature with NCEP *reanalysis* radiation, which is the natural reading of why its T2m is usable while its SW is not.

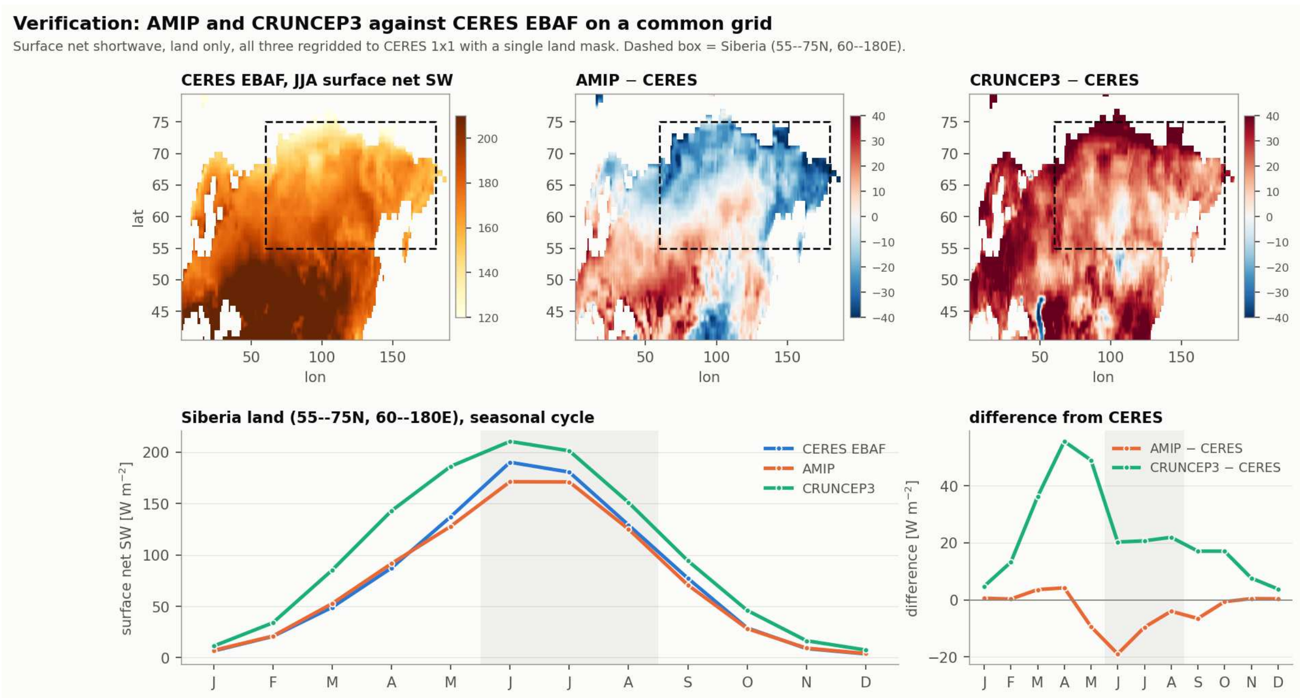


Figure 27: Surface net shortwave against CERES EBAF, all sources on a common 1° grid with one land mask. AMIP (orange) is $\approx 11\,W/m^2$ *dark* in JJA; the CRUNCEP forcing (green) is $\approx 21\,W/m^2$ *bright* in JJA and far more so in spring, where its fixed albedo does not mask snow. The two forcings differ from each other by $\approx 32\,W/m^2$ in the growing season — the transfer step of §14.

(iii) It is specific to boreal land. Over ocean in the same latitude band the cloud error nearly vanishes ($-2.2\,W/m^2$ CRE, $+1.6$ pp area) while over Siberian land it is -16.5 and $+8.2$. A *global* cloud knob would therefore move the ocean, where nothing is wrong, and spend TOA budget for no benefit. The error is over land, in summer — pointing at surface-driven convection and boundary-layer cloud rather than the large-scale cloud scheme.

Amount or opacity? Both, roughly half each. At the observed CRE per unit cloud area ($-75.1/0.696 \approx -108\,W/m^2$), the $+8.2$ pp excess accounts for ≈ -8.8 of the $-16.5\,W/m^2$ CRE error; the residual ≈ -7.7 is opacity or vertical placement. A cloud-*amount* lever alone therefore reaches at most half of it.

13.2 Does the cloud fix conflict with the energy target? Less than first thought

~~Burning off that cloud stops reflecting it at TOA, costing ≈ +0.18 W/m^2 on the annual global TOA — a 27 % worsening — so a pure cloud fix cannot meet a fixed-TOA constraint, and only a radiatively neutral surface-flux lever can.~~ Two things falsify that framing:

- **The target is the *surface*, not TOA** (§11.1), and the drift it controls is driven by the flux into the *ocean*. The cloud error is over *land* (§13.1), where annual-mean energy storage is ≈ 0 — land returns what it receives. A boreal-land cloud correction therefore barely touches the quantity that governs spin-up drift.
- **The error is land-specific**, so the correction need not be a global knob at all (§13.1) — which was the whole reason a TOA cost was incurred in the first estimate.

The two objectives are consequently far more separable than the original framing implied, and the cloud error can be attacked at its diagnosed cause rather than through a proxy.

13.3 The stomatal-resistance sensitivity test (experiment A)

A second, radiatively neutral lever was nonetheless identified and is being tested, because the measured Bowen ratio over Siberian JJA land is SH/LH $= -24.6/-56.8 = \mathbf{0.43}$, low against the general observed picture for boreal conifer forest. Minimum canopy resistance for boreal needleleaf (`susveg_mod.F90`:266–267) is 250 s/m, with 500 retained just above as a comment.

Provenance — and a correction. ~~500 is the IFS default and has been halved in this build.~~ Git refutes this: both the active 250 and the commented 500 are present in commit `cf82033`, "initial commit of 48r1 released version", and no AWI commit has ever touched `RVRSMIN`. 250 **is what ECMWF ships in 48r1**; 500 is the older HTESSEL-era value they themselves revised. Setting 500 moves *away* from the released configuration, so this is a **sensitivity test, not a default restoration**.

Status: `amip_expA_rvrsmin500`, 6 yr, paired against `amip_pi_base` on 1872–1875. Verified compiled in by inspecting the rebuilt object file (the value appears twice for types 3–4; 250 survives only for Desert and Mixed Forest). Note this tests *partitioning*, not the diagnosed cause: if the cloud error is real then a resistance change compensates for it, and with 11 W/m^2 less available energy there is also less to repartition. *Caveat:* "Bowen 0.43 is too low" is inferred from the literature's general picture, not measured — neither CRUNCEP3 nor CERES supplies SH/LH here, and FLUXNET boreal sites would settle it.

A connection worth testing in the same run. Excess summer evapotranspiration over boreal land would *both* cool the surface directly *and* moisten the boundary layer, feeding the excess low cloud (model `lcc`= 50.5 % over Siberian land in JJA). If cutting transpiration also reduces low cloud, the partitioning error and the cloud error are the same problem, and experiment A tests both at once.

14 Forcing transfer: the spin-up was grown under brighter, warmer conditions than the model delivers

The vegetation state every coupled run in rounds 08 and 09 restarts from is the standalone offline spin-up `LPJG-SPINUP_2000Y_TCO95_CORE3_CRUNCEPcalibrated_v3` (`/work/bb1469/a270270/runtime/lpjg-spinup/`), driven by `CRUNCEP_noLPJG_1d_1901-1910_TCO95_calibrated_v3.nc` (verified at `lpj_guess_levante_spinup_..._2000y.yaml`:38). **That forcing and the atmosphere the coupled model provides differ substantially, and in the same direction for both light and temperature.**

Siberia land, JJA	surface net SW [W/m^2]	vs CERES
CERES EBAF (observed)	166.7	—
CRUNCEP forcing (what LPJG grew under)	**187.6**	**+21** (+13 %)
AMIP (what our atmosphere delivers)	**155.8**	**−11** (−7 %)
gap between the two forcings		**32** W/m^2 (≈ 20 %)
plus 2 m temperature		**2.2** K (11.9 vs 9.7 °C)

The errors point *opposite* ways and therefore compound: CRUNCEP is too bright (the NCEP high-latitude brightness bias — its downward SW is 234 W/m^2 in July against CERES's 209, ≈ 12 % high), while our atmosphere is too dark by the cloud error of §13.1. Vegetation equilibrated to the first and then handed the second experiences a ≈ 20 % drop in growing-season radiation and a 2.2 K cooling — a step change in exactly the direction that pushes a boreal system from trees toward grasses. All numbers here are computed on a common CERES 1° grid with a single land mask (`verify_ceres_cruncep.py`); mask threshold 0.5 → 0.8 moves them by $< 1\,\mathrm{W/m^2}$.

14.1 Ruled out by measurement

Several components were suspected along the way and cleared. Recording them so the same ground is not re-covered:

suspect	the observation that cleared it
LPJ-GUESS core radiation handling	~~PAR is inflated because no albedo is applied.~~ Reading `rsns` as `NETSWRAD_TS` and ap input (`driver.cpp`:1435–1449); a second correction would double-count. Model beh
EC-Earth spin-up interface	Reads the right variable and sets the right type (`framework.cpp`:2501–2507). No fa
The forcing script's fixed albedo	~~`rsns=FSDS`×0.861 is a fixed 0.139 albedo, so spring PAR is inflated ×1.66 and the~~ True that the albedo is fixed (ratio 0.861 in every cell, April and July alike). But in 0.139 against CERES's **0.142** — right to 0.003. The large spring error (0.139 vs 0. is **biologically inert**: measured over 687 Siberian cells at equilibrium, April carrie GPP and April LAI is 0.02 (no canopy), while **JJA carries 80 %**. Weighted by wh is actually assimilated, the PAR excess is ≈ 13 %, not 66 %.
~~...and it is an unusual, campaign-specific defect~~	Standalone LPJ-GUESS assumes a constant albedo natively: `BETA`= 0.17 (`driver.c` in both the sunshine and `SWRAD` paths. The script's 0.139 is a slightly darker versio standard practice, and switching to its `bulk_albedo` option would give 0.195 in JJA *worse* than the fixed value. **Do not "fix" the script; it is not the problem.**

14.2 What survives, and how to settle it

What remains is not a defect in any component but a **transfer problem**: a ≈ 20 % growing-season radiation step and a 2.2 K temperature step between the forcing the vegetation equilibrated to and the forcing it is then asked to live in. **Whether LPJG's *parameters* were tuned against CRUNCEP — the "calibrated" in the directory name — is not established**; the 07-series did change `greff_min` and `pstemp_low`, but no evidence was found of what was calibrated against what, and the folder name is not proof.

It still weakens the "competition, not climate" verdict of §6.9, though less than first argued. That test moved a standalone re-spin from the CRUNCEP forcing to the OIFS–AMIP forcing and read the resulting under-vegetation as an LPJG-internal competition failure. But that move alone removes ≈ 20 % of the growing-season light and 2.2 K, so under-vegetation is partly what the forcing change predicts on its own. The verdict should be read as provisional. (The forcing used there was also the *superseded* AMIP build, §11.3.)

14.3 The transfer test, measured: forcing alone halves boreal tree cover

The test needs no knowledge of what was tuned — take the *same* LPJG parameters and spin to equilibrium under each forcing; the difference *is* the transfer penalty. **Both arms already existed** and required no new computation:

CRUNCEP arm	`lpjg-spinup/LPJG-SPINUP_2000Y_TCO95_CORE3_CRUNCEPcalibrated_v3`
AMIP arm	`lpjg-spinup-develop/LR_2000y_PIforcing_v3`

Controlled: `global.ins` **byte-identical** between the two trees, `firemodel "GLOBFIRM"` in both `.ins` files, `nyear_spinup`= 2000 / `freenyears`= 100, fixed 1850 CO_2/Ndep/LU, both reaching year 3900, and **10 043 gridcells in common** (99.8 % of the AMIP gridlist). The material difference is the forcing file.

region		CRUNCEP-forced	AMIP-forced	diff	%
Siberia	`TREEFPC`	0.357	**0.163**	−0.194	**−54**
	`GRASSFPC`	0.288	0.330	+0.042	+15
	`AGDD5`	776	645	−131	−17
E. Siberia	`TREEFPC`	0.359	**0.122**	−0.236	**−66**
	`GRASSFPC`	0.316	0.398	+0.082	+26
	`AGDD5`	748	568	−180	−24
NH 45°N+	`TREEFPC`	0.383	0.287	−0.096	−25

Changing nothing but the forcing halves the Siberian tree fraction and cuts East Siberia's by two-thirds, with grass taking up the space. For scale, the coupled 080a run lost `TREEFPC` 0.310 → 0.121 over Siberia in 50 years (§9.1); the forcing swap alone reproduces essentially that entire loss at equilibrium, with no parameter changed. `AGDD5` falls 17–24 % in the same direction, so this is not a light effect alone — the 2.2 K also does real work through the growing-degree-day gates.

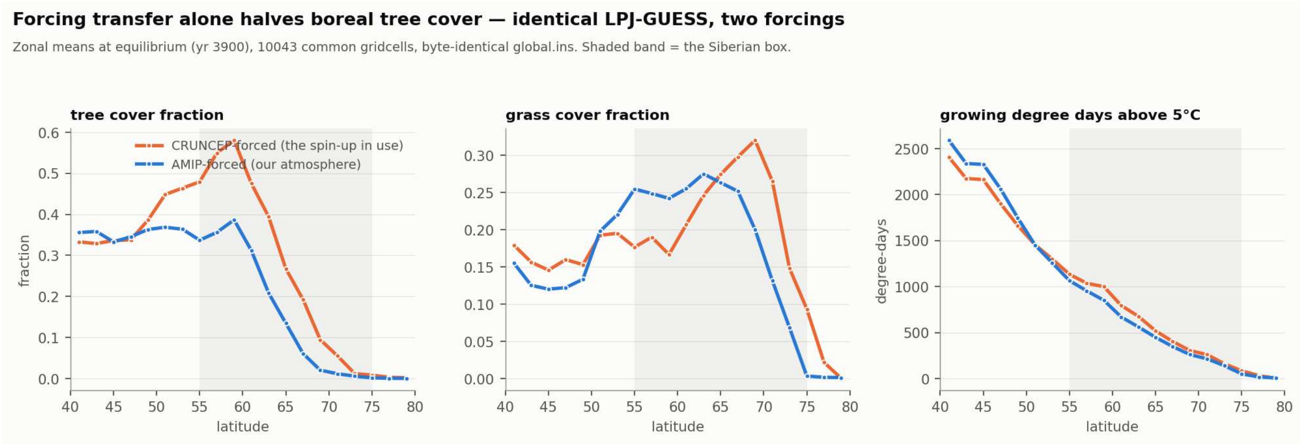


Figure 28: Zonal means at equilibrium (yr 3900), 10 043 common gridcells, byte-identical `global.ins`. South of ≈ 52°N the two arms are indistinguishable; northward they diverge sharply and the AMIP-forced treeline retreats several degrees south. The deficit the campaign has been attributing to competition is reproduced by the forcing change alone.

Three caveats, all material.

1. The arms were built from *different binaries* (a270092 vs a270270) of the same LPJG version with identical `global.ins`. A compiled difference cannot be excluded; re-running both arms from one build (≈ 3 h) would close it, and should be done before this is published.
2. The AMIP forcing is the *superseded* build (§11.3), ≈ 0.9 K warmer over Siberian JJA land than the corrected one — so this is a **lower bound** on the penalty against today's atmosphere.
3. ~~`FORESTFPC`= 0 in both arms, so the absence of closed canopy survives as a separate unexplained problem.~~ `FORESTFPC` is a *void diagnostic* in this configuration — it counts only the managed-forest land-use class, which is empty here, so it is identically zero regardless of what the vegetation does. There is no separate closed-canopy failure to explain.

15 Revised plan: tune the atmosphere on AMIP

Both errors that remain — ≈ 0.83 W/m^2 of TOA imbalance (§11.2) and ≈ 2.2 K of Siberian summer cold (§11.3) — are atmospheric, and both are now *measured* in a configuration where neither the ocean nor LPJ-GUESS can confound the answer. They should therefore be tuned on AMIP, not in coupled mode. This does not retire Phases 1A/1B of §10 — the LPJG competition fix and the stable-BL mixing fix still target the *cold-season* bias — but it inserts a cheaper, better-posed stage ahead of Phase 2.

Two objectives, now known to be largely separable (§13.2):

1. **Surface energy balance** → 0 to cut spin-up drift. This is an *ocean* surface-flux problem; the relevant number is F_{sfc} over ocean, which has *not* yet been computed — all figures quoted so far (+0.52 AMIP, +0.95 09C, +1.77 080a) are global. **Open item.**
2. **Boreal summer warmth** to restore GDD5 and let trees compete. This is a *land* problem, caused by an 11 W/m^2 cloud-driven SW deficit that is absent over ocean (§13.1).

	change	where	status / expectation
A	`RVRSMIN(3,4)`: 250 → 500	`susveg_mod.F90:266–267`	*running*; radiatively neutral; tests partitioning, not the diagnosed cause
B	`RVLAI(3,4)`: 5.0 → 4.0	`susveg_mod.F90:133–134`	same sign, smaller
C	~~`RCLDIFF` ↑, global~~	~~`NAMCLDP`~~	~~+surface SW, but +TOA~~
C′	a *land*-acting cloud lever	convection / BL cloud	the error is land-only, so a global knob would move the ocean where no
D	A + C′		joint target

Gates: Siberian JJA bias toward 0 *and* SW CRE toward CERES; ocean F_{sfc} not degraded; no new error outside the boreal (judge on the global `part8`/CMPI pattern, §4.3). Since the cloud error splits roughly half amount, half opacity (§13.1), a lever acting only on cloud amount can close at most half of it, and the diagnostic to watch is SW CRE against CERES rather than temperature alone.

Why AMIP is the right testbed. 2.3 node-hours per model year against 15.8 coupled — a $\approx 7\times$ saving — and 10 yr completes in one job. Measured interannual scatter of the boreal JJA mean is $\sigma = 0.25\,\mathrm{K}$ (Boreal), 0.59 K (Siberia), 0.72 K (E. Siberia), so with 8 usable years the standard error is 0.09–0.26 K against biases of 1–2 K: sampling is not the binding constraint. *Land spin-up is* — with `lresume: false` the deep soil (`stl4`, `swvl4`) carries initialisation shock, so discard the first 1–2 years. For paired tuning experiments 5 yr (1 discarded + 4 used) suffices, since identical prescribed SST makes most interannual variance cancel in the difference.

16 Round 10 design: a two-pronged, targeted attack (2026–07–28)

Original design as proposed. Operator notes to be appended in §16.4.

The structural fact the design rests on. The two largest regional radiation errors have *opposite* sign at TOA (§11.2.2, §13.1), so fixing them together is self-balancing, whereas fixing either alone is not:

	error	fixing it	ΔTOA
A1 Southern Ocean	too *little* cloud (−6.6 pp, SW CRE +7.8)	adds cloud, reflects more	**−0.5** to −0.6
A2 boreal land	too *much* cloud (+8.2 pp, surface SW −11 in JJA)	removes cloud, absorbs more	**+0.1**

16.1 Prong A — AMIP (atmosphere), A1 before A2

A1 first. It moves the global target in the right direction (+0.67 →≈ +0.1) *and* attacks the error the campaign has suspected since §5. A2 then costs back only ≈ 0.1 of the ≈ 0.55 that A1 buys, so the boreal fix becomes affordable — the "warm floor" logic of Part III, relocated from the vegetation to the atmosphere.

Critical: in AMIP the Southern-Ocean target is *radiation only*, never temperature. SST and sea ice are prescribed, so the SO surface temperature cannot respond to a radiation error — the excess absorbed shortwave is simply exported through the fixed boundary. **A1 must therefore be judged on SW CRE and cloud area against CERES, and it is meaningless to look for an SO temperature response in AMIP.** This is also why the SO *warm* bias appears only in AWI-CM3/AWI-ESM3: it is a *coupled* phenomenon, in which the same +5.8 W/m^2 of excess absorbed energy has somewhere to go and warms the ocean. The causal chain is then explicit and closes the compensating-error argument of §5:

AMIP SO cloud deficit (−6.6 pp) → +5.8 W/m^2 excess absorbed → *(coupled only)* SO warm SST bias → suppressed by the 06T/06V ocean-mixing levers

i.e. Part I was damping Weddell/Ross convection to compensate an *atmospheric* cloud error. It follows that the only test of whether A1 fixed the real problem is a *coupled* run — AMIP can confirm the radiation is right, but not that the SO warm bias is gone.

On levers. OpenIFS has no land-only cloud knob, and that is informative rather than obstructive: the land-specificity of A2 (−16.5 W/m^2 CRE error over Siberian land against −2.2 over ocean at the same latitude) is itself evidence that the cause is **land-surface moisture supply**, not the cloud scheme — which makes the `RVRSMIN` sensitivity test (§13.3) potentially the actual fix rather than a proxy for one. A1, being over ocean, is a genuine cloud-scheme / microphysics target.

Guardrails on every A-run — all four, every time:

1. global net TOA *and* surface flux (state which convention, §11.1);
2. SO 45–65°S SW CRE and cloud area against CERES — *radiation only*, since prescribed SST makes an SO temperature response impossible in AMIP by construction (above);
3. **the tropics** — 50 % of the globe and the largest single contribution (−1.26), and the term the 06O/06T/06V levers degraded (§4.3);
4. CMPI by *pattern*, not the global number.

16.2 Prong B — LPJ-GUESS, and it must go second

Do not tune competition parameters yet. The tree deficit is largely forcing-driven (−54 % Siberian `TREEFPC` from the forcing swap alone, §14.3), so tuning LPJG while its forcing is 11 W/m^2 dim and 2.2 K cold would bake the atmospheric error into vegetation parameters. That is the compensating-error trap, and it is what the 07-series was already doing — which is the natural reading of why it achieved nothing (§12).

1. Complete Prong A.
2. **Regenerate the LPJG forcing** from the corrected atmosphere. Enabling work required: `amip_pi_base` uses the default `file_def` and emits the evaluation fields but *not* the daily LPJG forcing set, so a merged `file_def` (or a second short run) is needed.
3. **Re-spin and re-derive the residual.** Whatever tree deficit survives against a corrected forcing is the real LPJG target. Only then are `greff_min` / `pstemp_low` legitimate levers.

Gates, replacing the void one. ~~FORESTFPC> 0~~ is unachievable by construction (§14.3) and must not be used. Use instead: `TREEFPC` against the CRUNCEP-arm equilibrium (0.357 Siberia, 0.359 E. Siberia); `AGDD5` recovery toward 776/748; and `cvh` toward the MODIS 0.4–0.7 closed-taiga range.

16.3 Ordering

A1 → A2 → regenerate forcing → re-spin LPJG → B. Prong B cannot be evaluated honestly until Prong A has fixed what feeds it.

16.4 Operator notes

(reserved — to be filled in)

17 Round 10 results (2026–07–30): 46-year runs rebuild the ranking

All 19 prong-A runs were extended from 6 to 46 years (evaluated 1872–1915, 44 yr) after a statistical-power check showed the original 4-year window could not resolve the boreal signal at all.

17.1 The 4-year window was resolving nothing in the boreal

A run × year ANOVA (`scripts/analysis/noise_floor.py`) decomposes each per-year value into a tuning signal, the SST-forced excursion common to every run, and internal noise. At 4 years the 95 % detection threshold on Siberian JJA T2m was ±0.89 K and **1 of 18 runs cleared it** — A1a, the overshoot already rejected. Siberian JJA surface SW: 1 of 18. Siberian JJA cloud area: **0 of 18**. The energy target was by contrast well resolved (±0.23 W/m^2 on global TOA, 9 of 18 significant), so the campaign's Southern-Ocean and global conclusions were safe while *every boreal ranking was noise.*

At 44 years the threshold falls to ±**0.240** K, and the ranking inverts:

run	4-yr Δ	**44-yr Δ**	t	verdict
B5 `RCAPDCYCL` 2 → 0	+0.001	**+0.407**	+3.32	**significant — best**
B2 `RCLDIFF_CONVI`= 25	+0.290	**+0.344**	+2.80	**significant**
B3 `RCLDIFF`= 1.5E-5	+0.366	**+0.306**	+2.50	**significant**
A1a `ovl`= 0.10	−1.149	−0.749	−6.11	significant, wrong way
B8 `RVLAMSK`= 5	**+0.502**	**−0.038**	−0.31	**noise**
AB, B7, C2, E1, A2, A1c, B6	—	≤ +0.12	—	noise

Two reversals matter. **B8, promoted to "best lever of the campaign" on its +0.502 K, is noise**; it had already been retracted on 07–29 when the noise floor was first measured, and the 46-year runs confirm that retraction. **B5, dismissed as "inert" at +0.001 K, is the strongest lever in the campaign.** The caution recorded at the time — *unresolved, not proven inert* — is what kept it in play.

Method note for future rounds: B8 was promoted and then built upon twice (ABB8, E1) before its uncertainty was ever estimated. Three of that round's four runs were spent following a result inside the noise. The noise floor cost one post-processing script and no node-hours.

17.2 Period offset, measured rather than estimated

`amip_presentday` is the same model and configuration run over 1989–2015, so the epoch offset is measurable directly instead of chained through observations (§11.3.1).

arm	Sib JJA T2m	vs ERA5 90–14	SO SW CRE	vs CERES	net TOA
control (1872–1915)	9.73	**−2.45**	−60.29	**+7.85**	0.64
presentday (1990–2014)	10.14	**−2.04**	−60.36	**+7.78**	2.20
measured offset		**+0.42**		**−0.07**	+1.56

Both 07–29 corrections were too large. The boreal offset is +0.42 K, not +1.12, so the period-clean bias is −**2.04** K; the SO SW CRE offset is −0.07, not −1.43, so that gap is +**7.78** — essentially the original +8.08, and the "≈ 6 W/m^2" claim is withdrawn.

The real finding is underneath the error. The model warms Siberian JJA by only +0.42 K between the epochs where observations imply $\approx +1.1$ K, so **the model under-warms the historical period by ≈ 0.7 K** — a *trend* error distinct from the mean-state bias. Taking observations as correct for the epoch change puts the 1870s bias at ≈ -1.3 K; taking the model's own offset puts it at ≈ -2.0 K. The honest target range is **−1.3 to −2.0 K**.

Caveat: `presentday` global net TOA is +2.20 against CERES's +0.97. Aerosols were verified correct (MACv2-SP transient, years 1989→2002 logged), so this is a genuine radiative bias rather than a missing-forcing artefact, consistent with the SO cloud deficit being identical across epochs.

17.3 The runs were never 1850 GHG — a dead namelist block

`NCMIPFIXYR: 1850` never took effect in any `oifsamip` run. All 24 runscripts hand-wrote a `&NAMECECMIP6` block; OpenIFS 48r1 renamed that namelist to `&NAMECECMIP` (`ece_cmip.F90`:68 `POSNAM`s `'NAMECECMIP'` only, and `NAMECECMIP6/SSPNA` appear *nowhere* in the source). The legacy block survives in esm-tools' `fort.4` template, so `f90nml` patched it without complaint and the generated namelist looked correct — while the model read only `NAMECECMIP` and left `NCMIPFIXYR` at its default −1. Since `LFIXYEAR = NCMIPFIXYR > 0`, forcing was **transient**. From the model's own output:

```
UPDRGAS: greenhouse gas concentrations for YEAR/MONTH  1870/01
CO2 = 286.94 -> 287.12 -> 287.31 -> 287.49 ...   (rising through the run)
SP_SETUP: IYR = 1870, 1871, 1872, ...            (MACv2-SP aerosols advancing)
```

So every run is a transient CMIP7 *historical* run at its actual calendar year, Krakatoa (1883: −0.98 K in Siberian JJA) included, not a fixed-1850 PI run.

The tuning results are unaffected: all runs share that identical forcing trajectory, the perturbations live in `NAMCLDP`/`NAMVDF`/source which *are* read, and the ANOVA absorbs the shared trend and the volcanoes in g[year]. **The absolute targets are affected**, since they are all read off the control: CO_2 drifts 287 → 301 ppm ($\approx +0.15\,\mathrm{W/m^2}$ mid-run vs 1850, partly offset by aerosols) and the volcanoes depress the 44-yr mean by ≈ 0.05 K. Small — but the global TOA target is only $+0.53\,\mathrm{W/m^2}$, so $\approx 0.1\,\mathrm{W/m^2}$ of spurious forcing is not negligible *there*. `amip_presentday` is unaffected: `historical` is *correct* for 1990–2014.

The fix is the setup's own switch, not a hand-patch: `oifs.scenario: "piControl"` routes through `configs/components/oif` which writes `NCMIPFIXYR` into the live block and sets the CMIP7 stratospheric-aerosol switches in `NAERAD`. The dead block has been removed from all 24 runscripts and `amip_picontrol` is running as a true PI reference; `amip_pi_base` is deliberately untouched because all 19 lever deltas reference it. **The CMIP7 production spin-up is not affected** — its `NCMIPFIXYR`= 1850 is in the live block and CO_2 is pinned at 284.30 ppm at calendar year 2018. *Residual code gap:* nothing pins MACv2-SP tropospheric aerosols (`updtim.F90`:812–818 takes the calendar year unconditionally).

17.4 Can the levers be combined? Not as they stand

Targets from the 44-year control, and the significant per-lever responses:

	SO SW CRE	Sib JJA T2m	global TOA	tropics TOA
required Δ	**−7.85**	**+1.3** to **+2.0**	**−0.64**	**+2.50**
A1a `ovl`= 0.10	**−6.46**	−0.749	−1.90	−1.07
A1b `ovl`= 0.35	−2.17	−0.135	−0.58	−0.29
B5	−0.23	**+0.407**	−1.00	**−1.94**
B2	+0.11	+0.344	+0.39	+0.65
B3	+0.67	+0.306	+1.34	**+2.02**

Note the tropics are $2.5\,\mathrm{W/m^2}$ *too low*, not too high, so B5's −1.94 is a real cost and only B3 helps there. Added linearly:

stack	SO	Sib	global TOA	tropics
A1b+B5	−2.40	+0.27	−0.94 ✗	40.4 ✗
A1b+B2+B5	−2.30	+0.62*	−0.55 ✗	41.0 ✗
A1b+B2+B3+B5	−1.63 ✗	+0.92*	+0.79 ✗	43.0
A1a+B2+B3+B5	**−5.91**	+0.31 ✗	−0.53	42.3 ✗

*optimistic: AB measured −0.053 K where additivity predicted +0.21, so A1b *cancels* B2's boreal gain.

Two structural walls, not bad luck.

- **The Southern Ocean is reachable only through A1-strength mixed-phase overlap, which cools Siberia.** That is the same process on both ends: more supercooled liquid brightens SO cloud *and* boreal mid-level cloud. A1a buys 82 % of the SO target and spends −0.75 K of Siberia doing it.
- **Siberia's levers are too small and do not add.** The three significant ones sum to +1.06 K against a +1.3–2.0 K need, before the non-additivity that AB demonstrated.

The LPJ-GUESS fallback is closed: §11.3.2 showed the forest is 75–85 % temperature-driven, so prong B cannot absorb a boreal residual the atmosphere does not deliver. The boreal problem must be solved in OpenIFS.

18 Round 11 design (2026–07–30): attack the mechanisms, not the parameters

Round 10 closed on two structural walls (§17.4). Round 11 therefore targets the underlying mechanisms. Two proposals were written and then **withdrawn for being tuning-to-fit**, and recording why is part of the result:

- ~~Land/sea split on `RCL_OVERLAPLIQICE`.~~ **Anti-physical.** That parameter is the sub-grid *overlap fraction of supercooled liquid with ice* (range $[0, 1]$, default 0.65, EC-Earth4 uses 0.1), entering $\texttt{ZDEPOS} = \max(\texttt{ZOVERLAP\_LIQICE} \cdot \texttt{ZA} \cdot (\texttt{ZINEW} - \texttt{ZICE0}), 0)$. The code's own comment states the intended selectivity — *"Reduce in shallow convection because assume SLW in active updraught is less overlapped with ice in less active part"* — i.e. convective segregation *lowers* overlap. But ocean mixed-phase cloud is predominantly stratiform (well mixed, so *high* overlap) and land cloud more convective (segregated, *low* overlap), the **opposite** of what the tuning wants.
- ~~`RCAPDCYCL` mode 3 = disable the CAPE diurnal-cycle correction over land.~~ **A regression.** That correction is Bechtold et al. (2014), which exists to fix the documented error of land convection triggering too early in the day.

18.1 D1 — `RCAPDCYCL` mode 4: reformulate the land closure

B5 was misdiagnosed. `RCAPDCYCL` is a *mode selector*, not a magnitude (`cumastrn.F90`:773,777): 1.0 = land only, scaled on `ZKHVFL` (surface kinematic virtual heat flux, `:464`); 2.0 = everywhere, land via $\texttt{ZCAPPBL} \cdot \texttt{ZTAU}/\texttt{ZTAURES}$ and ocean via a wind-based `ZTAUPBL`; 0.0 = off entirely. So B5's 2.0 → 0.0 **deleted the correction globally**: its +0.407 K boreal gain came from the land branch, its −1.94 W/m^2 tropical cost from losing the ocean branch. The two are separable.

The physical argument is in the scaling. Mode 1 is proportional to the *actual surface buoyancy flux* and vanishes where that forcing is weak; mode 2's land branch uses boundary-layer CAPE instead. Boreal summer has weak insolation, a shallow boundary layer and small surface fluxes, so the buoyancy-flux form self-limits while the BL-CAPE form need not — a real reason to suspect **mode 2's land branch is mis-scaled for weak-forcing high-latitude convection**, and it predicts the observed sign. **Mode 4** (added this build) is mode-1 land physics with mode-2's ocean branch verbatim; both are existing code. *Prediction:* recovers part of +0.407 K with little of the tropical cost. If the boreal gain vanishes too, the effect was never about the land closure formulation and the B5 result needs re-reading.

18.2 D2 — ice nuclei, not overlap

The Southern-Ocean supercooled-liquid deficit is an **ice-nucleation** problem, and the scheme has no aerosol dependence whatsoever:

$$\texttt{ZICENUCLEI} = 1000 \cdot \exp(12.96 \cdot \texttt{ZSUPERSATICE} - 0.639) \qquad (\texttt{cloudsc.F90} : 2516,\ : 2618)$$

That is **Meyers et al. (1992)** — a function of ice supersaturation *only*, documented to overestimate INP in clean environments, which is precisely the remote Southern Ocean; DeMott et al. (2010/2015) make INP a function of aerosol number instead. Reducing INP over ocean therefore has a real mechanism (INP sources are continental: mineral dust, biological) with the correct sign in both regions at once, and strong leverage since $\texttt{ZCVDS} \propto \texttt{ZICENUCLEI}^{0.666}$ — a factor 0.1 cuts deposition to ≈ 0.22.

Three better routes are closed. (i) `PNICE` is already a `cloudsc` argument (`:265`) and already used for autoconversion (`:2855`), but it is filled only when $\texttt{NAERCLD} > 0$ (`callpar.F90`:1155, else zeroed at `:1166`) and `NAERCLD` defaults to 0; that configuration is several times slower and **would not finish CMIP7 in time**. (ii) `onlinedust_v4_rmp.nc` is staged and *opened* unconditionally, but its arrays (`POTSRC`, `SOILTYPE`, `CULT`) are allocated only by `TM5M7_SRC_DUST_INIT` via `tm5m7_init.F90` — **only under m7** — so its contents are never populated here (`suecrad`'s `YDUSTCLIM` is a different object, dust *optical properties* for the radiation climatology). (iii) MACv2-SP plumes are anthropogenic sulphate/BC/OC: poor INP, sulphate coating *deactivates* mineral dust INP, and in a piControl anthropogenic aerosol is near zero, giving no land/ocean contrast.

18.2.1 Two physical caveats that shape the test

The tropics will pay, and "ocean" is the wrong discriminator. The scaling acts only inside the mixed-phase window ($\texttt{RTHOMO} \leq T < \texttt{RTT} - 5$, i.e. 235.15–268.15 K, with liquid present), which in the tropics sits at $\approx$ 500–300 hPa and *is* populated — deep convective updraughts carry supercooled liquid well above the freezing level, and anvil detrainment feeds it. Reducing INP there retains more supercooled liquid, brightening tropical mid-level cloud and pushing tropical TOA *further below* CERES. Expect a cost of order A1a's $-1.07\,\mathrm{W/m^2}$ or worse. More fundamentally, tropical and subtropical oceans are *not* dust-remote (Saharan dust over the tropical Atlantic, Asian dust over the North Pacific); the Southern Ocean is uniquely far from continental sources. The property that distinguishes it is **remoteness**, which a local land-fraction mask cannot encode.

Sea salt sets an INP floor. Sea salt itself is a poor INP (highly soluble, inefficient in the immersion mode at these temperatures), but sea spray carries **marine biogenic organics** — sea-surface microlayer material, phytoplankton exudates — which are a modest INP source (Wilson et al. 2015; DeMott et al. 2016). So the ocean INP factor **must not approach zero**: $\approx$ 0.1–0.3 is defensible, not 0.01. Sea salt's larger role is as CCN, which is moot here since `PCCN` is zeroed with `NAERCLD` = 0 and droplet number is the constant `RCL_KK_CLOUD_NUM_SEA`. A tempering caveat: marine biogenic INP peaks with the austral-summer bloom (DJF), exactly when SO insolation and our SW bias peak, so the SO is *least* INP-starved in the season we care most about.

18.3 The three runs

All on the same forcing as the 19 round-10 levers (`oifs.scenario` defaults to `historical`), so deltas are directly comparable to B5's; 46 yr, evaluated 1872–1915.

run	change	purpose
D1	`RCAPDCYCL`= 4 (source + namelist)	separate B5's boreal gain from its tropical cost
D2a	`ZICENUCLEI` $\times f$(`PLSM`), no remoteness	does the INP route have leverage at all?
D2b	tile-based two-term INP + remoteness gate	the physically honest version

D2a is run *knowing* it should damage the tropics: the question it answers is whether INP has Southern-Ocean leverage, which decides whether D2b is worth building. D2b computes the INP field in `callpar` from `SURFL%ZFRTI` (tile fractions, already used at `callpar.F90`:2006) and `PAUX%PGEMU` (sine of latitude, available at `:1801`), passing one new array to `cloudsc`:

$$\text{INP} \;\propto\; a \cdot [\text{bare soil (tile 8)} \times \text{wind}] \;+\; b \cdot [\text{low veg (4)} + \text{high veg (6)}]$$

Tile convention from `srfi_mod.F90`:52. The two terms matter: a *dust-only* proxy would give the **wrong sign** over boreal forest, which is a weak dust source — lowering its INP would brighten boreal cloud and cool it further. Vegetated land is instead a strong **biological** INP source in summer (pollen, bacteria, fungal spores, litter), peaking in exactly our season, while snow tiles (5/7) collapse that term in winter and leave the cold-season bias untouched.

All three are source edits on the single `oifsamip-cy48` tree and therefore **serialise**: edit, `esm_master recomp-oifsamip-cy48`, verify the binary md5 changed, submit, wait for staging, repeat.

19 Round 11 results: D2b adopted for the Southern Ocean, D1 falsified

	SO SW CRE	SO SW RMSE	Sib JJA T2m	global TOA	tropics
control	−60.29	6.88	9.73	0.64	42.61
piCTRL 1850	−60.65	6.83	9.74	**0.79**	42.69
D1 `RCAPDCYCL`= 4	−60.38	6.51	**9.56**	0.61	42.53
D2a INP, no gate	**−65.13**	5.07	9.56	−0.28	42.23
D2b INP + p700	−62.93	**4.96**	9.51	0.36	**42.70**

D2b is adopted as the Southern-Ocean lever. It gives the best SO surface-SW RMSE of any run (6.88 → 4.96, −28 %) at *zero* tropical cost (+0.09), beating both A1b (5.54) and the ungated D2a (5.07) — the latter despite a larger mean-CRE change, because D2b improves the spatial *pattern*, which is what the priority metric exists to measure. The ice-nucleation hypothesis (§18.2) is therefore supported, and the pressure gate did what it was designed to do: D2a's tropical cost (−0.38) vanishes (+0.09) while 55 % of the SO gain is retained.

D1's prediction is excluded, but as a NULL, not a sign reversal. It was predicted to recover part of B5's +0.407 K by switching the land CAPE closure to mode-1's buoyancy-flux scaling; it delivered **−0.170** K — which does *not* clear the ±0.242 K detection floor (§17.1), so the predicted recovery is excluded but the sign is unresolved. Quoting it as "the wrong sign" was the very error §17.1 exists to prevent. So B5's boreal gain requires *removing* the land correction, not rescaling it — the effect is not about which quantity the closure scales on. Stating the prediction in advance is what made the refutation unambiguous.

`amip_picontrol` settles two things. The energy target is +0.79 W/m^2, not +0.64 — slightly *larger*, so A1b does not overshoot. And Siberian JJA T2m differs by only +0.008 K, so the dead-namelist forcing bug (§17.3) never affected the boreal target and every round-10 boreal conclusion stands.

20 Round 12: boreal surface exchange, and the best lever of the campaign

Round 11 left the boreal unsolved. Round 12 changed target from convection and cloud to **boreal surface exchange**, anchored on one observation: the measured Siberian JJA Bowen ratio is **0.43** against ≈ 0.5–1.5 observed for boreal conifer, so the model puts too much surface energy into latent heat. Four levers, each raising Bowen by an independent route, all indexed on vegetation types 3/4 so that types 5/6/18 — and hence the tropics — are untouched *by construction.* That is exactly the property B5 lacks.

run	change	ΔT2m	*t*	verdict
F4	`RVRSMIN(3,4)` 250 → 1000	+**0.749**	+6.06	**significant, best ever**
F5	all four combined	+**0.746**	+6.03	significant, ≡ F4
B5	(previous best)	+0.407	+3.29	significant
F2	`RVLAI(3,4)` 5 → 3	+0.321	+2.59	significant
F1	`RVZOH(3,4)=z0m/10`	+0.185	+1.49	marginal
F3	`RVCOV(3,4)` 0.9 → 0.7	≈ 0	—	noise

F4 costs the tropics nothing (42.68 vs control 42.61) where B5 wrecked them (40.67), and it gives the best global T2m RMSE of any run (1.54). Bowen moves 0.44 → 0.60, into the observed range *without* overshooting — the risk named in advance did not materialise.

F5 equals F4 to 0.003 K. Against a naive sum of ≈ 1.25 K, adding F1, F2 and F3 on top of F4 contributes nothing. The advance prediction — F5 ≪ $\sum$, closer to the largest single lever — held, and more strongly than expected: near-total saturation of a single shared latent-heat pathway with a floor at zero. *Use F4 alone.*

A prediction that failed. F1 was called strongest a priori because $kB^{-1} = 0$ is the largest departure from observation (every closed-forest type sets $z_{0h} = z_{0m}$, i.e. $kB^{-1} = 0$, where observations give $kB^{-1} \approx 2$). It is marginal and barely moved Bowen (0.44 → 0.43). "Largest departure from observation" is not a reliable guide to leverage.

B5 carries a hidden −0.72 K DJF cooling (F4: +0.06), invisible for the whole campaign because only JJA was ever evaluated. Given that the coupled model's original complaint is a *cold-season* bias, that is disqualifying for B5, and DJF now belongs in the standard guardrails.

20.1 Is F4 fixing a cause or a symptom?

Two tests, both partly exonerating. *Seasonal selectivity:* DJF +0.061, MAM +0.047, JJA +**0.749**, SON +0.110 — exactly the signature a transpiration mechanism must have, since stomata are shut and the canopy snow-covered in winter. *Vertical targeting:* F4 warms 1000/925/850 hPa by +0.83/ + 0.84/ + 0.83, closing 56–83 % of the *Siberia-specific* layer bias (§21.1), and only +0.53 aloft.

What remains true is that `RVRSMIN` 250 → 1000 is an unanchored 4× excursion from what ECMWF ships, and that the vertical profiles show it removes moisture that is already deficient. **G1 = F4 + D2b** tested this: the first pairing whose components are disjoint in both process and geography, hence expected not to conflict — but AB and ABB8 both got superposition wrong in sign, so it was measured, not assumed. It superposes (§20.2).

20.2 G1 = F4 + D2b: the best configuration of the campaign

G1 is the first combination to improve *both* targets at once, and the first whose superposition is additive rather than wrong in sign.

metric	control	G1	Δ	target
SO SW RMSE (priority)	6.877	**4.809**	−2.067	—
SO TOA SW CRE [$\mathrm{W\,m^{-2}}$]	−60.29	−63.13	−2.84	−68.14
SO cloud area [%]	83.07	83.59	+0.52	89.72
Siberia JJA T2m [°C]	9.73	**10.25**	**+0.521** (t=4.22)	≈ 12.2
Siberia sfc net SW [$\mathrm{W\,m^{-2}}$]	153.78	159.31	+5.54	166.26
global net TOA [$\mathrm{W\,m^{-2}}$]	+0.64	+0.45	−0.195	∼ 0
tropics net TOA [$\mathrm{W\,m^{-2}}$]	42.61	42.71	+0.096	45.11
global T2m RMSE [K]	1.579	1.553	−0.026	—
Nordic Seas SW RMSE	9.058	9.358	**+0.300**	—

Superposition, measured: Siberia JJA T2m F4 +0.749 + D2b −0.222 = +0.527 predicted vs **+0.521** actual; SO SW RMSE −2.131 vs −2.067; SO CRE −2.719 vs −2.840; global net TOA −0.188 vs −0.195. Additive to within noise on every metric. The plausible reason it holds here and failed for AB/ABB8: F4 acts on a land surface flux in boreal summer, D2b on ice nucleation over ocean below 700 hPa, whereas AB and ABB8 combined levers modifying the *same* cloud scheme in the *same* regime.

The one cost is the Nordic Seas, +0.300 SW RMSE against a ±0.052 threshold, and worse than additive (+0.184 expected). No other lever in the campaign has ever moved Nordic RMSE; this one moves it the wrong way. Small box, not the deep-water priority, but it must not be allowed to grow.

G1 closes 26 % of the boreal bias and 36 % of the SO CRE gap. The SO *cloud-area* deficit of ∼6 pp remains untouched by every lever tried.

20.3 Seasonal audit: the noise floor is not the same in every season

The campaign evaluated JJA only for eleven rounds, while the coupled model's original complaint is a *cold-season* bias. `scripts/analysis/seasonal_by_run.py` applies the run×year ANOVA of §17.1 separately to each season. The detection threshold varies by a factor of 2.4 across the year:

season	control	sd(ϵ)	95 % threshold
DJF	−29.35 °C	1.407 K	±0.588 K
MAM	−9.28 °C	0.923 K	±0.386 K
JJA	+9.73 °C	0.580 K	±0.242 K
SON	−10.41 °C	1.031 K	±0.431 K

Applying the JJA threshold to a winter delta overstates significance by 2.4×, and a first pass through these results did exactly that, briefly recording D2b (−0.455) and B3 (−0.486) as winter-damaged. Both are *within* the DJF noise floor. Corrected:

- **B5 `capdcycl0` is the only lever with genuine winter damage** (DJF −0.720, clears ±0.588) and the only one that significantly warms JJA while significantly cooling DJF. Its rejection stands, now properly thresholded.
- **D2b's real seasonal cost is spring, not winter**: MAM −0.467 clears ±0.386; its DJF −0.455 does not clear ±0.588.
- **F4 is seasonally clean** — DJF +0.061, MAM +0.047, SON +0.110, all within noise, the entire signal in JJA. A further argument for F4 over the cloud levers.
- **G1 is seasonally clean**: DJF +0.070, MAM −0.296, SON +0.280, none significant; D2b's MAM cooling is diluted below threshold.
- A1a warms DJF/SON and cools MAM/JJA by large amounts — a global cloud change, not a boreal lever. B6, B7 and F3 cool MAM significantly; F5 inherits F3's MAM −0.507.

21 Vertical structure vs ERA5: where the biases live

21.1 The Siberia-specific cold bias is confined below 850 hPa

Model minus ERA5 on the model's 19 pressure levels, `amip_presentday` vs ERA5 1990–2014 so the comparison is period-clean. ERA5 comes from the DKRZ pool; all 19 model levels exist exactly in ERA5's 37, so no interpolation is needed.

hPa	Siberia	Global	**Siberia-specific**
1000	−1.90	−0.73	**−1.17**
925	−2.17	−0.65	**−1.52**
850	−1.99	−0.99	**−1.00**
700	−1.22	−1.15	−0.07
500	−1.67	−1.49	−0.18
300	−2.32	−2.22	−0.10

This separates two problems that had been conflated. Above 700 hPa Siberia merely shares a **global tropospheric cold bias of** 0.7–2.9 **K** that no boreal lever will touch; the boreal-specific excess lives entirely in the bottom three levels, which is the signature a surface-exchange problem should have and which validates the F-series framing.

The excess boreal cloud is thermally driven, not moisture-driven. Siberian RH is +5.6 to +8.2 % too high while q is *too low* (−0.2 to −0.3 g/kg): a 2 K cold bias cuts q_{sat} by ≈ 14 % while q falls only ≈ 6 %. So a large F-response must not be read as confirming the cause.

Independent support for D2: the Southern Ocean has RH +5.8 to +6.6 % too high through the mixed-phase layer while reflecting *too little* SW — humidity present, liquid absent, exactly the supercooled-liquid deficit the INP hypothesis predicts.

Broken diagnostic, recorded: the model's pressure-level relative humidity `r` is **identically zero in every run checked** (0 nonzero of 17 510 400 values) while `q` is fine. The XIOS `r` output yields silent garbage and must not be used; RH above is recomputed from `t` and `q`.

21.2 Extending the column downward: the soil is the least-biased part

§21.1 stops at 1000 hPa. Continuing into the skin and the four soil layers (`scripts/analysis/vertical_bias_column.py`), JJA, model `amip_presentday` minus ERA5. Low-lying land (< 500 m) is shown alongside because the all-land figure is inflated by orographic mismatch (§21.3):

JJA	global land	land < 500 m	Siberian box
700 hPa	−0.83	−0.89	−1.22
850 hPa	−0.79	−0.92	−1.99
925 hPa	−0.82	−1.02	−2.17
2 m	−1.34	**−1.00**	−2.04
skin	−0.83	**−0.53**	−1.66
soil L1 (7 cm)	−0.40	**−0.36**	−1.73
soil L2 (28 cm)	−0.42	−0.38	−1.90
soil L3 (1 m)	−0.78	−0.72	−2.14
soil L4 (2.9 m)	−1.09	−0.89	−1.02

Over ordinary low-lying land the **soil and skin are the least-biased parts of the entire column**, and 2 m sits about 0.5 K *below* the skin under it. The ground is not what holds the near-surface cold; if anything it is doing better than the air above it. Siberia is the exception that §21.1 already identified — there the soil is cold too (−1.7 to −2.1 K), which is the surface-confined excess the F-series was built to attack.

Caveats. The 1000 hPa row is unreliable over land, where much of that level is below ground and both datasets extrapolate; 925/850 are the lowest honest levels. ERA5's soil and skin are HTESSEL output, so those rows are model-versus-sibling rather than observation — the 2 m and pressure-level rows are the observation-constrained ones.

21.3 The near-surface cold is not boreal: every surface type is cold

The campaign has only ever scored one box, so it had never checked whether its levers address a local fault or a global one. Resolving the 2 m bias by dominant surface type (`scripts/analysis/bias_by_tile.py`) gives an unambiguous answer: **all 20 surface types are cold, in every season. Not one is warm.**

dominant type	area	DJF	MAM	JJA	SON
bare soil	25.3 %	−1.86	−2.12	−2.16	−2.03
Evgr Broadleaf (tropical)	8.7 %	−1.78	−1.79	−1.46	−1.42
Tundra	5.2 %	−2.51	−0.18	−1.43	−1.84
Evgr Needleleaf	4.7 %	−2.00	−1.84	−1.16	−1.05
Dec Needleleaf	3.0 %	−1.22	−1.55	−1.87	−1.69
ALL LAND	100 %	**−1.73**	**−1.60**	**−1.35**	**−1.39**

Two controls bound how much of this is real. The land figure is *elevation-dependent* — −0.70 K below 200 m rising to −2.43 K above 2000 m, the expected signature of the model's smoothed orography against ERA5's 25 km — so the all-land number overstates it and the low-lying value is the honest one. And over **prescribed-SST ocean** 2 m is −0.72 K: about 0.7 K of near-surface cold exists where the land surface is absent entirely.

Consequence for the campaign. Siberia's JJA −2.58 K decomposes roughly as ∼0.7 K universal, ∼0.3 K orographic, and 1.3–1.6 K boreal-specific. G4 has taken ∼0.95 K of that boreal budget — which is a far better explanation of the F-series saturation (F5 = F4 to within noise, §20) than any parameter limit. *The boreal budget is nearly spent, and the remainder is not reachable by a boreal-indexed lever.*

One benign consequence. Because every type is cold, a lever indexed by vegetation type does not damage the regions it leaks into: the non-boreal areas of types 4 and 9 are cold too (JJA −0.68 and −1.06). That is what licenses the round-17 restriction to those two types (§28).

22 Surface albedo: half the Siberian SW deficit is not cloud

Every boreal lever so far has attacked cloud. Decomposed against CERES on the identical box and mask, the 12.5 W/m^2 Siberian JJA surface-SW deficit is only half cloud: **7.0** W/m^2 from too little SW reaching the surface, and **5.5** W/m^2 from **surface albedo too high** — 0.1753 model versus 0.1461 CERES, entirely independent of cloud. F4 does not touch it (0.1749).

22.1 Not a global albedo error, and not the forest

Temperate and tropical land is essentially perfect (Europe, Amazon, Great Plains, steppe, Australia all within ±0.004), so the albedo *scheme* is sound and a global fix would be wrong. *This held up, with one amendment.* §29.1 later measured land albedo globally and by type with snow masked per cell per month: every forest type is inside +0.006 and tropical broadleaf is at −0.0003, confirming the scheme — but a residual +0.0080 does exist on crops, semidesert and bare soil, so "a global fix would be wrong" is right about the *vegetation table* and too strong about the *soil background*. The error is concentrated at high latitude: Siberian tundra +0.094 (−16.0 W/m^2), Canada boreal +0.032, Siberia boreal +0.029, Sahara +0.020 — but **Fennoscandia, the same needleleaf type, is correct at** +0.002. Splitting Siberia by latitude gives +0.011 in the forest core and +0.087 at 70–75N.

22.2 How the albedo is built, and two hypotheses that died

All configurations — AMIP, coupled round-09, *and the CMIP7 production piControl* — run `LRDALB = .false.` with `NALBEDOSCHEME = 3`, which **discards the MODIS albedo present in the input file** and constructs albedo each step (`surfbc_ctl_mod.F90`:512) as $\alpha = \texttt{RVVEGALB}(t_{vl})c_{vl} + \texttt{RVVEGALB}(t_{vh})c_{vh} + \alpha_{soil}(1 - c_{vl} - c_{vh})$.

~~Hypothesis 1: the soil-albedo field dominates tundra.~~ The bare fraction is only 0.01–0.03, so the soil term is negligible. ~~Hypothesis 2: the soil fields are corrupt.~~ GRIB 117–120 do carry an inconsistent header (`N=128` with a 256-row `pl` but only 40 320 values, the O96 count), which **breaks `cdo` and `grib_get_data`** on those fields and should be reported upstream — but the values are O96-ordered and correct: zero over 99.2 % of ocean cells, corr(α_{soil}, lsm) = +0.894. Not our bias.

22.3 The real signature is snow-melt timing

The blend at 70–75N yields 0.158 while the model runs at 0.303; the extra ≈ 0.145 is snow. Monthly, the model is **too dark in April** (−0.06 to −0.07), **near-perfect in July/August** when snow-free (+0.004), and **too bright in June** (+0.086 to +0.116) **and September** (+0.08 to +0.10). So `RVVEGALB` is essentially right and the *snow season is too long at both ends.*

Cause and effect. Through February–April the model is too cold *and* too dark simultaneously; if bright snow caused the cold those signs would agree, so the winter cold is **not** albedo-driven. The albedo bias turns positive only in June, after the surface crosses freezing. But the June snow is a genuine defect in its own right: **June is**

$+6.2\,°$**C with** $213\,$**W/m**^{2} **of insolation and** $0.020\,$**m w.e. still lying**, and the implied snow-cover fraction is $\approx 17\,\%$ where CERES implies $\approx 2\,\%$. That points at the **snow-cover-fraction formulation**, not snow albedo (too *low* in April) and not melt energy (ample).

22.4 Growing season: June is albedo, July is cloud

Siberia 55–75N, W/m^2	Jun	Jul	Aug	JJA
cloud	6.9	**9.9**	4.4	7.0
albedo / snow	**13.4**	1.7	1.3	5.5
total	20.2	11.6	5.7	12.5

Almost cleanly separated in time. The residual snow term is $5.5\,\mathrm{W/m^2}$ on the JJA mean, essentially all in June, worth ≈ 0.2–$0.7\,$K depending on which sensitivity applies — potentially another F4-sized gain, and *orthogonal* to F4, which fixes SW-down while this fixes the absorbed fraction.

But albedo does not explain the land cold bias. Insolation-weighted over land 60S–75N excluding Greenland, the mean albedo-induced SW loss is $-3.00\,\mathrm{W/m^2}$ against a mean T2m bias of $-1.35\,$K, with spatial correlation $r = +0.145$ and slope $+0.033\,$K per W/m^2. Decisively, **cells where the model is too *dark* are still** $-1.16\,$**K cold**. A $\approx 1.2\,$K land-wide cold bias therefore has no albedo signature at all, consistent with the global tropospheric cold bias of §21.1. Fixing albedo is worth doing; it is not the root cause.

23 Round 13: snow cover fraction, the June lever

23.1 Design and prediction

The albedo decomposition (§22) splits the $12.5\,\mathrm{W\,m^{-2}}$ Siberian JJA surface-SW deficit into $\sim$7.0 cloud and $\sim$5.5 **surface albedo**, with the albedo half concentrated almost entirely in **June** (§22.4; bias +0.086, July and August near-perfect). That is a snow-*melt-timing* signature, not a snow-*brightness* one. Cause and effect were established before touching anything: winter cold is *not* albedo-driven, since February–April is simultaneously cold and dark (§22.3), which rules out the albedo → cold direction for those months.

Snow cover fraction (`surfbc_ctl_mod.F90:317-322`) is

$$\mathrm{ZCVS} \;=\; \min(1,\ d_{\mathrm{snow}}[\mathrm{cm}]\cdot \texttt{RQSNCR})\,, \qquad \texttt{RQSNCR} = 1/10 \ \text{ under } \texttt{LESN09=T}.$$

A grid box is therefore fully snow-covered at **10 cm**, linearly below that, with no dependence on vegetation height or orographic roughness. For boreal forest this is wrong in an obvious direction: 10 cm of snow does not hide a spruce canopy, and the observed masking depth over forest is several times larger. The H-series sets `RQSNCR` $= 1/30$.

Why the ice sheets are safe: Antarctica and Greenland carry snow far deeper than 30 cm, so $\min(1,\cdot)$ saturates at 1 for any value in this range and they are untouched. Only shallow, marginal, melting snow responds — which is the target.

H1 applies the change alone against control (clean attribution); **H2** applies it on top of G1's F4 source change and D2b namelist (the deployable stack). H2 – G1 measured against H1 – control gives the interaction directly.

Falsifiable prediction, recorded before the runs finished: ~~H1 recovers a good part of June's 13.4 W m^{-2} albedo term, worth $\sim +0.2$ to $+0.7\,$K on Siberian JJA T2m, and does little in July and August.~~ **Falsified** (§23.2): JJA came out at $+0.020\,$K, $t = 0.17$. The failure mode was not the anticipated one — it did not warm July as much as June, it failed to warm *any* summer month.

23.2 Results: prediction falsified, lever rejected

Predicted $+0.2$ **to** $+0.7\,$**K on Siberian JJA T2m; measured** $+0.020\,$**K,** $t = 0.17$**.** And the lever inflicts the largest winter cold bias in the campaign, $-1.233\,$K DJF against a ± 0.588 threshold, beating the -0.720 for which B5 was rejected.

	DJF	MAM	JJA	SON
H1 `snowcr30`	**−1.233***	−0.473*	**+0.020**	+0.428
H2 = G1 + snow	−0.973*	−0.342	+0.662*	+0.863*
(G1, reference)	+0.070	−0.296	+0.521*	+0.280

Why it failed: the mechanism was pointed at the wrong month. The lever did exactly what it was built to do to the albedo — but not where the reasoning assumed. H1 all-sky surface albedo change (×100, * = clears that month's threshold):

	Jan	Feb	Mar	Apr	May	**Jun**	Jul	Aug	**Sep**	**Oct**	**Nov**	Dec
H1	−0.51*	−0.21	−0.42*	−0.84*	−1.43*	**−0.95***	−0.10*	−0.32*	**−3.08***	**−8.16***	**−4.66***	−1.60*

The response concentrates in **September–November**, the *snow-onset* season, and is small in June. The design error is plain in hindsight: ZCVS = $\min(1, d_{\rm cm} \cdot \texttt{RQSNCR})$ can only respond where snow is **shallow**, and snow is shallow while it is *accumulating in autumn*, not while it is melting in June — by June the pack either still saturates $\min(1, \cdot)$ under both settings or is already gone. **Shallow snow is an autumn phenomenon, not a melt phenomenon.** Consistently, June/July/August T2m move $-0.076/-0.067/+0.205$ (none significant) while **October T2m is** $+1.135$ **K**.

The winter cooling is not radiative. February and March are dark yet cool by −1.65 and −1.19 K. Reducing the snow-covered fraction moves area from the exposed and sheltered **snow tiles (5, 7)** onto the vegetation tiles, and the snow tile is what buffers the surface against radiative cooling in polar night. The lever trades a small autumn warming for a large winter cooling through **tile fractions, not albedo**.

Verdict: rejected, and `RQSNCR` reverted. Three things survive:

1. The June surface-albedo bias is **not** dominated by the snow-cover-fraction formulation. Its cause — snow-albedo decay, vegetation masking of snow albedo, or the snow-free surface albedo itself — remains unidentified.
2. `RQSNCR` has large leverage on *autumn* albedo (−8 points in October). If an autumn or snow-onset bias is ever diagnosed, this is the knob.
3. H2 posts the campaign's best SO SW RMSE (4.787), best subpolar N Atl (4.713) and best global T2m RMSE (1.530) — but that is D2b's doing, and the winter damage disqualifies the combination. **G1 remains the configuration to carry forward.**

24 Round 14 (2026–08–03): the June albedo bias is a SPRING MELT bias

Round 13 established what the June albedo bias is *not* (§23.2): not the snow-cover formulation. This section identifies what it is, entirely from output already on disk plus ERA5 — no new integration was needed to get here.

24.1 Against ERA5: too much snow, and snow that is too DARK

`scripts/analysis/albedo_decompose.py`, model `amip_presentday` vs ERA5 1990–2014, both land-masked to the Siberian box, June:

June	model	ERA5	Δ
surface albedo `fal`	0.2147	0.1730	+**0.042**
snow cover fraction	0.380	0.261	+**0.119**
snow albedo `asn`	0.658	0.759	−**0.102**

Both `f_snow` columns are the *same* HTESSEL formula $\min(1, d_{\rm cm}/10)$ applied to each dataset's snow mass, so the gap is **snow AMOUNT, not the cover formula** — which is precisely why round 13, which changed only the formula, could not move it. The model carries roughly twice ERA5's June snow mass.

Corroborated independently by CERES (§25.1), which puts the June excess slightly higher at +0.046; and the cover excess is larger still against satellite, +0.176 rather than +0.119 (§25.2).

And the model's snow is too dark, not too bright (−0.10 in June, −0.11 in May). Making snow albedo more realistic would make June *worse*. Any lever that brightens snow is pushing the wrong way; the excess reflection comes from snow *extent*, not snow *brightness*.

Two things this analysis does not claim. The naive two-term split $\alpha_{\rm box} = f\,\alpha_{\rm snow} + (1-f)\,\alpha_{\rm free}$ does **not** close: backing out $\alpha_{\rm free}$ gives negative values over the box, because HTESSEL does not apply `asn` to the whole snow fraction — tile 7 is snow *sheltered* under high vegetation and carries a canopy-shaded albedo far below `asn`. That the residual is large and negative in exactly the melt months is direct evidence the canopy-masking term already does most of the work there, so no snow-free albedo is reported. Separately, ERA5's albedo scheme is a relative of HTESSEL, so `asn` agreement is not independent evidence; only ERA5's snow *extent* is observationally constrained, via assimilated IMS snow cover.

24.2 Accumulation is right; the melt is a month late and 30 % too slow

Snow water equivalent over the box, model vs ERA5 [mm], with the month-to-month increment:

	model	ERA5	Δ	δ(model)	δ(ERA5)
Jan	89.8	82.3	+7.5		
Feb	106.5	97.6	+9.0	+16.7	+15.3
Mar	122.1	114.5	+7.7	+15.6	+16.9
Apr	131.6	112.8	+18.7	**+9.4**	**−1.6**
May	98.1	65.3	**+32.8**	**−33.5**	**−47.5**
Jun	14.1	9.6	+4.5	−84.0	−55.7
Nov	49.2	41.5	+7.7	+25.9	+25.4
Dec	70.7	63.8	+6.9	+21.5	+22.3

Accumulation is essentially correct. Through the accumulation season the model is only 7–9 % high and the monthly increments track ERA5 almost exactly (+16.7 vs +15.3, +15.6 vs +16.9, +25.9 vs +25.4, +21.5 vs +22.3). The divergence is entirely in spring:

- **April**: the model is still *gaining* (+9.4 mm) while ERA5 has already peaked and begun to melt (−1.6). **The snowpack peaks a month late.**
- **May**: the model melts 33.5 mm against ERA5's 47.5 — **30 % too slow** — leaving it with 50 % more snow by the end of May.

Both figures come from ERA5. The satellite record confirms the timing directly and makes it worse: melt-out is 13 days late, and June cover is nearly double observed (§25.2).

24.3 May is albedo, not cloud — and it is a runaway

Model `amip_presentday` vs CERES on the same box:

	SW_{net} mod	SW_{net} CERES	Δ	$SW_{\downarrow}$ mod	$SW_{\downarrow}$ CERES	Δ
Apr	87.8	86.6	+1.2	176.9	187.7	−10.8
May	122.5	136.4	**−13.9**	220.7	224.2	**−3.5**
Jun	167.6	189.7	−22.2	215.7	228.2	−12.5
Jul	165.9	180.4	−14.5	193.8	208.7	−14.9

In May the downwelling flux is nearly right (−3.5) while the net is −13.9: the May deficit is *almost pure surface albedo*, i.e. the excess snow reflecting. June is roughly half cloud (−12.5 of downwelling) and half albedo. July is essentially all cloud (−14.9 down, −14.5 net), consistent with §22.4.

This closes a **self-reinforcing loop**: late melt → more snow in May → higher albedo → $-13.9\,\mathrm{W\,m^{-2}}$ less absorbed → less melt energy → snow survives into June. It also explains why the model melts slower *despite* having darker snow than ERA5: the brightness term is working against the bias, and the extent term is winning.

24.4 The gap F4 leaves: tundra is the largest cover type in the box

Area-weighted vegetation cover over land in the evaluation box, from the model's own `tvh`/`tvl`/`cvh`/`cvl` output:

	type	cover	
low	9 Tundra	**25.6 %**	**F4 does not touch this**
high	4 Deciduous Needleleaf	19.2 %	F4
high	18 Mixed Forest	5.2 %	
low	13 Bogs/Marshes	3.9 %	
high	5 Deciduous Broadleaf	3.6 %	
high	19 Interrupted Forest	3.5 %	
high	3 Evergreen Needleleaf	3.4 %	F4
high	16 Evergreen Shrubs	3.0 %	

Tundra is the single largest cover type in the box, and the campaign's best lever misses it entirely. F4 reaches about 24 % of the box (types 3, 4); tundra is another 26 %.

Its stomatal resistance is a table-internal outlier. `RVRSMIN(9)` $= 80\,\mathrm{s\,m^{-1}}$ is the *lowest value of any vegetated type* in HTESSEL:

```
crops 100  short grass 100  tall grass 100  semidesert 150  decid. broadleaf 175
interrupted forest 175  evergreen shrubs 225  decid. shrubs 225  evergr. broadleaf 240
bogs/marshes 240  desert 250  needleleaf 250  mixed forest 250  TUNDRA 80
```

The model therefore has arctic tundra transpiring *more freely than tropical grassland.* Tundra's closest physiognomic analogues in the same table — evergreen and deciduous shrubs — are both 225, and bogs/marshes, which co-occur with tundra in this very box, are 240. This is a **table-consistency** argument, checkable against the table itself and independent of our bias; it is not a value chosen because it helps. Contrast the standing criticism of F4, whose 250 → 1000 remains an unanchored 4× excursion.

24.5 Round 14 runs, and their predictions

All three are **namelist-only** (§24.7) — one shared binary, no rebuild, no serialisation against the model tree. That is what the `NAMSURFTUNE` work bought.

- **G2** — `RVRSMIN(3,4)` = 500. Half-strength F4.
- **G3** — `RVRSMIN(3,4)` = 2000. Double-strength F4.

  G2/G3 bracket G1 and map the response curve, answering the standing criticism that 1000 was never chosen from a measurement. *Prediction:* G2 keeps > 60 % of G1's gain (≥ +0.31 K) ~~and G3 adds little (< +0.15 K beyond G1)~~. First half held (64 %); second half **failed** — G3 added +0.355 (§24.6). *The response is a ramp, which was flagged in advance as an argument* **against** *the approach, not for a larger number.*
- **G4** — G1 plus `RVRSMIN(9)` 80 → 225 (tundra), keeping 1000 for types 3 and 4. *Prediction:* +0.2 to +0.5 K beyond G1's +0.521 (held: +0.431), ~~accompanied by a measurable drop in May/June snow water equivalent~~ — **falsified**, §24.6. As stated in advance, **T2m rose with no change in May–June snow mass, so the melt mechanism is unsupported** and the gain is the plain sensible-heat route. Guardrails: tundra is confined to high latitudes so tropical net TOA should stay near 42.6; check DJF against its own ±0.588 K threshold and the Nordic Seas SW RMSE that G1 already degraded.

24.6 Round 14 results: G4 adopted, mechanism falsified, `RVRSMIN` capped

G4 is the best configuration of the campaign and closes 48 % **of the boreal bias**, against G1's 26 % — and its stated mechanism is falsified. Both matter.

metric	control	G1	G2 (500)	G3 (2000)	**G4**
Siberia JJA T2m [°C]	9.73	+0.521	+0.336	+0.876	**+0.952** (t=7.68)
Siberia sfc net SW	153.78	+5.54	+3.21	+7.27	**+8.10**
SO SW RMSE	6.877	4.809	4.816	4.903	**4.800**
subpolar N Atl SW RMSE	5.007	4.872	4.866	4.728	**4.738**
Nordic Seas SW RMSE	9.058	+0.300	+0.098	+0.304	**+0.089**
tropics net TOA	42.61	+0.096	+0.041	+0.082	+0.124
global T2m RMSE [K]	1.579	1.553	1.558	1.546	**1.543**
DJF	—	+0.070	−0.325	**−0.798***	−0.385
MAM / SON	—	−0.296/+0.280	−0.462*/−0.030	−0.217/−0.012	−0.360/+0.185

G4 is seasonally clean on every season's own threshold, and it **largely repairs G1's Nordic Seas damage** (+0.300 → +0.089) — which nothing predicted and nothing yet explains. An unexplained improvement is as much a loose end as an unexplained degradation.

The melt mechanism is falsified

The prediction on record was a measurable drop in May–June snow water equivalent. Siberian land box SWE [mm], with run×year ANOVA thresholds:

	Mar (±3.53)	Apr (±3.85)	May (±5.25)	Jun (±2.72)
control	121.98	131.07	97.12	15.39
G1	+0.88	+0.37	**+5.58***	**+5.68***
G4	−1.44	−2.39	+2.75	+1.60

G4's snow mass did not move. T2m rose +0.952 K with May and June SWE unchanged within noise, so the gain is the **plain sensible-heat route**, exactly as for F4 — *not* the melt-timing route that motivated the run. The tundra lever stands on its independent table-consistency justification (§24.4) and it works; the mechanism claimed for it does not.

And F4 makes the snow bias worse. G1 *significantly increases* May–June SWE (+5.58, +5.68, both clearing threshold). The campaign's best boreal lever improves temperature while **aggravating** the very snow bias it was meant to address. Leading suspect: cutting evapotranspiration also cuts **sublimation**, a genuine snowpack sink. Untested.

`RVRSMIN` does not saturate — cap it at 1000

500/1000/2000 give $+0.336/+0.521/+0.876$: increments per doubling of +0.185 then +**0.355**, i.e. *accelerating*. **There is no knee.** 1000 is defensible only as *the largest value before the winter damage starts* — G3 at 2000 costs DJF −0.798, clearing ±0.588 and joining B5 and H2 in the "warms JJA, cools DJF" list. F4's magnitude is a fitted ramp, not a corrected bias, and §24.5 flagged this outcome in advance as an argument against pushing it further.

Two mechanisms for the June albedo bias are now dead: the snow *cover formulation* (round 13) and the snow *melt rate* (round 14). Untried candidates: the snow-albedo decay timescale; the canopy-masking tile-7 formulation, which the round-13 residual showed already does most of the work in the melt months; and the `RVVEGALB` vegetation albedo table, which the campaign has never touched.

24.7 Tuning without rebuilding: `NAMSURFTUNE`

The ECMWF `surf` library deliberately ships no namelist, so every HTESSEL lever in this campaign was a source edit plus a rebuild. That serialises experiments against the single model tree, and it puts AWI tuning inside upstream ECMWF files where it collides with EC-Earth's tuning of the same tables at every merge.

`surf/module/surfece.F90` is already an AWI module *inside* `surf` with its own namelist, so the overrides live there: `ECE_TUNE_RVRSMIN`, `_RVLAI`, `_RVCOV`, `_RVZ0H`, `_RVLAMSK` (all 0:20, by vegetation type) and scalar `ECE_TUNE_RQSNCR` — covering the whole F-series, the H-series and the skin-conductivity levers. They are applied by `SURFECE_APPLY_TUNING` from `susurf.F90` *after* `SUSURF_CTL` has filled the tables; nothing in the setup path derives from these entries, so a late override is equivalent to editing the table in place. Every element defaults to a sentinel meaning "keep the as-released value", so a run setting none of them is bit-identical to untouched code, and every applied override is logged to `NULOUT`.

They occupy their own `&NAMSURFTUNE` group, *not* `NAMECECFG`: `NAMECECFG` is read twice from the same `fort.4`, by `ECE_CONFIG` in `arpifs/ecearth.F90` and by `SURFECE_CONFIG` in `surf`, and a Fortran namelist read aborts with *invalid reference to variable* on any name the reading module does not declare — so adding the tune entries there killed the arpifs read at `su0yoma.F90:152`. Found by a one-day test run, which is the argument for always doing one. Exactly **one line** changes in an ECMWF file. G1 is now

```
NAMCLDP: RCL_INPSEA=0.2, RCL_INPPMIN=70000.0   NAMSURFTUNE: ECE_TUNE_RVRSMIN(3)=ECE_TUNE_RVRSMIN(4)=1000.0
```

25 Observational grounding: what ERA5 can and cannot settle

Rounds 13 and 14 leaned on ERA5 for the snow diagnosis. That is a methodological weakness worth stating plainly: **ERA5's land surface is the same HTESSEL family as ours**, so comparing our snow and flux fields to ERA5's is partly comparing a scheme against a sibling of itself. This section replaces ERA5 with genuine observations wherever one exists, and marks the places where none does.

25.1 Surface albedo: CERES corroborates ERA5, and the model bias is real

CERES supplies `sfc_sw_up` and `sfc_sw_down` separately, so their ratio is a surface albedo derived from a *radiative retrieval* rather than a land model. Siberian land box, 1990–2014, against model `amip_presentday` and ERA5 `fal`:

	CERES	ERA5	model	ERA5−CERES	**model−CERES**	$SW_\downarrow$
Mar	0.5540	0.5325	0.5055	−0.021	−**0.049**	109
Apr	0.5386	0.5045	0.4990	−0.034	−**0.040**	188
May	0.3918	0.3722	0.4285	−0.020	+**0.037**	224
Jun	0.1685	0.1730	0.2147	+0.005	+**0.046**	228
Jul	0.1357	0.1362	0.1487	+0.001	+0.013	209
Oct–Feb	—	—	—	+0.10 . . . + 0.13	—	6–42

ERA5 and CERES agree to ±0.02 in every month with enough sun to retrieve (Mar–Sep), which is what licenses ERA5's albedo for this purpose. The June excess is confirmed *independently* at +0.046 — slightly larger than the +0.042 measured against ERA5 (§24.1). The two diverge badly Oct–Feb, but at 6–42 W m^{-2} downwelling no albedo retrieval is trustworthy at these latitudes, so that is expected rather than disqualifying.

25.2 Snow cover and melt date: satellite, and ERA5 was flattering us

The Rutgers NH 24 km Weekly Snow Cover Extent CDR (NSIDC G10035, 1980–2024; NOAA visible-band snow charts to 1999, NOAA/NIC IMS thereafter) gives snow *extent* and *melt timing* from observation. `scripts/analysis/snowcove`

snow cover fraction	**Rutgers (sat.)**	ERA5	model	**G4**
Apr	0.950	0.991	0.964	0.961
May	**0.647**	0.796	0.890	0.893
Jun	**0.203**	0.261	0.380	**0.447**
Sep	0.075	0.178	0.257	0.249
Oct	0.612	0.755	0.825	0.834
melt-out (50 % crossing)	**11 May**	∼17 May	**24 May**	**28 May**

Three consequences.

(i) ERA5 was flattering the model. ERA5 itself runs +0.058 high in June and +0.149 in May against satellite, so the model excess quoted in §24.1 as +0.119 is really +**0.176** — June snow cover is close to *double* the observed value.

(ii) The model melts about 15 days late (control; `amip_presentday` is 13), now measured against an observation rather than inferred from a sibling model. *Note the two runs differ* — June cover 0.413 / melt-out 26 May for the control, 0.380 / 24 May for `amip_presentday` — and earlier drafts of this report quoted them interchangeably. This is round 14's "peaks a month late" (§24.2) confirmed independently and given a number.

(iii) G4 makes the snow field worse. June cover 0.380 → 0.447; melt-out 24 May → 28 May. The campaign's best *temperature* configuration pushes the spring snow cycle *further* from satellite, consistent with G1/F4 significantly increasing May–June SWE (§24.6). **This matters for the coupled goal:** a later melt is a later growing-season onset for LPJ-GUESS, which is a plausible route to losing boreal forest even while AMIP T2m improves. It should be weighed before G4 is committed to the PI spin-up.

Comparability caveat: Rutgers SCE is a **binary** classification per 24 km cell, whereas HTESSEL's `ZCVS` is a *sub-grid* fraction within a ∼100 km cell. The two agree on "how much of the region is snow" but differ in how a partially covered cell is counted, which matters most mid-melt. The **crossing date** is the robust comparison; absolute mid-melt fractions are not over-read here.

25.3 Where no observation exists: the turbulent fluxes

A claim is withdrawn here. The surface energy budget (`scripts/analysis/snow_energy.py`) appeared to show the model venting 3–5 W m^{-2} more upward sensible heat than ERA5 in February–April, which would have made turbulent exchange over snow the round-15 target. ~~The model loses its March–April melt energy to the atmosphere as sensible heat.~~ **That comparison has no observational authority**: ERA5's `sshf`/`slhf` are pure HTESSEL output, unconstrained by any assimilated observation, so it compares our scheme to a sibling of itself. It survives only as a hypothesis. The same caveat applies to the cold-content test on `tsn` (model within a few tenths of a K of ERA5), which is likewise model-versus-model — though as a *null* result it is the less dangerous of the two.

No satellite product measures surface turbulent fluxes. Only eddy-covariance towers can, and the Siberian candidates (Spasskaya Pad, ZOTTO) are point sites, not a gridded field. Any round-15 lever aimed at turbulent exchange over snow would therefore be tested against its *consequences* — melt date and snow cover, both now observationally anchored — rather than against the flux itself.

25.4 Summary: what each reference can support

quantity	reference	status
surface albedo	CERES up/down	**observational**; confirms model +0.046 June
snow cover extent, melt date	Rutgers/IMS	**observational**; model 13 days late
TOA and surface SW fluxes	CERES	**observational**
snow water equivalent	ERA5	partly constrained (SYNOP, IMS); forest bias
snowfall, snowmelt	ERA5	model output, weakly constrained
`sshf`, `slhf`, `str`, `tsn`	ERA5	**unconstrained — HTESSEL output**

ERA5 assimilates IMS snow cover, so ERA5 agreeing with Rutgers on extent is *expected* and is not independent corroboration — but it is what justifies having treated extent as ERA5's trustworthy field throughout.

26 Route B resolved: HTESSEL has no sub-grid snow depletion

Rounds 13 and 14 both failed on Route B, and the observational work (§25) sharpened the target without naming the cause. It is now named, and it also explains *why both rounds failed in the way they did.*

26.1 The arithmetic that settles it

Snow cover fraction is set from the **cell-mean** snow depth alone (`surfbc_ctl_mod.F90:317-322`):

$$\mathrm{ZCVS} = \min\bigl(1,\ \bar{d}_{\mathrm{cm}} \cdot \mathtt{RQSNCR}\bigr)\,, \qquad \mathtt{RQSNCR} = 1/10.$$

So cover can only fall below 1 once the *cell mean* drops under 10 cm. The model's actual depth distribution over the box:

	mean d	p_{10}	p_{50}	p_{90}	frac < 10 cm	ZCVS
Mar	51.1	31.4	48.7	75.6	0.037	0.964
Apr	48.9	24.8	47.2	75.8	0.044	0.964
May	28.8	4.5	**27.1**	56.5	0.181	**0.890**
Jun	4.8	0.0	2.7	13.5	0.842	0.380

The median May cell still holds 27 cm, so the formula returns essentially full cover — while the satellite record shows 0.647. Real 100 km regions at 27 cm mean depth are patchy: ridges blow bare, hollows hold metre drifts, canopies intercept. The model has no representation of that variance.

26.2 Why both previous attempts had to fail

This is the part that makes the diagnosis convincing rather than merely plausible.

- **Round 13 was doomed by arithmetic.** Changing `RQSNCR` 1/10 → 1/30 moves the threshold from 10 to 30 cm — which does nothing at all where depths are 27–51 cm. It could only bite where depths sit *near* the threshold, i.e. 13–27 cm, which is October and November. That is precisely where the response appeared (−8.2 albedo points in October, −0.95 in June, §23.2). The lever was not weak; it was arithmetically incapable of acting in spring.
- **The commented-out `tanh` and `sqrt` variants** at `surfbc_ctl_mod.F90:325` share the defect — still functions of the cell-mean depth. $\tanh(27/10) = 0.99$.
- **Snow brightness is the wrong direction** (model snow is darker than both ERA5 and CERES) and **energy supply is not short** in Mar–Apr (§25.1).

26.3 The fix, validated offline before implementation

A depth-only function cannot work; the depletion must respond to something that tracks the *state* of the pack. Niu & Yang (2007), as used in Noah-MP and CLM:

$$\mathrm{SCF} = \tanh\left(\frac{d}{2.5\, z_{0g}\, (\rho_{\mathrm{bulk}}/\rho_{\mathrm{new}})^m}\right)$$

The density ratio supplies exactly the missing behaviour: aged, melting snow is denser, so it patches out at a given depth. It needs **no new prognostic** — depth and density are already carried.

Tested on existing output *before* writing any code, Siberian box, `amip_presentday` vs Rutgers 1990–2014:

	satellite	as-released	$z_0 = 0.01$	$z_0 = 0.02$
Jan–Mar	0.999	0.964	0.964	0.952
Apr	0.950	0.964	0.948	0.893
May	**0.647**	0.890	0.772	**0.588**
Jun	**0.203**	0.380	0.321	**0.180**

$z_0 = 0.02$ matches the melt season closely at a small cost in midwinter cover; $z_0 = 0.01$ is conservative, leaving winter untouched and closing about half the May–June gap.

Provenance. The problem and the two standard remedies are well established in the land-surface literature: Liston (2004) sets out why a cell-mean depth cannot represent depletion; Niu & Yang (2007) give the density-dependent tanh form used here, calibrated against North American basin observations and adopted in Noah-MP and CLM; Swenson & Lawrence (2012) give the main alternative, separate accumulation and depletion curves driven by a snow-depth standard deviation, which is more faithful but needs an extra prognostic; Roesch et al. (2001) take the third route of adding sub-grid *orographic* variance in a GCM. Douville et al. (1995) is the classic depth-threshold form that HTESSEL's `RQSNCR` formulation resembles, and Dutra et al. (2010) documents the `LESN09` scheme this campaign has been modifying. That CMIP-class models share the bias found here is documented by Thackeray et al. (2015), and its consequence for the snow-albedo feedback spread by Qu & Hall (2014) and Loranty et al. (2014). Betts & Ball (1997) remains the reference for boreal canopy masking, which is why the tile-7 residual behaves as it does (§24.1).

Implemented as `ECE_SNOW_SCF` in `&NAMSURFTUNE` (§24.7), with `ECE_SNOW_SCF_Z0`, `_M`, `_RHONEW`. **Default is** 0 **= the as-released formula**, so the change is inert unless switched on, consistent with every other entry in that namelist. Verified by a 1-day run: the switch logs to `NULOUT`, no namelist errors, the model steps normally.

26.4 What this still does not tell us

- **The temperature leverage remains unproven** (§42.4). G4 gained a full kelvin with snow mass unchanged and cover *worse*. Fixing the cover may buy much less T2m than its $-5.6\,\mathrm{W\,m^{-2}}$ suggests. Where it should matter is *coupled*, through LPJ-GUESS growing-season onset.
- **The March–April mass deficit is a separate, weaker claim.** It rests on ERA5 SWE (forest-biased) and may be a second effect or partly an artifact (§24.2).
- **Comparability.** Rutgers is a *binary* 24 km classification, so its 0.647 is "65 % of cells classed snow", not a true areal fraction. The melt-out *date* is the robust comparison; mid-melt absolutes are softer, and the z_0 choice should not be over-fitted to them.

27 Round 15 (2026–08–04): the snow fix works, buys no summer, and cools winter

Three runs, all namelist-only on one binary: **I1** = G4 + depletion, **I2** = depletion alone (isolating the snow route from the vegetation route), and **I3** = G4 + the SDOR-scaled variant. Sections §27.1–§27.5 were written on I1 and I2 alone; **I3 is evaluated in §27.6**, and it changes the round's verdict — the snow route *does* buy summer temperature, but only in the scale-aware formulation, and it presents a bill in the Arctic energy budget.

27.1 The scheme does exactly what it was built to do

Siberian snow cover	satellite	control	G4	**I1**	**I2**
Apr	0.950	0.962	0.961	0.931	0.931
May	0.647	0.891	0.893	**0.703**	**0.721**
Jun	0.203	0.413	0.447	**0.160**	**0.175**
melt-out	**11 May**	26 May	28 May	**12 May**	**13 May**

Melt-out went from **15 days late to 1 day late**, against the satellite record.

27.2 And it delivers no summer temperature

	surface SW gain	JJA T2m	K per $\mathrm{W\,m^{-2}}$
G4 (vegetation route)	+8.10	**+0.952**	0.12
I2 (snow route alone)	+2.91	**+0.094** (noise)	**0.03**

This settles §42.4: the link is weak. Route B was a genuine, satellite-measured radiative bias with almost no temperature leverage — per watt it is **four times less efficient** than the vegetation route. The reading is clean: energy added over a *melting* snowpack goes into melt, not sensible heat, which is precisely what it did. Fixing it remains right — the snow field is now correct, which is what matters for LPJ-GUESS growing-season onset — but Route B is not where the remaining boreal kelvin lives.

27.3 The winter response is a compensating error being removed

Scored against ERA5 rather than against the control (period-clean, `amip_presentday`):

bias vs ERA5 [K]	DJF	MAM	JJA	SON
control	−1.95	−1.97	−2.58	−2.69
G4	−2.33	−2.33	**−1.63**	−2.51
I1	**−4.71**	−2.28	**−1.33**	−3.46
I2	−4.63	−2.36	−2.49	−3.77

The control is too cold in every season, so there was never a warm bias to protect — scoring against observations rather than against the control is what makes that visible, and eleven rounds had not done it.

What this does NOT establish. ~~The excess snow cover had been propping DJF up by ∼2.7 K, masking a −4.7 K bias that was there all along.~~ That reading was asserted here and is **not supported**: winter cover is *unchanged* between the two schemes (0.963 vs 0.964, both saturating at 43 cm, §27.4), so excess *winter* cover cannot have been doing the propping. What is measured is only that **I1 cools DJF by −2.76 K by an unidentified route**. Whether that unmasks a pre-existing bias or creates a new one is **untested**.

The point is sharpened by a precedent this report got the other way round: H1 produced the same signature (−1.233 K DJF by cutting snow-tile fraction, §23.2) and was *rejected* for it. Identical evidence cannot be disqualifying in one run and revelatory in another. Until the route is identified, the honest position is that the snow fix and the winter response are two findings, not one story — which is also why round 16 tests the winter on its own terms rather than assuming it is recoverable.

27.4 Ruled out by measurement: it is not a radiation-supply problem

Downward longwave over the box, model vs CERES [W m^{-2}]: Oct −**12.9**, Nov −**11.9**, Dec −3.1, Jan +0.1, Feb +5.6 — **DJF mean** +0.9**, essentially exact**. That kills the Pithan et al. (2014) supercooled-liquid route for midwinter. The Oct/Nov deficit is real and is a *separate* autumn problem. The +30 pp polar-night cloud excess against CERES is retrieval failure over snow at night, not evidence.

Still unresolved: winter cover is **unchanged** (0.963 vs 0.964 — both formulas saturate at 43 cm) and winter SWE is slightly *higher*, so the DJF response is not a direct cover or mass effect. `str` and `sshf` shifts are the surface being colder, i.e. responses. ~~`stl1` and `skt` are unusable (∼ 0.07 K — the same dead-field trap as pressure-level `r`).~~ **That was wrong**: both fields are sound (210–319 K). The 0.07 was 273/3600 — an analysis-side bug that divided an *instantaneous* temperature by the accumulation period. Fixed in `boxcache.py`; unlike pressure-level `r`, which really is identically zero, these fields were never broken. Using them (§27.5) shows I1 and I2 cool `stl4` by ∼2.1 K, so the unexplained winter signal reaches the *deep soil* — which matters because LPJ-GUESS is fed `ST1L`–`ST4L`. H1 cooled DJF by −1.233 through a different route to the same tile fractions, so this is the second time cutting snow-tile fraction has cooled winter hard.

27.5 Does the warming reach the skin and the soil?

A lever that raises 2 m air temperature by mixing heat *away* from the surface would improve the metric we score while degrading the surface itself. That is not a cosmetic concern: **LPJ-GUESS is fed `ST1L`–`ST4L_LPJ` (soil temperature, layers 1–4) and `SM1L`–`SM4L_LPJ`** alongside `T2M_LPJG` (`lpj_guess/framework/OasisCoupler.cpp:17-95`), so a warmer-air/colder-ground configuration would hand the vegetation model an incoherent state. Skin temperature itself is not coupled.

Siberian land box, JJA, deltas against control [K]:

	T2m	`skt`	`stl1`	`stl4`	Δ(`skt`−T2m)
control (absolute)	9.73	10.18	9.31	−1.31	0.45
G4	+0.952	**+0.990**	**+0.883**	+0.157	+0.037
I1	+1.250	+1.318	+1.160	**−2.092**	+0.068
I2	+0.094	+0.100	+0.046	**−2.190**	+0.006

G4's warming is genuine, not a redistribution: skin and layer-1 soil warm *as much as* the air and the skin-to-2 m gradient barely moves (+0.037 K). What LPJ-GUESS would receive from G4 is coherent.

But I1 and I2 cool the deep soil by ∼2.1 K, I2 doing so with essentially no surface change at all. The unexplained winter response of the snow scheme therefore reaches `stl4`, which *is* a coupled field. That is a second, independent reason to keep the snow-depletion scheme out of a coupled run until round 16 explains the winter behaviour.

This subsection exists because the diagnostic was wrongly believed impossible: the soil and skin fields had been written off as dead (§27.4) on the basis of an analysis-side scaling bug.

27.6 I3 (2026–08–04): the strongest summer lever, and the Arctic bill

I3 = G4 + mode 2, the SDOR-scaled depletion, calibrated to match I1 *over the Siberian box* so that any remaining difference would be spatial structure (§26.3). It does not behave like I1, and the differences are the point.

What it wins.

	control	G4	I1	**I3**	CERES
Siberian JJA T2m [°C]	9.728	+0.952	+1.250	**+1.322** (t=10.61)	—
Siberian surface net SW [W m^{-2}]	153.78	+8.10	+12.77	**166.58**	**166.26**
global T2m RMSE vs ERA5 [K]	1.579	−0.036	**+0.048**	**−0.023**	—
global net TOA [W m^{-2}]	0.643	−0.128	−0.021	+0.056	—
SO SW RMSE	6.877	−2.076	−1.906	−2.065	—

+1.322 K is the **largest JJA gain of 36 evaluated tuning runs**, and the Siberian surface SW deficit that §22 opened this whole thread with **closes to within** 0.3 **W m^{-2}** of CERES. Global TOA does not move, and where I1 *degrades* global T2m RMSE (+0.048), I3 improves it. Per season it dominates I1 outright — more JJA, 0.32 K less DJF damage, and I1's significant autumn penalty (−0.766*) collapses to a null:

∆ vs control [K]	DJF	MAM	JJA	SON
threshold (own season)	±0.601	±0.394	±0.244	±0.441
G4	−0.385	−0.360	+0.952*	+0.185
I1	−2.759*	−0.306	+1.250*	−0.766*
I3	**−2.439***	−0.214	**+1.322***	**−0.105**

What only the full guardrail set revealed. The Siberian-box scripts alone would have promoted I3. They were run first, and the verdict "best lever of the campaign" was drafted on them — prematurely, and caught only because the guardrails were demanded before writing. Four guardrails move together, all one way:

guardrail	control	I1	**I3**	CERES
net TOA 60–90N [W m^{-2}]	−97.74	+0.87	**−96.02** (**+1.72**)	**−97.98**
net TOA NH [W m^{-2}]	0.280	+0.245	**0.952** (**+0.672**)	**0.08**
NH−SH albedo	−0.003	−0.002	**−0.007** (**−0.004**)	—
Nordic Seas SW RMSE	9.058	+0.196	**9.557** (**+0.499**)	—

This is the mechanism *working as designed*: removing high-latitude snow cover admits more shortwave at high latitude. The Arctic did not have that energy spare. The control sat almost exactly on the observation at 60–90N (−97.74 vs −97.98); I3 leaves it 1.96 away. The NH net TOA change (+0.672) and the NH−SH albedo change (−0.004) are each the **largest of all 37 runs**, and Nordic Seas is the **only significant SW RMSE degradation in the campaign** — `rmse_significance.py` flags exactly two runs there and I3 is one (+0.07 against a ±0.051 threshold).

Caveat against these four, stated so it is not over-read: these are PI runs scored against present-day CERES, and this report's own rule (§42.7) is that the PI target is net TOA ≈0, **not** the CERES column. That weakens the "away from CERES" reading of the two regional TOA rows. It does *not* weaken the NH−SH albedo or Nordic Seas results, which are model-versus-model and model-versus-RMSE.

Why this matters more for the coupled run than for AMIP. All four are Arctic energy-*gain* guardrails, and AMIP structurally hides their consequence: prescribed SST and prescribed sea ice absorb 1.7 W m^{-2} without responding. In the coupled PI spin-up — the configuration I3 is a candidate for — they would not. This is the same structural blindness as §42.5, in the opposite direction.

27.7 The winter penalty is not radiative

§27.4 left the DJF route unidentified. It can now be excluded as radiative. Siberian land, DJF, all four configurations:

Siberian DJF	snow albedo	SWE [m]	absorbed SW [W m^{-2}]
control	0.6789	0.6933	8.16
G4	0.6760	0.6901	8.18
I1	0.6800	0.6926	8.16
I2	0.6796	0.6958	8.16
I3	0.6797	0.6944	8.22

Albedo, snowpack and absorbed shortwave are **unchanged to three decimals**, and there are only ~8 W m^{-2} of insolation in the box in midwinter regardless. **A −2.4 K penalty cannot be constructed from that budget.** This generalises the finding that killed the compensating-error story (§27.3, winter cover 0.963 vs 0.964) from cover to the whole radiative chain, and it holds for I2, which carries the penalty with no vegetation change at all. The route must be **thermal — the snow/exposed-tile coupling** — which is precisely what round 16 (§28) tests, so the diagnosis and the running experiment agree.

27.8 Direct observations at last: RIHMI-WDC station soil temperature (2026–08–05)

Every soil claim in this report had rested on ERA5, whose land surface *is* HTESSEL — the same snow scheme, the same soil thermal conductivity — so it is a sibling, not a witness. **RIHMI-WDC (Sherstyukov v3)** ends that: daily soil temperature at Russian meteorological stations, 1963–2024, twelve depths from 2 to 320 cm, measured **under natural cover** so the snowpack is intact. 43 of the 110 stations lie in the campaign box. Model and ERA5 are sampled *at the station points*, not box-averaged, which is the one representativeness control available without land-cover weighting.

What the archive cannot do: depths 2/5/10/15 cm carry *no* winter data — the Russian network withdraws the shallow Savinov thermometers for the cold season. So `stl1` (0–7 cm), where the damage is largest, **has no observational counterpart**; `stl2` is the shallowest verifiable layer and bounds it. Only `tsoil_qc == 0` is used.

	obs depth	OBS	model	bias	OBS offset	model offset
`stl2`	20 cm	−6.07	−5.03	**+1.04**	**+17.78**	+20.11
`stl3`	40/80 cm	−3.58	−3.19	**+0.39**	**+20.28**	+21.95
`stl4`	120–240 cm	+0.34	−0.23	**−0.57**	**+24.18**	+24.85

Model is `amip_presentday` (1990–2014), period-clean. Offset = soil minus 2 m air, each dataset using its *own* air, which is what makes it insensitive to the 2.8 K air-temperature change the I-series also carries.

Two results, both DJF-only. In *winter* the control soil is correct to ~1 K at every verifiable depth — there was no pre-existing *winter* soil bias to remove — and the model is mildly *over*-insulated, by +0.7 to +2.3 K of offset.

~~The control soil is correct to ~1 K at every verifiable depth.~~ **That was measured in DJF only and does not hold in summer** (§27.9): the same control is −3.4 to −5.4 K too cold in JJA, a bias *larger* than the 2.58 K Siberian air bias this campaign has been chasing. The winter statement stands; the unqualified one does not.

~~The winter soil-air offset under a deep Siberian snowpack is +10 to +20 K.~~ That was quoted from memory of the permafrost literature and is too low: the station record gives +17.8 to +24.2 K. The correction strengthens rather than weakens the conclusion below.

Scored on offset (the tuning runs are 1872–1915, so absolutes are not period-comparable):

	`stl1`	`stl2`	`stl3`	`stl4`
control	19.87	20.39	22.24	25.13
G4	20.27	20.81	22.72	25.82
I1	**1.53**	**2.61**	**6.36**	**15.25**
I3	1.41	2.54	6.46	15.63
OBSERVED	—	**+17.78**	**+20.28**	**+24.18**

I1 is −15.2 **K wrong at 20 cm**, −13.9 at 40–80 cm, −8.9 at 1.2–2.4 m. In raw terms its soil sits at −25.8 °C where stations measure −6.1 °C. The snowpack has stopped insulating at every depth.

27.9 The summer soil bias, and why it dwarfs the air bias

The DJF validation above was extended to JJA at the same 43 stations, and it changes the picture completely.

`amip_presentday`	OBS	DJF model	bias	OBS	**JJA** model	**bias**
boreal `stl2` 20 cm	−6.07	−5.03	+1.04	13.68	10.33	**−3.35**
boreal `stl3` 40/80 cm	−3.60	−3.19	+0.42	10.17	5.84	**−4.33**
larch `stl2` 20 cm	−11.05	−7.76	+3.29	12.52	8.33	**−4.19**
larch `stl3` 40/80 cm	−7.80	−5.54	+2.26	7.64	2.29	**−5.35**

The soil is 3–5 K too cold in summer while being slightly too warm in winter, and the summer soil bias is *larger than the 2.58 K Siberian air bias* the campaign has spent 45 runs on. It was never looked at because no soil diagnostic was scored until §27.8.

It is far worse coupled. LPJ-GUESS's `soiltemp25cm` is the OIFS soil temperature received through OASIS, so it measures the same quantity:

larch zone, ≈25 cm	DJF	**JJA**
RIHMI obs (20/40 cm bracket)	−11.4 / −8.3	**+12.5** / + **10.0**
06 Baseline	−15.96	+1.80
080a CRUNCEP	−16.35	+1.38
09A newSeaIce	−18.09	+0.72
10A (G4)	−21.04	**−0.37**
10B (G4 + snow depletion)	−20.85	+**2.31**

Against an observed ≈ + 11.8 °C at 25 cm, every coupled run sits at +0.7 to +2.3 °C: a ∼10 **K summer soil cold bias**. **Larch cannot build cover in soil near freezing all summer**, which makes this a more plausible driver of the forest collapse than either the evergreen gate or grass competition (§31) — those describe *which* PFT wins, this describes why none of them grow.

Representativeness caveat, and it is a real one. RIHMI plots are open, bare or short-grass sites in full sun; a forested cell's summer soil is legitimately cooler under canopy shade. Part of AMIP's −4 K is therefore representativeness and cannot be separated without land-cover weighting. What is *not* representativeness is the ∼6 K gap between AMIP and coupled at the same stations with the same comparison — that is real degradation contributed by the coupled vegetation feedback.

27.10 The snow scheme, reconsidered on coupled evidence (2026–08–05)

~~`ECE_SNOW_SCF` is switched off and stays off; no production run should set it.~~ **Retracted the same day.** That verdict was reached on AMIP alone, and **AMIP was the wrong testbed for it**: it has no vegetation response, which is precisely the thing the scheme was helping. On the coupled 10A/10B pair — identical binaries, identical settings (`ECE_SNOW_SCF=1`, `Z0=0.016`), differing *only* in the scheme:

larch zone	BNS LAI	C3G LAI	grass : larch	TREEFPC
09C reference	0.096	0.400	4.2×	0.048
10A (G4)	**0.019**	0.388	**20.4**×	**0.019**
10B (G4 + scheme)	**0.134**	0.830	6.2×	**0.065**

10B carries 7× **the larch and** 3.4× **the tree cover of 10A**, and G4 alone (10A) is *worse than the 09C reference* — it actively degrades the forest. NH total LAI bias against the Lund CRUNCEP reference halves, −0.62 (−28.1 %) → −0.28 (−12.9 %).

And the winter soil damage does not appear coupled. 10A and 10B are −21.04 and −20.85 in DJF — identical — while JJA goes −0.37 → +2.31. The ∼5 K winter cooling relative to the 06/070/080a baselines came in with G4 + CERES-init, *not* with the scheme. And 10A's summer soil is the outlier against every earlier baseline (+1.4 to +3.0); **the scheme restores the baseline range rather than departing from it**, recovering 2.7 K of the ∼12 K deficit.

~~Standing position: the scheme is not rejected. The AMIP soil error is real, but it is not disqualifying for coupling, where the scheme is the difference between a badly degraded and a substantially recovered boreal forest.~~ **RETRACTED the same day — see §32.** The claim that the winter damage does not appear coupled was an *artefact of where I sampled*: I measured only the larch zone, where 10A is already saturated near −21 °C, so both runs looked alike. Across the full station network 10B is −16.2 K in DJF and −18.3 K in January. The AMIP testbed had predicted exactly this.

Why. Verified against *direct observations* — RIHMI-WDC station soil temperature, 43 stations in the box, under natural cover (§27.8) — the scheme destroys the winter snow insulation. Observed thermal offset (soil minus 2 m air) is +17.8 K at 20 cm, +20.3 at 40–80 cm, +24.2 at 1.2–2.4 m. The control reproduces those to within 0.7–2.3 K. I1 gives +2.6, +6.4, +15.3: an error of −15 K at 20 cm. The control soil itself is correct to ∼1 K at every depth the Russian network reports in winter, so this is a **new error, not the removal of an old one**. Since `ST1L–ST4L` feed LPJ-GUESS, that alone disqualifies it for coupling.

What we still do not know, and it matters. The mechanism is *unidentified.* The monthly decomposition rules out the obvious candidates: Siberian snow cover changes by −0.02 in October and SWE not at all, so it is neither a cover nor a mass effect. The soil *leads* the air (October: soil −4.19 K, air −0.02), so it is a surface-coupling change rather than an atmospheric one. Winter snow density falls 20–27 kg m^{-3}, which with unchanged SWE means a *deeper, less dense* pack — that insulates *better*, so density is a consequence of a cold pack, not its cause. ~~The tanh formula bites hardest in October when snow is shallow, cutting cover and exposing the soil.~~ Measured: cover is essentially unchanged. That explanation was written into `soil_temp_vs_obs.py` before it was checked and is wrong.

Reactivation therefore needs three things, in order:

1. **Diagnose `ZCVS`.** It is not in the output stream, which is why the mechanism is still open — every field we do write exonerates itself. Nothing should be rebuilt before the snow-tile fraction can actually be seen.
2. **Accumulation/ablation hysteresis.** Our implementation applies one depletion curve year-round. Both Niu & Yang (2007) and Swenson & Lawrence (2012) use *different* curves for accumulation and ablation, and that omission is the most likely structural defect. It would also match the measured timing: the benefit is May–June, the damage begins in October, so the two are cleanly separable in season.
3. **Re-verify on the SOIL OFFSET, not on snow cover.** The scheme was accepted in round 15 because it fixed melt-out to within a day of satellite. That was true and it was not sufficient. §27.8 is the test any successor must pass.

And weigh the prize before spending the effort. I2 — the scheme alone, no vegetation levers — bought +0.094 **K JJA, inside the noise floor**. The value of the snow fix is not temperature; it is that the snow field becomes correct, which matters for LPJ-GUESS growing-season onset. That is a real but narrow benefit, and it does not justify a source change plus a run family while a 20 K soil error remains unexplained.

28 Round 16 (2026–08–04): the snow-tile skin conductivity

`vlamsk_mod.F90` hardcoded the snow-tile skin conductivities, so they were untunable. λ_{sk} sets how tightly the skin couples to the medium below it; a small value lets it radiatively decouple and crash in polar night. Physically $\lambda_{sk} \approx k_{\text{snow}}/d_{\text{skin}}$ with k_{snow} 0.1–0.4 W m^{-1} K^{-1} and d_{skin} 0.01–0.05 m, so **roughly 2–40**. As released:

constant	value	tile
`ZSNOW`	**7**	5, **exposed snow** — the weakly coupled one, and the tile covering tundra (25.6 % of the box)
`ZSNOW_GLACIER`	8	5, where SWE > 9000 mm
`ZSNOWHVEG`	20	7, snow under high vegetation
`ZLARGESN`	50	5, when melting

The as-released 7 sits at the decoupled end of the physical range, on precisely the tile whose *fraction* round 15 corrected. Now exposed as `ECE_LAMSK_SN/_SNGL/_SNHV/_SNMLT`, all defaulting to the shipped values.

run	setting	prediction
J1	I1 + `ECE_LAMSK_SN: 15`	DJF +1 to +3 K, JJA barely moves
J2	I1 + `ECE_LAMSK_SN: 25`	same, stronger

Falsifier on record: if DJF does not respond, skin conductivity is not the route, and the winter bias lies in boundary-layer mixing instead (Holtslag et al. 2013; Sandu et al. 2013). Also watched: melt-out and May/June cover must *stay* fixed — drift back toward the control would mean the two changes interact and the pair is not separable.

28.1 Round 16 result (2026–08–05): falsified, and the falsifier was on record

Δ vs control [K]	DJF	JJA	SON	global T2m RMSE
I1 (the base)	−2.759*	+1.250*	−0.766*	1.627
J1 (λ_{sk} 15)	**−2.834***	+1.313*	−1.132*	1.581
J2 (λ_{sk} 25)	**−3.226***	+1.246*	−0.879*	1.597
control	—	—	—	1.579

The prediction was **DJF +1 to +3 K**. Measured: DJF got **worse**, and *monotonically with the knob* — −2.834 at λ_{sk}=15, −3.226 at 25, against I1's −2.759. Both runs carry a global T2m RMSE *worse than the control.*

The pre-registered falsifier fires: skin conductivity is not the winter route. By §27.7 the route is non-radiative, and λ_{sk} was the leading non-radiative candidate; with it excluded, **boundary-layer mixing** (Holtslag et al. 2013; Sandu et al. 2013) is what remains. The round is doubly moot in practice, since its base I1 is now deactivated (§27.10) — but the falsification stands on its own and the diagnosis it leaves behind is the useful part.

Both J runs were built on I1, before §27.6 showed I3 to be the better base in every season. That was the correct decision with the information available and the round is still diagnostic — it tests whether λ_{sk} is the winter route at all, and §27.7 shows I1 and I3 share the same non-radiative penalty, so an answer on I1 transfers. But if λ_{sk} does prove to be the route, **the production pair must be re-run on I3**, and the Arctic guardrails of §27.6 re-checked on the combination rather than assumed to superpose — superposition has been wrong in sign twice in this campaign (§42.7).

29 Land albedo (2026–08–04): half of it is the snow tile

With the boreal budget nearly spent (§21.3) the target moved to the ∼0.7 K *universal* land bias, and §21.1 localised its energy to a clear-sky shortwave deficit — cloud is the wrong sign. Decomposing that deficit assumption-free (each dataset divided by its *own* downward flux) gives an all-sky land surface albedo +0.0154 too high against CERES.

29.1 Snow is a tile, not a vegetation type

The first attribution split that +0.0154 by *dominant vegetation type*, built from `tvh/cvh/tvl/cvl`. That map has no snow in it: in HTESSEL snow is a separate tile (5 on low vegetation, 7 under high) that **overlies** every vegetation type. So each type was part-time snow, and the ranking the split produced was largely a map of *snow duration* — evergreen needleleaf is snow-covered 53.9 % of the year, tundra 63.2 %, "bare soil" 49.2 % (a bucket holding the Sahara and high-Arctic bare ground together). An annual-mean $T_{2m} > 5°$C filter does not fix this: a cell averaging 6°C still has months of snow.

Masking snow **per cell per month** (SWE < 1 mm, sunlit months only) splits the error almost exactly in half:

Δ albedo vs CERES	all sunlit months	snow-free only
ALL LAND	**+0.0154**	**+0.0080**
⇒ *snow contributes* +0.0074*, the snow-free surface* +0.0080		

and the per-type ranking *inverts*:

type	snow-covered	all months	**snow-free**
Evgr Needleleaf	53.9 %	+0.0250	**+0.0032**
Bogs/Marshes	37.1 %	+0.0143	+0.0039
Evgr Shrubs	41.2 %	+0.0150	+0.0058
Evgr Broadleaf (tropical)	6.9 %	+0.0017	**−0.0003**
Crops	25.9 %	+0.0215	**+0.0204**
Semidesert	22.5 %	+0.0342	**+0.0169**
bare soil	49.2 %	+0.0240	**+0.0145**
Irrig Crops	24.6 %	+0.0253	**+0.0138**
Tundra	63.2 %	+0.0064	**+0.0105**

Every forest type falls inside +0.006 once snow is removed, tropical broadleaf to −0.0003. **This vindicates §22.1 and §22.3**: `RVVEGALB`'s high-vegetation entries are sound and the snow season really is the story. What is *new* is the residual +0.0080, which sits on the sparse and cultivated surfaces where the soil background shows through — and on tundra, which gets **worse** under masking (+0.0064 → +0.0105) because snow was hiding a genuine snow-free summer error. Nothing in 41 runs has touched that half.

29.2 The snow half already has a lever

Land albedo over snow-covered sunlit months, model-internal:

run	snow-month albedo	vs G4	all-land
G4	0.5185	—	0.2586
I1 = G4 + mode 1	0.5011	−0.0174	−0.0024
I2 = mode 1 alone	0.4996	−0.0189	−0.0018
I3 = G4 + mode 2	**0.4689**	**−0.0496**	**−0.0052**

I3 removes ∼70 % of the snow half, I1 ∼32 % — and I3 was calibrated to match I1 *in the Siberian box*, so globally it is twice as strong as intended. That is new information about mode 2 and it is the same over-reach that §27.6 bills at 60–90N: the SDOR scaling does most of its work outside the box it was tuned in.

29.3 Why the two halves cannot simply be stacked

The obvious next move — keep I3 and add a vegetation-albedo correction on top — is **not additive in the direction that matters**. A darker land surface at high latitude adds absorbed shortwave to precisely the region I3 has already pushed 1.72 W m^{-2} past CERES. The two changes compete for one energy budget. Consequences for the plan:

- **The mid-latitude and subtropical entries are the safe ones.** Crops (+0.0204), semidesert (+0.0169), bare soil (+0.0145) and irrigated crops (+0.0138) lie outside the Arctic, so correcting them attacks the ∼0.7 K *universal* bias without adding to the 60–90N surplus.
- **Tundra (**+0.0105**) is the dangerous one.** It is the single entry that *would* stack with I3, being boreal and in-season — which makes it both the most attractive for the summer target and the one most likely to break the Arctic guardrails.
- **The snow-free half is not reachable by `RVVEGALB` alone** where bare fraction dominates; there the background soil albedo climatology carries it, and that is an ancillary field, not a namelist parameter (§22.2).

Assets: `land_albedo_snow_split.py`, chained into `evaluate.sh -obs`.

30 Round 17 (2026–08–05): K1 adopted, and land albedo closed out

Two families on the G4 base, run concurrently and deliberately orthogonal: **L** puts a moss/organic layer on the boreal types, **K** corrects the snow-free land albedo diagnosed in §29.

30.1 L: the moss proxy is falsified, monotonically the wrong way

λ_{sk} 10 → 2.9 (L1) and 10 → 1.7 (L2) on types 4 (larch) and 9 (tundra) only — the two most boreal-confined types, 62 % and 70 % of their area poleward of 50N. Gaillard (2025) attributes most of a >2 K high-latitude summer gain to a surface organic layer, which HTESSEL lacks entirely.

	JJA	vs G4	verdict (threshold ±0.244)
G4	+0.952*	—	the base
L1 (λ_{sk} 2.9)	+0.784*	−0.168	inside → null
L2 (λ_{sk} 1.7)	+0.638*	**−0.314**	**outside → significant degradation**

More moss is *worse*, the opposite of the prediction, and the monotonic ordering makes a noise explanation implausible. **Caveat that limits the claim**: λ_{sk} is only a *proxy* for an organic layer — it changes the skin-to-substrate coupling but not the heat capacity, not the moisture retention, and not the summer evaporative regime. This falsifies *the proxy*, not necessarily the Gaillard mechanism. A real organic layer would need a soil-column change, which is a much larger piece of work.

30.2 K: adopted

K1 = G4 + `RVVEGALB` on crops (×0.887), irrigated crops (×0.910) and semidesert (×0.906) + `ECE_TUNE_SOILALB` 0.95. K2 is the soil background *alone*, isolating the ∼25 % of land that `RVVEGALB` cannot reach. **Tundra was deliberately excluded** so K stayed orthogonal to L and did not add absorbed shortwave to the Arctic (§29.3).

	Siberian JJA	DJF	net TOA 60–90N	**global T2m RMSE**
control	—	—	−97.737	1.579
G4	+0.952*	−0.385	−97.603	1.543
I3	+1.322*	−2.439*	**−96.017**	1.556
K1	**+1.036***	−0.496	−97.463	**1.524**
K2	+0.913*	−0.144	−97.525	**1.518**
CERES	—	—	−97.98	—

Both are clean — every season inside its own threshold — and they are the **two best global T2m RMSE of the campaign**. The tundra exclusion did its job: 60–90N stays near the control, against I3's $+1.72\,\mathrm{W\,m^{-2}}$ excursion. In the Siberian box neither K is significant against G4 (+0.084, −0.039, both inside ±0.244), *which was predicted in advance* — K's mass is crops, semidesert and the subtropical desert belt.

30.3 What K bought, measured on the metric it was designed for

global LAND, vs G4	land T2m	land SW_{net}	land albedo
K1	**+0.089** K	$+0.77\,\mathrm{W\,m^{-2}}$	−0.0059
K2	+0.076 K	+0.48	−0.0038

The lever did what it was built to do. The snow-free residual was +0.0080; K1 removed −0.0059 of it, **74 %**. The measured sensitivity is 0.089/0.77 = **0.116** K per $\mathrm{W\,m^{-2}}$ against the 0.12 assumed — essentially exact. The geography is right too: land equatorward of 60N warms +0.109 K while land poleward of 60N *cools* 0.055 K, exactly what excluding tundra was meant to achieve.

~~Predicted ∼1.1 W m⁻² and 0.1–0.2 K of land warming.~~ Measured $0.77\,\mathrm{W\,m^{-2}}$ and 0.089 K — the SW gain was 70 % of the prediction and the temperature at or below its low end. The sensitivity assumption was right; the forcing estimate was optimistic.

THE NEGATIVE RESULT IS WORTH MORE THAN THE 0.089 **K.** Three quarters of a *directly measured* land albedo bias has now been removed, and it recovered 0.089 K of the ∼0.7 K universal bias. Extrapolating, a *perfect* albedo fix buys ∼0.12 K — about **one sixth** of the target. **Land surface albedo is therefore closed out as the explanation for the universal cold bias**, which is a real constraint on where the remaining ∼0.6 K lives.

~~K3 should push the soil-albedo scale further, since K2 alone gives the best global RMSE.~~ **Retracted before it was run.** The remaining 26 % of the residual is worth roughly 0.03 K, and the 0.95 scale was calibrated to the *measured* −0.0163 bias — pushing past it means tuning *against* the observation, which is the same overfitting-to-present-day trap that ruled out editing the soil ancillary (§29.3). **K1 is adopted and albedo tuning stops here.**

31 The boreal forest fails by TWO different routes (2026–08–05)

The report has carried a single story for the missing forest: the winter evergreen cold gate (§6.6). Evaluating the `Tuning_test_07*/08*` LPJ-GUESS families (`lai.out`, `fpc.out`, `est_limits.out`) shows that story is **correct but only half the picture**, and that it does not apply at all to the region that matters most — NE Siberia, which is ∼80 % larch.

31.1 BNE is gate-excluded; BNS is permitted and outcompeted

Larch zone, 60–72°N × 100–160°E, 2520 cells, last decade:

	080a (CRUNCEP base)	09A (newSeaIce)
`mTmin20`	−38.94 °C	−41.89 °C
BNE gate `tcmin_est`= −30	**0.7 % clear**	**0.0 % clear**
BNS cold gate	**none, 100 %**	**none, 100 %**
`agdd5` vs BNS `gdd5min_est`= 350	**83.6 % clear**	79.0 %
BNE LAI	**0.000**	**0.000**
BNS LAI	0.129	0.103
C3G LAI	0.751	0.510
grass : larch	**5.8**×	5.0×
cells with C3G > BNS	65 %	60 %

The two PFTs fail for completely different reasons.

- **BNE/BINE: a climate gate.** 0.7 % of larch-zone cells clear `tcmin_est`= −30, and their LAI is *exactly* 0.000. The winter-cold story of §6.6 is right, and this is where it applies — the evergreen zone of west Eurasia and North America, *not* NE Siberia.
- **BNS (larch): no gate at all, and it loses.** It has no cold limit and clears its `gdd5min_est`= 350 on 84 % of cells, so it is **climatically permitted** across the zone — yet grass carries 5–6× its LAI, grass exceeds larch on 65 % of cells, and larch is absent outright from half the cells where it is allowed. **Larch is not excluded; it is outcompeted in the growing season.**

This is a correction to this report, not a new hypothesis. §6.6 already states that BNS has no cold gate and is "*already permitted*" — and then never explains why it is absent anyway. A reader was left applying the winter-gate mechanism to a zone where it explains nothing, with only the roughness leg (§6.5) for the 80 %-larch region.

31.2 Reconciling this with the falsified "competition, not climate"

§44 carries *"the boreal failure is an LPJG-internal competition error"* as dead, killed by the forcing-transfer test reproducing most of the loss with byte-identical parameters. **That falsification stands and is not weakened here**, because it and the result above are different claims:

claim	status	
competition is an LPJG *parameter error*	**dead**	forcing reproduces the loss unchanged
competition is the *pathway* by which climate removes larch	**supported**	permitted on 84 %, loses 5.8:1

Forcing is the *driver*; grass-versus-larch is the *mechanism*. Both hold at once. As written, the report recorded only the falsification, so a reader would reasonably conclude that growing-season competition had been ruled out — it had not been tested.

No contradiction with §6.4 either, though the two look opposed. That section measures vegetation→climate (the LPJG cold effect on T2m, which vanishes in July–August). This measures climate→vegetation (how the growing season sets larch cover). Opposite directions of the same coupling; both can be true.

Weak evidence, flagged as such. Weakening grass (`C3G pstemp_low`= 12, run 08G) moves BNS 0.129 → 0.156 and TREEFPC 0.060 → 0.072 while C3G falls 0.751 → 0.583 — directionally exactly what competition predicts. But the 08 runs sit on *different atmospheric bases* (06T vs 06V), so this is not a clean attribution and should not be quoted as one. The gate/permission evidence above is the load-bearing part; the knob evidence is suggestive only. On the old forcing, **07A (BNS `greff_min` relaxed) did not recover BNS at all** (0.024 → 0.021), which is why a forcing-consistent re-spin remains the precondition for any competition tuning (§14.3).

32 The snow scheme, settled on coupled soil (2026–08–05)

A colleague's figure — OpenIFS `stl2` against RIHMI-WDC at 105 stations, coupled gen-10 runs — overturned §27.10. Reproduced independently to within rounding:

stl2 − RIHMI 20 cm [K]	DJF	MAM	JJA	SON	annual
AMIP K1 (adopted, tuned)	+0.49	−1.60	−1.44	−1.30	−0.96
COUPLED 10A (G4)	−4.33	**−6.57**	−3.62	−3.61	−4.53
COUPLED 10B (+scheme)	**−16.19**	−6.04	**−2.44**	−5.95	−7.66

January reaches −18.3 K. The scheme buys 1.2 K of summer and pays 12 K of winter. **It is rejected for coupling.**

Why the earlier verdict was wrong, recorded because the error is repeatable. I sampled the larch zone alone (60–72°N, 100–160°E), where 10A already sits near −21 °C in DJF — *saturated.* Both runs looked identical there. The damage is only visible across the wider, more southerly station network where 10A still retains its snow insulation. **I measured in the one place the signal was flat.** Worse, the AMIP run had already predicted it: the I1 delta is DJF −13.57 K, composing to ≈ − 13.3 against the −16.2 measured. When the coupled LAI figure appeared I treated it as overturning the AMIP evidence instead of asking why the two disagreed.

32.1 Reconstructing `ZCVS`, and what it excludes

`ZCVS` is not an output field, and neither are the tile fractions (`ZCVS` = `FRTI(5)`+`FRTI(7)` exactly, from `surfbc_ctl_mod.F90:421,4` since the three cover fractions sum to one). But it is a pure function of `sd` and `rsn`, both of which *are* written, so it can be reconstructed offline from each run's own snowpack:

Siberian land	Sep	**Oct**	Nov	Dec	Jan	May
ΔZCVS	+0.026	**−0.075**	−0.008	−0.001	−0.000	−0.188
Δsoil [K]	+0.0	**−4.2**	−16.2	−22.2	−23.8	+2.4

Midwinter cover is identical (−0.0004 in January): at 0.43 m depth the tanh is saturated. So the −19 K January soil **cannot be a cover effect**. The cover difference *peaks in October exactly as the soil begins to diverge*, then vanishes while the soil runs on to −24 K — autumn seeding plus thermal memory, with nothing in midwinter needing to differ at all.

A mechanism proposed and falsified within the hour. ~~The snow fraction is applied twice on the conduction path: ZHFLUX is already area-weighted while ZSNCONDH is multiplied by PFRSN again, so the pack drains through a conductance scaled by area a second time.~~ **Wrong.** `ZHFLUX` is W per m^2 of *grid box* (which is why the `/ZFRSN` at `srfsn_driver_mod.F90`:271 is commented out), `ZDSN` uses *grid-mean* SWE, and `ZSNCONDH` is therefore also per m^2 of grid box. Flux, conductance and heat capacity are all in the same units. There is no second division for the factor to pair with.

Supporting evidence for the surviving reading: the snowpack cools only 3.9 K and the air 2.6 K while the soil cools 23.8 K. In January the control holds a +21.7 K gradient between snow surface and soil; I1 holds +1.8 K. The insulation has gone, with SWE identical and density *lower* — deeper, less conductive snow. No physical field changed in a way that could do it, which is what pointed at autumn seeding rather than a winter process.

32.2 Rounds 19: what is running, and what would falsify it

Four AMIP runs on the K1 base at **full campaign length** (1870–1916, evaluate 1872–1915), with daily `sd/rsn/tsn/asn/stl1/stl2` added through a per-run `add_config_sources` override — so all four difference directly *and* are scorable against the standard thresholds. An earlier 10 yr version was cancelled: at that length they could not be judged against the 44-yr detection floors, which would have made them incomparable with the other 43 runs.

run	role	z_0	ρ_{new}	m	SCF Oct	SCF May
N1	reference, scheme off	—	—	—	—	—
N2	current parameters	0.016	100	1.6	0.894	0.817
O1	re-parameterised	0.018	170	**4.0**	**0.995**	**0.495**
O2	re-parameterised, milder	0.018	170	3.0	0.995	0.764

The O series rests on one piece of arithmetic. With SCF = $\tanh(d/L)$ and $L = 2.5z_0(\rho/\rho_{new})^m$,

$$\frac{x_{Oct}}{x_{May}} = \frac{d_{Oct}}{d_{May}}\left(\frac{\rho_{May}}{\rho_{Oct}}\right)^m = 0.469 \times 1.853^{\,m}.$$

Depth works *against* us — October's pack is half as deep. Density works for us, and $m > 1.23$ is needed merely for October to hold more cover than May. $m = 1.6$ **clears that bar but gives a ratio of only** 1.23, hence a separation of 0.077: the −0.075 October deficit that seeds the winter collapse buys a spring depletion only 0.08 deeper. $m = 4$ gives 5.5. Raising m alone makes October *worse* (0.383), because $\rho_{Oct} = 170$ already exceeds $\rho_{new} = 100$ — so ρ_{new} must be recentred on autumn density first.

~~Density cannot substitute for melt state; the two regimes are inseparable by density at their respective depths.~~ **Wrong, corrected the same day**: that was argued from the current parameters rather than from what the formula can do.

Why this replaced the hysteresis it was meant to be. A ripe/not-ripe state test is *not* hysteresis — it is reversible, so cover would flip back every night and every cold snap, injecting a spurious diurnal oscillation into the tile fractions. Real melt-out is irreversible within a season. Swenson & Lawrence (2012) avoid this with a prognostic seasonal SWE_{max}, which is a new surface prognostic. **Density sidesteps the problem entirely**: it is a genuine physical memory variable, ripening monotonically and not walking backwards. And it is namelist-only.

Pre-registered falsifiers. (i) Winter soil must return to the N1 reference; if O1 still shows the −16 K DJF collapse the density route is dead and only the SWE_{max} rewrite remains. (ii) May/June cover must still deplete, or the summer gain is lost and O1 is all cost. *And this is curve-fitting until the daily data supports it*: $m = 4$ is far from Niu & Yang's calibrated 1.6 and was derived from two box-mean monthly numbers, while a steep exponent amplifies cell-to-cell spread.

33 Round 19 results and Round 20 (2026–08–06): the `tanh` is falsified on observations, and the curve is rebuilt

The O series failed its own pre-registered falsifier, and so did every mechanism proposed to explain the failure. Three hypotheses died in sequence before the real one was found, and *two of the deaths were caused by my own analysis errors* — both recorded in §33.4 because both are easy to repeat.

33.1 The measurement: cover is identical, the soil is 23 K apart

All four runs at 46 yr on the K1 base, Siberian box, differenced against N1:

	Δcover DJF	ΔSWE DJF	Δdepth DJF	**Δsoil DJF**	Jan f_{full}
N1 reference	—	—	—	—	**0.960**
N2 current m=1.6	−0.002	+0.65	+0.064	**−22.77**	**0.773**
O1 m=4	+0.001	−0.60	+0.047	−18.72	0.947
O2 m=3	+0.001	+0.12	+0.043	−16.93	0.958

DJF snow cover is identical to within 0.002 **across all four runs while the soil differs by 23 K.** October soil damage is $-4.94/-5.02/-4.95$ K — identical to 0.08 K — while October cover spans 0.10. Mean cover is not the winter channel, and no re-parameterisation of a cover curve could ever have been.

33.2 The mechanism: same mean, opposite state

The exposed area is $1-\overline{\text{cover}}$ by definition, so it is *identical* between the schemes. Where it sits is not, and the soil response is nonlinear in it.

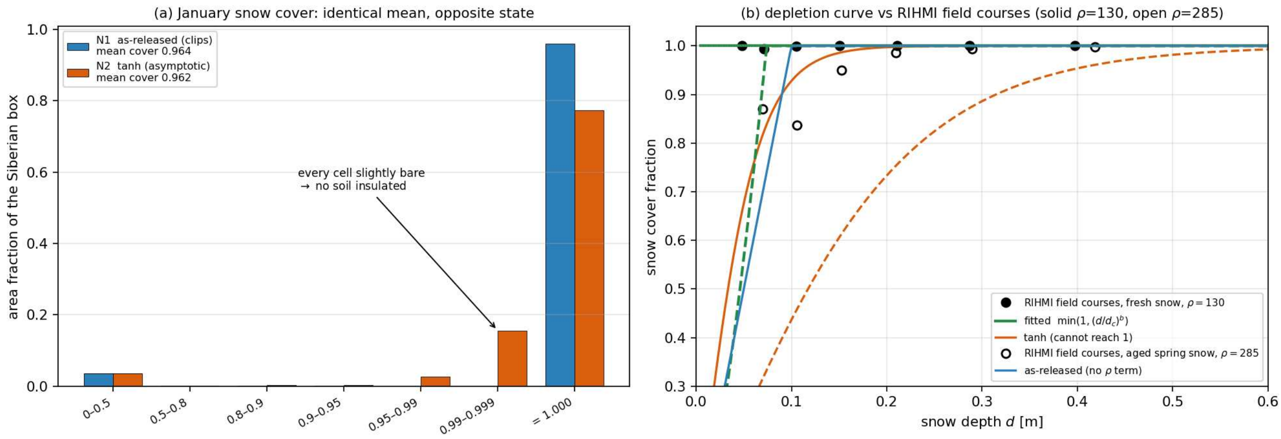


Figure 29: **(a)** January snow cover across the Siberian box. N1 and N2 have the same box-mean cover to within 0.002 and the same total exposed area, but N1 (clipping ramp) puts 93.5% of the box at cover *exactly* 1.0 while N2 (`tanh`) dumps a quarter of it into 0.99–0.999. A sliver of bare ground in *every* cell instead of a few genuinely bare cells: each sliver couples that cell's soil to the air through exposed tiles at $\lambda_{sk} = 10\,\mathrm{W\,m^{-2}\,K^{-1}}$, about 16× stiffer per unit area than the whole snow column (0.61), so no soil is left insulated. **(b)** The depletion curve against 36 492 RIHMI snow-course surveys. `tanh` has range $(0,1)$ *open at 1* and so cannot represent complete cover at any parameters; the as-released ramp reaches 1 but ignores density and so has no spring depletion; the fitted form does both.

The signature in the model is unmistakable: **N1 holds a 22 K gradient from soil (265.0 K) to snowpack (242.7 K); N2 holds 2 K** (240.4/238.6). A 2 K gradient under half a metre of snow cannot be produced by conduction — N2's soil sits at air temperature under a deep pack. `RTHRFRTI` = `EPSILON` (`su0phy.F90:1308`), so no tile is ever culled and every sliver is fully active.

f_{full}, the area fraction at cover ≥ 0.999, is the only quantity found that predicts the damage: per-cell correlation with Δsoil is $+0.75/+0.52/+0.52$ against $+0.32/+0.19/+0.10$ for mean cover.

33.3 RIHMI-WDC settles it: the as-released cover was right

Snow-course transects (`snmar`, 1–2 km, the direct observational analogue of `ZCVS`): **14 331 of 14 369 Siberian DJF surveys report cover of exactly** 10/10 — 99.74%, mean 0.9995. Complete cover is real, not an artefact of clipping. Soil temperature (`tpg`, 43 stations, 174 676 DJF observations at 0.2 m against `stl2`):

	f_{full} DJF	soil at 0.2 m	**bias**
observed	**0.997**	**−6.1** °C	—
N1, scheme off	0.990	−5.3	**+0.8**
N2, `tanh`	**0.678**	−26.3	**−20.2**
O1	0.932	−22.2	−16.1
O2	0.965	−20.8	−14.7

Field January f(complete cover): **observed** 0.996**,** **`tanh`** 0.290. Period caveat: observations are 1963–2024 against a PI model, so the PI-equivalent observation is nearer −9 °C, making N1 ∼ +3 K warm — the scheme is still −13 to −18 K wrong. **This retracts §27.3**: for soil temperature the as-released cover was correct and the scheme replaced a good field with a broken one.

33.4 Two analysis errors, and four dead mechanisms

~~The soil collapse is seeded by an October cover deficit and carried by thermal memory.~~ Killed by its own pre-registered test: O1/O2 produced *positive* autumn cover anomalies and still lost 18–20 K.

~~A melt-state (ripe/not-ripe) gate can separate spring from autumn.~~ May ripe-day fraction is 0.04 in every run; the gate would fire in June, after melt-out.

~~The residual exposed fraction at saturation explains it — `tanh` never reaches 1 while the ramp clips to exactly 1.~~ Right about the codomain, wrong about the box: N1's own January cover is 0.963, not 1.0, so both schemes leave the *same* exposed area. The distribution, not the total, is what differs.

~~Drying the soil before freeze-up removes the latent-heat buffer.~~ The freeze-up buffer differs by only $-0.51/-0.72/-0.68$ K against 17–25 K to explain, and is not ordered with the damage — O1 has the largest water deficit and not the largest cooling.

~~More cover means a colder soil, because `ZSNCONDH` ∝ `PFRSN`.~~ $\mathrm{d}G/\mathrm{d}f = 0.61 - 10 \approx -9.4$: the exposed route swamps the snow route, so more cover means a *warmer* soil. The blanket intuition holds; the exposed fraction is what matters.

Both of my errors were the same error: averaging before applying a nonlinear function. `snow_daily_diag.py` averaged depth and density over 40 years and *then* applied the `tanh`; its Δcover numbers are void and the conclusions drawn from them were withdrawn. The first monthly fit check evaluated $\mathrm{SCF}(\overline{d}, \overline{\rho})$ against $\overline{\text{cover}}$, which made a good fit look like it saturated in May. $\mathrm{SCF}(\overline{x}) \neq \overline{\mathrm{SCF}(x)}$ — **apply the formula per sample, always.**

33.5 Round 20: `ECE_SNOW_SCF` = 3, fitted to observations

A source change this time (`surfbc_ctl_mod.F90` branch, `surfece.F90` parameters); modes 1/2 retained so N/O stay reproducible.

$$\mathrm{SCF} = (1 - c_{vh}) \min\bigl(1, (d/d_{cl})^{b_l}\bigr) + c_{vh} \min\bigl(1, (d/d_{ch})^{b_h}\bigr), \qquad d_c = \min\bigl(D_{max},\ S \cdot D \cdot (\rho/\rho_{ref})^M\bigr)$$

$\min(1, \cdot)$ **reaches 1 exactly** — that is the entire fix; everything else is calibration. It nests the as-released ramp at $D = 0.1$, $M = 0$, $b = 1$. Split by vegetation **type and fraction only** (via `PCVH`), never gridpoint or hemisphere, so it stays portable to other climate states and follows the vegetation if LPJ-GUESS moves the treeline. **No new prognostic and no new input**: `PSNM1M`, `PRSNM1M` and `PCVH` are already arguments of `SURFBC_CTL`, and **nothing knows the date** — the seasonal cycle is an output of the snow physics, not an input to the formula.

Parameters are fitted to 36 492 snow-course surveys, each measuring depth, density and covered fraction on the same transect, split by course type (1 = *pole*/field → low vegetation, 2 = *les*/forest → high). Constrained fit: hold Oct–Feb saturated, then minimise Mar–May error.

	Oct	Nov	Dec	Jan	Feb	Mar	Apr	**May**
field observed	0.998	0.999	0.999	0.999	0.999	0.994	0.974	**0.914**
field fit	0.999	1.000	0.998	1.000	1.000	0.996	0.970	**0.915**
forest observed	0.999	1.000	1.000	1.000	1.000	1.000	0.987	**0.931**
forest fit	0.999	1.000	1.000	1.000	1.000	1.000	0.987	**0.931**

Mar–May RMSE 0.0025/0.0001 against 0.0229/0.0156 (as-released) and 0.1207/0.0964 (`tanh`). **The frontier is flat** — the same parameters win at every winter floor from 0.000 to 0.998, so the winter/spring trade-off that dominated rounds 15–19 was an artefact of the `tanh` being unable to do either, not a real tension. Kept: $D_{cl} = 0.014$, $D_{ch} = 0.026$ m at $\rho_{ref} = 200$, $M = 4.70$, $b_l = 1.46$, $b_h = 0.40$, $D_{max} = 0.30$.

The physical reading is that d_c runs from ∼2 mm at $\rho = 130$ to 8–14 cm at $\rho = 285$: fresh low-density snow drapes over everything, while old snow is dense *because* it has been wind-redistributed and melt-metamorphosed, and that same history is what leaves bare patches. **Density is the proxy for age and melt state** — the hysteresis the campaign spent two rounds trying to build, delivered by a state variable.

33.6 The runs, and what would falsify them

P1 (`amip_P1_scffit`, $S = 1$) takes the observations literally. P2 (`amip_P2_scffit_x3`, $S = 3$) scales d_c for 100 km sub-grid variance. Both on the K1 base at 46 yr with the daily snow/soil output retained so f_{full} can be measured directly.

S is the one uncalibrated knob and is labelled as such. A snow course is 1–2 km; a TCO95 box is ∼100 km, and sub-grid variance grows with scale, so course-fitted d_c is a *lower bound* and $S > 1$ is the physically expected direction. At the Siberian box-mean May state ($d = 0.31$ m, $\rho = 313$) $S = 1$ gives $d_c = 0.118$ m, i.e. $d/d_c = 2.6$ and **no depletion at all** — P1 is expected to behave much like as-released in spring. Rutgers/IMS 24 km cover aggregated to the model grid is what should set S, and **that calibration is not done**; P2 is an exploratory bracket, not a candidate for adoption.

Pre-registered falsifiers. (i) The DJF soil must return to the N1/K1 reference; if it does not, the distribution diagnosis is wrong and this whole section fails. (ii) January f_{full} must return to ∼0.96 from N2's 0.773. (iii) If the scaled variant recovers the JJA gain *but* reopens the winter soil bias, then spring depletion and winter insulation are genuinely coupled at grid scale and the flat frontier found on station data does not survive.

33.7 P1 crashed: the fit extrapolated where it had no data (2026–08–06)

P1 aborted at 1888–03–22 with `ABOR1: Very snow cold temperature` (`srfsn_webal_mod.F90:451`, the PTSN < 100 K sanity guard):

```
Tsn 91.93613   SWE-1 5.6058891E-02   SWE 0.0   Snow frac 1.000000   heat 10.88281
```

A pack of $0.056\,\mathrm{kg\,m^{-2}}$ — half a millimetre — was given *full* snow cover. The snow tile then absorbed the entire grid-box flux into essentially zero heat capacity and the snow temperature collapsed. The cause is that the fitted power law was extrapolated below any depth the snow courses ever sampled: there is no survey under ∼5 cm, and at $\rho = 100$ the formula gives $d_c = 0.014(0.5)^{4.7} = 5.4 \times 10^{-4}$ m, so a 0.56 mm dusting reaches $d/d_c = 1.04$ and saturates. The governing ratio is SCF/SWE: as-released gives 0.100 at that state, mode 3 unfloored gives 17.86 — **180× worse**. I had written down that the low-depth end was unconstrained and then extrapolated into it with no guard.

The fix is `ECE_SNOW_SCF_SWEMIN`, which floors d_c at SWEMIN/ρ: a minimum snow *mass* before cover may saturate, which is the right currency because the abort is a heat-capacity failure, not a depth one. SWEMIN = $3\,\mathrm{kg\,m^{-2}}$ is the *smallest* value that beats the as-released margin (SCF/SWE = 0.053 against 0.100) while barely touching the calibration — October field cover 0.999 → 0.996 against an observed 0.998, and Dec–May unchanged to three decimals. SWEMIN = 5, which I had picked by guess first, would have cost 0.983 in October for no extra safety that matters.

P2 was cancelled un-crashed and is also withdrawn: SCALE = 3 only makes d_c 1.6 mm instead of 0.5 mm, so it carries the same defect and its output is TAINTED wherever thin snow occurs, which includes every autumn onset and spring melt-out. **P3** (SCALE = 1) and **P4** (SCALE = 3), both with SWEMIN = 3, supersede them.

~~P1 was stopped by an unexplained `scancel` from the account UID rather than by a model failure.~~ Wrong, and the reasoning behind it was worse than the conclusion: the model aborted normally on `PROC=48` and esm_tools tore the job down afterwards, exactly as designed. I searched `NODE.001_01` (rank 1 only) for `forrtl/nan/ abort` and for the esm_tools trigger strings, none of which match `ABOR1` or `MPL_ABORT`, concluded "no model error", and then constructed an elaborate infrastructure story on top of that gap. **Method note: an OpenIFS abort is `ABOR1`/`MPL_ABORT` and can land on any rank — grep the whole compute log for those tokens before concluding a job died externally.**

One tension, found while making Figure 29 and not resolved. The constrained fit optimises the *monthly climatology*, but the model applies the curve *pointwise*. Scored on the pointwise (d, ρ) bin surface instead, the adopted field parameters give RMSE 0.0624 against 0.0384 for a fit made directly to that surface ($b \approx 0.1$) — though both beat as-released (0.0771). For forest the adopted parameters are better on *both* metrics (0.0420 vs 0.0483; as-released 0.0644). So the field curve is a compromise: optimal on the seasonal cycle, second-best on the surface, and an improvement on the incumbent either way. Whether the climatology or the surface is the right objective is not obvious — the model integrates the curve over a cell distribution, which is closer to the climatology aggregation — and it should be settled alongside the Rutgers calibration of S.

Residual hysteresis is real but small. At matched (d, ρ) the ablation season is lower in *24 of 24* non-zero bins — systematic, $p \approx 10^{-4}$ — but the weighted mean is only -0.005 (field) and -0.001 (forest). Capturing it would mean plumbing `PWSNM1M` through the caller chain for $\sim$0.005 of cover; not done. The matched test can only compare bins populated in both seasons, so spring's extreme states ($\rho > 300$) are structurally untestable this way.

34 Round 20 results (2026–08–07): P3 adopted, and the fix costs the autumn

Both pre-registered falsifiers passed, and by more than they required.

	DJF soil vs N1	Jan f_{full}	soil→snow gradient, Jan
N1, scheme off	—	0.961	22.4 K
N2, `tanh`	**−22.71** K	0.755	**1.8** K (impossible)
P3	**+0.30** K	**0.965**	**22.9** K
P4	+0.16 K	0.965	21.8 K

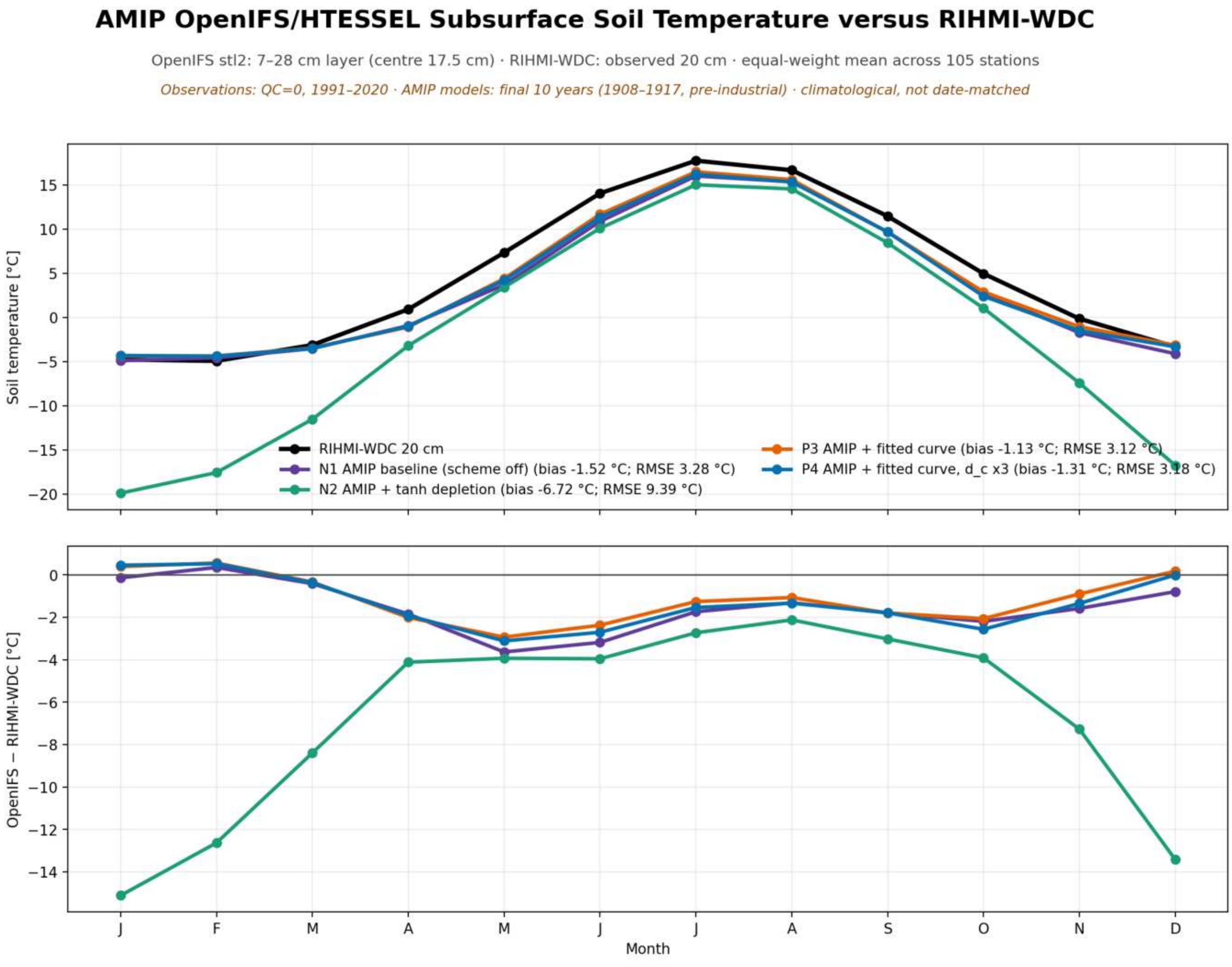


Figure 30: AMIP soil temperature against 105 RIHMI-WDC stations, built with the same methodology as the coupled 09C/10A/10B figure so the two are directly comparable (`stl2` 7–28 cm vs observed 20 cm, QC=0, 1991–2020, final model decade). The `tanh` run (green) peels away from October and reaches −15 °C in January — the same catastrophe 10B shows coupled. **P3 and P4 lie on top of the scheme-off baseline all winter, and slightly above it.** P3 beats that baseline on every metric: bias −1.13 vs −1.52, RMSE 3.12 vs 3.28, JJA −1.56 vs −2.08. The residual bias is now entirely a growing-season deficit that no snow scheme can touch. Absolute offsets are not comparable with the coupled figure (prescribed vs interactive SST); the spread between lines is.

Seasonal ANOVA against the control, each season on its own threshold: P3 −0.584 / +1.341* and P4 −0.132 / +1.092* (DJF / JJA). **Neither appears on `evaluate.sh`'s "warms JJA significantly but cools DJF**

significantly" list, which contains I1, I3, J1, J2, N2, O1 and O2. The winter penalty that has followed this scheme since round 15 is gone.

Marginal over the K1 base, which is the honest accounting: P3 +0.305 JJA / −0.088 DJF; P4 +0.056 / +0.364. Most of the +1.3 is K1's. P3 beats P4 despite SCALE = 1 being the conservative choice, which *falsifies my own prediction* that P3 would show no spring depletion — cutting cover *slows* box-mean melt, because the snow tile receives energy in proportion to its area.

~~SCALE is the parameter that controls the spring depletion, and the uncalibrated one that matters.~~ Varying it 1 → 4 moves the offline Sep–May RMSE by 0.002. P4 was a wasted run; `SWEMIN` is the whole lever.

34.1 Rutgers: the fix made the autumn 2.4× worse

Rutgers 24 km, Siberian box, area-weighted, model minus observed:

	Sep	Oct	Jan	Apr	May	Jun
observed (absolute)	0.067	0.599	0.999	0.956	0.655	0.202
N1 as-released	+0.107	+0.103	−0.035	−0.012	+0.118	−0.011
P3	**+0.259**	**+0.249**	−0.034	−0.019	+0.074	−0.035

The fitted d_c for fresh low-density snow is ∼2 mm, so a dusting saturates. That is *correct* at 1–2 km, which is what the snow courses measure, and wrong at grid scale, where September Siberia is 7 % covered rather than 33 %. The uniform −0.035 in midwinter is the model's permanent lake/water bare fraction, present in every run including scheme-off — not the scheme.

An offline `SWEMIN` sweep scored over the whole Sep–May season gives an *interior* optimum at 30 (RMSE 0.045 against P3's 0.139); past it the September gain is paid for by November and December going deficient. **P5** (SWEMIN = 15) and **P6** (= 30) bracket the albedo→melt→SWE feedback the offline sweep holds fixed, so those numbers are lower bounds on the sensitivity.

34.2 Round 21: resolution awareness, and a cap that disabled the floor

Three source changes (`562df81`), one of them a real bug.

`DCMAX` was capping the mass floor. The order was $d_c = \min(D_{max}, \max(\text{fitted}, \text{floor}))$, so the 0.30 m cap killed the floor above SWEMIN $= D_{max}\rho$ — $30\,\mathrm{kg\,m^{-2}}$ at $\rho = 100$. Every grid coarser than TCO95 therefore got no floor beyond the reference value, and it is *also* part of why the sweep flattened past 30 and reversed past 40, which I had attributed entirely to November. Now $d_c = \max(\min(D_{max}, \text{fitted}), \text{floor})$: the cap guards the steep $M = 4.7$ extrapolation of the *fit*, the floor is a separate physical statement about how much box-mean mass fills a box of this size. The two were nested and should not have been.

A resolution preset. `ECE_SNOW_SCF_DXKM` rescales `SWEMIN` from its TCO95 reference, $\text{SWEMIN}_{\text{eff}} = \text{clamp}(\text{SWEMIN} \cdot (\Delta x/100)^{0.624}, 3, 120)$, off by default so every earlier run reproduces. The exponent $\ln 10/\ln 40$ comes from the only two *measured* anchors: 3.0 at 2.5 km (the snow courses are 1–2 km transects, i.e. the high-resolution limit observed directly) and 30 at 100 km (Rutgers). Presets: TL21 94, TL63 48, TL255 26, TCO95 30, TCO399 13, TCO1279 6, TCO4000 3.

Below 2 km the scheme disables itself, falling back to the as-released ramp with a loud log line. At that scale the model resolves individual patches, so a subgrid curve double-counts them, and 2 km is the finest scale any calibration exists for.

~~The scale dependence can be measured as a power law by coarse-graining Rutgers and the model together.~~ It cannot, and the attempt is kept because the negative result is the useful part. The ladder confirms the *sign* — `SWEMIN` grows with box size, 41.5 at 66 km against ∼114 at ≥ 133 km — but the values **plateau** beyond 133 km rather than following a power law; the per-cell fit implies ∼80 at 100 km where the seasonal fit gives 30, a 3× disagreement that is the same climatology-versus-pointwise tension as the (d, ρ) surface fit; its own extrapolation to 2.5 km gives 9.4 against the course anchor of 3; and absolute skill is poor at every scale (RMSE 0.20–0.25 on a 0–1 field), which indicts the functional form rather than the parameter. **The preset is an interpolation between two trusted anchors, not a measured law, and a new resolution needs validation rather than a `DXKM` value.**

35 Round 18 results (2026–08–07): the aerosol term is six times smaller than inferred

M1 removes the anthropogenic MACv2-SP plumes entirely; M2 gives the same CAMS mass its 3D vertical distribution. Both sit on the present-day base, so the comparison is against `amip_presentday` over the same years.

Units first, because they invalidated the first pass. IFS TOA fluxes are *accumulated* $\mathrm{J\,m^{-2}}$ over the 3600 s step, not $\mathrm{W\,m^{-2}}$; the raw numbers come out ∼3600× too large. Dividing by the timestep reproduces the campaign's own recorded decomposition almost exactly — global +2.68 clear-sky and −1.18 cloud against the recorded +2.68 / −1.17 — which is the check that the pipeline is right.

band	M1 clear-sky	t	M1 **cloud**	t
60–90N	+0.381	+3.8*	**−1.014**	−7.7*
30–60N	+0.699	+9.4*	**−1.397**	−13.3*
tropics	−0.037	−2.0	−0.439	−5.9*
SO 45–65S	**−0.116**	**−18.9***	+0.022	+0.1
60–90S	+0.010	+0.3	+0.079	+1.0
GLOBAL	+0.126	+11.1*	**−0.542**	−8.9*

Paired annual differences over 25 yr; the runs share prescribed SSTs and years, so the pairing removes the forcing variability. * clears two-sided 95 % ($|t| > 2.06$).

~~Aerosol contributes ∼0.7 W m⁻², 13 % of the Southern Ocean's +5.10 clear-sky excess, inferred from a +0.033 AOD excess against MISR.~~ **Measured, it is −0.116 $\mathbf{W\,m^{-2}}$ ($t = -18.9$) — six times smaller.** Removing *all* anthropogenic aerosol moves SO clear-sky reflection by a tenth of a $\mathrm{W\,m^{-2}}$. Aerosol is ∼2 % of the SO excess, not 13 %, and **the unexplained residual grows from ∼1.3 to ∼1.9 $\mathbf{W\,m^{-2}}$.**

The two reflection columns are not separable. M1's effect is overwhelmingly on *cloud* — −1.40 against +0.70 at 30–60N, −0.54 against +0.13 globally — through `LMACV2SP_CCNF`. Any future aerosol lever must be scored on both columns; a clear-sky-only argument is invalid. M2 moves the tropical clear-sky column by +0.553 ($t = +79$) at *fixed column mass*, so where the aerosol sits is a first-order term of its own.

Open and untraced: the M1 clear-sky deltas are mostly *positive* — removing aerosol *increased* clear-sky reflection (+0.126 global, $t = +11.1$). That is backwards for a direct effect and is significant, so it is not noise; most plausibly a surface-albedo or water-vapour adjustment to the changed surface SW, but it has not been chased down.

36 Round 21 results (2026–08–07): P5 adopted, and the summer ceiling

run	DJF	MAM	JJA	SON
K1 (base)	−0.496	−0.265	+1.036*	+0.238
N2 `tanh`	−2.783*	−0.599*	+1.350*	−0.888*
P3 SWEMIN = 3	−0.584	−0.333	+1.341*	+0.217
P5 SWEMIN = 15	**−0.060**	**−0.140**	**+1.149***	**+0.004**
P6 SWEMIN = 30	**−0.850***	−0.288	+1.096*	**+0.487***

P5 is the cleanest run in the campaign — every season inside its own threshold bar a significant summer gain. Against Rutgers it cuts the autumn excess P3 introduced by ≈40 % (Sep +0.259 → +0.151, Oct +0.249 → +0.148) while leaving Nov–Jan intact, and its DJF soil is +0.22 K against the scheme-off reference.

~~Raising SWEMIN cannot reopen the winter, because at midwinter SWE≈93 the floor is 0.15 m against a 0.43 m pack — safe by construction.~~ True in *January*, and I never checked **November**, where the pack is still shallow. P6's November f_{full} falls to 0.752 against N1's 0.884 and its DJF soil to −1.00 K. The floor binds in early winter; that is the interior optimum the offline sweep predicted, arriving by the mechanism I had dismissed.

36.1 The JJA ceiling: the lever was nearly exhausted before round 15

Marginal over the K1 base, against the ±0.242 threshold: P3 +0.305 (marginal), P5 +0.113 (noise), P6 +0.060 (noise). Only P3 buys a measurable summer gain, and P3 is the version with the worst autumn cover. The reason is structural. May cover against Rutgers' 0.655:

	May cover		vs observed
N1 as-released	0.773	**+0.118**	(the entire available error)
N2 `tanh`	0.615	**−0.040**	(overshot *past* observations)
P3 / P5	0.729 / 0.732		+0.074 / +0.077

The whole spring cover error is 0.118**.** P5 removes 0.044; a perfect correction removes 0.118, worth roughly +0.2 K of JJA over K1 on a crude scaling of N2's response. **And N2's larger gain was never legitimate** — it took 0.158 out of a 0.118 error, overshooting into a −0.040 deficit, so that extra warmth came from having too little May snow. It is the same over-depletion that destroyed the winter soil, and any comparison against N2's +1.350 is a comparison against an error.

Honest accounting for the whole depletion line: worth ≈0.1–0.3 K of Siberian JJA, not the ≈1 K the early rounds implied. What it delivered instead is a scheme that is no longer *wrong*. **The missing summer is elsewhere**: every AMIP run including scheme-off tracks RIHMI soil through Dec–Mar and falls 2–3.5 °C short from April to September (Fig. 30), a deficit common to all of them, and that is the global tropospheric cold bias of §21.1 — still untouched after 50 runs, and now the sharpest remaining target. **Further `SWEMIN` optimisation is not worth runs**: P3 and P5 differ by 0.19 K in JJA, at or below noise.

37 The coupled radiation objective, and why the goal moves to zero (2026–08–07)

The 10 series branches from the same point as round 09 and runs twenty years longer, so it extends the campaign's central radiation figure rather than replacing it. 10A is 09C plus G4; 10B adds the `tanh` depletion since falsified (§34).

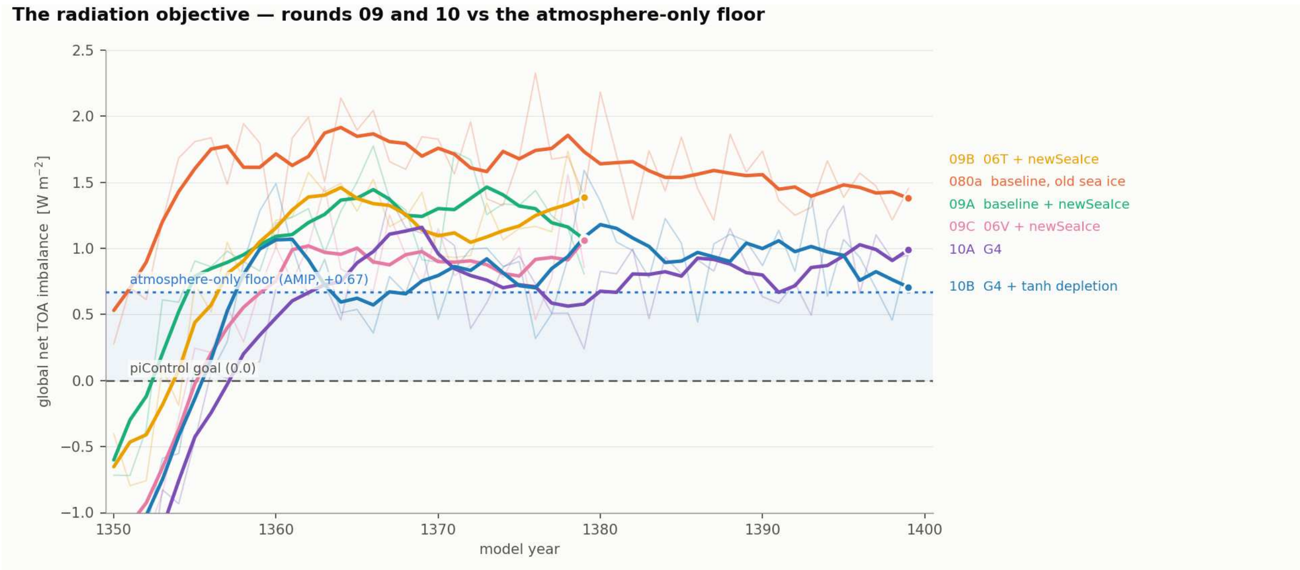


Figure 31: Global-mean net TOA by model year, annual (thin) and 5-yr running mean (bold). **G4 is worth ≈0.25 W m^{-2} at top of atmosphere**: 10A settles at +0.854 against 09C's ≈1.06, the best of the previous round, and both 10-series runs spend 1362–1380 at or below the +0.67 atmosphere-only floor. **10A and 10B are indistinguishable** (+0.854 vs +0.899, curves interleaving throughout), so the depletion scheme that destroyed the winter soil by 20 K cost essentially nothing at TOA — consistent with its damage being a distributional surface effect invisible to box-mean energy diagnostics. The dashed goal is now 0, not the inherited −0.16; the shaded band is the part of the gap ocean-side tuning could still close, and everything above the dotted line is atmospheric. 11C is absent: it aborted in its first leg on the crash of §33.7.

37.1 Why the goal is 0 and not −0.16

−0.16 was never derived. It is whatever the HR piControl settled at, adopted as a matching target. A target displaced from zero is only defensible if the atmosphere fails to conserve energy: with a spurious atmospheric source S, equilibrium gives SFC $=$ TOA $+ S$, so zero *ocean* drift would require TOA $= -S$. Whether the two conditions differ is measurable, not arguable.

Measured on 10A over 1390–1399, global, cos-latitude weighted, using the release tool's own formula (`ssr` + `str` + `sshf` + `slhf` − `sf` $\cdot 3.3355 \times 10^8$, each divided by the accumulation period):

term	$\mathrm{W\,m^{-2}}$
`ssr` + `str` + `sshf` + `slhf`	+1.843
snow enthalpy (`sf` = 1.042×10^{-5} m/step)	−0.966
net surface	**+0.877**
net TOA (`tsr` +238.749, `ttr` −237.895)	**+0.854**
residual	**−0.023**

The atmosphere conserves to 0.02 $\mathbf{W\,m^{-2}}$**.** There is no leak for an offset target to absorb, so "the ocean is not drifting" and "the system is not drifting" are the same condition, and both put the goal at **0**.

Snow enthalpy is doing nearly all of the reconciliation — 0.97 $\mathrm{W\,m^{-2}}$ of a 0.99 discrepancy. Omitted, the surface reads +1.84 against a TOA of +0.85, and that artefact is exactly the kind of apparent imbalance an offset target gets invented to absorb. Anyone comparing a raw surface flux against TOA without it will see a spurious ≈1 $\mathrm{W\,m^{-2}}$ gap.

Re-framing the campaign arithmetic against the corrected goal:

	net TOA	= atmosphere +	coupled-state	gap to 0
080a (untuned coupled)	1.38	0.67 +	**0.71**	1.38
09C (06V + newSeaIce)	1.06	0.67 +	**0.39**	1.06
10A (+G4)	**0.85**	0.67 +	**0.18**	**0.85**
10B (+`tanh`)	0.90	0.67 +	0.23	0.90

The conclusion is unchanged in kind and smaller in size: the ocean-side coupled-state term is now only 0.18 $\mathrm{W\,m^{-2}}$ in 10A, so **almost the entire remaining imbalance is the atmosphere** — the same +0.67 the AMIP testbed measures directly. Chasing −0.16 would have meant over-tuning by that margin against a target with no physical basis in this configuration.

~~The HR piControl goal of −0.16 is the target for the coupled spin-up.~~ Inherited, not derived; the budget closes to 0.02 $\mathrm{W\,m^{-2}}$, so the goal is 0.

38 Round 22 (2026–08–09): the energy target re-based, and three targets relocated

Four diagnostics, no new model time except the coupled runs a colleague had already finished. Three of them move a target, and two of them falsify something this report was recommending.

38.1 Coupled round 11: the first runs carrying G4 *and* D2b

The report has said the coupled branch carries only G4 out of the whole AMIP campaign. **That is out of date.** `11A`, `11B` and `11D` all carry `ECE_TUNE_RVRSMIN` 1000/1000/225 *and* `RCL_INPSEA`= 0.2 with `RCL_INPPMIN`= 70000 — G4 plus D2b — on top of a new `useIFSsoiltemp` coupling. `110Baseline` is 09C plus that coupling change alone and is therefore the correct control; comparing against 10A would confound the snow scheme with the soil-temperature coupling.

Δ vs 110Baseline	net TOA	glob T2m	Sib JJA	Sib DJF	**soil DJF**	SO cloud
11A +G4+D2b	−0.233	−0.150	−0.16	+0.13	−0.41	+0.84
11B +`tanh` snow	−0.060	−0.019	+1.81	−1.62	**−19.40**	+0.76
11D +fitted, `SWEMIN`= 30	+0.023	−0.031	+0.16	+0.05	**−0.45**	+0.41

The `tanh` falsification reproduces coupled on a clean pair: 11B minus 11A is −18.99 K of Siberian DJF soil, against the −13 to −16 K the AMIP run predicted. This is a better-controlled version of the 10B result, because 11A and 11B differ *only* in the snow scheme.

11D is not catastrophic, but it has not tested what it looks like it tested. 11D runs the fitted scheme at SWEMIN = 30 — that is P6, the setting AMIP *rejected* (DJF −0.850*, November f_{full} 0.752, soil −1.00 K), not the adopted P5 at 15. Against 11A it gives soil −0.04 K and DJF T2m −0.08 K. Before reading that as vindication, the detection threshold was measured from the interannual scatter of these very runs:

	interannual sd	95 % threshold, 10-yr pair
Siberian DJF soil (`stl2`)	0.80 K	±0.75 K
Siberian DJF T2m	1.33 K	±1.25 K

So the soil half of the P6 falsifier (-1.00 K) is marginally testable and comes out null; **the DJF T2m half (-0.850 K) is not testable at all on ten coupled years**, and a null there was the expected outcome whether the scheme is safe or not. What 11D *does* establish is that this diagnostic has ample power for damage of the kind that mattered — it sees 11B's -19 K without difficulty. **The outstanding coupled run is 11D at** SWEMIN = 15 **plus K1**, which would bring the coupled configuration level with the atmosphere-only one for the first time.

38.2 The energy target, period-clean

Every band target in this report was measured on `amip_pi_base` against CERES 07/2005–06/2015 — an 1870s atmosphere against a present-day satellite record. `amip_presentday` overlaps CERES directly and its forcing is *correct* for 1990–2014 (unlike the PI arm, whose `NCMIPFIXYR` never took effect, §17.3), so the epoch offset is measurable per band rather than caveated. Band means over full latitude circles are exact on each source's own grid, so nothing is remapped. The CERES read is validated against the published EBAF globals first: ASR 241.18 (published ∼240.5), OLR 240.21 (∼240.2), net +0.970.

band	PI vs CERES	**PD vs CERES**	offset	verdict
90S–65S	+7.24	+8.11	−0.87	robust to the epoch
SO 65S–45S	+6.32	**+7.46**	−1.15	**robust, and larger**
45S–30S	−4.46	−2.49	−1.97	partly epoch
tropics 30S–30N	−2.51	**−0.67**	−1.83	**epoch-dominated**
30N–45N	+0.81	+2.08	−1.27	partly epoch
45N–65N	+4.40	+5.35	−0.96	robust to the epoch
65N–90N	+0.39	+1.53	−1.14	partly epoch
GLOBAL	−0.33	**+1.23**	−1.56	

Two targets move. **The tropics are epoch-dominated** — period-clean the band is only $-0.67\,\mathrm{W\,m^{-2}}$ from CERES, not -2.51, so most of what §42.8 has carried as a tropical deficit was never model error. **The Southern Ocean survives the correction and grows**, +6.32 → +7.46, and its contribution to the global mean rises to $+0.735\,\mathrm{W\,m^{-2}}$. It is now unambiguously the largest single regional term.

A small discrepancy, recorded. §11.2.2 quotes +5.81 for this band against a CERES value of −55.3; reading the gridded climatology directly gives −55.81 and hence +6.32 on the same PI arm. The new number comes with a validated global check and is preferred, but the difference is not chased further — it does not change any conclusion.

The OLR test, which does not close. A cold troposphere under-emits, so the measured OLR deficit and the measured cold bias should be quantitatively consistent. Period-clean, ΔOLR $= -2.529$ and ΔASR $= -1.299\,\mathrm{W\,m^{-2}}$, which is where the +1.229 net comes from. But the effective emitting temperature from CERES OLR is 255.1 K, giving a Planck slope of $3.77\,\mathrm{W\,m^{-2}\,K^{-1}}$, and the model is -1.62 K through 700–300 hPa: a purely radiative response would be $-6.10\,\mathrm{W\,m^{-2}}$. **Measured minus predicted is $+3.57\,\mathbf{W\,m^{-2}}$**, and §38.4 identifies what is offsetting it.

38.3 The Southern Ocean deficit is cloud AMOUNT — and no lever moves it

The band error has been tracked by CRE, which cannot distinguish too little cloud from cloud that is not bright enough. Splitting it at the *observed* CRE per unit area (`amip_presentday` vs CERES, period-clean):

SO 65–45S	model	CERES		diff
cloud area [%]	83.06	89.49		**−6.43**
SW CRE [$\mathrm{W\,m^{-2}}$]	−60.39	−67.84		**+7.45**
CRE per unit area	−72.71	−75.80		+3.10
decomposition of the +7.45*:*				
AMOUNT (missing area × observed CRE/area)			**+4.87**	(65.5 %)
OPACITY (residual)			+2.57	(34.5 %)

Amount-dominated, and **zonally uniform** — Atlantic −5.40, Indian −6.82, Australian −6.55, Pacific −6.81 pp, a sector spread of 1.42 pp, which points at microphysics rather than circulation or the ice edge. It is worst in austral spring and summer (SON −7.03, DJF −6.94 pp), when there is shortwave to reflect: DJF alone carries $+11.67\,\mathrm{W\,m^{-2}}$ of net TOA error in the band. That seasonality is what the supercooled-liquid hypothesis predicts.

Then the part that falsifies the plan. Scored on cloud *area* for the first time, against the +6.43 pp that needs closing:

lever (44-yr AMIP, Δ vs control)	Δ area [pp]	Δ SW CRE	
D2a `inpsea.2`	+0.63	−4.80	9.7 % of the gap
D2b `inp+p700`	+0.43	−2.61	6.7 %
A1a `ovl=0.10`	+0.62	−6.44	9.7 %
G4 **`tundra225`**	+0.48	−2.73	**a LAND lever**
K1 **`landalb`**	+0.45	−2.62	**a LAND lever**
B7 `rvice.22`	−0.60	+0.66	wrong sign

~~G4 and K1 are boreal-land levers that cannot physically change Southern Ocean cloud area, and they move it by +0.48 and +0.45 pp, so ≈0.5 pp is the noise level of this diagnostic and every INP run sits inside it.~~ **Wrong, and corrected the same day.** G4 and K1 are not independent of D2b — they *contain* it (G4 = F4 + D2b + tundra, K1 = G4 + land albedo), so their +0.48/+0.45 is D2b's +0.43 inherited, not a null. Differenced against D2b the pure land contribution is +0.050 and +0.022 pp, which is the clean control and is genuinely negligible.

The real threshold, measured. Interannual scatter of SO cloud area in the 44-year control is 0.232 pp, giving a 95 % detection threshold of ±0.098 pp on a paired difference — five times smaller than the figure asserted above. **Every lever in the table is therefore statistically significant.** The conclusion survives, but it rests on MAGNITUDE rather than on significance:

lever	Δarea	% of the 6.43 gap	ΔCRE	of which opacity
D2a `inpsea 0.2`	+0.626	10 %	−4.80	90 %
A1a `ovlliqice 0.10`	+0.623	10 %	−6.44	93 %
D2b `inp+p700`	+0.432	7 %	−2.61	87 %
A1b `ovlliqice 0.35`	+0.331	5 %	−2.17	88 %
G4 minus D2b	+0.050	—	−0.12	*the land control*

The best lever in the campaign closes one tenth of the area gap and takes ≈90 % of its CRE from opacity. Two independent mixed-phase routes agree — ice nuclei (D2a/D2b) and the liquid/ice deposition overlap (A1a/A1b, the latter pushed to EC-Earth4's 0.1) — so this is a property of the mixed-phase branch, not of one parameter. **The amount deficit, two thirds of the error, is 90 % intact.**

~~The Southern Ocean cloud-area deficit should be attacked by pushing the INP branch harder, since D2b is the only lever that has moved the SO.~~ Written into the plan on 2026–08–09 and falsified the same day. The INP branch acts on opacity, which is 34.5 % of the error; its area response closes only one tenth of the gap, and the same is true of the deposition-overlap route even at EC-Earth4's setting. A cloud-*amount* knob is required, and it is a different knob.

One more pointer. CERES puts the SO effective cloud pressure at 656 hPa, i.e. just *above* the 700 hPa ceiling of D2b's pressure gate, with optical depth 7.41. The gate was designed to protect the tropics and it does, but it may also be excluding the layer where the observed cloud actually sits.

38.4 The cold bias is extratropical and UTLS — and the atmosphere is dry as well as cold

§21.1 established that above 700 hPa Siberia merely shares a global cold bias. Where that bias lives has never been asked. `amip_presentday` vs ERA5 1990–2014, annual, all levels exact in both so nothing is interpolated:

hPa	90S–60S	60S–30S	**tropics**	30N–60N	60N–90N	GLOBAL
850	+0.02	−1.17	−1.10	−0.90	−0.87	−0.99
700	−0.53	−1.40	−1.23	−1.00	−0.86	−1.15
500	−1.81	−1.75	−1.31	−1.56	−1.57	−1.49
300	**−3.65**	−2.48	**−1.72**	−2.34	**−3.43**	−2.22
200	**−4.20**	−3.28	**−2.09**	−3.53	**−4.45**	−2.87
100	+0.25	−1.52	−2.07	−1.30	+0.26	−1.51

The tropics are the *least* biased band, −1.72 at 300 hPa against an extratropical mean of −2.97. That excludes the hypothesis a reader would reach for first: a cold tropical upper troposphere over prescribed, correct SST would have meant a convective heating-profile error propagating globally. It is not that. The bias peaks at 200 hPa over *both* poles (−4.20 and −4.45) and **reverses sign at 100 hPa** (+0.25, +0.26) — a tropopause-region signature, not a tropospheric one. The seasonal tables behave the same way, so it is not a summer or winter effect.

And the atmosphere is dry, increasingly so with height:

hPa	ΔT [K]	q model	q ERA5	$\Delta q/q$ [%]
850	−0.99	5.7843	5.9887	−3.4
700	−1.15	2.9203	3.1596	−7.6
500	−1.49	1.0082	1.1018	−8.5
300	−2.22	0.1454	0.1695	**−14.2**
200	−2.87	0.0208	0.0260	**−20.2**
100	−1.51	0.0014	0.0026	−46.9

This is the missing 3.57 W m^{-2} of §38.2. The two errors oppose each other in OLR: the cold depresses emission, while the dryness lowers the effective emission level into warmer air and raises it. Over 700–300 hPa the model is −1.62 K and −10.1 % relative humidity content — both large, both real, and their sum is small.

The consequence is a constraint on all future radiation tuning. Fixing either one alone moves net TOA by several W m^{-2} in the *wrong* direction. The global budget that §11.2.1 found "right by cancellation" is cancelling in the vertical as well as in the horizontal, and any lever scored on a global mean is being scored on top of that. *Caveat:* ERA5 upper-tropospheric humidity is itself weakly constrained by observation, so the 100 hPa row is indicative only; the 300–200 hPa rows carry the argument.

38.5 What round 22 changes about the plan

- **The Southern Ocean needs a cloud-*amount* lever**, not more ice nucleation. Two thirds of the error is missing area; the INP family moves area at the noise level. Whatever is chosen must be scored on area, and the D2b gate ceiling should be revisited against the observed 656 hPa cloud pressure.
- **The tropical TOA deficit is largely not model error** and should come off the open list at its old size.
- **The cold-bias investigation starts at the extratropical tropopause**, and it must treat cold-and-dry as one problem. A lever that warms without moistening will make the energy target worse.
- **The next coupled run is 11D at** SWEMIN = 15 **plus K1.** Ten years cannot resolve a 1 K DJF signal, so if the P6-versus-P5 question is to be settled coupled at all, it needs either a longer run or the AMIP answer taken on trust — and the AMIP answer says 15.

39 Round 23 (2026–08–09): the reference check, and a round withdrawn

Round 23 was designed to close the Southern Ocean cloud-*amount* deficit. Three of its four runs were withdrawn before completion — one as a duplicate, two on a design error — and the round's lasting product is a reference check that decides which observation the campaign is allowed to tune against.

39.1 Prior work this campaign had not incorporated

Two AWI-ESM project issues (#169, #170, 2025–03 to 2026–04) cover the same Southern Ocean warm bias from the coupled side and contain results this report was working without.

- **The origin is cy48's moist-physics upgrade.** Cycle 48r1 raised high cloud cover everywhere relative to 43r3, strongest in the SH high latitudes, and lowered SO low cloud (#170). The exact diff is the ECMWF Newsletter 164 moist-physics patch, which was evaluated mainly at ∼5 km and mainly in low latitudes.
- **The lead/lag test, which orders the causal chain.** Scored on *year one* rather than the 50-year mean, the hcc offset is already present while the T2m bias and the SO lcc dip are *not.* So hcc leads and lcc lags: `fix hcc` → `reduces t2m` → `fixes lcc`. In the coupled model the SO low-cloud deficit is partly a *consequence* of the warm bias.
- **`RVICE` is already the hcc lever.** The v3.4 tuning raised it from the 0.13 source default to 0.16–0.19 (TUNE1–TUNE5, with `NDECOLAD` 2 → 0 and `RALBSEAD` 0.06 → 0.05) precisely to suppress SO high cloud. Every run in this campaign carries `RVICE`= 0.16 for that reason.
- **The compensating-bias trap is documented.** "All further tests regarding reducing HCC resulted in Arctic and Tropics becoming too cold", and `RALBSEAD` was found "not a strong enough counter tuning method for the combined `RVICE` + `ENTSTPC3`".

~~Lowering RVICE below 0.16 would add tropical cirrus and warm the tropics, providing the counter-lever the Southern Ocean levers need.~~ Mechanically correct — B7 at 0.22 gives Δhigh −2.31 pp and ΔLW CRE −1.35 — but it would undo the v3.4 SO tuning, which raised `RVICE` for exactly this reason. `RVICE` is global and the two regions need opposite signs, so it cannot serve both.

39.2 Four-way attribution: what kind of error each band has

Net TOA cannot distinguish a surface-albedo error from a cloud-reflection one. Splitting each band against CERES into clear/cloud × SW/LW, period-clean, positive = the model gains energy:

band	net	SW clr	**SW cld**	LW clr	**LW cld**	contribution
90S–65S	+8.11	+1.53	+3.98	+4.35	−1.75	+0.390
SO 65S–45S	+7.46	−3.79	**+7.45**	+4.21	−0.40	**+0.735**
45S–30S	−2.49	−2.54	−2.82	+4.66	−1.79	−0.257
tropics	−0.67	−1.39	−0.40	+3.28	**−2.16**	−0.336
30N–45N	+2.08	−4.90	+4.67	+4.36	−2.05	+0.215
45N–65N	+5.35	−2.67	+2.45	+5.11	+0.45	+0.527
65N–90N	+1.53	−3.79	−1.45	+6.30	+0.46	+0.073

The two problems are different cloud types. The Southern Ocean loses its energy through **SW cloud** (+7.45): the low cloud does not reflect enough. The tropics lose theirs through **LW cloud** (−2.16): the high cloud does not trap enough. With SST prescribed and correct, a tropical LW-cloud deficit is an anvil/cirrus problem — too low, too thin, or radiatively too weak — not a surface or a convective-heating one.

That is why no single microphysics knob has ever satisfied both, and it is also the shape of a solution: a lever that makes ice cloud *fewer but optically stronger* would move the Southern Ocean and the tropics in opposite, both-correct directions. `RVICE` changes amount alone, which is why it trades one for the other.

Scanning all 15 NAMCLDP-family levers for tropical response, only three warm the tropics at all: B3 `clddiff`= 1.5e-5 (+2.02, via SW not LW), B2 `clddiff_convi` (+0.65) and K1 (+0.37, clear-sky, its subtropical crop/semidesert albedo). **Nothing in 50 runs warms the tropics through the longwave**, which is the term that is actually deficient.

39.3 Which reference is right about Southern Ocean cloud?

The round-22 amount deficit rests on one source. Bringing ERA5 in appeared to contradict it outright, so before spending runs the references were tested against each other.

SO 65–45S	model	ERA5	CERES	MODIS
cloud area [%]	83.1	81.6	**89.5**	**89.3**
SW CRE [$\mathrm{W\,m^{-2}}$]	−60.39	−61.52	**−67.84**	—
vs CERES	**+7.45**	**+6.31**	—	—

The test is what the cloud DOES, not how much is counted. Cloud area is not one quantity: CERES `cldarea` is a MODIS *mask* fraction that flags optically thin cloud in full, while ERA5 and model `tcc` are radiative covers built by overlapping layer fractions. Comparing them directly is a category error. Shortwave CRE is a difference of two broadband fluxes and *is* identically defined across all three.

If ERA5's lower area were a definition artefact, ERA5 would reflect like CERES. **It does not: ERA5 has low area *and* low reflection, in proportion** — 6.31 $\mathrm{W\,m^{-2}}$ short where the model is 7.45 short. ERA5 is reproducing the model's bias because ERA5 *is* IFS. **So CERES is the arbiter and the Southern Ocean amount deficit is real.**

~~The Southern Ocean cloud-area deficit may be an artefact of comparing a satellite mask against a model `tcc`, since ERA5 puts SO cloud *below* the model's.~~ Raised as a concern and refuted the same day by the CRE test above. **ERA5 cannot be used as a cloud reference for an IFS-derived model at all** — its global cloud area is 62.8 against CERES's 67.4, and in the tropics it reads 55.3 where the model's 61.6 matches CERES's 61.7 almost exactly. On cloud, ERA5 is the outlier, not the model.

A limit on CERES, in the other direction. At 90S–65S CERES reads 67.5 and MODIS 59.8 — a 7.7 pp disagreement between two products of the same instrument. Passive cloud masking over bright ice is unreliable. ~~The Arctic has +11.49 pp too much cloud (§38.2 band table).~~ Not supportable: that number is a cloud mask over sea ice and snow, exactly where the two retrievals disagree most. **CERES cloud area is trustworthy over dark ocean and should not be used over ice.**

39.4 The runs, and why three were withdrawn

Four levers were designed on the P5 base (which already carries G4, K1 and D2b):

run	lever	outcome
S1	`RCLDIFF` 6e-6 → 3e-6	**cancelled** — would cool the tropics
S2	`RCL_OVERLAPLIQICE` 0.65 → 0.35	**withdrawn** — this is A1b, already run
S3	`RCLCRIT_SEA` 2.5e-4 → 6e-4	**cancelled** — same reason as S1
S4	`RCL_INPPMIN` 70000 → 50000	**running**

~~`RCL_OVERLAPLIQICE` 0.65 → 0.35 is an untested mixed-phase lever.~~ It is **A1b**, run for 44 years in round 3; A1a is the same knob at EC-Earth4's 0.1. The A-series label "ovl" denotes the liquid/ice *deposition* overlap, not the radiative cloud overlap, and it was read as the latter. The duplicate is not a total loss: A1a/A1b give a *second, independent* mixed-phase route, and it behaves exactly like the INP route (+0.62 pp of area for −6.44 of CRE, i.e. 93 % opacity), which is far stronger evidence that opacity-not-amount is a property of the whole mixed-phase branch than one parameter could be.

S1 and S3 were cancelled for the tropics, not for the Southern Ocean. Both add low cloud, and B3 — the same parameter as S1 moved the other way — *warms* the tropics by $+2.02\,\mathrm{W\,m^{-2}}$. S1 would therefore cool a band that is period-cleanly only −0.67 from CERES, overshooting it roughly threefold. The deficit they were built to close is real; a *global* low-cloud knob is the wrong instrument for it. Their runscripts are kept under `runscripts/oifsamip/withdrawn/`.

`RCLCRIT_SEA` and `RCLCRIT_LAND` were exposed to `NAMCLDP` for S3 (`sucldp.F90`: moved from the `ASSOCIATE` block onto the `POINTER` mechanism every other `NAMCLDP` variable uses, with the association placed before the defaults are assigned so a silent namelist still reproduces). The exposure is kept — it works, and the lever may be wanted later with better targeting.

39.5 What round 23 changes about the plan

- **The Southern Ocean deficit is confirmed and stays the largest single term**, but it needs a *regime-selective* reflection lever. Global low-cloud knobs overshoot the tropics; the D2b pressure gate is the template for what selective looks like. *Answered in §40*: marine CCN is that lever, it is selective by construction, and it reaches the opacity third only.
- **The tropics need longwave cloud *effectiveness*, not more cloud.** The deficit is $-2.16\,\mathrm{W\,m^{-2}}$ of LW CRE, and no lever in 50 runs has moved the tropics through the longwave at all.
- **Reference hygiene is now a stated rule.** ERA5 is not an independent cloud reference for this model; CERES cloud area is valid over dark ocean and not over ice.
- **Check the prior issues before designing a round.** Both errors this round — the A1b duplicate and the `RVICE` proposal — were already answered in #169/#170.

40 Round 24 (2026–08–09): the marine CCN path, traced end to end

No runs. The question was whether cloud condensation nuclei offer the *regionally selective* Southern Ocean lever round 23 concluded was needed. The answer is a qualified yes, by a route nobody had looked at, and the trace corrected one of my own claims on the way.

40.1 The path, and where it is decided

Marine CCN reaches cloud albedo through four hops, each gated by a switch whose *runtime* value had to be read from `fort.4` rather than a source default:

where	what	runtime state
`surfrad_ctl_mod.F90:772`	Genthon (1992) sea salt from wind	`LCCNO`=**.TRUE.**
`liquid_effective_radius.F90:153`	CCN → N_d → r_{eff}	`NRADLP`=**2**
`cloudsc.F90:1023`	MACv2-SP CDNC factor	`LMACV2SP_CCNF`=**.true.**
`cloudsc.F90:2050`	CCN → autoconversion (lifetime)	`LAERLIQAUTOLSP`=**.FALSE.**

With LCCNO true the code takes the PCCN_SEA branch, so **RCCNSEA is inert** — its own source comment ("now irrelevant") is correct, but only because of a switch three files away. The live formula is

$$Q = e^{0.16u+1.44}, \quad \mathrm{CCN} = 10^{\,1.2+0.5\log_{10} Q}, \quad N_d = -1.15{\times}10^{-3}\mathrm{CCN}^2 + 0.963\,\mathrm{CCN} + 5.30.$$

~~Marine CCN is pinned at a uniform 125 cm^{-3} because NAERCLD= 0 switches the aerosol–cloud path off.~~ Read from a *default assignment* in sucldp.F90 and reported as the runtime state, while fort.4 carries LMACV2SP_CCNF=.true., PCCN had already been printed as a 2-D field, and §35 of this report already documented that CCN path. Three independent disproofs, two of them on screen at the time.

40.2 The correction that matters: N_d is right

Evaluating Genthon on *monthly-mean* winds gives a Southern Ocean wind of 6.54 m s^{-1} and $N_d = 55$, which against an observed 60–120 cm^{-3} looks like a clear deficit. **That was wrong.** Monthly *vector* averaging cancels the transient eddies that are the entire Southern Ocean wind field, and the parameterisation acts on instantaneous speed. On 6-hourly output:

SO 65–45S	wind [m s^{-1}]	CCN	N_d
monthly-mean wind (wrong)	6.54	55.6	55.3
6-hourly wind	**10.28**	**77.6**	**73.5**

~~Southern Ocean droplet number is too low, so the cloud is too dark, and raising marine CCN is the opacity lever.~~ $N_d \approx 73$ cm^{-3} sits squarely inside the observed 60–120. **The mean is correct.** This is the same error class as $\mathrm{SCF}(\bar{d}) \neq \overline{\mathrm{SCF}(d)}$ from round 19 — a nonlinear operator applied to a mean — committed hours after writing it into a skill as a rule.

What survives is the phase. The model's seasonal cycle is inverted:

SO 65–45S	DJF	MAM	JJA	SON
N_d [cm^{-3}]	**68.5**	74.9	**76.1**	74.5
SW CRE error [W m^{-2}]	**+10.86**	+3.71	+2.09	**+13.11**

Model N_d is *lowest* in DJF because it is wind-driven and Southern Ocean winds peak in winter. Observations put the maximum in DJF, driven by biogenic sulphate. So the model has its fewest droplets exactly when the sun is up and the measured error is largest. Crudely — $\tau \propto N_d^{1/3}$, two-stream albedo $A \approx \tau/(\tau + 7.7)$ on CERES's $\tau = 7.41$ — lifting DJF N_d from 68.5 to an observed ∼95 is worth about −6 W m^{-2} of DJF SO CRE, roughly half that month's error.

40.3 The amount two-thirds is unreachable from here

N_d acts through r_{eff}, i.e. **opacity**, which round 23 measured as 34.5 % of the SO error. The cloud-*lifetime* route that would reach the amount two-thirds runs through LAERLIQAUTOLSP/LAERLIQCOLL, and both are .FALSE.

Enabling them would divide by zero. Traced: callpar.F90:1165 sets AUXL%ZCCN= 0 when NAERCLD≤ 0; cloud_layer.F90:191 passes that array to CLOUDSC as PCCN; cloudsc.F90:2050 then evaluates (RCCN/PCCN)$^{0.333}$. Using the lifetime effect requires NAERCLD> 0 and prognostic aerosol, which also changes the direct effect — so it stops being one lever. **Recorded so nobody enables the switch.**

40.4 The missing term, and that the data already exists

Genthon is sea-salt-from-wind with *no biogenic term*, and MACv2-SP — the only other live CCN path — is anthropogenic plumes only. Natural marine biogenic sulphate is absent from this model by construction, which is exactly why the seasonal phase is wrong.

The input already ships with OpenIFS and nothing reads it:

climate.v020/95_4/month_dms	emdms, paramId **210043**
units	kg m^{-2} s^{-1} (DMS surface *emission*)
grid	N96 reduced Gaussian, 40 320 points — **TCO95's own grid**
time	12 monthly fields
land	−99 **sentinel**

It exists at every resolution from TL21 to TCO1279. It is not ingested: `ICMCL` carries only `al`, `aluvp/d`, `alnip/d`, `lai_lv/hv`, and OpenIFS references DMS only in the TM5/M7 chemistry paths, which need `NACTAERO`$\neq 0$.

Route, via ocp-tool rather than `mkclimr.sh`. ocp-tool already owns `ICMCL` (`config.py`:180–186, `paleo_input.py`:278–320, `run_ocp_tool.py`:189 step 14b) and, decisively, already applies a changed land–sea mask to it. `emdms` is an *ocean* field with the opposite land convention to albedo and LAI, so under a moved coastline new ocean points need infill and new land points need −99. `mkclimr.sh` cannot do that; ocp-tool can, which is what makes the field paleo-safe.

1. ocp-tool: append `emdms` to `ICMCL` in deterministic order — `ece_updclie_climr.F90`:233 aborts on "Inconsistent order of fields".
2. ocp-tool: extend step 14b for the inverted land convention.
3. Reader: one new `paramId` case (210043) beside the existing 174 / 15–18 / 66–67 branches. `max_fields`= 10 **against 7 in use**, so there is headroom for an eighth field without touching the limit.
4. Physics: an additive biogenic term on `PCCNO` in the `LCCNO` branch.

The interface is a DMS surface flux in $\mathbf{kg\,m^{-2}\,s^{-1}}$, which is exactly what a REcoM DMS module would produce. So a prescribed climatology now and a prognostic source later are a drop-in swap at the ocp-tool boundary, with no change to items 3–4. That is a better seam than a chlorophyll relation, which would need its own empirical fit on both sides.

Unpriced, all cheap: the provenance of ECMWF's `emdms` (Kettle & Andreae 2000 or Lana et al. 2011 — matters for citation); the flux→CCN relation, which is the one genuinely scientific choice and needs a published basis, not a fitted curve; and whether −99 is the sentinel everywhere or only where it was sampled.

Why this is not scheduled first: the SO band is 9.8 % of the globe, so closing the opacity third perfectly is $2.57\,\mathrm{W\,m^{-2}}$ in-band but only $\approx 0.25\,\mathrm{W\,m^{-2}}$ global, against a +1.23 error. 11E is a namelist edit on a lever already validated over 44 AMIP years. Do the cheap decisive run first.

~~Prognostic marine biogenic CCN needs the `-cc` setup with REcoM ocean biogeochemistry.~~ REcoM carries no DMS tracer; the only DMS in the tree is Icepack's sea-ice biogeochemistry (`dmspp, dmspd, dms`), which is DMS *within* sea ice and is in a component this configuration does not run (`whichevp`= 1, FESOM's native EVP). And there is no coupling field or atmospheric sulphur chemistry to receive it. The prescribed `emdms` route needs none of that.

40.5 T1 read test: passed, and the `emdms` data characterised

The 1-year read test completed with no abort and the reader's own diagnostics confirm every pass criterion: `Found 12 dates`, `Found 8 fields`, `emdms` discovered as the eighth GRIB message matching `NCLIGC(8)`, and `CLIMR` slot 8 filled. Slot 8 takes **twelve distinct values through the model year**, following a clean seasonal march on rank 1's northern high-latitude strip: 1.13 in January, peaking at 3.04 in June, minimum 0.30 in October, returning to 1.13. **The DMS ingest works.**

The data, characterised at full precision (source file and built `ICMCL` are identical, so the ocp-tool step is also verified):

land sentinel (−99)	29.9 % of points
ocean values	min 0.043, median 2.05, max 53.25
exact zeros	240 of 339 360 (0.1 %)
SO 65–45S, September	0.52
SO 65–45S, December	**5.06**
austral seasonal ratio	**9.8**×
tropics, all months	2.0–3.1 (flat)

The phase is exactly the missing term. Model marine CCN is wind-driven and so minimises in DJF (N_d = 68.5) and peaks in JJA (76.1), anti-phased with observation. `emdms` peaks in DJF at 9.8× its September value — largest precisely when and where the SO cloud error is worst (DJF −6.94, SON −7.03 pp).

~~`emdms` is a DMS surface *emission flux* in kg m⁻² s⁻¹, which is exactly what a REcoM DMS module would produce, so a prescribed climatology now and a prognostic source later are a drop-in swap.~~ The GRIB carries that unit label, but values of $\sim 2\,\mathrm{kg\,m^{-2}\,s^{-1}}$ of DMS are physically absurd. A global mean of 2.36 with maxima of 53 matches **DMS seawater *concentration* in $\mathbf{nmol\,L^{-1}}$** — Lana et al. (2011) put the global surface mean at ≈2.3. **The unit label is wrong.** That inserts a step the round-24 design did not price: a sea–air flux parameterisation (wind-dependent piston velocity, Nightingale or Wanninkhof) is needed *before* any flux→CCN relation. The interface argument survives — REcoM would also produce a concentration — but the flux calculation belongs on the atmosphere side.

Two diagnostic traps recorded, both mine. The reader printed paramIds with an `i5` format, so the 6-digit 210043 came out as `*****`; widened to `i8`. And the field was twice declared broken from too small a sample — once from an `awk` column error that reported the ocean maximum as 0, once from reading four *consecutive* timesteps, which all fall in the same month, and concluding the field never varied. Both were wrong; the sampling was.

40.6 T2: the sea–air flux, validated in the model

§40.5 established that `emdms` is a *concentration*, so a flux has to be formed before the field can drive anything — and the flux is what must be validated first. Implemented as a diagnostic in `surfrad_layer.F90`, where the DMS field, the model's own instantaneous wind and the surface temperature are all already in scope:

$$\begin{aligned}
\mathrm{Sc} &= 2674.0 - 147.12\,T + 3.726\,T^2 - 0.038\,T^3 && \text{Saltzman et al. (1993)}\\
k_{600} &= 0.222\,u^2 + 0.333\,u \quad [\mathrm{cm\,hr^{-1}}] && \text{Nightingale et al. (2000)}\\
k &= k_{600}\sqrt{600/\mathrm{Sc}}\,/\,360000 \quad [\mathrm{m\,s^{-1}}] &&\\
F &= k\,C_w \times 10^{-6} \quad [\mathrm{mol\,m^{-2}\,s^{-1}}] &&
\end{aligned}$$

with $1\,\mathrm{nmol\,L^{-1}} = 10^{-6}\,\mathrm{mol\,m^{-3}}$. The wind is the model's instantaneous lowest-level speed, *not* a monthly mean — k is quadratic in u, and §40.2 showed monthly vector averaging halves the Southern Ocean wind.

The pass criterion was set before running. Hand-evaluating the chain at $C_w = 2\,\mathrm{nmol\,L^{-1}}$, $u = 7\,\mathrm{m\,s^{-1}}$, $T = 15\,°\mathrm{C}$ gives $k_{600} = 13.2$, $\mathrm{Sc} = 1177$, $k = 2.6 \times 10^{-5}\,\mathrm{m\,s^{-1}}$ and $F = 5.2 \times 10^{-11}\,\mathrm{mol\,m^{-2}\,s^{-1}}$; scaled by an ocean area of $3.6 \times 10^{14}\,\mathrm{m^2}$, $3.156 \times 10^7\,\mathrm{s}$ and $32.06\,\mathrm{g\,mol^{-1}}$ that is $\approx 19\,\mathrm{Tg\,S\,yr^{-1}}$, against a published 18–34 (Lana et al. 2011 give 28.1). So the model should print a mean flux of order 10^{-11}–$10^{-10}\,\mathrm{mol\,m^{-2}\,s^{-1}}$.

The run completed a full year, 8760 steps, no abort, 70 088 diagnostics (8 blocks × 8760).

flux, minimum	3.78×10^{-13}
flux, median	2.33×10^{-11}
flux, mean	$\mathbf{3.52 \times 10^{-11}}$
flux, maximum	3.31×10^{-10}
predicted window	$\mathbf{10^{-11}}$–$\mathbf{10^{-10}}$
indicative global budget	$12.8\,\mathrm{Tg\,S\,yr^{-1}}$
published	18–34 (Lana et al. 2011: 28.1)

The unit chain is correct. The mean flux sits inside the predicted window, and it varies while C_w is uniform on a block — so the variation is coming from wind and temperature through k, which is the intended behaviour.

Three limits, stated so the number is not over-read. The diagnostic covers rank 1's blocks, a northern high-latitude strip, not the globe. Land points enter as zero because the flux clamps C_w at 0 to reject the −99 sentinel, which drags the sampled mean down. And the 12.8 is that non-global mean scaled as though it applied over all ocean. Landing within a factor of ≈2 of 28.1 on so crude an extrapolation is a *pass* for a unit test — the purpose was to catch an error of orders of magnitude, not to measure the budget. A real budget needs a global reduction the diagnostic does not do.

The run changes no physics. The flux is computed and printed, deliberately not wired into `PCCNO`, because the flux→CCN step is a separate scientific choice that needs a published relation rather than a fitted constant.

40.7 T3: DMS-Rev3 ingested through ocp-tool

The shipped `month_dms` is Lana et al. (2011). DMS-Rev3 (Hulswar et al. 2022) rests on 865 109 observations against L11's 47 313, and the paper states the largest revisions are in the polar oceans — the only region this campaign is looking at. Measured on the model's own grid after regridding, SO 65–45S:

SO 65–45S	L11	DMS-Rev3	change
annual	1.93	2.24	+16 %
DJF	4.23	4.36	+3 %
SON	**1.34**	**2.02**	**+51 %**
November	**2.18**	**4.24**	**+95 %**
global maximum	53.25	26.56	−50 %

Rev3 does not raise the summer peak, it broadens it — the bloom starts a month earlier — and it halves the maximum, which is the paper's "reduced patchiness". The model's SO shortwave CRE error is worst in SON (−7.03 pp) and DJF (−6.94), so the upgrade lands in the season the old climatology was weakest.

Units derived, not read. `DMS_REV3.nc` carries no attributes at all. Integrating the shipped `DMSflux` globally gives 28.19 $\mathrm{Tg\,S\,yr^{-1}}$ (`DMSfluxSI` 28.36) against a published 27.1–28.1, which fixes the flux as $\mathrm{mol\,m^{-2}\,d^{-1}}$ and the concentration as $\mathrm{nmol\,L^{-1}}$; every other assumption is out by orders of magnitude. Only the *concentration* is ingested — the shipped flux uses Rev3's own wind climatology, whereas the model must form the flux online from its own wind, which is quadratic in speed. Their flux is retained as a validation target.

Built through ocp-tool (`dms_rev3_to_grib.py`), which is what makes it paleo-safe. Nearest *valid* neighbour in 3-D — 1° source against ≈0.9° target is matched resolution, so nothing is invented, no land NaN bleeds into coastal ocean, and the dateline and poles cannot yield a spurious neighbour. Values are overwritten into the shipped messages, so grid, `dataDate`, `paramId` and all headers stay byte-identical to what the reader accepts. The land sentinel is taken from the template, preserving the model's own mask; a Rev3 hole at a template ocean point would read as land, so L11 is retained there and counted — for TCO95 that count is **zero**, with 12 040 land points held at −99. `icmcl_dms.py` then interleaves it: 84 → 96 messages, 12 dates × 8 fields.

T3, one year, verified. `Found 12 dates`, `Found 8 fields`, no ordering abort, full year, no abort. Against T2 on the identical wind field:

	C_w [$\mathrm{nmol\,L^{-1}}$]	flux mean	indicative global
L11	1.825	3.515×10^{-11}	12.8 $\mathrm{Tg\,S\,yr^{-1}}$
DMS-Rev3	**2.044**	$\mathbf{4.218 \times 10^{-11}}$	15.4
ratio	1.120	**1.200**	

The flux rises by 20 % **where the concentration rises by only** 12 %**.** That is not an inconsistency — the flux is $\overline{k\,C_w}$, not $\bar{k}\,\overline{C_w}$, and k varies spatially, so the ratio of means need only agree if the spatial pattern is unchanged. It means **Rev3 places more DMS where the wind is stronger**, a positive C_w–k covariance relative to L11, and the flux is what feeds CCN.

Caveat on the 12.8 → 15.4. That is rank 1's blocks — a northern high-latitude strip with land entering as zero — not a global budget. Rev3 and L11 have near-identical *global* fluxes (28.19 against a published 28.1), so the +20 % is regional sampling, consistent with the polar oceans being where Rev3 revised most, and is not a global statement.

40.8 Screening at one year: measured, with the threshold that makes it safe

Project issue #170 found the cy48 high-cloud offset already present in year 1 while the T2m bias and the SO low-cloud dip were not, and concluded that one-year runs would suffice to test tuning parameters. That claim is about `hcc`; the campaign's Southern Ocean metric is SW CRE, a different quantity with different noise, so the transfer was measured rather than assumed.

lever	44-yr Δ	**year-1 Δ**	ratio	screens at 1 yr?
A1a `ovlliqice 0.10`	−6.44	**−6.56**	1.02	**yes**
D2a `inpsea 0.2`	−4.80	**−5.23**	1.09	**yes**
D2b `inp+p700`	−2.61	**−2.01**	0.77	**yes**
A1b `ovlliqice 0.35`	−2.17	−1.74	0.80	no
B3 `clddiff 1.5e-5`	+0.67	−0.17	**−0.25**	no — *wrong sign*
B7 `rvice 0.22`	+0.66	+1.22	1.85	no

Control interannual scatter in SO SW CRE is 0.712 $\mathrm{W\,m^{-2}}$, so a *pair* of single years resolves ±1.97 against ±0.30 for a 44-year pair. **The claim holds, with a threshold: a lever screens at one year if its effect exceeds ≈2 $\mathbf{W\,m^{-2}}$, and is unreliable below it** — B3 comes out with the *opposite sign* at one year.

Why this is worth having. Closing the SO amount deficit requires ≈4.9 $\mathrm{W\,m^{-2}}$, so any lever large enough to matter would show at one year. A 1-year AMIP leg is ≈11 minutes against 9.5 hours for the 44-year standard, so candidate triage is ≈50× cheaper and only survivors need full length. Round 23 spent four 48-year submissions on levers that could have been screened in 45 minutes together.

What it does NOT license. Temperature. #170's own lead/lag result is that T2m and the SO low-cloud dip lag the cloud change, and the campaign's seasonal thresholds (DJF ±0.588 K on 44 years) are far beyond a 1-year window. **Screen the radiative response at one year; never adopt on one.**

41 Round 25 (2026–08–09): the tropopause is not displaced, and cold-and-dry share one entry point

No runs. Round 22 found the cold bias extratropical, peaking at 200 hPa over both poles and reversing sign at 100 hPa, and read that shape as a displaced tropopause. It is not, and what is there instead is better news.

41.1 The tropopause height is right

WMO lapse-rate tropopause, model and ERA5 on identical levels, annual, period-clean:

band	model [hPa]	ERA5 [hPa]	diff
90S–60S	250	250	0
60S–30S	200	200	0
tropics	100	100	0
30N–60N	200	200	0
60N–90N	250	250	0

~~The sign reversal at 100 hPa is a displaced tropopause: the model is still on a tropospheric lapse rate at 200 hPa where the real atmosphere has turned isothermal.~~ **Falsified.** The tropopause sits on the same level in every band. The vertical resolution here is one level, so a displacement smaller than ~50 hPa would not be seen — but a displacement large enough to produce a 4 K bias would be.

What the lapse rates show instead is a *sharper* transition, not a moved one: the model cools faster than ERA5 through 300–250 hPa (+1.76 and +1.78 K km^{-1} at the two poles, against +0.21 in the tropics) and then slower than ERA5 above 200 hPa (−1.49, −1.85). Same tropopause, too much curvature at it.

41.2 The cold point, and one caveat on reading it

band	T model	T ERA5	ΔT	p model	p ERA5
90S–60S	208.77	212.03	−3.26	250	70
60S–30S	211.80	213.32	−1.52	100	100
tropics	**193.44**	**195.51**	**−2.07**	100	100
30N–60N	211.54	212.84	−1.30	100	100
60N–90N	213.21	217.55	−4.33	250	30

The two polar rows are not like-for-like and must not be quoted as 4 K cold points. Model and ERA5 locate their profile minimum at different levels there (250 against 70 and 30 hPa), because at high latitudes the upper stratosphere is near-isothermal and the minimum is ambiguous. What those rows actually say is a structural difference: the model's polar stratosphere is too *cold* at 250 hPa and too *warm* above 100 (§38.4 gives +0.25 and +0.26 there). The three rows where both datasets pick the same level are clean, and the tropical one carries the next result.

41.3 Cold and dry are one problem with one entry point

Stratospheric water vapour is set by the temperature of the tropical cold point, which freeze-dries air entering the stratosphere; the Brewer–Dobson circulation then carries that air poleward. So a cold-biased tropical cold point dries the stratosphere *everywhere*, including over the poles. Testing it quantitatively with Clausius–Clapeyron over ice (Murphy & Koop 2005) on the saturation mixing ratio at each dataset's own cold point:

tropical cold point, model	193.44 K at 100 hPa
tropical cold point, ERA5	195.51 K at 100 hPa
saturation mixing ratio	3.575 vs 4.998 (ppmv-mass)
predicted entry-air drying	**−28.5 %**
measured drying at 100 hPa	**−46.9 %**
share explained	**61 %**

A 2 K cold bias at the tropical cold point accounts for three fifths of a 47 % stratospheric dry bias. That reframes §38.4, which recorded cold and dry as a compensating *pair* — two independent errors whose opposing OLR effects happen to cancel. They are not independent. They share an entry point, and the polar UTLS dryness is the tropical cold point seen downstream.

Why that is much better news. Two independent cancelling errors would have to be fixed together or not at all, and fixing either alone moves net TOA several $\mathrm{W\,m^{-2}}$ the wrong way (§38.4). One error with one entry point means **a lever that warms the tropical cold point should move both**, in the right direction, at once. The remaining 39 % of the drying is a second source — transport, or the model's own UTLS microphysics — and is not yet identified.

Caveat carried forward: ERA5's upper-level humidity is weakly constrained by observation, so the −46.9 % denominator is the softest number here. The −2.07 K cold point and the −28.5 % it implies rest only on temperature, which ERA5 constrains well.

42 Synthesis: the causal web, and what is still dark

This section is the campaign's current best account of *why* the boreal cold bias exists, assembled from every round. It is written to be falsifiable: each link carries a status marker, and the unknowns are listed rather than glossed.

✓OBS	established against observation (CERES, Rutgers/IMS satellite)
●MOD	established only model-vs-model (ERA5 = HTESSEL sibling); suggestive, not authoritative
✗DEAD	tested and falsified
?	unknown — no evidence either way

42.1 The picture

```
                  ROUTE A  (transpiration - cloud)        ROUTE B  (snow - albedo)
                  ===============================         ========================

 RVRSMIN too low                                        ?  unknown Mar-Apr energy sink
 (tundra 80 s/m,                                           |
  lowest in table)  OBS-adjacent: table-internal            v
        |                                               melt-out 13 days late      OBS
        v                                                   |
 over-transpiration                       MOD                v
        |                                               Jun snow cover 2x observed OBS
        +--> too much latent cooling                         |
        |                                                    v
        +--> moist boundary layer                       Jun surface albedo +0.046  OBS
                   |                                         |
                   v                                         v
           Siberian JJA cloud +8.5 pp     OBS           -5.6 W/m2 surface SW       OBS
                   |                                         |
                   v                                         |  ? does this reach T2m?
           -7.0 W/m2 surface SW           OBS                 |    G4 gained +0.95 K with
                   |                                         |   snow UNCHANGED, so the
                   v                                         v   link may be weak
           ================  Siberian JJA T2m ~2.0 K too cold  ================
                                        |
                                        |  (AMIP: severed deliberately)
                                        v
                            boreal forest fails to establish
                                        |
                                        v
                            tundra replaces forest -> brighter, smoother
                                        |
                                        +-------> colder  (runaway, coupled only)

 G4 CLOSES ROUTE A:  cloud term -7.04 -> +1.20 W/m2, T2m +0.952 K, 48% of the bias
 ROUTE B UNTOUCHED:  albedo term -5.45 -> -5.59 W/m2, and G4 makes the snow WORSE
```

42.2 Route A — transpiration to cloud: closed

Status: the strongest *measured* result of the campaign — but the mechanism is only partly established. The chain as proposed is: excess stomatal conductance cools the surface latently *and moistens the boundary layer*; the moisture becomes low cloud; the cloud removes shortwave; the surface cools.

The moisture step is contradicted by this report's own profiles. §21.1 finds the excess boreal cloud is *thermally* driven, not moisture-driven: Siberian RH is +5.6 to +8.2 % too high while q is *too low* (−0.2 to $-0.3\,\mathrm{g\,kg^{-1}}$). What was measured is the *correlation* F4 → less cloud → warmer; the moisture pathway is [?], not ✓OBS.

- ✓ **OBS** Siberian JJA cloud fraction was 78.1 % against CERES 69.6 %, and the cloud share of the SW deficit was $-7.04\,\mathrm{W\,m^{-2}}$ (§24.6).
- ✓ **OBS** G4 removes it: cloud term $-7.04 \to +1.20\,\mathrm{W\,m^{-2}}$, worth +0.952 K.
- The lever is defensible on *table-internal* grounds for tundra (`RVRSMIN(9)`= 80, the lowest of any vegetated type, §24.4) but **not** for needleleaf, where 250 → 1000 remains an unanchored 4× excursion that does not saturate (§24.6).
- **?** G4 slightly *overshoots* in August ($+3.19\,\mathrm{W\,m^{-2}}$ cloud term). A new bias in the making; watch it.

42.3 Route B — snow to albedo: open, and the interesting one

- ✓ **OBS** June surface albedo +0.046 too high (CERES, §25.1); June snow cover 0.380 vs satellite 0.203; melt-out 13 days late (§25.2).
- ● **MOD** Snow *accumulation* is right; the loss term is mistimed — too weak in Mar–Apr, too strong in May (§24.2).
- ✗ **DEAD** It is not the snow-*cover formulation*: round 13 changed `RQSNCR` and moved the albedo response into Sep–Nov instead (§23.2).
- ✗ **DEAD** It is not an energy *supply* problem: the model absorbs *more* net SW than CERES in Mar–Apr (+2.5, +1.2), because its spring albedo is too *low* (§25.1).
- ✗ **DEAD** It is not snow *brightness*: model snow is *darker* than both ERA5 and CERES. Brightening snow would worsen June.
- ● **MOD** It is probably not cold content: `tsn` is within a few tenths of a K of ERA5 — but this is model-vs-model (§25.3).
- ✓ **RESOLVED for the cover half** — see §26: HTESSEL has no sub-grid snow depletion, so cover stays pinned near 1 while the median cell still holds 27 cm in May. That explains the late melt-out and the June albedo, and explains why round 13 could not have worked.
- **? The March–April mass deficit remains open.** ERA5-only, The absorbed shortwave is present and the snow does not melt. Candidates: turbulent flux to the atmosphere (*unconstrained by any observation*); conduction/refreeze within the pack; or the canopy-shaded snow tile (7) behaving differently from exposed snow — the round-13 residual showed canopy masking already dominates in the melt months (§24.1).

42.4 The link that may not exist

The most consequential uncertainty in this report. Route B is a well-measured *radiative* bias, but its effect on *temperature* is unproven:

> G4 raised Siberian JJA T2m by +0.952 K while May–June snow mass did *not* change (§24.6), and while snow cover and melt date got *worse* (§25.2).

Temperature moved a full kelvin with the snow term stationary or degrading. Either the June albedo error is weakly coupled to T2m — a real radiative bias that the atmosphere largely exports — or the two routes offset each other in a way we have not separated. **Until this is resolved, the size of the prize in fixing Route B is unknown**, and it is possible that closing it buys far less than its $-5.6\,\mathrm{W\,m^{-2}}$ suggests. A clean test would be a run that changes the snow term alone with Route A held fixed; round 13 attempted exactly that and failed for an unrelated reason (it moved the wrong season).

42.5 The coupled amplifier, deliberately severed

The AMIP configuration exists to cut the vegetation–albedo feedback so the root cause can be hunted with the loop open. In coupled mode with LPJ-GUESS the chain closes: cold summers → boreal forest fails to establish → tundra replaces forest → higher albedo and lower roughness → colder. That runaway is why a 2 K AMIP bias becomes forest loss in the coupled model, and it is why 48 % closure may or may not be sufficient — **a question AMIP structurally cannot answer.**

A new risk sits precisely here. G4 delays melt-out from 24 to 28 May (§25.2). Later melt is later growing-season onset for LPJ-GUESS. G4 may therefore improve AMIP T2m while *harming* forest establishment through a pathway AMIP cannot see. This should be weighed before G4 is committed to the PI spin-up, and it is an argument for running coupled with G4 sooner rather than later.

42.6 The energy target, now decomposed PER RUN

Where the imbalance lives was settled long ago: §11.2.2 showed the global budget is right *by cancellation*, with the Southern Ocean carrying $+5.8\,\mathrm{W\,m^{-2}}$ and contributing $+0.57$ on its own, from a cloud deficit — the Hyder et al. signature. Nothing below revises that.

What was missing is that the decomposition existed **only for the control**. `amip_toa_decomp.py` never used the `RUNS` list, so across the tuning runs the energy target was tracked by a single global number with no idea which band any lever was moving. That is now part of the standard table and produced for every run automatically (§42.7). The new information is the **response**:

Δ net TOA vs control [$\mathrm{W\,m^{-2}}$]	60–90S	30–60S	tropics	30–60N	60–90N	global
control, absolute	−90.44	−22.24	+42.61	−21.78	−97.74	+0.64
CERES	−98.82	−21.63	+45.11	−24.29	−97.98	—
gap to close	**+8.38**	−0.61	−2.50	+2.51	+0.24	
A1a `ovl=0.10`	−3.33	−2.80	−1.06	−2.44	−2.75	−1.90
D2a `inpsea.2`	**−2.83**	−1.91	−0.38	−0.79	−0.81	−0.93
D2b `inp+p700`	−1.76	−0.83	+0.09	−0.19	−0.34	−0.28
B3 `clddiff`	−0.11	+0.89	**+2.02**	+0.86	+0.25	+1.34
B4 `entshalp3`	−0.56	−2.00	−1.55	−0.75	−0.24	−1.33
B5 `capdcycl0`	−0.47	+0.06	**−1.93**	−0.14	+0.18	−1.00
F4 `rsmin1000`	−0.40	+0.04	+0.07	+0.30	+0.38	+0.10
G1 = F4+D2b	−2.03	−0.83	+0.10	+0.26	−0.02	−0.20
G4 = G1+tundra	−1.98	−0.65	+0.12	+0.28	+0.13	−0.13

Three things this view gives that the global number could not:

(i) The 60–90S band is the most responsive in the model, moving −0.4 to −3.3 under levers that were designed for entirely different purposes. It is not inert; it has simply never been *targeted*.

(ii) The G-series buys about −2.0 **of the** +8.4 **needed** there, all of it from D2b, and D2a alone buys −2.8 — but D2a costs −0.93 globally and was rejected on other grounds. *Roughly a quarter of the Southern Ocean TOA gap is already closed as a by-product*, and the remainder is the same ~ 6 pp cloud-area deficit that no lever has moved (§42.8).

(iii) The surface levers are confirmed regionally clean. F4 moves the tropics by +0.07 and G4 by +0.12, against B3's +2.02 and B5's −1.93. The tropical guardrail was being checked as a single number; it is now visible as a band, and it shows the F/G family does what it was designed to do and nothing else.

One trap the guardrail table invited. The PI target is **net TOA** ≈ 0, *not* the CERES column. CERES is present-day (2005–2015) and carries the real planetary warming imbalance, so it reads $\sim +1.1\,\mathrm{W\,m^{-2}}$ globally: the control's +0.64 is *too positive* even though it sits *below* CERES. Per-band CERES differences are the right way to use it — for locating error, never as a global tuning target.

42.7 One command: `evaluate.sh`

The protocol had grown to five scripts run from memory, and skipping one has repeatedly produced wrong conclusions here — a lever promoted on 4-year noise (B8), a winter penalty unnoticed for eleven rounds (B5), and a TOA decomposition done only for the control. `scripts/analysis/evaluate.sh` now runs the whole protocol, with the TOA decomposition included in the main table rather than as a separate invocation. The way to make a guardrail effective is to make skipping it require effort.

42.8 The open list

1. **?** Where the March–April melt energy goes. Unconstrained; no satellite product measures turbulent flux.
2. **?** Why F4/G1 *increases* May–June snow mass (+5.58, +5.68 mm, both significant). Sublimation was the suspect and is *not* implicated in Mar–Apr; May snowfall rises significantly instead, for reasons unknown.
3. **?** Whether the June albedo bias reaches T2m at all (§42.4).
4. **?** Why G4 *repairs* G1's Nordic Seas SW RMSE damage (+0.300 → +0.089). Adding tundra resistance should not act on a Nordic ocean box. An unexplained improvement is as much a loose end as an unexplained degradation.
5. **?** The Southern Ocean cloud-*area* deficit (~ 6 pp), untouched by every lever in all 34 evaluated runs.
6. **?** Tropics net TOA $2.4\,\mathrm{W\,m^{-2}}$ too low.

7. ✓ **but unaddressed** Global net TOA $+0.52\,\mathrm{W\,m^{-2}}$ against a ~ 0 target — *one of the two original problems*, barely moved by the whole campaign (control +0.64). For a PI spin-up this drift matters more than the last tenths of Siberian warming.

43 State of the campaign (2026–08–09)

A snapshot, not a story. Numbers are 44-year means over 1872–1915 unless stated.

43.1 Proven

Added 2026–08–09 (round 25, §41).

- **The tropopause is not displaced.** Model and ERA5 put the WMO lapse-rate tropopause on the same level in all five bands. The 200/100 hPa sign reversal is a *sharper* transition, not a moved one — the model cools $+1.8\,\mathrm{K\,km^{-1}}$ faster than ERA5 through 300–250 at the poles and -1.5 to -1.9 slower above 200.
- **Cold and dry share one entry point.** A -2.07 K tropical cold point predicts -28.5 % of entry-air drying by Clausius–Clapeyron against a measured -46.9 % at 100 hPa, i.e. **61 %**. They are not the independent compensating pair §38.4 recorded, so a lever that warms the tropical cold point should move both the right way (§41.3).

Added 2026–08–09 (round 24, §40).

- **Southern Ocean droplet number is correct in the mean** — $N_d \approx 73\,\mathrm{cm^{-3}}$ against an observed 60–120, evaluated on 6-hourly wind. The marine CCN path is mapped end to end and `RCCNSEA` is inert because `LCCNO=.TRUE.`
- **But the seasonal phase is inverted**: model N_d is lowest in DJF (68.5) and highest in JJA (76.1), because it is wind-driven, where observations peak in DJF from biogenic sulphate. The measured SO CRE error is worst in exactly those months.
- **The cloud-lifetime CCN route would divide by zero** at `NAERCLD`= 0, so the amount two-thirds is unreachable from the CCN path (§40.3).
- **A DMS emission climatology already ships at TCO95 and nothing reads it** — `emdms`, paramId 210043, on the model's own N96 grid, with `ICMCL` having headroom for an eighth field (§40.4).

Added 2026–08–09 (round 23, §39).

- **The Southern Ocean amount deficit is confirmed against the right reference.** CERES and MODIS put SO cloud area at 89.5 and 89.3 % against the model's 83.1; ERA5 reads 81.6 but reflects $6.31\,\mathrm{W\,m^{-2}}$ less than CERES against the model's 7.45, i.e. it has low area *and* low reflection in proportion. **ERA5 reproduces the model's bias because ERA5 is IFS**, so it is not an independent cloud reference (§39.3).
- **The Southern Ocean and the tropics fail through different cloud types.** SO loses energy via SW cloud (+7.45); the tropics via LW cloud (-2.16). No global microphysics knob can serve both, which is why `RVICE` trades one for the other (§39.2).
- **Nothing in 50 runs warms the tropics through the longwave**, the term that is actually deficient there. The only tropics-warming levers on record act in the shortwave.
- **CERES cloud area is valid over dark ocean, not over ice** — CERES and MODIS disagree by 7.7 pp at 90S–65S.

Added 2026–08–09 (round 22, §38).

- **The period-clean radiative error is $+1.23\,\mathrm{W\,m^{-2}}$, about twice what the PI framing reports.** `amip_presentday` reads net TOA +2.20 against CERES's +0.97 (§38.2).
- **The tropical TOA deficit is mostly the reference period** — -2.51 PI-vs-CERES against -0.67 period-clean. The Southern Ocean is the opposite: robust to the epoch and *larger* period-clean, $+7.46\,\mathrm{W\,m^{-2}}$ and +0.735 of the global mean, now unambiguously the largest single regional term.

- **The SO deficit is cloud AMOUNT,** 65.5 % **of it**, zonally uniform to 1.42 pp and worst in austral spring and summer — and **no lever in the campaign moves cloud area**: D2a +0.63 pp and D2b +0.43 pp of a 6.43 pp gap, against +0.48 and +0.45 pp from two boreal-*land* levers that cannot physically act there (§38.3).
- **The global cold bias is extratropical and lives at the tropopause**, peaking at 200 hPa over both poles (−4.20, −4.45) and reversing sign at 100 hPa, with the tropics the *least*-biased band. The convective-heating hypothesis is excluded (§38.4).
- **The atmosphere is cold AND dry, and the two cancel in OLR.** −1.62 K and −10.1 % over 700–300 hPa; a Planck response to the cold alone predicts $-6.10\,\mathrm{W\,m^{-2}}$ of OLR against −2.53 measured. Fixing either alone moves net TOA several $\mathrm{W\,m^{-2}}$ the wrong way.
- **The `tanh` falsification reproduces coupled on a controlled pair**: 11B minus 11A is −18.99 K of Siberian DJF soil (§38.1).
- **The coupled branch already carries D2b**, not just G4 — 11A/11B/11D all set `RCL_INPSEA` with the 700 hPa gate. K1 and P5 are what remain missing.

Added 2026–08–07.

- **The `tanh` depletion is falsified on observations, not on preference.** Its range is open at 1, so complete snow cover is unrepresentable at any parameters, while Russian snow courses report exactly 10/10 in 99.74 % of 14 369 Siberian DJF surveys. Cost: −20.2 K against 174 676 station soil observations where the scheme *off* is right to +0.8 K (§33.3).
- **The damage was distributional, not a matter of mean cover.** DJF cover is identical to 0.002 across N1/N2/O1/O2 while the soil differs by 23 K; the clipping ramp leaves 93.5 % of the box perfectly insulated, the `tanh` smears the same deficit into every cell (§33.2).
- **P3 is adopted** — both falsifiers passed, and it beats the scheme-off baseline on bias, RMSE and JJA against 105 RIHMI stations (§34).
- **Aerosol is ∼2 % of the Southern Ocean clear-sky excess, not 13 %**, measured rather than inferred, and the two reflection columns are not separable (§35).

- **K1 is the carried-forward configuration (adopted 2026–08–05)** = G4 + the snow-free land albedo correction. Siberian JJA +1.036 K (t=8.29), global T2m RMSE 1.579 → **1.524** (the campaign's best bar K2), every season inside its own threshold, and no Arctic energy penalty — 60–90N net TOA −97.463 against a control −97.737 and CERES −97.98 (§30.2). Namelist-only on top of G4.
- **Land surface albedo is CLOSED as an explanation for the universal bias.** Removing 74 % of a directly measured bias bought 0.089 K of ∼0.7 K; a perfect fix buys ∼0.12 K, one sixth of the target (§30.3).
- **G4 is the base K1 is built on**: Siberian JJA T2m +0.952 K (t=7.68). *Closure depends on a denominator this report does not pin down*: against the ≈ 2.0 K working figure it is 48 %, but §17.2 gives an honest range of −1.3 to −2.0 K and §27.3 measures −2.58 K against ERA5, so the defensible span is **37–73 %**, SO SW RMSE 6.88 → 4.80, subpolar N Atl 5.01 → 4.74, tropics untouched (+0.124), every season inside its own threshold. Namelist-only (§24.7).
- **The boreal budget is nearly spent.** Siberia's JJA −2.58 K is ∼0.7 universal (present over prescribed-SST ocean too), ∼0.3 orographic and 1.3–1.6 boreal-specific; G4 has taken ∼0.95 of the last (§21.3). The remainder is not reachable by a boreal-indexed lever, and this — not a parameter limit — is why F5 = F4.
- **The boreal forest fails by TWO routes, not one** (§31.1). BNE/BINE are *gate-excluded* — 0.7 % of NE-Siberian cells clear `tcmin_est`= −30 and their LAI is exactly 0.000. BNS (larch) has *no* cold gate, clears its GDD gate on 84 % of cells, and is *outcompeted*: grass carries 5.8× its LAI. Since NE Siberia is ∼80 % larch, the winter-gate mechanism does not explain the region that matters most.
- **The near-surface cold is global, and the soil is not the cause.** All 20 surface types are cold in every season, and over low-lying land the soil and skin are the *least*-biased parts of the column (§21.2, §21.3).
- **The boreal cold bias had two routes.** Route A (over-transpiration → low cloud → SW loss) is **closed**: the Siberian JJA cloud term went $-7.04 \to +1.20\,\mathrm{W\,m^{-2}}$. Route B (late melt → June albedo) is **diagnosed and fixable** but carries almost no temperature leverage (§27.2).
- **The snow field can be made correct, but not at this price**: melt-out 12 May against a satellite 11 May (§27.1) — and the scheme that does it costs −16.2 K of coupled DJF soil for +1.2 K of summer (§32). **Rejected**; round 19 tests a re-parameterisation (§32.2).

- **The winter damage is seeded in AUTUMN, not winter.** Midwinter snow cover is identical between the schemes (Δ`ZCVS` $= -0.0004$ in January); the deficit peaks in October (-0.075) exactly as the soil begins to diverge, then vanishes while the soil runs to -24 K (§32.1).
- **The control soil is right, verified against direct observations.** RIHMI-WDC station soil temperature under natural cover puts the control within $\sim$1 K at every depth the Russian network reports in winter, and its thermal offset within 0.7–2.3 K of the observed +17.8 to +24.2 K (§27.8). The campaign no longer depends on ERA5 — an HTESSEL sibling — for any soil claim.
- **I3 is the strongest summer lever built**: Siberian JJA +1.322 K (t=10.61), and it closes the Siberian surface SW deficit to 166.58 against CERES 166.26 — the deficit §22 opened the albedo thread with. It is **not adopted**: it buys that with +1.72 W m^{-2} at 60–90N, the largest NH−SH albedo shift of any run, and the campaign's only significant Nordic Seas RMSE degradation (§27.6).
- **The Siberian winter penalty is not radiative.** DJF snow albedo, SWE and absorbed SW are identical across control/G4/I1/I2/I3 to three decimals, on an 8 W m^{-2} midwinter budget that cannot build -2.4 K (§27.7). The route is thermal.
- **Half the land albedo excess is the snow tile.** Of +0.0154 against CERES, +0.0074 is snow and +0.0080 the snow-free surface; every forest type is inside +0.006 once snow is masked per cell per month (§29.1).
- `RVRSMIN` **does not saturate**; 1000 is a cap set by winter damage, not a knee (§24.6).
- **The energy target is Antarctic/Southern Ocean**, and the two campaign targets share that root cause (§11.2.2, §42.6).
- **Tuning is namelist-driven and mergeable with EC-Earth** — one added line in an ECMWF file (§24.7).

43.2 Open, in priority order

Re-ranked after round 24 (§40), which answers round 23's question. Round 23 asked for a *regionally selective* Southern Ocean reflection lever. Round 24 found one and priced it: marine CCN is selective by construction, but it reaches only the **opacity third** — 2.57 W m^{-2} in-band, ≈0.25 global — and the amount two-thirds is unreachable from the CCN path because the lifetime route divides by zero at `NAERCLD`= 0. The list is therefore:

1. **11E coupled** (11D at SWEMIN = 15 plus K1). A namelist edit on a lever already validated over 44 AMIP years, and the only route to three questions AMIP cannot price. Cheapest decisive run available.
2. **The tropical longwave cloud deficit**, -2.16 W m^{-2}, untouched by any lever in 50 runs and the one term that is deficient in the direction the tropics need.
3. **The tropical cold point**, -2.07 K, which §41 shows is the shared entry point for the cold bias and 61 % of the stratospheric dry bias. The tropopause *height* is correct, so this is a cold-point temperature problem, not a displacement one, and one lever should move both errors.
4. **Marine biogenic CCN via `emdms`** — real missing physics with the input already on disk, but ≈0.25 W m^{-2} global and it needs a published flux→CCN relation, so it follows the cheaper items.
5. **The SO cloud-*amount* two-thirds** stays open with no identified lever.

Re-ranked again after round 23 (§39). The Southern Ocean stays first but the requirement has sharpened: it needs a *regime-selective* reflection lever, because every global low-cloud knob overshoots a tropical band that is only -0.67 W m^{-2} from CERES. Second is the tropical *longwave* cloud deficit, untouched by any lever. Third is the extratropical tropopause. Fourth is 11E coupled.

Re-ranked 2026–08–09 after round 22 (§38). The top of this list is now: (i) **the Southern Ocean cloud-*amount* deficit**, 65.5 % of a +7.46 W m^{-2} band error, untouched by every lever and *not* reachable by the INP branch (§38.3); (ii) **the extratropical tropopause cold-and-dry pair**, which is the dominant land-T2m term and simultaneously the reason the global energy budget cancels (§38.4); (iii) **11D re-run at** SWEMIN = 15 **plus K1**, the first coupled configuration level with the AMIP one.

Two entries below are **superseded** and struck rather than deleted: ~~The snow-free land albedo residual +0.0080 is untouched by all 41 runs.~~ K1 is exactly that lever and removed 74 % of it for +0.089 K; the entry predates K1's adoption and survived unedited (§30.3, which also retracted K3 before it ran). ~~Tropics net TOA 2.4 W m^{-2} too low.~~ Period-clean it is 0.67 (§38.2).

Added 2026–08–07. (i) **The autumn cover excess P3 introduced** — +0.259 in September against Rutgers, where as-released was +0.107. P5/P6 test `SWEMIN` = 15 and 30; the offline optimum is 30 (§34.1). (ii) **The**

Southern Ocean unexplained residual has grown to $\sim$1.9 $\mathbf{W\,m^{-2}}$ now that the aerosol term is measured (§35). (iii) **Scale awareness is a two-anchor interpolation, not a law** — a new resolution needs validation (§34.2). (iv) The M1 clear-sky sign anomaly is significant and untraced.

1. **The blocker: a** -4.7 **K Siberian winter bias** accompanying the snow fix (§27.3). Not a radiation-supply problem (§27.4) and now positively excluded as radiative at all (§27.7), so the route is thermal. Round 16 tests skin conductivity; if that fails, boundary-layer mixing. *J1/J2 are built on I1, not I3* — diagnostic either way, but the production pair needs re-running on the better base (§28).
2. **Decide the coupled base: I3 or I1.** I3 gives +0.072 K more JJA and a better global T2m RMSE, but carries the Arctic energy bill of §27.6, which **AMIP structurally cannot price** — prescribed SST and ice absorb 1.7 $\mathrm{W\,m^{-2}}$ without responding, and the coupled spin-up will not. Open question: whether a recalibrated mode 2 keeps the +1.3 K without the 60–90N surplus, since the SDOR scaling does most of its work outside the box it was tuned in (§29.2).
3. **The snow-free land albedo residual** +0.0080 — crops +0.0204, semidesert +0.0169, bare soil +0.0145, tundra +0.0105 — is untouched by all 41 runs and is the one lever aimed at the $\sim$0.7 K *universal* bias rather than the boreal one. It **must not simply be stacked on I3** (§29.3): the mid-latitude entries are safe, the tundra entry competes for the same Arctic energy budget.
4. **Global net TOA** +0.52 $\mathbf{W\,m^{-2}}$ against a PI target of ~ 0 — one of the two original faults, barely moved by the whole campaign. Note this sits oddly beside a cold troposphere: in AMIP the prescribed SST absorbs the imbalance without warming, so the two are not contradictory, but the pairing is unexplained.
5. **The global tropospheric cold bias** of §21.1 (-0.9 K at 700–850 hPa, -2.8 to -3.0 K at 200 hPa) is now the dominant remaining term for land 2 m temperature and has never been worked on. It is out of reach of every lever this campaign has built.
6. **SO cloud-area deficit** $\sim$**6 pp**, untouched by all 34 evaluated levers, and now known to drive the energy target too.
7. -12 $\mathbf{W\,m^{-2}}$ **downward-LW deficit in Oct–Nov**, unexplained.
8. Tropics net TOA 2.4 $\mathrm{W\,m^{-2}}$ too low; Nordic Seas SW RMSE moved by G1 and partly repaired by G4, both unexplained.
9. Whether 48 % of the boreal bias suffices for forest survival — **AMIP structurally cannot answer this**; it needs the coupled tests.

44 Falsified claims, kept

Every mechanism this campaign has killed, with the measurement that killed it. Retained because the history has to stay honest, and because three of them were killed only after being built on.

claim	what killed it	where
~~The SO cloud deficit should be attacked by pushing the INP branch harder, since D2b is the only lever that has ever moved the Southern Ocean~~	Scored on cloud *area* rather than CRE: D2a +0.63 pp and D2b +0.43 pp of a 6.43 pp gap, against +0.48 and +0.45 pp from G4 and K1 — boreal-land levers that cannot physically act there. The INP area response is at the diagnostic's noise level; it works on opacity, which is 34.5 % of the error.	§38.3
~~Tropics net TOA 2.4 W m^{-2} too low~~	Measured on a PI arm against present-day CERES. Period-clean, on `amip_presentday`, the band is −0.67 — most of it was reference period, not model error.	§38.2
~~The snow-free land albedo residual +0.0080 is untouched by all 41 runs~~	K1 *is* that lever (`RVVEGALB` on crops/irrigated/semidesert + `ECE_TUNE_SOILALB` 0.95) and removed 74 % of it. The open-list entry predates K1's adoption and was never updated; acting on it would have re-bought the remaining 26 %, worth ∼0.03 K.	§30.3
~~The boreal cold bias above 700 hPa might be a tropical convective-heating error propagating globally~~	The tropics are the *least*-biased band at −1.72 K against an extratropical mean of −2.97. The bias is extratropical, peaks at 200 hPa over both poles and reverses at 100 hPa.	§38.4
~~Marine CCN is a uniform 125 cm^{-3}; the aerosol–cloud path is off~~	Read from a default in `sucldp.F90`. `fort.4` carries `LMACV2SP_CCNF=.true.`, `PCCN` is a field, and §35 already documented that path.	§40.1
~~Southern Ocean N_d is too low, so raising marine CCN is the opacity lever~~	$N_d = 55$ came from applying Genthon to *monthly-mean* wind, which cancels the transient eddies. On 6-hourly wind it is 73.5, inside the observed 60–120. A nonlinear operator applied to a mean — the round-19 error again.	§40.2
~~Prognostic marine biogenic CCN needs -cc with REcoM~~	REcoM has no DMS tracer; the only DMS is Icepack's sea-ice biogeochemistry, in a component this setup does not run. The prescribed `emdms` climatology needs none of it.	§40.4
~~`RCL_OVERLAPLIQICE` 0.65 → 0.35 is an untested mixed-phase lever~~	It is A1b, already run for 44 years; A1a is the same knob at 0.1. The A-series "ovl" label means the liquid/ice *deposition* overlap, not radiative cloud overlap. Run cancelled.	§39.4
~~Lowering `RVICE` would warm the tropics and pay for the SO levers~~	Mechanically true, but `RVICE` was *raised* from 0.13 to 0.16 in the v3.4 tuning precisely to suppress SO high cloud (#169/#170). It is global; the two regions need opposite signs.	§39.1
~~The SO cloud-area deficit may be a CERES-vs-`tcc` definition artefact~~	Refuted by the CRE test: ERA5 has low area *and* low reflection in proportion, so it shares the model's bias rather than measuring a different quantity.	§39.3
~~The Arctic carries +11.49 pp too much cloud~~	A passive cloud mask over sea ice, where CERES and MODIS disagree by 7.7 pp at 90S–65S. Not supportable.	§39.3
~~The 200/100 hPa sign reversal is a displaced tropopause~~	The WMO lapse-rate tropopause is on the same level as ERA5 in all five bands. It is a sharper transition, not a moved one.	§41.1
~~Cold and dry are independent errors that happen to cancel in OLR~~	A −2.07 K tropical cold point explains 61 % of the 100 hPa drying by Clausius–Clapeyron. One entry point, not two.	§41.3
~~≈0.5 pp is the noise floor of the SO cloud-area diagnostic, set by land levers that cannot act there~~	G4 and K1 *contain* D2b (G4 = F4+D2b+tundra), so their +0.48/+0.45 pp was D2b's own signal read back as a null. The measured threshold is ±0.098 pp and every lever clears it; the conclusion rests on magnitude — 10 % of the gap — not on significance.	§38.3
~~Aerosol contributes ∼0.7 W m^{-2} (13 %) of the SO clear-sky excess~~	Inferred from a +0.033 AOD excess vs MISR. Measured by removing all anthropogenic aerosol: −0.116 W m^{-2}, $t = -18.9$ — six times smaller. The unexplained residual grows to ∼1.9.	§35
~~`SCALE` is the parameter that controls spring depletion~~	Varying it 1 → 4 moves the offline Sep–May RMSE by 0.002; P3 and P4 came out near-identical. `SWEMIN` is the whole lever and P4 was a wasted run.	§34
~~The scale dependence of `SWEMIN` can be measured as a power law~~	The ladder confirms the sign but the values plateau beyond 133 km, the per-cell fit disagrees 3× with the seasonal fit, and its own extrapolation to 2.5 km gives 9.4 against a measured anchor of 3. Kept as an interpolation, labelled as such.	§34.2
~~P1 was stopped by an unexplained `scancel`, not a model failure~~	The model aborted normally on `PROC=48`; I searched rank 1 only, for tokens that do not match `ABOR1`/`MPL_ABORT`, and built an infrastructure story on the gap.	§33.7
~~B8 (`RVLAMSK` →5) is the best boreal lever, +0.502 K~~	44-year re-evaluation: −0.038 K, pure noise. Two runs had already been built on it.	§17.1
~~The band contributions show no region carrying an anomalous TOA imbalance~~	Artefact of comparing bands to each other, not to CERES. Differenced against CERES the Southern Ocean carries most of it.	§11.2.2
~~About half the boreal bias is the reference period (+1.12 K)~~	Indirect HadCRUT5 chain; direct measurement gives +0.42 K, wrong by 2.7×.	§17.2
~~D1: reformulating the `RCAPDCYCL` land closure will warm the boreal~~	Measured −0.170 K, inside the ±0.242 floor — recovery excluded, sign unresolved.	§19
~~Round 13: raising the snow-cover threshold 10 → 30 cm recovers June's albedo term~~	JJA +0.020 K (t=0.17). Arithmetically incapable of acting in spring — it can only bite where depths are near the threshold, i.e. autumn, which is what it did.	§23.2, §26.2
~~G4's gain comes via the melt-timing route~~	May–June SWE did not move (+2.75, +1.60, both inside noise) while T2m rose +0.952 K. The gain is the plain	§24.6

Two further items are *provisional* rather than dead: the March–April mass deficit rests on ERA5 SWE (forest-biased), and mode 2's scale-awareness carries only ~16 % of the length scale over this box (§26.4).

A Parameter catalogue (distinguishing change per run)

Run	Change (file/namelist)	Baseline → New	Process
06A	`albpnd` (`namelist.ice`)	0.2 → 0.28	melt-pond albedo
06B	+ `rfracmax`	1.0 → 0.75	pond water retention
06C	+ `pndaspect`	0.8 → 1.3	pond depth:area (=common pond)
06D *HRlike*	(only common pond shift)	–	no other diff found
06G	`albpnd/rfracmax/pndaspect`	0.35/0.6/1.6	stronger meltpond
06L	`albpnd/rfracmax/pndaspect`	0.4/0.55/1.8	extra-strong meltpond
06H	strong pond + `RVICE` (`&NAMCLDP`)	0.16 → 0.18	meltpond + cloud-ice fall speed
06Q	`albw/albocn`	0.1 → 0.045	open-water/ocean albedo
06R	`albw/albocn`	0.1 → 0.075	open-water/ocean albedo
06S	`whichevp` (`&ice_dyn`)	1 → 0	sea-ice rheology mEVP→EVP
06E	`RVICE+GGAUSSB` (`&NAMGWWMS`)	0.18; −0.5 → −0.6	cloud-ice + gravity-wave
06F	`LRDALB` (`NALBEDOSCHEME=3`)	F → T	diagnosed surface albedo
06P	`ENTSTPC3` (`&NAMCUMF`)	absent → 1	deep-convection entrainment
06N	`use_momix` (`namelist.tra`)	F → T	near-surface ocean mixing (mospp)
06M	OASIS coupling period + LAG	7200 → 3600 s	atmos↔ocean 2h→1h (1hcpl)
06O	1hcpl + mospp	–	coupling + mixing
06O4	O + `momix_kv`	0.01 → 0.012	+ larger M-O mixing
06O5	O + `momix_kv`	0.01 → 0.02	+ even larger mixing
06T	O + KPP-low (`kpp_av0/kv0` 0.005 → 0.003, bckg ↓ 10×)	–	+ weaker KPP mixing
06U	T + open-water albedo 0.075	–	combo
06V	T + `ENTSTPC3=1`	–	combo (most aggressive)

B Implementation & lab-notebook detail

Operational detail moved here to keep the work-plan science legible. Retained for reproducibility.

A.1 esm-tools fixes for the AMIP–noLPJG generator (§10.2). Model tree `oifsamip-cy48` git-aligned to `awiesm3-develop` (oifs `local_combined_fixes`@1303a90, oasis `afac6a3`, xios `2.5.3`) + bundle/field patches. Three bugs fixed, committed & pushed (commit `9921d95b`, branch `development_interactiv_mesh_awiesm3`, and a follow-up for the WAM recipe):

- `xios.yaml` dereferenced the `fesom` namespace via `choose_fesom.*` and crashed *any* FESOM-less setup — now gated on `"$(( 'fesom' in config ))"`.
- `oifsamip.yaml` never overrode the OIFS `comp_command`, so the binary compiled *without* XIOS or coupling despite `LXIOS:TRUE` — now adds `-with-cplng2 -with-oifs-xios -build-type=release` (like awicm3/awiesm3).
- WAM aborted (`CDATEA≠CDATEF`) at any start year ≠1990 because the OIFS `preprocess` step (`change_icm_date.sh`, which re-dates ICMGG/ICMSH/wave init files) never ran: `oifsamip.yaml` lacked `prepcompute_recipe: ${oifs.prepco` and fell back to the default recipe with no preprocess step. Added (matches `awicm3.yaml`).

Runscript `oifsamip-cy48-levante-TCO95L91_lpjgforcing.yaml` overrides XIOS output to `file_def_lpjg_spinup` + `field_def_lpjg_safe`, emitting daily `tas,tasmin,tasmax,pr,rsns,rlns,sfcWind` + `tdps,ps,psl,clt,prsn,evspsbl,p` *esm_master gotcha:* the single-component target is `recomp-<setup>/oifs-48r1` (the *component* name); the alias `/oifs` silently falls back to the setup and doubles the path (`oifsamip-cy48/oifsamip-cy48`).

A.2 Flux-accumulation fix (§10.2), at root. The de-accumulation divisor in `field_def_lpjg_safe.xml` was the hard-coded `/21600` (6 h). It is now templated: `field_def_lpjg_safe.xml.j2` uses `/{{oifs.flux_accumulation_period}` with `oifs.flux_accumulation_period` defaulting to `${oifs.time_step}` (= NFRHIS×TSTEP; with NFRHIS=1 this is the timestep). A validated ×6 scalar patch (`AMIP_noLPJG_1d_1870-1879_TCO95_PI_fluxfix.nc`) corrected the existing forcing without a rerun.

Controlled temperature test now ready. Because the AMIP–CRUNCEP comparison shows meaningful forcing sensitivity but the CERES replacement rules out net radiation as its dominant source, two additional AMIP forcing cycles were created for the next attribution step. They add exactly +1 or +2 K to `tas`, `tasmin` and `tasmax` at all TCO95 cells north of or at 45°N; every field south of 45°N and every non-temperature variable is unchanged. Their NH-land forcing-cycle means are −3.57 and −2.57 °C, compared with −4.57 °C in the source AMIP file. These are *prepared inputs, not spin-up results*. Running them with unchanged LPJG parameters will quantify whether a uniform boreal temperature increment crosses a vegetation threshold or merely produces a modest continuous response.

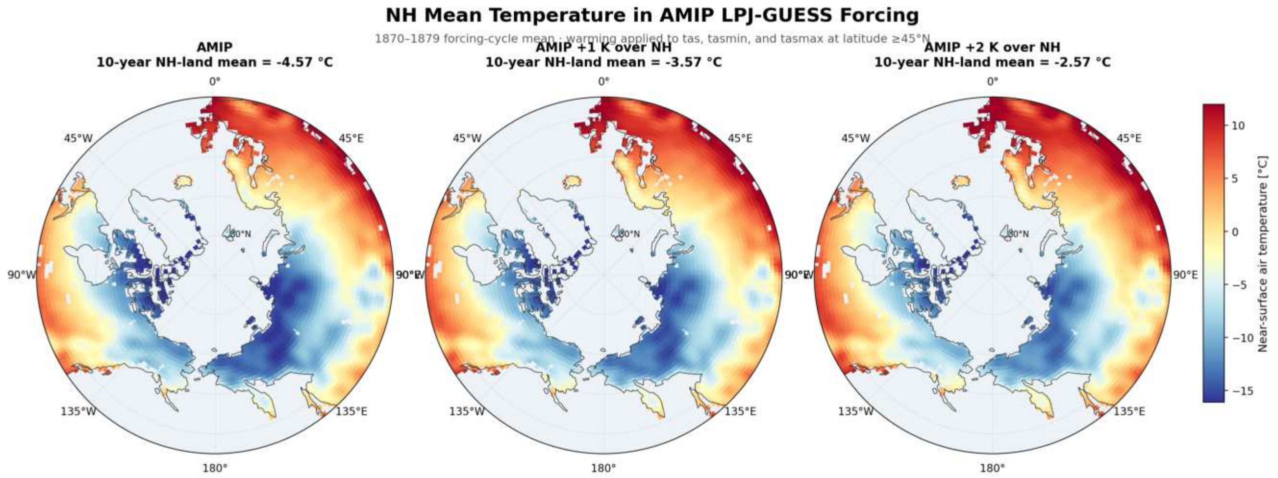


Figure 32: Input check for the planned AMIP temperature sensitivity. Ten-year NH-land mean near-surface temperature in the original, +1 K and +2 K forcing cycles. The increments were applied consistently to `tas`, `tasmin` and `tasmax` at latitude $\geq$ 45°N; no vegetation response is shown because these two spin-ups have not yet been run.

A.3 CERES forcing construction and validation (§7). The direct source is `/work/ab0246/a270092/obs/CERES/CERES_EBAF_Ed4.1_Subset_200003-202106.nc`; the climatological source is `/work/ab0246/a270092/obs/CERES/CERES_EBAF_Ed4.1_Subset_CLIM01-CLIM12.nc`. The output files are `/work/ab0995/a270270/input/cruncep_v7/CRUNCEP_noLPJG_1d_1901-1910_TCO95_CERES_rsns_rlns.nc` and `/work/ab0995/a270270/input/cruncep_v7/CRUNCEP_noLPJG_1d_1901-1910_TCO95_CERESclim_daily_variability.nc`. Both contain 3652 daily timesteps and 40 320 TCO95 cells. Coordinates and all non-radiation fields are unchanged from CRUNCEP v3. The direct file has `rsns` range 0–358.23 and `rlns` range −197.30–34.76 W m^{-2}. The daily-variability file has ranges 0–455.61 and −206.92–93.85 W m^{-2}; every mapped CERES cell-month target is reproduced within 8×10^{-6} W m^{-2}. Construction and preanalysis scripts, cached intermediates and the detailed provenance document are retained under `/work/ab0995/a270270/analysis/Offline_LPJ_GUESS_analysis/`.

A.4 Phase-1B OpenIFS source edits (§10.3). Paths under `oifs-48r1/ifs-source/arpifs/phys_ec/`:

knob	file:line	current	proposed	effect
`ZLMIN`	`vdfexcu.F90`:201	30 m	40–50 m	min stable mixing length
`ZCB`	`vdfexcu.F90`:190	5.0	3.0–4.0	LTG b: lower ⇒ longer tail ⇒ more K

Secondary companions if needed: the 0.1 / 300 m stable mixing-length limits (`vdfexcu.F90`:374,376) and the surface-layer constants `RCHETA`/`RCHETB`/`RCHBHDL` (`surf/module/suvexcs_mod.F90`, mirror in `suvdfs.F90`).

References

Alessandri, A. et al. (2017). Multi-scale enhancement of climate prediction over land by increasing the model sensitivity to vegetation variability in EC-Earth. *Clim. Dyn.* 49, 1215–1237.

Balsamo, G. et al. (2009). A revised hydrology for the ECMWF model: verification from field site to terrestrial water storage and impact in the ECMWF-IFS. *J. Hydrometeorol.* 10, 623–643. *(HTESSEL.)*

Beljaars, A.C.M. & Viterbo, P. (1998). Role of the boundary layer in a numerical weather prediction model. In *Clear and Cloudy Boundary Layers*, KNAW, 287–304.

Betts, A.K. & Ball, J.H. (1997). Albedo over the boreal forest. *J. Geophys. Res.* 102, 28901–28909. *(BOREAS observations; canopy masking of snow albedo.)*

Betts, R.A. (2000). Offset of the potential carbon sink from boreal forestation by decreases in surface albedo. *Nature* 408, 187–190.

Bonan, G.B. (2008). Forests and climate change: forcings, feedbacks, and the climate benefits of forests. *Science* 320, 1444–1449.

Bonan, G.B., Pollard, D. & Thompson, S.L. (1992). Effects of boreal forest vegetation on global climate. *Nature* 359, 716–718.

Chapin, F.S. et al. (2005). Role of land-surface changes in Arctic summer warming. *Science* 310, 657–660.

de Lavergne, C. et al. (2014). Cessation of deep convection in the open Southern Ocean under anthropogenic climate change. *Nat. Clim. Change* 4, 278–282.

Douville, H., Royer, J.-F. & Mahfouf, J.-F. (1995). A new snow parameterization for the Météo-France climate model. *Clim. Dyn.* 12, 21–35. *(The classic depth-threshold cover form.)*

Döscher, R. et al. (2022). The EC-Earth3 Earth system model for CMIP6. *Geosci. Model Dev.* 15, 2973–3020.

Dutra, E. et al. (2010). An improved snow scheme for the ECMWF land surface model: description and offline validation. *J. Hydrometeorol.* 11, 899–916. *(The `LESN09` scheme whose `RQSNCR` this campaign modifies.)*

Essery, R. et al. (2013). A comparison of 1701 snow models using observations from an alpine site. *Adv. Water Resour.* 55, 131–148.

Estilow, T.W., Young, A.H. & Robinson, D.A. (2015). A long-term Northern Hemisphere snow cover extent data record for climate studies and monitoring. *Earth Syst. Sci. Data* 7, 137–142. *(The Rutgers CDR used here.)*

Helfrich, S.R. et al. (2007). Enhancements to, and forthcoming developments in the Interactive Multisensor Snow and Ice Mapping System (IMS). *Hydrol. Process.* 21, 1576–1586.

Herzschuh, U. (2020). Legacy of the Last Glacial on the present-day distribution of deciduous versus evergreen boreal forests. *Glob. Ecol. Biogeogr.* 29, 198–206.

Heuzé, C. (2021). Antarctic bottom water and North Atlantic deep water in CMIP6 models. *Ocean Sci.* 17, 59–90.

Heuzé, C. et al. (2013). Southern Ocean bottom water characteristics in CMIP5 models. *Geophys. Res. Lett.* 40, 1409–1414.

Holtslag, A.A.M. et al. (2013). Stable atmospheric boundary layers and diurnal cycles: challenges for weather and climate models. *Bull. Am. Meteorol. Soc.* 94, 1691–1706.

Hyder, P. et al. (2018). Critical Southern Ocean climate model biases traced to atmospheric model cloud errors. *Nat. Commun.* 9, 3625.

Jiang, Y. et al. (2012). [Boreal tree/grass fraction spread across dynamic global vegetation models.] *Ecol. Evol.* (structural-uncertainty reference; details to verify).

Kato, S. et al. (2018). Surface irradiances of Edition 4.0 CERES EBAF data product. *J. Climate* 31, 4501–4527. *(The surface SW up/down used for the observational albedo check.)*

Körner, C. & Paulsen, J. (2004). A world-wide study of high altitude treeline temperatures. *J. Biogeogr.* 31, 713–732.

Liston, G.E. (2004). Representing subgrid snow cover heterogeneities in regional and global models. *J. Climate* 17, 1381–1397. *(Why cell-mean depth is insufficient.)*

Loeb, N.G. et al. (2018). Clouds and the Earth's Radiant Energy System (CERES) Energy Balanced and Filled (EBAF) Top-of-Atmosphere (TOA) Edition-4.0 data product. *J. Climate* 31, 895–918.

Loranty, M.M. et al. (2014). Vegetation controls on northern high-latitude snow-albedo feedback. *Glob. Change Biol.* 20, 594–606.

Louis, J.-F. (1979). A parametric model of vertical eddy fluxes in the atmosphere. *Bound.-Layer Meteorol.* 17, 187–202.

Medeiros, B. et al. (2011). Arctic inversion strength in climate models. *J. Climate* 24, 4733–4740.

Mohrmann, M., Heuzé, C. & Swart, S. (2021). Southern Ocean polynyas in CMIP6 models. *The Cryosphere* 15, 4281–4313.

NASA Langley Research Center (2019). CERES Energy Balanced and Filled (EBAF) Edition 4.1, Level-3B monthly product. DOI: 10.5067/TERRA-AQUA/CERES/EBAF_L3B004.1.

Niu, G.-Y. & Yang, Z.-L. (2007). An observation-based formulation of snow cover fraction and its evaluation over large North American river basins. *J. Geophys. Res.* 112, D21101. ***(The depletion formulation implemented as `ECE_SNOW_SCF`.)***

Pithan, F., Medeiros, B. & Mauritsen, T. (2014). Mixed-phase clouds cause climate model biases in Arctic wintertime temperature inversions. *Clim. Dyn.* 43, 289–303.

Qu, X. & Hall, A. (2014). On the persistent spread in snow-albedo feedback. *Clim. Dyn.* 42, 69–81.

Robinson, D.A., Estilow, T.W. & NOAA CDR Program (2012). NOAA Climate Data Record (CDR) of Northern Hemisphere (NH) Snow Cover Extent (SCE), Version 1. NOAA NCEI. DOI: 10.7289/V5N014G9.

Roesch, A., Wild, M., Gilgen, H. & Ohmura, A. (2001). A new snow cover fraction parametrization for the ECHAM4 GCM. *Clim. Dyn.* 17, 933–946. *(Adds sub-grid orography; the other main GCM approach.)*

Sandu, I. et al. (2013). Why is it so difficult to represent stably stratified conditions in numerical weather prediction models? *J. Adv. Model. Earth Syst.* 5, 117–133.

Sitch, S. et al. (2003). Evaluation of ecosystem dynamics, plant geography and terrestrial carbon cycling in the LPJ dynamic global vegetation model. *Glob. Change Biol.* 9, 161–185.

Smith, B. et al. (2001). Representation of vegetation dynamics in the modelling of terrestrial ecosystems. *Glob. Ecol. Biogeogr.* 10, 621–637.

Smith, B. et al. (2014). Implications of incorporating N cycling and N limitations on primary production in an individual-based dynamic vegetation model. *Biogeosciences* 11, 2027–2054.

Swenson, S.C. & Lawrence, D.M. (2012). A new fractional snow-covered area parameterization for the Community Land Model and its effect on the surface energy balance. *J. Geophys. Res.* 117, D21107. *(Separate accumulation/depletion curves; the alternative to Niu & Yang.)*

Thackeray, C.W., Fletcher, C.G. & Derksen, C. (2015). Quantifying the skill of CMIP5 models in simulating seasonal albedo and snow cover evolution. *J. Geophys. Res. Atmos.* 120, 5831–5849. *(CMIP5 models show the same late-melt/high-albedo bias found here.)*

Zhang, Y. et al. (2011). On the computation of planetary boundary-layer height / polar inversion strength in climate models. *J. Climate* 24, 5167–5186.

Citations added in response to internal review (rev. 2026-07-06); bibliographic details for Beljaars & Viterbo 1998, Jiang et al. 2012, Medeiros et al. 2011 and Zhang et al. 2011 to be page-verified. Snow-cover references added 2026-08-03: volume/page numbers for Essery et al. 2013, Helfrich et al. 2007 and Thackeray et al. 2015 are from memory and should be page-verified before external use; the Niu & Yang 2007, Swenson & Lawrence 2012, Roesch et al. 2001, Liston 2004, Dutra et al. 2010 and Betts & Ball 1997 entries are the load-bearing ones for §26.